

Cisco Certified Network Associate (CCNA)

Number: 200-301
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File Version: 7.6



Cisco Certified Network Associate (CCNA 200-301)

Exam A

QUESTION 1

What are two recommendations for protecting network ports from being exploited when located in an office space outside of an IT closet? (Choose two.)

- A. configure ports to a fixed speed
- B. configure static ARP entries
- C. shut down unused ports
- D. implement port-based authentication
- E. enable the PortFast feature on ports

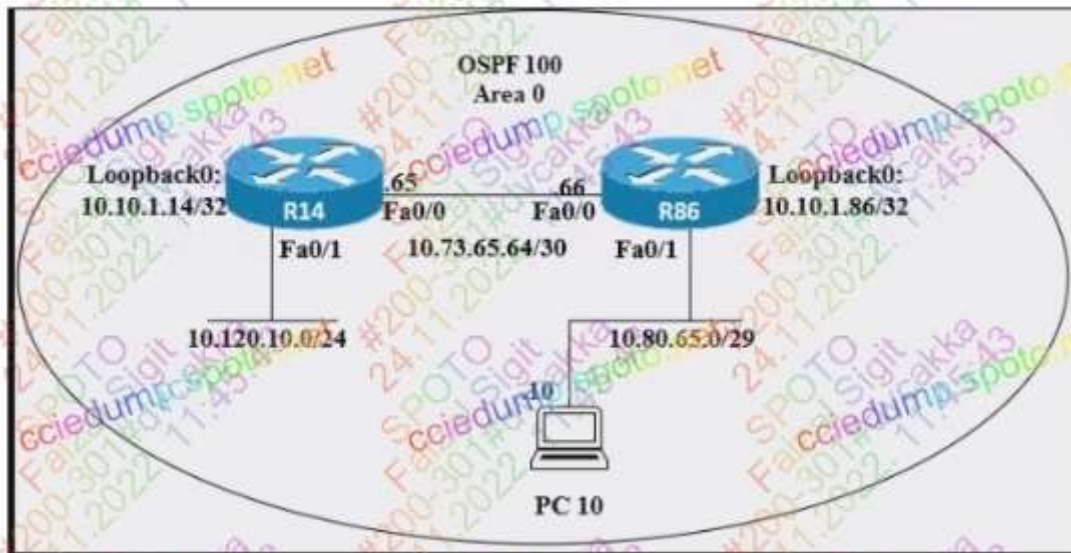
Correct Answer: CD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 2



An engineer just installed network 10.120.10.0/24. Which configuration must be applied to the R14 router to add the new network to its OSPF routing table?

- A.

```
router ospf 100
network 10.120.10.0 255.255.255.0 area 0
```
- B.

```
router ospf 100 area 0
network 10.120.10.0 0.0.0.255
```
- C.

```
router ospf 100
network 10.120.10.0 0.0.0.255 area 0
```
- D.

```
router ospf 120
network 10.120.10.0 255.255.255.0 area 0
ip route 10.120.10.0 255.255.255.0 fa0/1
```

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 3

```
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.1.1.0/30 is directly connected, GigabitEthernet0/0
L 10.1.1.2/32 is directly connected, GigabitEthernet0/0
S 192.168.0.0/20 [1/0] via 10.1.1.1
S 192.168.1.0/30 is subnetted, 1 subnets
S 192.168.1.0/30 [1/0] via 10.1.1.1
S 192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
S 192.168.2.0/28 [1/0] via 10.1.1.1
S 192.168.2.0/29 [1/0] via 10.1.1.1
```

Refer to the exhibit. An engineer is checking the routing table in the main router to identify the path to a server on the network. Which route does the router use to reach the server at 192.168.2.2?

- A. S 192.168.1.0/30 [1/0] via 10.1.1.1
- B. S 192.168.2.0/28 [1/0] via 10.1.1.1
- C. S 192.168.0.0/20 [1/0] via 10.1.1.1
- D. S 192.168.2.0/29 [1/0] via 10.1.1.1

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 4

What is the role of the root port in a switched network?

- A. It replaces the designated port when the root port fails.
- B. It is the best path to the root from a nonroot switch.
- C. It is administratively disabled until a failover occurs.
- D. It replaces the designated port when the designated port fails.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 5

When the active router in an HSRP group fails, what router assumes the role and forwards packets?

- A. Forwarding
- B. Backup
- C. Listening
- D. Standby

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 6

OSPF must be configured between routers R1 and R2. Which OSPF configuration must be applied to router R1 to avoid a DR/BDR election?

- A.

```
router ospf 1
network 192.168.1.1 0.0.0.0 area 0

interface e1/1
ip address 192.168.1.1 255.255.255.252
ip ospf cost 0
```
- B.

```
router ospf 1
network 192.168.1.1 0.0.0.0 area 0
hello interval 15

interface e1/1
ip address 192.168.1.1 255.255.255.252
```
- C.

```
router ospf 1
network 192.168.1.1 0.0.0.0 area 0

interface e1/1
ip address 192.168.1.1 255.255.255.252
ip ospf network point-to-point
```
- D.

```
router ospf 1
network 192.168.1.1 0.0.0.0 area 0

interface e1/1
ip address 192.168.1.1 255.255.255.252
ip ospf network broadcast
```

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 7

```
R_1# show ip route
....
D 192.168.20.0/26 [90/24513456] via 10.10.10.1
D 192.168.20.0/24 [120/5] via 10.10.10.2
O 192.168.0.0/19 [110/219414] via 10.10.10.13
B 192.168.0.0/16 is variably subnetted, 4 subnets, 4 masks
D 192.168.20.0/27 [90/4123710] via 10.10.10.12
D 192.168.20.0/25 [90/14464211] via 10.10.10.11
S 0.0.0.0/0 [1/0] via 10.10.10.14
```

Refer to the exhibit. Packets are flowing from 192.168.10.1 to the destination at IP address 192.168.20.75. Which next hop will the router select for the packet?

- A. 10.10.10.1
- B. 10.10.10.14
- C. 10.10.10.12
- D. 10.10.10.11

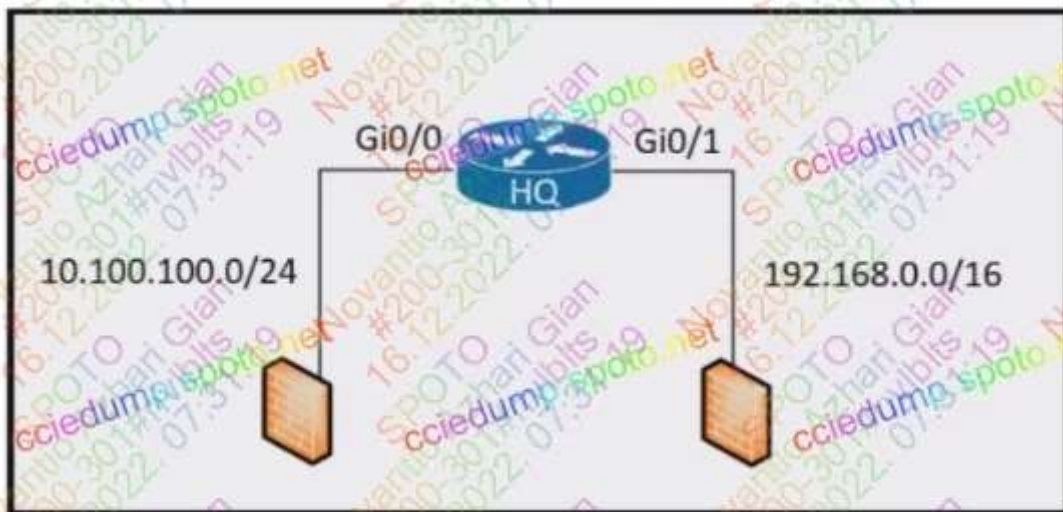
Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 8



Refer to the exhibit. An access-list is required to permit traffic from any host on interface Gi0/0 and deny traffic from interface Gi0/1 .Which access list must be applied?

- A.

```
ip access-list standard 199
permit 10.100.100.0 0.0.0.255
deny 192.168.0.0 0.0.255.255
```
- B.

```
ip access-list standard 199
permit 10.100.100.0 0.0.0.255
deny 192.168.0.0 0.255.255.255
```
- C.

```
ip access-list standard 99
permit 10.100.100.0 0.0.0.255
deny 192.168.0.0 0.0.255.255
```
- D.

```
ip access-list standard 99
permit 10.100.100.0 0.0.0.255
deny 192.168.0.0 0.255.255.255
```

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 9

What are two advantages of implementing a controller-based architecture instead of a traditional network architecture? (Choose two.)

- A. It allows for seamless connectivity to virtual machines.
- B. It supports complex and high-scale IP addressing schemes.
- C. It increases security against denial-of-service attacks
- D. It provides increased scalability and management options.
- E. It enables configuration task automation.

Correct Answer: DE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 10

Which CRUD operation corresponds to the HTTP GET method?

- A. Delete
- B. Create
- C. Read
- D. Update

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 11

When deploying syslog, which severity level logs informational messages?

- A. 2
- B. 0
- C. 4
- D. 6

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 12

Current Neighbor Relationship					
Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.1.1	1	FULL/DR	00:00:33	192.168.1.1	GigabitEthernet0/0
Desired Neighbor Relationship					
Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.1.1	0	FULL/-	00:00:31	192.168.1.1	GigabitEthernet0/0

Refer to the exhibit. How must OSPF be configured on the GigabitEthernet0/0 interface of the neighbor device to achieve the desired neighbor relationship?

- A. Router(config)# interface GigabitEthernet 0/0
Router(config-if)# ip ospf cost 5
- B. Router(config)# interface GigabitEthernet 0/0
Router(config-if)# ip ospf priority 1
- C. Router(config)# interface GigabitEthernet 0/0
Router(config-if)# ip ospf network point-to-point
- D. Router(config)# interface GigabitEthernet 0/0
Router(config-if)# ip ospf 1 area 2

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 13

```
RI# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate
       default
       U - per-user static route, o - ODR
Gateway of last resort is not set
C    10.0.0.0/8 is directly connected, Loopback0
O    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O    10.0.1.3/32 [110/100] via 10.0.1.100, 00:39:08, Serial0
C    10.0.1.0/24 is directly connected, Serial0
O    10.0.1.5/32 [110/5] via 10.0.1.50, 00:39:08, Gigabit Ethernet 0/0
O    10.0.1.4/32 [110/10] via 10.0.1.4, 00:39:08, Gigabit Ethernet 0/0
```

Refer to the exhibit .What does route 10.0.1.3/32 represent in the routing table?

- A. all hosts in the 10.0.1.0 subnet
- B. the 10.0.0.0 network
- C. the source 10.0.1.100
- D. a single destination address

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 14

When configuring a WLAN with WPA2 PSK in the Cisco Wireless LAN Controller GUI, which two formats are available to select? (Choose two.)

- A. hexadecimal
- B. base64
- C. ASCII
- D. binary
- E. decimal

Correct Answer: AC

Section: (none)

Explanation

Explanation/Reference:

QUESTION 15

What is the effect when loopback interfaces and the configured router ID are absent during the OSPF Process configuration?

- A. The router ID 0.0.0.0 is selected and placed in the OSPF process.
- B. No router ID is set, and the OSPF protocol does not run.
- C. The highest up/up physical interface IP address is selected as the router ID.
- D. The lowest IP address is incremented by 1 and selected as the router ID.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 16

When a WPA2-PSK WLAN is configured in the Wireless LAN Controller, what is the minimum number of characters that is required in ASCII format?

- A. 8
- B. 18
- C. 6
- D. 12

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 17

```
R1#show ip ospf interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Internet address is 192.168.1.2/24, Area 0
Process ID 1, Router ID 192.168.1.1, Network Type POINT-TO-POINT, Cost: 1
Transmit Delay is 1 sec, State POINT-TO-POINT,
Timer intervals configured, Hello 15, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:08
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Suppress hello for 0 neighbor(s)

R2#show ip ospf interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Internet address is 192.168.1.1/24, Area 0
Process ID 1, Router ID 10.1.1.1, Network Type POINT-TO-POINT, Cost: 1
Transmit Delay is 1 sec, State POINT-TO-POINT,
Timer intervals configured, Hello 15, Dead 45, Wait 15, Retransmit 5
Hello due in 00:00:11
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Suppress hello for 0 neighbor(s)
```

Refer to the exhibit. The network engineer is configuring router R2 as a replacement router on the network. After the initial configuration is applied, it is determined that R2 failed to show R1 as a neighbor. Which configuration must be applied to R2 to complete the OSPF configuration and enable it to establish the neighbor relationship with R1?

- A. R2(config)#router ospf 1
R2(config-router)#network 192.168.1.0 255.255.255.0 area 2
- B. R2(config)#router ospf 1
R2(config-router)#router-id 192.168.1.2
- C. R2(config)#interface g0/0/0
R2(config-if)#ip ospf dead-interval 40
- D. R2(config)#interface g0/0/0
R2(config-if)#ip ospf hello-interval 10

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 18

What is a difference between RADIUS and TACACS+?

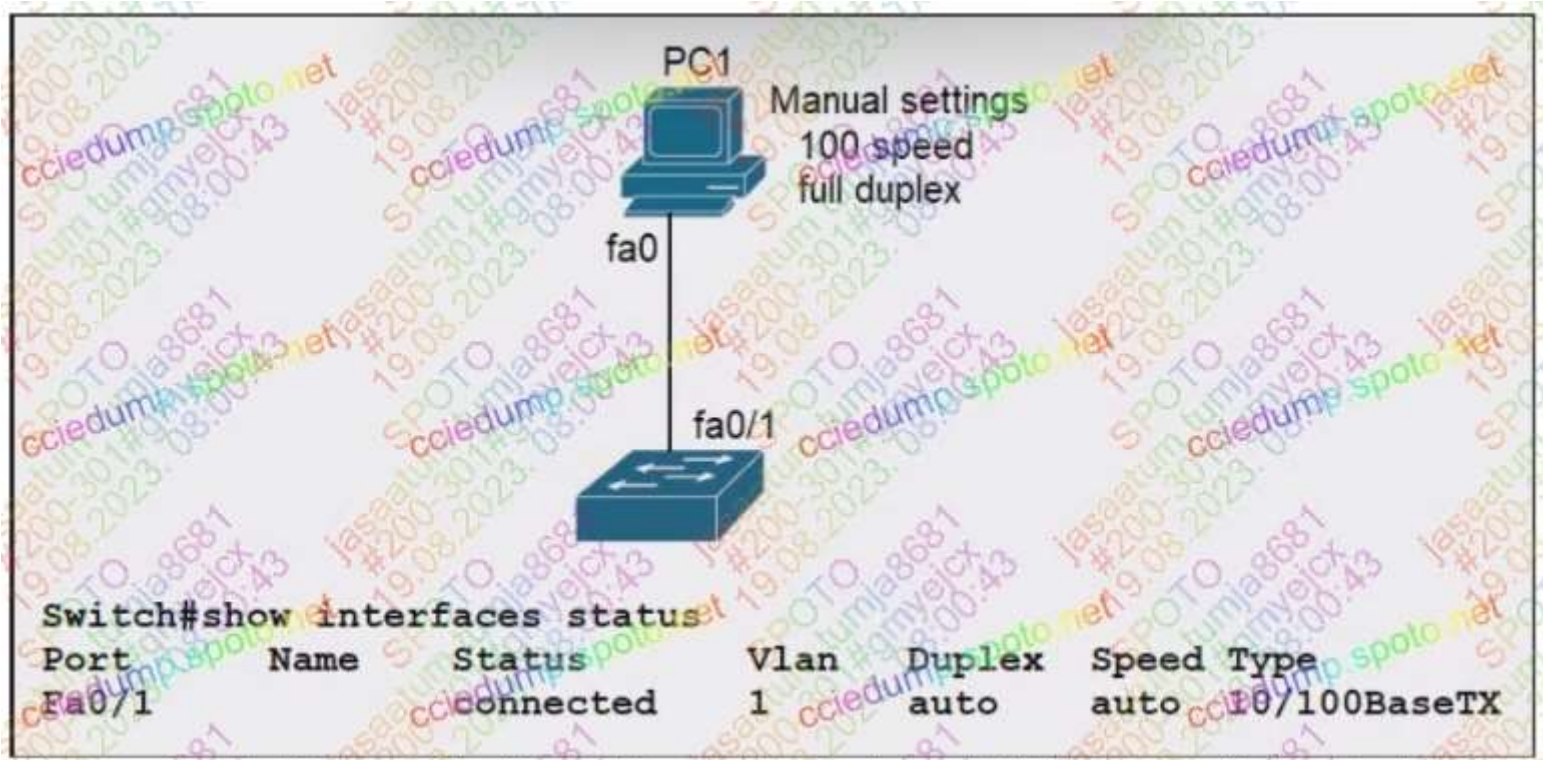
- A. RADIUS logs all commands that are entered by the administrator, but TACACS+ logs only start, stop, and interim commands.
- B. TACACS+ encrypts only password information, and RADIUS encrypts the entire payload.
- C. RADIUS is most appropriate for dial authentication, but TACACS+ can be used for multiple types of authentication.

D. TACACS+ separates authentication and authorization, and RADIUS merges them.

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 19



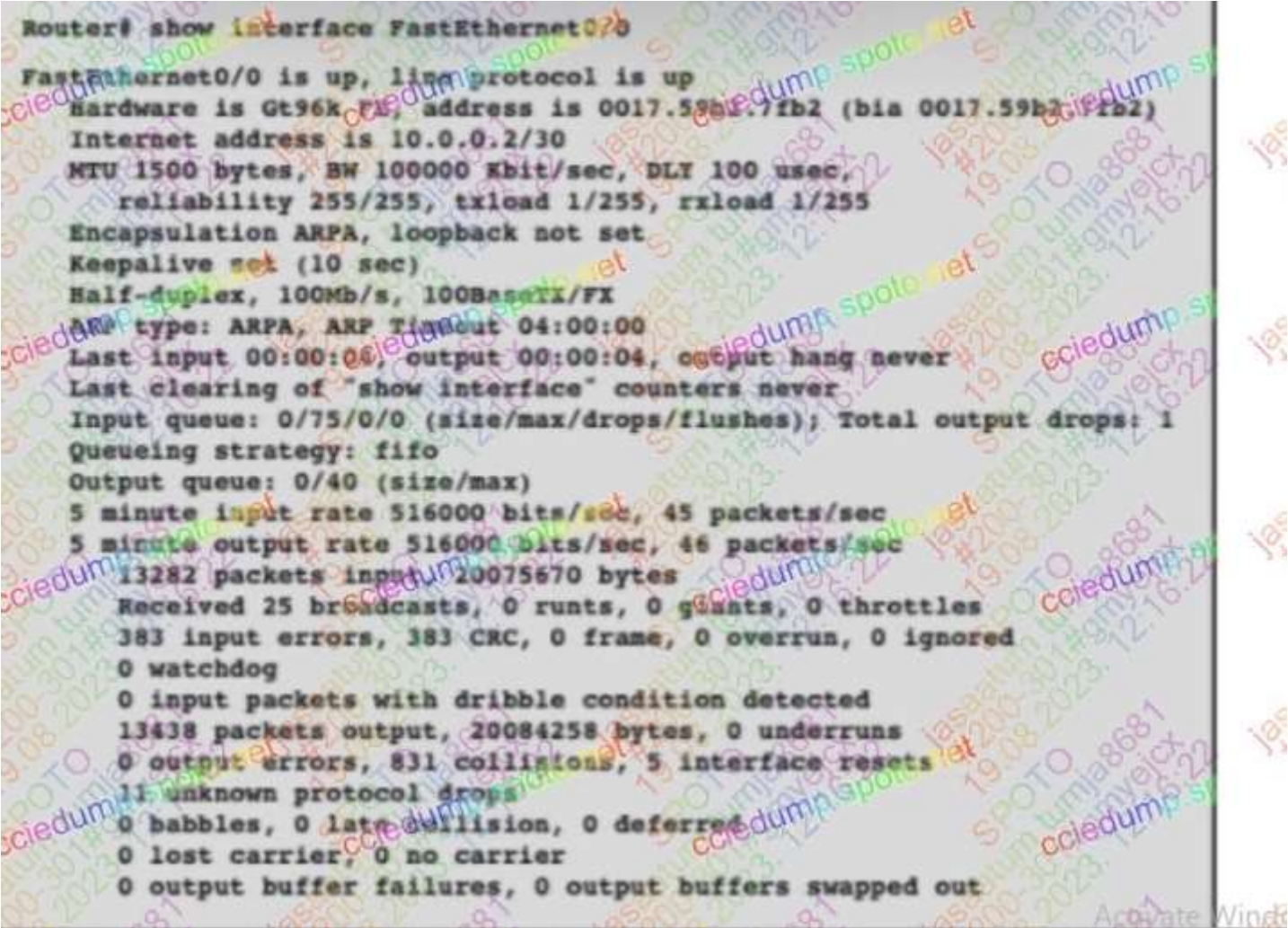
Refer to the exhibit. The link between PC1 and the switch is up, but it is performing poorly. Which interface condition is causing the performance problem?

- A. There is an interface type mismatch.
- B. There is a duplex mismatch on the interface.
- C. There is a speed mismatch on the interface.
- D. There is an issue with the fiber on the switch interface.

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 20



Refer to the exhibit. Users at a branch office are experiencing application performance issues, poor VoIP audio quality, and slow downloads. What is the cause of the issues?

- A. overutilization
- B. broadcast storm
- C. interface configuration
- D. QoS queuing

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 21

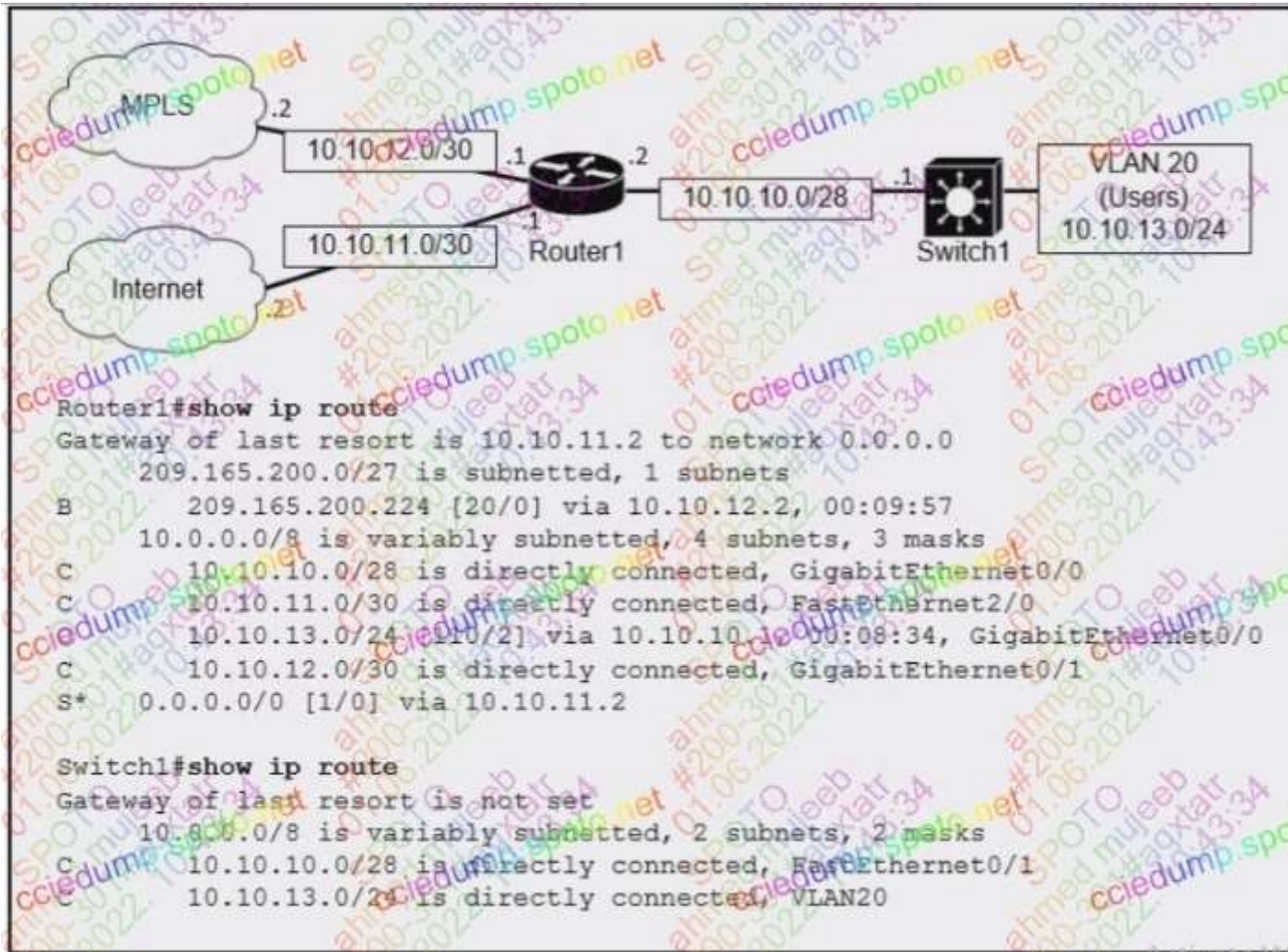
After installing a new Cisco ISE server, which task must the engineer perform on the Cisco WLC to connect wireless clients on a specific VLAN based on their credentials?

- A. Disable the LAG Mode on Next Reboot
- B. Enable the Allow AAA Override.
- C. Enable the Event Driven RRM
- D. Enable the Authorize MIC APs against auth-list or AAA.

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 22



Refer to the exhibit. Which path is used by the router for internet traffic?

- A. 209.165.200.0/27
- B. 0.0.0.0/0
- C. 10.10.10.0/28
- D. 10.10.13.0/24

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 23

A user configured OSPF and advertised the Gigabit Ethernet interface in OSPF. By default, which type of OSPF network does this interface belong to?

- A. point-to multipoint
- B. broadcast
- C. point-to-point
- D. nonbroadcast

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 24

Which type of port is used to connect to the wired network when an autonomous AP maps two VLANs to its WLANs?

- A. EtherChannel
- B. Trunk
- C. Access
- D. LAG

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 25

How do TCP and UDP fit into a query-response model?

- A. TCP encourages out-of-order packet delivery, and UDP prevents re-ordering
- B. TCP establishes a connection prior to sending data, and UDP sends immediately.
- C. TCP uses error detection for packets, and UDP uses error recovery.
- D. TCP avoids using sequencing, and UDP avoids using acknowledgments.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 26



Refer to the exhibit. An architect is managing a wireless network with APs from several branch offices connecting to the WLC in the data center. There is a new requirement for a single WLAN to process the client data traffic without sending it to the WLC. Which action must be taken to complete the request?

- A. Enable local HTTP profiling.
- B. Enable FlexConnect Local Switching.
- C. Enable local DHCP Profiling.
- D. Enable Disassociation Imminent.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 27

What is the difference between IPv6 unicast and anycast addressing?

- A. Unlike an IPv6 anycast address, an IPv6 unicast address is assigned to a group of interfaces on multiple nodes.
- B. An individual IPv6 unicast address is supported on a single interface on one node, but an IPv6 anycast address is assigned to a group of interfaces on multiple nodes.
- C. IPv6 unicast nodes must be explicitly configured to recognize the unicast address, but IPv6 anycast nodes require no special configuration.
- D. IPv6 anycast nodes must be explicitly configured to recognize the anycast address, but IPv6 unicast nodes require no special configuration.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 28

What is the primary difference between AAA authentication and authorization?

- A. Authentication verifies a username and password, and authorization handles the communication between the authentication agent and the user database.

- B. Authentication identifies and verifies a user who is attempting to access a system, and authorization controls the tasks the user can perform.
- C. Authentication identifies a user who is attempting to access a system, and authorization validates the user's password.
- D. Authentication controls the system processes a user can access, and authorization logs the activities the user initiates.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 29

What is a difference between local AP mode and FlexConnect AP mode?

- A. Local AP mode creates two CAPWAP tunnels per AP to the WLC.
- B. FlexConnect AP mode fails to function if the AP loses connectivity with the WLC.
- C. FlexConnect AP mode bridges the traffic from the AP to the WLC when local switching is configured.
- D. Local AP mode causes the AP to behave as if it were an autonomous AP.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 30

An engineer is configuring remote access to a router from IP subnet 10.139.58.0/28. The domain name, crypto keys, and SSH have been configured. Which configuration enables the traffic on the destination router?

- A.

```
interface FastEthernet0/0
ip address 10.122.49.1 255.255.255.252
ip access-group 105 in

ip access-list standard 105
permit tcp 10.139.58.0 0.0.0.7 eq 22 host 10.122.49.1
```
- B.

```
interface FastEthernet0/0
ip address 10.122.49.1 255.255.255.248
ip access-group 10 in

ip access-list standard 10
permit udp 10.139.58.0 0.0.0.7 host 10.122.49.1 eq 22
```
- C.

```
interface FastEthernet0/0
ip address 10.122.49.1 255.255.255.252
ip access-group 110 in

ip access-list extended 110
permit tcp 10.139.58.0 0.0.0.15 host 10.122.49.1 eq 22
```
- D.

```
interface FastEthernet0/0
ip address 10.122.49.1 255.255.255.240
access-group 120 in

ip access-list extended 120
permit tcp 10.139.58.0 255.255.255.248 any eq 22
```

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 31

```
Gateway of last resort is 172.16.2.2 to network 0.0.0.0
10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
  10.10.100.0/26 is directly connected, GigabitEthernet0/0/6
C    10.10.10.0/24 is directly connected, GigabitEthernet0/0/0
L    10.10.10.3/32 is directly connected, GigabitEthernet0/0/0
  172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
S    172.16.1.33/32 is directly connected, GigabitEthernet0/0/1
C    172.16.2.0/23 is directly connected, GigabitEthernet0/0/1
L    172.16.2.1/32 is directly connected, GigabitEthernet0/0/1
S*  0.0.0.0/0 [1/0] via 172.16.2.2
```

Refer to the exhibit. A packet sourced from 10.10.10.32 is destined for the Internet. What is the administrative distance for the destination route?

- A. 0
- B. 32
- C. 2
- D. 1

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 32

What differentiates device management enabled by Cisco DNA Center from traditional campus device management?

- A. centralized
- B. device-by-device hands-on
- C. per-device
- D. CLI-oriented device

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 33

Which benefit does Cisco DNA Center provide over traditional campus management?

- A. Cisco DNA Center leverages APIs, and traditional campus management requires manual data gathering.
- B. Cisco DNA Center leverages SNMPv3 for encrypted management, and traditional campus management uses SNMPv2.
- C. Cisco DNA Center automates SSH access for encrypted entry, and SSH is absent from traditional campus management.
- D. Cisco DNA Center automates HTTPS for secure web access, and traditional campus management uses HTTP.

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 34

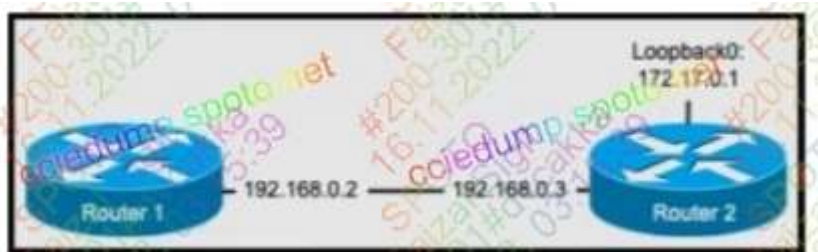
Which protocol requires authentication to transfer a backup configuration file from a router to a remote server?

- A. FTP
- B. DTP
- C. SMTP
- D. TFTP

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 35



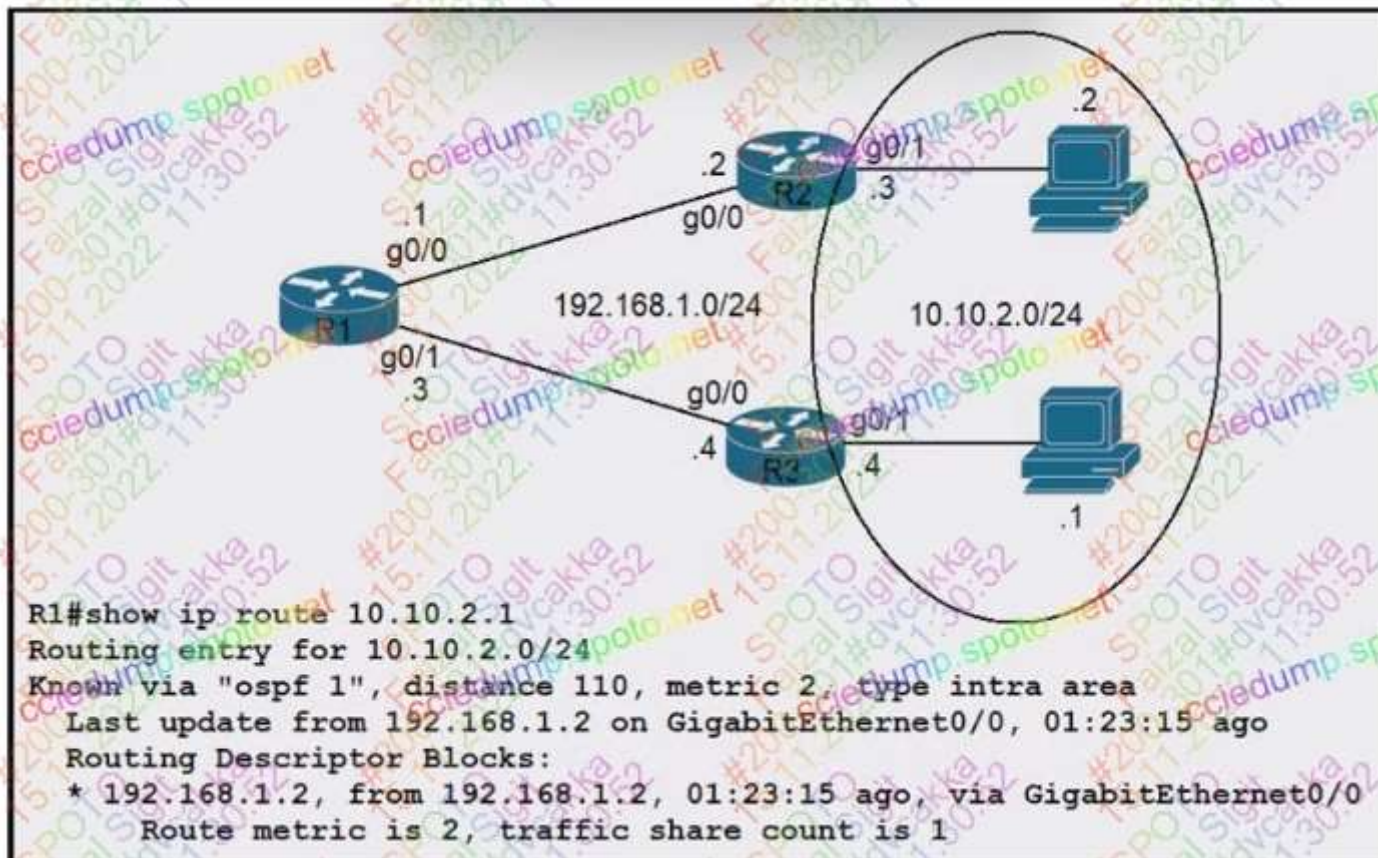
Refer to the exhibit. The `ntp server 192.163.0.3` command has been configured on router 1 to make it an NTP client of router 2. Which command must be configured on router 2 so that it operates in server-only mode and relies only on its internal clock?

- A. Router2(config)#ntp passive
- B. Router2(config)#ntp server 192.168.0.2
- C. Router2(config)#ntp server 172.17.0.1
- D. Router2(config)#ntp master 4

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 36



Refer to the exhibit. Traffic from R1 to the 10.10.20/24 subnet uses 192.168.1.2 as its next hop. A network engineer wants to update the R1 configuration so that traffic with destination 10.10.2.1 passes through router R3, and all other traffic to the 10.10.2.0/24 subnet passes through R2. Which command must be used?

- A. ip route 10.10.2.1 255.255.255.255 192.168.1.4 100
- B. ip route 10.10.2.0 255.255.255.0 192.168.1.4 115
- C. ip route 10.10.2.0 255.255.255.0 192.168.1.4 100
- D. ip route 10.10.2.1 255.255.255.255 192.168.1.4 115

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 37

What are two benefits of controller-based networking compared to traditional networking? (Choose two.)

- A. controller-based inflates software costs, while traditional decreases individual licensing costs
- B. controller-based reduces network configuration complexity, while traditional increases the potential for errors
- C. controller-based increases network bandwidth usage, while traditional lightens the load on the network
- D. controller-based allows for fewer network failures, while traditional increases failure rates
- E. controller-based provides centralization of key IT functions, while traditional requires distributed management functions

Correct Answer: BE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 38

```

OldR#show ip ospf interface
GigabitEthernet0/0/0 is up, line protocol is up
Internet address is 192.168.1.2/24, Area 0
Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DROTHER, Priority 1
Designated Router (ID) 192.168.1.1, Interface address 192.168.1.1
Backup Designated Router (ID) 192.168.1.1, Interface address 192.168.1.1
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:01
Index 1/1, flood queue length 0
Neighbor Count is 1, Adjacent neighbor count is 2

R2#show ip ospf interface
GigabitEthernet0/0/0 is up, line protocol is up
Internet address is 192.168.1.1/24, Area 0
Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DROTHER, Priority 1
Designated Router (ID) 192.168.1.1, Interface address 192.168.1.2
Backup Designated Router (ID) 192.168.1.1, Interface address 192.168.1.2
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:02
Index 2/2, flood queue length 0
Neighbor Count is 1, Adjacent neighbor count is 2

```


Refer to the exhibit. Router OldR is replacing another router on the network with the intention of having OldR and R2 exchange routes. After the engineer applied the initial OSPF configuration, the routes were still missing on both devices. Which command sequence must be issued before the clear IP ospf process command is entered to enable the neighbor relationship?

- A. OldR(config)# interface g0/0/0
OldR(config-if)# ip ospf dead-interval 1
- B. OldR(config)# interface g0/0/0
OldR(config-if)# ip ospf hello-interval 15
- C. OldR(config)# router ospf 1
OldR(config-router)# no router-id 192.168.1.1
- D. OldR(config)# router ospf 1
OldR(config-router)# network 192.168.1.0 255.255.255.0 area 2

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 39

R1#show ip interface brief					
Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	NVRAM	administratively down	down
GigabitEthernet1/0	192.168.0.1	YES	NVRAM	up	up
GigabitEthernet2/0	10.10.1.10	YES	manual	up	up
GigabitEthernet3/0	10.10.10.20	YES	manual	up	up
GigabitEthernet4/0	unassigned	YES	NVRAM	administratively down	down
Loopback0	172.16.15.10	YES	manual		

Refer to the exhibit. What does router R1 use as its OSPF router-ID?

- A. 10.10.10.20
- B. 10.10.1.10
- C. 192.168.0.1
- D. 172.16.15.10

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 40

A port security violation has occurred on a switch port due to the maximum MAC address count being exceeded. Which command must be configured to increment the security-violation count and forward an SNMP trap?

- A. switchport port-security violation shutdown
- B. switchport port-security violation protect
- C. switchport port-security violation restrict
- D. switchport port-security violation access

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 41

A device detects two stations transmitting frames at the same time. This condition occurs after the first 64 bytes of the frame is received. Which interface counter increments?

- A. Collision
- B. CRC
- C. Runt
- D. late collision

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

- Collision errors
 - Reducing the number of devices per collision domain will usually solve the problem
 - You can do this by segmenting your network with a router, a bridge, or a switch
 - Late collision
 - Occurs when two stations transmit more than 64 bytes of data frames before detecting a collision

QUESTION 42

R1 has learned route 10.10.10.0/24 via numerous routing protocols. Which route is installed?

- A. route with the lowest administrative distance
- B. route with the next hop that has the highest IP
- C. route with the shortest prefix length
- D. route with the lowest cost

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 43

```
Router#show run
Building configuration...
Current configuration : 1530 bytes
!
! Last configuration change at 11:32:53 UTC Sat Oct 10 2020
upgrade fpd auto
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname Router
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
no ip icmp rate-limit unreachable
!
--More--
```

Refer to the exhibit. Which minimum configuration items are needed to enable Secure Shell version 2 access to R15?

- A. Router(config)# hostname R15
R15(config)# crypto key generate rsa general-keys modulus 1024
R15(config-line)# line vty 0 15
R15(config-line)# transport input ssh
R15(config)# ip ssh source-interface Fa0/0
R15(config)# ip ssh stricthostkeycheck
- B. Router(config)# hostname R15
R15(config)# ip domain-name cisco.com
R15(config)# crypto key generate rsa general-keys modulus 1024
R15(config)# ip ssh version 2
R15(config-line)# line vty 0 15
R15(config-line)# transport input ssh
- C. Router(config)# ip domain-name cisco.com
Router(config)# crypto key generate rsa general-keys modulus 1024
Router(config)# ip ssh version 2
Router(config-line)# line vty 0 15
Router(config-line)# transport input all
Router(config)# ip ssh logging events

D. Router(config)# crypto key generate rsa general-keys modulus 1024
Router(config)# ip ssh version 2
Router(config-line)# line vty 0 15
Router(config-line)# transport input ssh
Router(config)# ip ssh logging events
R15(config)# ip ssh stricthastkeycheck

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 44

Which two QoS tools provide congestion management?

- A. PQ
- B. CAR
- C. FRTS
- D. CBWFQ
- E. PB

Correct Answer: AD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 45

What are two characteristics of a controller-based network? (Choose two.)

- A. It moves the control plane to a central point.
- B. The administrator can make configuration updates from the CLI
- C. It uses northbound and southbound APIs to communicate between architectural layers.
- D. It decentralizes the control plane, which allows each device to make its own forwarding decisions
- E. It uses Telnet to report system issues.

Correct Answer: AC

Section: (none)

Explanation

Explanation/Reference:

QUESTION 46

```
ip arp inspection vlan 5-10
interface fastethernet 0/1
switchport mode access
switchport access vlan 5
```

Refer to the exhibit. What is the effect of this configuration?

- A. All ingress and egress traffic is dropped because the interface is untrusted.
- B. The switch discards all ingress ARP traffic with invalid MAC-to-IP address bindings
- C. Egress traffic is passed only if the destination is a DHCP server.
- D. All ARP packets are dropped by the switch.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 47

```
R1# show ip route | begin gateway
Gateway of last resort is 209.165.200.246 to network 0.0.0.0
S* 0.0.0.0/0 (1/0) via 209.165.200.246, Serial0/1/0
    is directly connected, Serial0/1/0
O 172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
  L 172.16.3.0/24 [1/0] via 207.165.200.250, Serial0/0/0
  O 172.16.3.0/28 [110/84437] via 207.165.200.254, 00:00:28, Serial0/0/1
    207.165.200.0/24 is variably subnetted, 6 subnets, 2 masks
  C 207.165.200.244/30 is directly connected, Serial0/1/0
  L 207.165.200.245/32 is directly connected, Serial0/1/0
  C 207.165.200.248/30 is directly connected, Serial0/0/0
  L 207.165.200.249/32 is directly connected, Serial0/0/0
  C 207.165.200.252/30 is directly connected, Serial0/0/1
  L 207.165.200.253/32 is directly connected, Serial0/0/1
```

Refer to the exhibit. A packet is being sent across router R1 to host 172.16.3.14. To which destination does the router send the packet?

- A. 207.165.200.254 via Serial0/0/1
- B. 207.165.200.250 via Serial0/0/0
- C. 207.165.200.254 via Serial0/0/0
- D. 207.165.200.246 via Serial0/1/0

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 48

```
R1# show ip route
....
D 172.16.32.0/27 [90/25597172] via 20.1.1.10
O 172.16.32.0/19 [110/292094] via 20.1.1.10
R 172.16.32.0/24 [120/2] via 20.1.1.3
```

Refer to the exhibit. Router R1 is running three different routing protocols. Which route characteristic is used by the router to forward the packet that destination IP 172.16.32.1?

- A. longest prefix
- B. administrative distance
- C. cost
- D. metric

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 49

Which function does an SNMP agent perform?

- A. It coordinates user authentication between a network device and a TACACS+ or RADIUS server.
- B. It requests information from remote network nodes about catastrophic system events.
- C. It manages routing between Layer 3 devices in a network.
- D. It sends information about MIB variables in response to requests from the NMS.

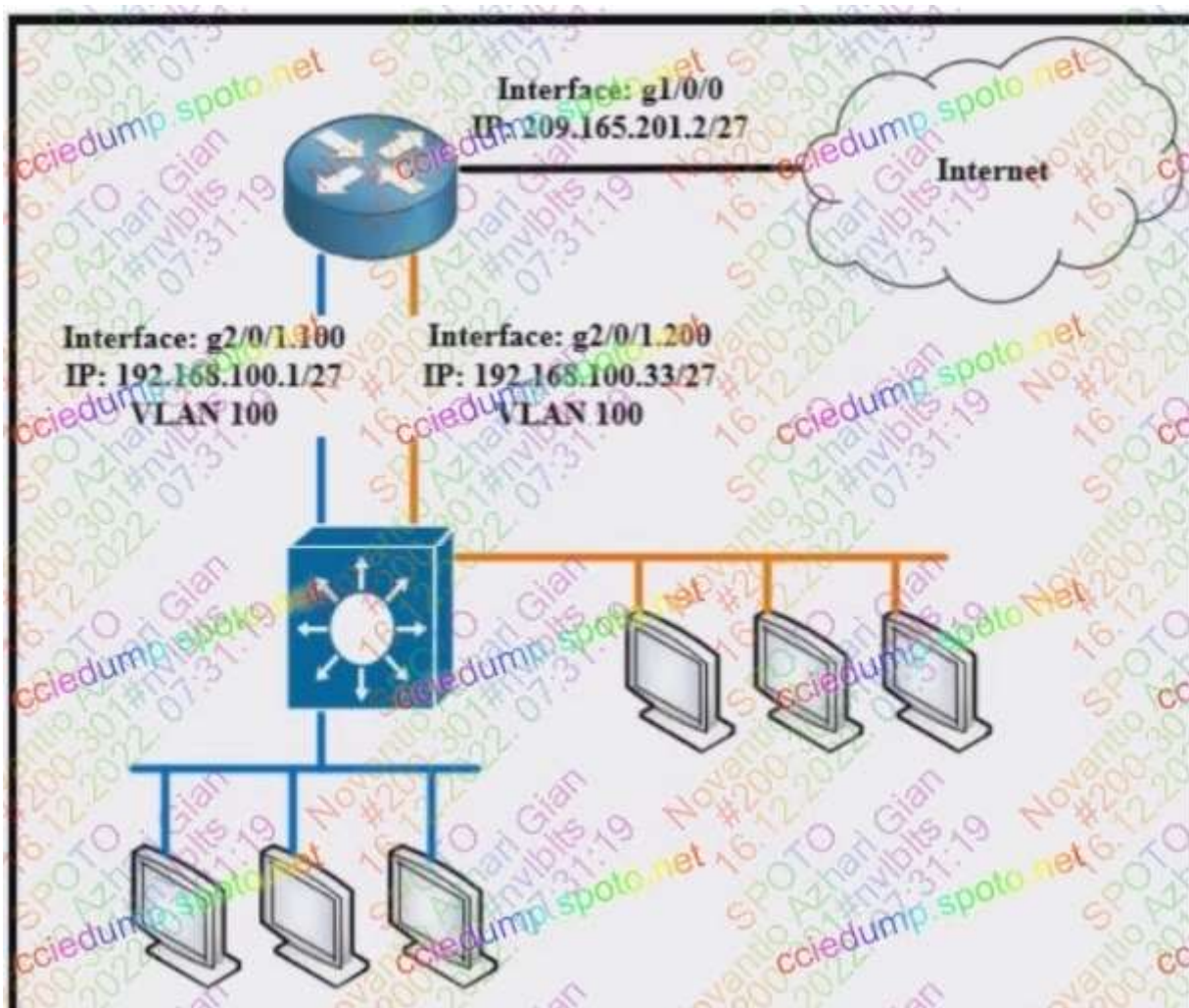
Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 50



Refer to the exhibit. Which configuration must be applied to the router that configures PAT to translate all addresses in VLAN 200 while allowing devices on VLAN 100 to use their own IP addresses?

- A. Router1(config)#access-list 99 permit 192.168.100.0 0.0.0.255
Router1(config)#ip nat inside source list 99 interface gi2/0/0 overload
Router1(config)#interface gi2/0/1.200
Router1(config-if)#ip nat inside
Router1(config)#interface gi1/0/0
Router1(config-if)#ip nat outside

- B. Router1(config)#access-list 99 permit 209.165.201.2 0.0.0.0
Router1(config)#ip nat inside source list 99 interface gi1/0/0 overload
Router1(config)#interface gi2/0/1.200
Router1(config-if)#ip nat inside
Router1(config)#interface gi1/0/0
Router1(config-if)#ip nat outside
- C. Router1(config)# access-list 99 permit 209.165.201.2 255.255.255.255
Router1(config)#ip nat inside source list 99 interface gi1/0/0 overload
Router1(config)#interface gi2/0/1.200
Router1(config-if)#ip nat inside
Router1(config)#interface gi1/0/0
Router1(config-if)#ip nat outside
- D. Router1(config)#access-list 99 permit 192.168.100.32 0.0.0.31
Router1(config)#ip nat inside source list 99 interface gi1/0/0 overload
Router1(config)#interface gi2/0/1.200
Router1(config-if)#ip nat inside
Router1(config)#interface gi1/0/0
Router1(config-if)#ip nat outside

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 51

Which type of IPv6 address is publicly routable in the same way as IPv4 public addresses?

- A. multicast
- B. unique local
- C. global unicast
- D. link-local

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 52

Which two components are needed to create an Ansible script that configures a VLAN on a switch? (Choose two.)

- A. Cookbook
- B. Model
- C. Playbook
- D. Recipe
- E. Task

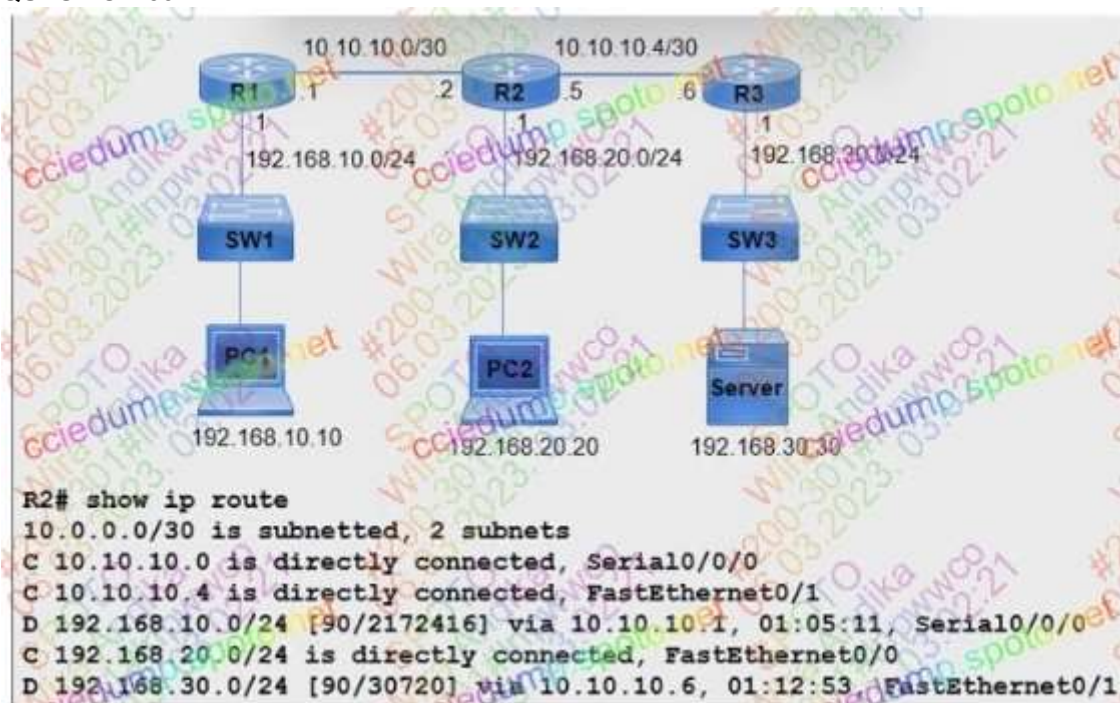
Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 53



Refer to the exhibit. What is the next-hop IP address for R2 so that PC2 reaches the application server via EIGRP?

- A. 192.168.30.1
- B. 10.10.10.6
- C. 192.168.20.1
- D. 10.10.10.5

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 54

What is a function of TFTP in network operations?

- A. transfers files between file systems on a router
- B. transfers a backup configuration file from a server to a switch using a username and password
- C. transfers IOS images from a server to a router for firmware upgrades
- D. transfers a configuration files from a server to a router on a congested link

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 55

Which characteristic differentiates the concept of authentication from authorization and accounting?

- A. service limitations
- B. identity verification
- C. consumption-based billing
- D. user-activity logging

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 56

A wireless administrator has configured a WLAN, however, the clients need access to a less congested 5-GHz network for their voice quality. What action must be taken to meet the requirement?

- A. enable DTIM
- B. enable AAA override
- C. enable Band Select
- D. enable RX-SOP

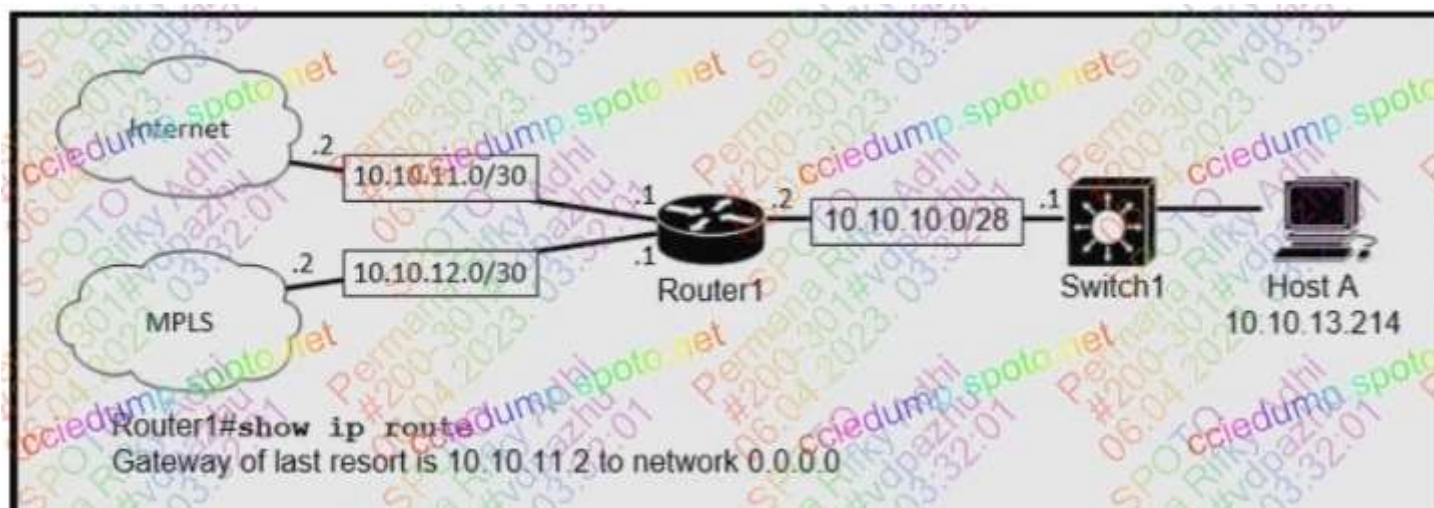
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 57



```

209.165.200.0/27 is subnetted, 1 subnets
B   209.165.200.224 [20/0] via 10.10.12.2, 03:22:14
209.165.201.0/27 is subnetted, 1 subnets
B   209.165.201.0 [20/0] via 10.10.12.2, 02:26:33
209.165.202.0/27 is subnetted, 1 subnets
B   209.165.202.128 [20/0] via 10.10.12.2, 02:26:03
10.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
C   10.10.10.0/28 is directly connected, GigabitEthernet0/0
C   10.10.11.0/30 is directly connected, FastEthernet2/0
C   10.10.12.0/30 is directly connected, GigabitEthernet0/1
O   10.10.13.0/25 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O   10.10.13.128/28 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O   10.10.13.144/28 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O   10.10.13.160/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O   10.10.13.208/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
S*  0.0.0.0/0 [1/0] via 10.10.11.2

```

Refer to the exhibit. Which prefix does Router1 use for traffic to Host A?

- A. 10.10.13.208/29
- B. 10.10.30.0/25
- C. 10.10.10.0/28
- D. 10.10.13.144/28

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 58

```

R1#show ip route
#output suppressed
Gateway of last resort is 192.168.14.4 to network 0.0.0.0

C   172.16.1.128/25 is directly connected, GigabitEthernet1/1/0
C   192.168.12.0/24 is directly connected, FastEthernet0/0
C   192.168.13.0/24 is directly connected, FastEthernet0/1
C   192.168.14.0/24 is directly connected, FastEthernet1/0
C   172.16.16.1 is directly connected, Loopback1
192.168.10.0/24 is variably subnetted, 3 subnets, 3 masks
O   192.168.10.0.24 [110/2] via 192.168.14.4, 00:02:01, FastEthernet1/0
O   192.168.10.32/27 [110/11] via 192.168.13.3, 00:00:52, FastEthernet0/1
O   192.168.0.0/16 [110/2] via 192.168.15.5, 00:05:01, FastEthernet1/1
D   192.168.10.1/32 [90/52778] via 192.168.12.2, 00:03:44, FastEthernet0/0
O*E2 0.0.0.0/0 [110/1] via 192.168.14.4, 00:00:10, FastEthernet1/0

```

Refer to the exhibit. If R1 receives a packet destined to 172.16.1.1, to which IP address does it send the packet?

- A. 192.168.14.4
- B. 192.168.15.5
- C. 192.168.13.3
- D. 192.168.12.2

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 59

Which 802.11 frame type is Association Response?

- A. action
- B. control
- C. protected frame
- D. management

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 60

R1 has learned route 192.168.12.0/24 via IS-IS, OSPF, RIP, and Internal EIGRP. Under normal operating conditions, which routing

protocol is installed in the routing table?

- A. RIP
- B. OSPF
- C. IS-IS
- D. Internal EIGRP

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 61

```
R1# show ip route
Codes:
C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP, D -
EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA
external type 1, N2 - OSPF NSSA external type 2, E1 - OSPF external type
1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default,
U - per-user static route, o - ODR
Gateway of last resort is not set
C 10.0.0.0/8 is directly connected, Loopback0
O 10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O 10.0.1.3/32 [110/100] via 10.0.1.100, 00:39:08, Serial0
C 10.0.1.0/24 is directly connected, Serial0
O 10.0.1.5/32 [110/5] via 10.0.1.50, 00:39:08, Serial0
O 10.0.10.0/24 [110/10] via 10.0.1.4, 00:39:08, Gigabit Ethernet 0/0
D 10.0.10.0/24 [90/10] via 10.0.1.5, 00:39:08, Gigabit Ethernet 0/1
```

Refer to the exhibit. Web traffic is coming in from the WAN interface. Which route takes precedence when the router is processing traffic destined for the LAN network at 10.0.10.0/24?

- A. via next-hop 10.0.1.4
- B. via next-hop 10.0.1.5
- C. via next-hop 10.0.1.50
- D. via next-hop 10.0.1.100

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

the same two 10.0.10.0/24 route in routing table?

QUESTION 62

What are two reasons that cause late collisions to increment on an Ethernet interface? (Choose two.)

- A. when one side of the connection is configured for half-duplex
- B. when the sending-device waits 15 seconds before sending the frame again
- C. when a collision occurs after the 32nd byte of a frame has been transmitted
- D. when the cable length limits are exceeded
- E. when Carrier Sense Multiple Access/Collision Detection is used

Correct Answer: AD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 63

When a site-to-site VPN is configured, which IPsec mode provides encapsulation and encryption of the entire original IP packet?

- A. IPsec tunnel mode with ESP
- B. IPsec transport mode with AH
- C. IPsec transport mode with ESP
- D. IPsec tunnel mode with AH

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 64

A network analyst is tasked with configuring the date and time on a router using EXEC mode. The date must be set to January 1,2020 and the time must be set to 12:00 am. What command should be used?

- A. clock summer-time date
- B. clock timezone
- C. clock summer-time recurring
- D. clock set

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 65

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       E - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
Gateway of last resort is 192.168.30.10 to network 0.0.0.0
192.168.30.0/29 is subnetted, 2 subnets
C    192.168.30.0 is directly connected, FastEthernet0/0
C    192.168.30.8 is directly connected, Serial0/0.1
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
O IA  192.168.10.32/28 [110/193] via 192.168.30.10, 00:18:49, Serial0/0.1
O IA  192.168.10.0/27 [110/192] via 192.168.30.10, 00:18:49, Serial0/0.1
192.168.20.0/30 is subnetted, 1 subnets
O IA  192.168.20.0 [110/128] via 192.168.30.10, 00:18:49, Serial0/0.1
192.168.50.0/32 is subnetted, 1 subnets
C    192.168.50.1 is directly connected, Loopback0
O*IA  0.0.0.0/0 [110/84] via 192.168.30.10, 00:10:36, Serial0/0.1
```

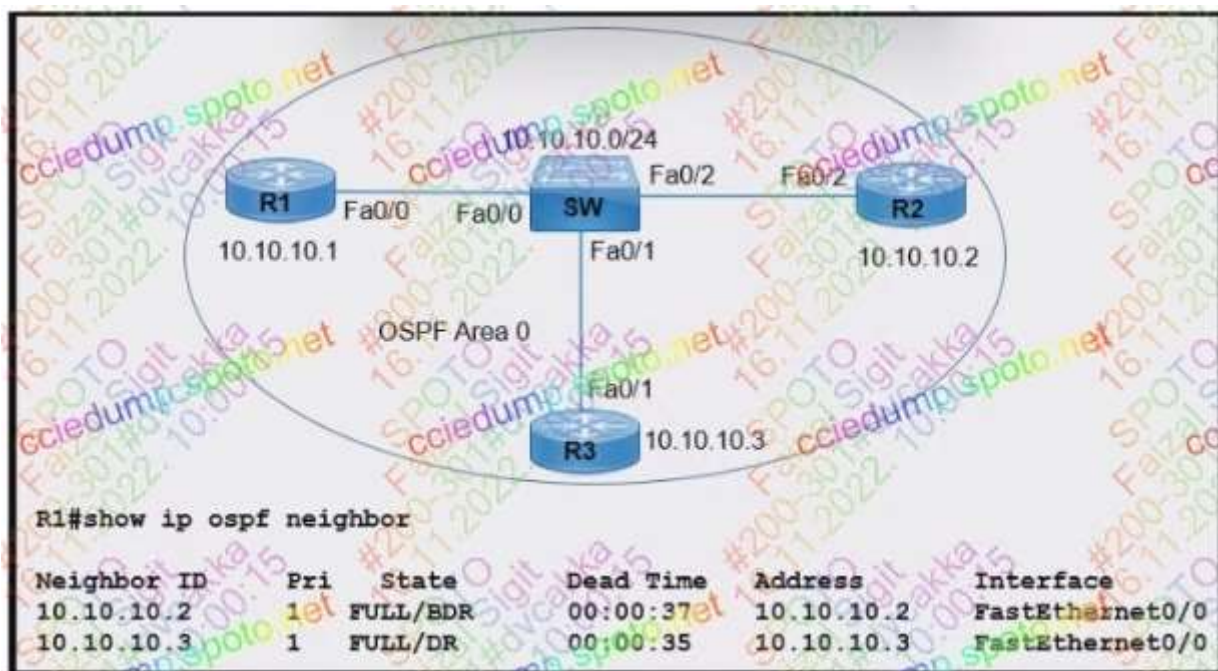
Refer to the exhibit. What is the metric of the route to the 192.168.10.33 host?

- A. 193
- B. 84
- C. 192
- D. 110

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 66



Refer to the exhibit. R1 has taken the DROTHER role in the OSPF DR/BDR election process. Which configuration must an engineer implement so that R1 is elected as the DR?

- A. R1(config)# interface FastEthernet 0/0
R1(config-if)# ip ospf priority 1
R1# clear ip ospf process
- B. R1(config)# interface FastEthernet 0/0
R1(config-if)# ip ospf priority 200
R1# clear ip ospf process
- C. R2(config)# interface FastEthernet 0/2
R2(config-if)# ip ospf priority 1
R2# clear ip ospf process
- D. R3(config)# interface FastEthernet 0/1
R3(config-if)# ip ospf priority 200
R3# clear ip ospf process

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 67

Which two outcomes are predictable behaviors for HSRP? (Choose two.)

- A. The two routers negotiate one router as the active router and the other as the standby router.
- B. Each router has a different IP address, both routers act as the default gateway on the LAN, and traffic is load-balanced between them.
- C. The two routers share the same interface IP address, and default gateway traffic is load-balanced between them.
- D. The two routers share a virtual IP address that is used as the default gateway for devices on the LAN.
- E. The two routers synchronize configurations to provide consistent packet forwarding.

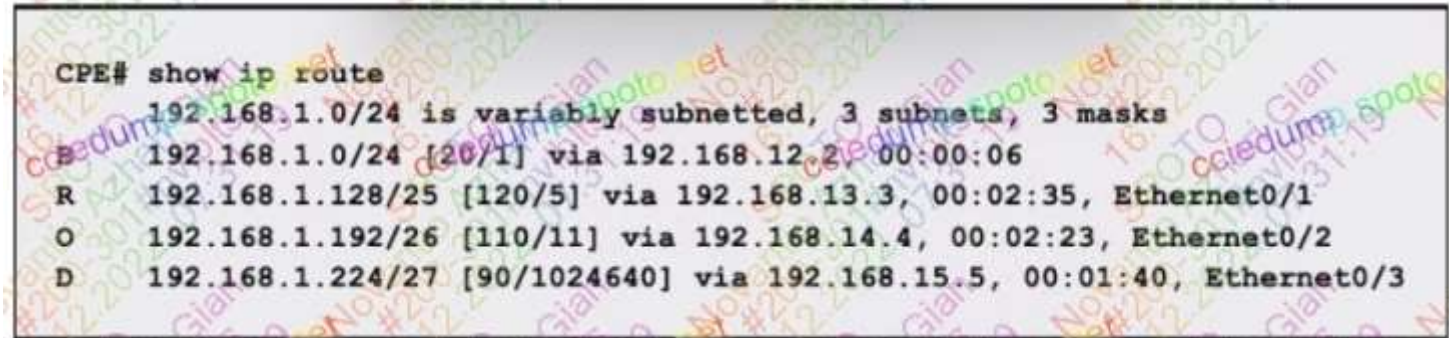
Correct Answer: AD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 68



Refer to the exhibit All traffic enters the CPE router from interface Serial0/3 with an IP address of 192.168.50.1. Web traffic from the WAN is destined for a LAN network where servers are load-balanced. An IP packet with a destination address of the HTTP virtual IP of 192.168.1.250 must be forwarded. Which routing table entry does the router use?

- A. 192.168.1.0/24 via 192.168.12.2
- B. 192.168.1.128/25 via 192.168.13.3
- C. 192.168.1.192/26 via 192.168.14.4
- D. 192.168.1.224/27 via 192.168.15.5

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 69

A network administrator is setting up a new IPv6 network using the 64-bit address 2001:0E88:00C1:2200:0001:0000:0000:0331/64. To simplify the configuration, the administrator has decided to compress the address. Which IP address must the administrator configure?

- A. ipv6 address 2001:EB8:C1:2200:1::331/64
- B. ipv6 address 2001:EB8:C1:2200:1:0000:331/64
- C. ipv6 address 21:EB8:C1 :2200:1::331/64
- D. ipv6 address 2001:EB8:C1:22:1::331/64

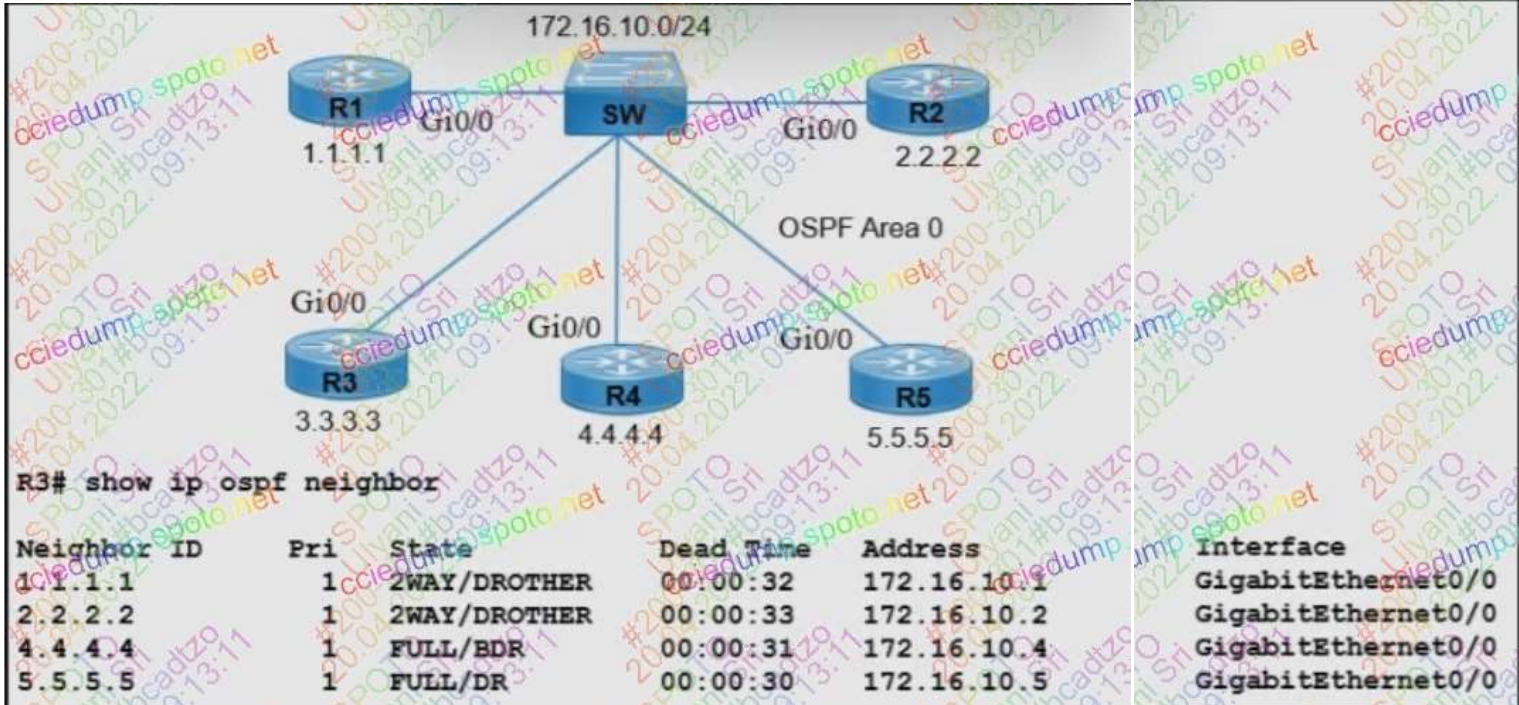
Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 70



Refer to the exhibit. R5 is the current DR on the network, and R4 is the BDR. Their interfaces are flapping, so a network engineer wants the OSPF network to elect a different DR and BDR. Which set of configurations must the engineer implement?

- A. R3(config)# interface gi0/0
R3(config-if)# ip ospf priority 255
- R2(config)# interface gi0/0
R2(config-if)# ip ospf priority 240
- B. R4(config)# interface gi0/0
R4(config-if)# ip ospf priority 20
- R5(config)# interface gi0/0
R5(config-if)# ip ospf priority 10
- C. R2(config)# interface gi0/0
R2(config-if)# ip ospf priority 259
- R3(config)# interface gi0/0
R3(config-if)# ip ospf priority 256
- D. R5(config)# interface gi0/0
R5(config-if)# ip ospf priority 120
- R4(config)# interface gi0/0
R4(config-if)# ip ospf priority 110

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 71

RIP	10.1.1.16/28[120/5]	via	F0/0
OSPF	10.1.1.0/24[110/30]	via	F0/1
OSPF	10.1.1.0/24[110/40]	via	F0/2
EIGRP	10.1.0.0/26[90/20]	via	F0/3
EIGRP	10.0.0.0/8 [90/133]	via	F0/4

Refer to the exhibit. Packets received by the router from BGP enter via a serial interface at 209.165.201.1. Each route is present within the routing table. Which interface is used to forward traffic with a destination IP of 10.1.1.19?

- A. F0/3
B. F0/0
C. F0/4
D. F0/1

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 72

A network engineer is configuring a switch so that it is remotely reachable via SSH. The engineer has already configured the host name on the router. Which additional command must the engineer configure before entering the command to generate the RSA key?

- A. password *password*
B. crypto key generate rsa modulus 1024
C. ip ssh authentication-rabies 2
D. ip domain-name domain

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 73

What is the role of SNMP in the network?

- A. to collect telemetry and critical information from network devices using an SSH underlay that operates on the network layer
B. to monitor network devices and functions using a TCP underlay that operates on the presentation layer
C. to collect data directly from network devices using an SSL underlay that operates on the transport layer
D. to monitor and manage network devices using a UDP underlay that operates on the application layer

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 74

Which two values or settings must be entered when configuring a new WLAN in the Cisco Wireless LAN Controller GUI? (Choose two.)

- A. management interface settings
- B. QoS settings
- C. SSID
- D. profile name
- E. IP address of one or more access points

Correct Answer: CD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 75

Refer to the exhibit. Router R1 is added to the network and configured with the 10.0.0.64/26 and 10.0.20.0/24 subnets. However, traffic destined for the LAN on R3 is not accessible. Which command when executed on R1 defines a static route to reach the R3 LAN?

- A. ip route 10.0.15.0 255.255.255.192 10.0.20.1
- B. ip route 10.0.15.0 255.255.255.0 10.0.20.3
- C. ip route 10.0.15.0 255.255.255.0 10.0.20.1
- D. ip route 10.0.0.64 255.255.255.192 10.0.20.3

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 76

Which IPsec transport mode encrypts the IP header and the payload?

- A. pipe
- B. transport
- C. tunnel
- D. control

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 77

What does a switch use to build its MAC address table?

- A. VTP
- B. ingress traffic
- C. egress traffic
- D. DTP

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 78



Refer to the exhibit. Which action must be taken to ensure that router A is elected as the DR for OSPF area 0?

- A. Configure router B and router C as OSPF neighbors of router A
- B. Configure the router A interfaces with the highest OSPF priority value within the area.
- C. Configure the OSPF priority on router A with the lowest value between the three routers.
- D. Configure router A with a fixed OSPF router ID.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 79



Refer to the exhibit. An engineer must configure router R2 so it is elected as the DR on the WAN subnet. Which command sequence must be configured?

- A.

```
interface gigabitethernet0/0
ip address 10.0.0.34 255.255.255.224
ip ospf priority 100
```
- B.

```
interface gigabitethernet0/0
ip address 10.0.0.34 255.255.255.248
ip ospf priority 0
```


- C. `interface gigabitethernet0/0`
`ip address 10.0.1.1 255.255.255.0`
`ip ospf priority 255`
- D. `interface gigabitethernet0/0`
`ip address 10.0.1.1 255.255.255.224`
`ip ospf priority 98`

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 80

To improve corporate security, an organization is planning to implement badge authentication to limit access to the data center. Which element of a security program is being deployed?

- A. user awareness
- B. vulnerability verification
- C. user training
- D. physical access control

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 81

What must be considered before deploying virtual machines?

- A. whether to leverage VSM to map multiple virtual processors to two or more virtual machines
- B. resource limitations, such as the number of CPU cores and the amount of memory
- C. support for physical peripherals, such as monitors, keyboards, and mice
- D. location of the virtual machines within the data center environment

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 82

A network engineer must configure the router R1 GigabitEthernet1/1 interface to connect to the router R2 GigabitEthernet1/1 interface. For the configuration to be applied, the engineer must compress the address 2001:0db8:0000:0000:0500:000a:400F:583B. Which command must be issued on the interface?

- A. `ipv6 address 2001:db8:0::500:a:4F:583B`
- B. `ipv6 address 2001:0db8::5:a:4F:583B`
- C. `ipv6 address 2001::db8:0000::500:a:400F:583B`
- D. `ipv6 address 2001:db8::500:a:400F:583B`

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 83

```
R1# show ip route | begin Gateway
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
S* 0.0.0.0/0 is directly connected, Serial0/0/1
L 172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
  C 172.16.2.0/24 is directly connected, GigabitEthernet0/0
  L 172.16.2.2/32 is directly connected, GigabitEthernet0/0
  C 172.16.4.0/21 is directly connected, Serial0/0/1
  L 172.16.8.2/26 is directly connected, Serial0/0/1
```

Refer to the exhibit. What is the subnet mask for route 172.16.4.0?

- A. 255.255.248.0
- B. 255.255.255.192
- C. 255.255.254.0
- D. 255.255.240.0

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 84

What is a reason why an administrator would choose to implement an automated network management approach?

- A. limit recurrent management costs
- B. enable "box by box" configuration and deployment
- C. reduce inconsistencies in the network configuration.
- D. decipher simple password policies.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 85

```

R1# show ip route
D    192.168.16.0/26 [90/2679326] via 192.168.1.1
R    192.168.16.0/24 [120/3] via 192.168.1.2
    192.168.16.0/21 [110/2] via 192.168.1.3
L L1 192.168.16.0/27 [115/30] via 192.168.1.4

```

Refer to the exhibit. Which route does R1 select for traffic that is destined to 192.168.16.2?

- A. 192.168.16.0/27
- B. 192.168.16.0/24
- C. 192.168.16.0/21
- D. 192.168.26.0/26

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 86

What is a requirement for nonoverlapping Wi-Fi channels?

- A. discontinuous frequency ranges
- B. unique SSIDs
- C. different transmission speeds
- D. different security settings

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 87

```

R1# show ip route | begin gateway
Gateway of last resort is 209.165.200.254 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 209.165.200.254, Serial0/0/1
    is directly connected, Serial0/0/1
C 172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
C 172.16.1.0/24 is directly connected, FastEthernet0/0
L 172.16.1.1/32 is directly connected, FastEthernet0/0
R 172.16.2.0/24 [120/2] via 207.165.200.250, 00:00:25, Serial0/0/0
O 192.168.1.0/24 [110/4437] via 207.165.200.254, 00:00:17, Serial0/0/1
D 192.168.2.0/24 [90/84437] via 207.165.200.254, 00:00:15, Serial0/0/1
    207.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
S 207.165.200.244/30 [1/1] via 207.165.200.254, Serial0/0/1
C 207.165.200.248/30 is directly connected, Serial0/0/0
L 207.165.200.249/32 is directly connected, Serial0/0/0
C 207.165.200.252/30 is directly connected, Serial0/0/1
L 207.165.200.253/32 is directly connected, Serial0/0/1

```

Refer to the exhibit. Which network prefix was learned via EIGRP?

- A. 207.165.200.0/24
- B. 192.168.1.0/24
- C. 192.168.2.0/24
- D. 172.160.0/16

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 88

Which protocol uses the SSL?

- A. HTTPS
- B. Telnet

- C. SSH
- D. HTTP

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 89



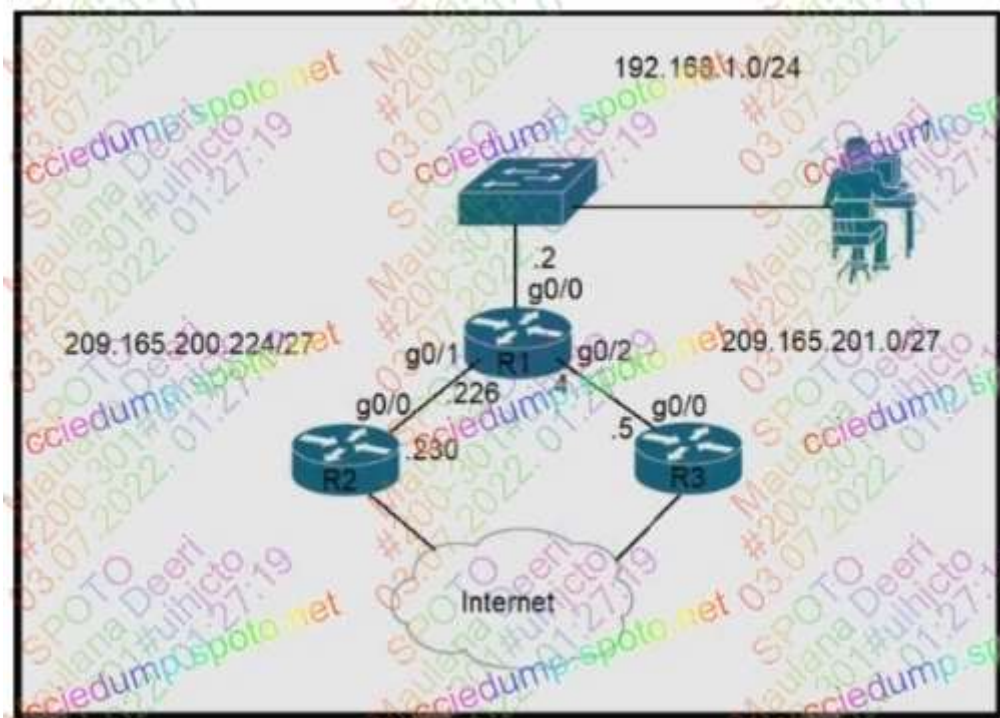
Refer to the exhibit. The New York router must be configured so that traffic to 2000::1 is sent primarily via the Atlanta site, with a secondary path via Washington that has an administrative distance of 2. Which two commands must be configured on the New York router? (Choose two.)

- A. ipv6 route 2000::1/128 2012::1
- B. ipv6 route 2000::1/128 2023::3 2
- C. ipv6 route 2000::1/128 2023::2 5
- D. ipv6 route 2000::1/128 2012::1 5
- E. ipv6 route 2000::1/128 2012::2

Correct Answer: AB
Section: (none)
Explanation

Explanation/Reference:

QUESTION 90



Refer to the exhibit. Router R1 currently is configured to use R3 as the primary route to the Internet, and the route uses the default administrative distance settings. A network engineer must configure R1 so that it uses R2 as a backup, but only if R3 goes down. Which command must the engineer configure on R1 so that it correctly uses R2 as a backup route, without changing the administrative distance configuration on the link to R3?

- A. ip route 0.0.0.0 0.0.0.0 g0/1 6
- B. ip route 0.0.0.0 0.0.0.0 209.165.201.5 10
- C. ip route 0.0.0.0 0.0.0.0 209.165.200.226 1
- D. ip route 0.0.0.0 0.0.0.0 g0/1 1

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 91

```

Gateway of last resort is 172.16.2.2 to network 0.0.0.0
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   10.10.8.0/28 is directly connected, GigabitEthernet0/0/2
C   10.10.10.0/24 is directly connected, GigabitEthernet0/0/0
L   10.10.10.3/32 is directly connected, GigabitEthernet0/0/0
172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
S   172.16.1.33/32 is directly connected, GigabitEthernet0/0/1
C   172.16.2.0/23 is directly connected, GigabitEthernet0/0/1
L   172.16.2.1/32 is directly connected, GigabitEthernet0/0/1
S*  0.0.0.0/0 [1/0] via 172.16.2.2

```

Refer to the exhibit. A packet sourced from 10.10.10.1 is destined for 10.10.8.14. What is the subnet mask of the destination route?

- A. 255.255.255.240
- B. 255.255.254.0
- C. 255.255.255.248
- D. 255.255.255.252

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 92

In QoS, which prioritization method is appropriate for interactive voice and video?

- A. traffic policing
- B. round-robin scheduling
- C. low-latency queuing
- D. expedited forwarding

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 93

```

Gateway of last resort is 0.0.0.0 to network 0.0.0.0
10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
S   10.0.0.0/8 is directly connected, GigabitEthernet0/0
C   10.1.1.0/24 is directly connected, GigabitEthernet0/0
L   10.1.1.1/32 is directly connected, GigabitEthernet0/0
S   10.10.0.0/22 is directly connected, GigabitEthernet0/0
S   10.10.10.0/28 is directly connected, GigabitEthernet0/0
S   10.10.10.1/32 is directly connected, GigabitEthernet0/0
S*  0.0.0.0/0 is directly connected, GigabitEthernet0/0

```

Refer to the exhibit. Which IP route command created the best path for a packet destined for 10.10.10.3?

- A. ip route 10.0.0.0 255.0.0.0 g0/0
- B. ip route 10.10.0.0 255.255.252.0 g0/0
- C. ip route 10.10.10.0 255.255.255.240 g0/0
- D. ip route 10.10.10.1 255.255.255.255 g0/0

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 94

```

R1# show ip route
D   192.168.10.0/24 [90/2679326] via 192.168.1.1
R   192.168.10.0/27 [120/3] via 192.168.1.2
O   192.168.10.0/23 [110/2] via 192.168.1.3
i   192.168.10.0/13 [115/3] via 192.168.1.4

```

Refer to the exhibit. How does router R1 handle traffic to 192.168.10.16?

- A. It selects the RIP route because it has the longest prefix inclusive of the destination address.
- B. It selects the IS-IS route because it has the shortest prefix inclusive of the destination address.
- C. It selects the OSPF route because it has the lowest cost.
- D. It selects the EIGRP route because it has the lowest administrative distance.

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 95

Which two conditions must be met before SSH operates normally on a Cisco IOS switch? (Choose two.)

- A. The switch must be running a k9 (crypto) IOS image.
- B. IP routing must be enabled on the switch.
- C. A console password must be configured on the switch.
- D. The ip domain-name command must be configured on the switch.
- E. Telnet must be disabled on the switch.

Correct Answer: AD
Section: (none)
Explanation

Explanation/Reference:

QUESTION 96

A network engineer is replacing the switches that belong to a managed-services client with new Cisco Catalyst switches. The new switches will be configured for updated security standards, including replacing Telnet services with encrypted connections and doubling the modulus size from 1024. Which two commands must the engineer configure on the new switches? (Choose two.)

- A. transport input ssh
- B. crypto key generate rsa general-keys modulus 1024
- C. crypto key generate rsa usage-keys
- D. crypto key generate rsa modulus 2048
- E. transport input all

Correct Answer: AD
Section: (none)
Explanation

Explanation/Reference:

QUESTION 97

An engineer is configuring switch SW1 to act as an NTP server when all upstream NTP server connectivity fails. Which configuration must be used?

- A.

```
SW1# config t
SW1(config)#ntp backup
SW1(config)# ntp server 192.168.1.1
```
- B.

```
SW1# config t
SW1(config)# ntp master
SW1(config)# ntp server 192.168.1.1
```
- C.

```
SW1# configt
SW1(config)# ntp server 192.168.1.1
SW1(config)# ntp access-group server accesslist1
```
- D.

```
SW1# config t
SW1(config)# ntp peer 192.168.1.1
SW1(config)# ntp access-group peer accesslist1
```

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 98

How does HSRP provide first hop redundancy?

- A. It uses a shared virtual MAC and a virtual IP address to a group of routers that serve as the default gateway for hosts on a LAN
- B. It load-balances Layer 2 traffic along the path by flooding traffic out all interfaces configured with the same VLAN.
- C. It load-balances traffic by assigning the same metric value to more than one route to the same destination in the IP routing table.
- D. It forwards multiple packets to the same destination over different routed links in the data path.

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 99

What is a benefit for external users who consume public cloud resources?

- A. all hosted on physical servers
- B. accessed over the Internet
- C. located in the same data center as the users

D. implemented over a dedicated WAN

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 100

Which unified access point mode continues to serve wireless clients after losing connectivity to the Cisco Wireless LAN Controller?

- A. flexconnect
- B. sniffer
- C. local
- D. mesh

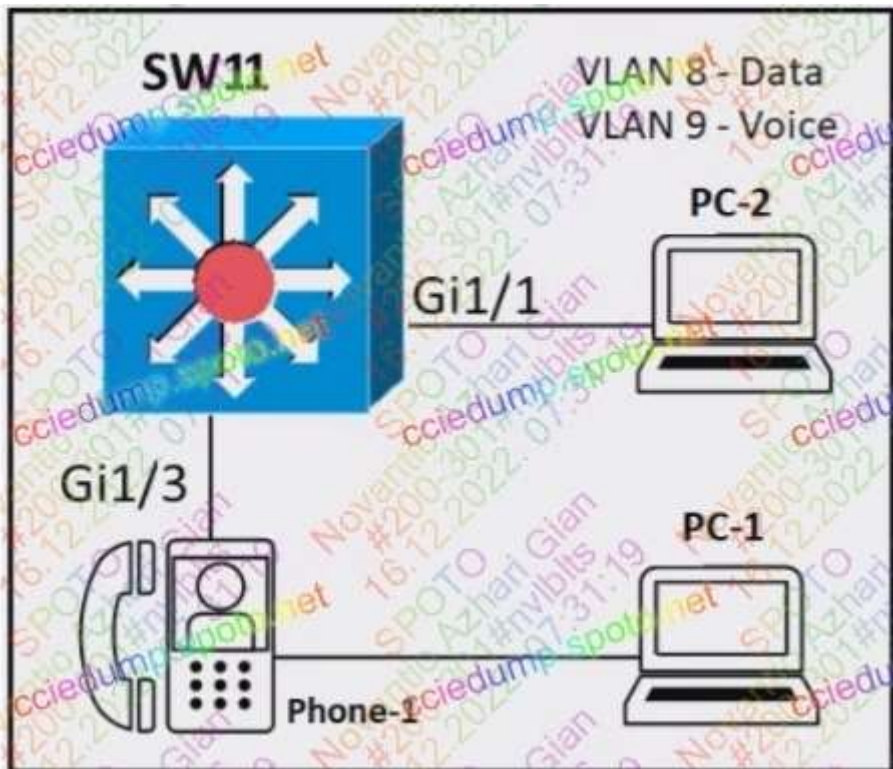
Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 101



Refer to the exhibit. An administrator must configure interfaces Gi1/1 and Gi1/3 on switch SW11. PC-1 and PC-2 must be placed in the Data VLAN, and Phone-1 must be placed in the Voice VLAN. Which configuration meets these requirements?

- A.

```
interface qigabitethernet1/1
switchport mode access
switchport access vlan 9
!
interface gigabitethernet1/3
switchport mode trunk
switchport trunk vlan 8
switchport trunk vlan 9
```
- B.

```
interface gigabitethernet1/1
switchport mode access
switchport access vlan 8
!
interface gigabitethernet1/3
switchport mode access
switchport voice vlan 8
switchport access vlan 9
```
- C.

```
interface gigabitethernet1/1
switchport mode access
switchport access vlan 8
!
interface qigabitethernet1/3
switchport mode access
switchport access vlan 8
switchport voice vlan 9
```
- D.

```
interface qicqabitethernet1/1
switchport mode access
switchport access vlan 8
!
interface gigabitethernet1/3
switchport mode trunk
switchport trunk vlan 8
switchport voice vlan 9
```

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 102

What is the maximum bandwidth of a T1 point-to-point connection?

- A. 43.7 Mbps
- B. 1.544 Mbps
- C. 2.048 Mbps
- D. 34.368 Mbps

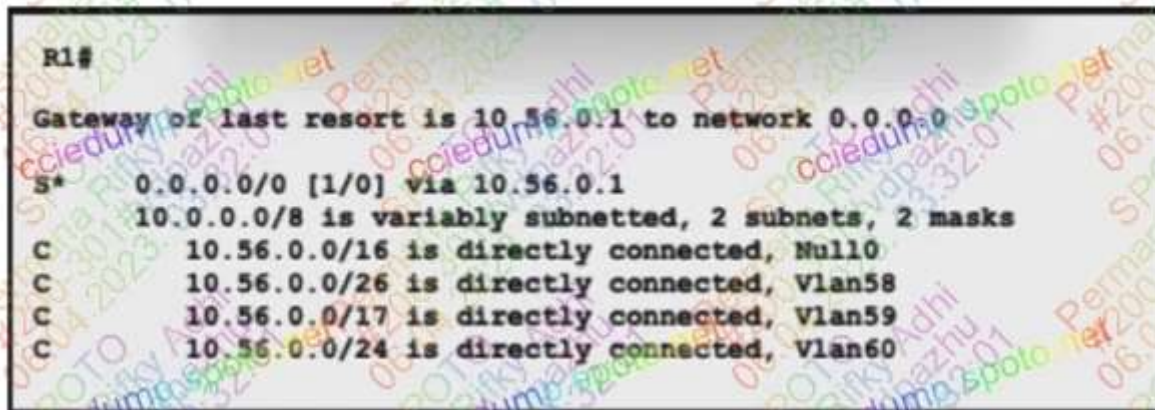
Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 103



Refer to the exhibit. When router R1 receives a packet with destination IP address 10.56.0.62, through which interface does it route the packet?

- A. Vlan60
- B. Null0
- C. Vlan58
- D. Vlan59

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 104

A network engineer must migrate a router loopback interface to the IPv6 address space. If the current IPv4 address of the interface is 10.54.73.1/32, and the engineer configures IPv6 address 0:0:0:0:ffff:a00:0000, which prefix length must be used?

- A. /128
- B. /64
- C. /96
- D. /124

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 105

Which value is the unique identifier that an access point used to establish and maintain wireless connectivity to wireless network devices?

- A. RFID
- B. SSID
- C. VLAN ID
- D. WLAN ID

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 106

```

R1# show ip route
Codes: C - connected, S - static, I - IGRP, E - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default
       U - per-user static route, o - ODR
Gateway of last resort is not set
C    172.16.0.0/16 is directly connected, Loopback0
O    172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
O    172.16.1.3/32 [110/100] via 10.0.1.100, 00:39:08, Serial0
O    172.16.1.9/32 [110/5] via 172.16.1.50, 00:43:01, Gigabit Ethernet 0/0
D    172.16.1.4/30 [90/7445] via 172.16.9.5, 00:39:08, Gigabit Ethernet 0/0
      [90/7445] via 172.16.4.4, 00:39:08, Gigabit Ethernet 0/4

```

Refer to the exhibit. How does router R1 handle traffic to the 172.16.1.4/30 subnet?

- A. It sends all traffic over the path via 10.0.1.100.
- B. It load-balances traffic over 172.16.9.5 and 172.16.4.4.
- C. It sends all traffic over the path via 172.16.4.4.
- D. It sends all traffic over the path via 172.16.9.5 using 172.16.4.4 as a backup.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 107

An engineer configured an OSPF neighbor as a designated router. Which state verifies the designated router is in the proper mode?

- A. 2-way
- B. Init
- C. Full
- D. Exchange

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 108

An organization secures its network with multi-factor authentication using an authenticator app on employee smartphones. How is the application secured in the case of a user's smartphone being lost or stolen?

- A. The application verifies that the user is in a specific location before it provides the second factor.
- B. The application requires the user to enter a PIN before it provides the second factor.
- C. The application requires an administrator password to reactivate after a configured interval.
- D. The application challenges a user by requiring an administrator password to reactivate when the smartphone is rebooted.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 109

A packet from a company's branch office is destined to host 172.31.0.1 at headquarters. The sending router has three possible matches in its routing table for the packet prefixes 172.31.0.0/16, 172.31.0.0/24, and 172.31.0.0/25. How does the router handle the packet?

- A. It sends the traffic via prefix 172.31.0.0/25.
- B. It sends the traffic via the default gateway 0.0.0.0/0.
- C. It sends the traffic via prefix 172.31.0.0/24.
- D. It sends the traffic via prefix 172.31.0.0/16.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 110

What configuration management mechanism uses TCP port 22 by default when communicating with managed nodes?

- A. Python
- B. Ansible
- C. Chef
- D. Puppet

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 111

How do servers connect to the network in a virtual environment?

- A. a cable connected to a physical switch on the network
- B. a virtual switch that links to an access point that is physically connected to the network
- C. wireless to an access point that is physically connected to the network
- D. a software switch on a hypervisor that is physically connected to the network

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 112

```
Router# show interface gi0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Hardware is ISR4331-3x1GE, address is 5486.bc25.1f70 (bia 5486.bc25.1f70)
  Description: << WAN Link >>
  Internet address is 192.0.2.2/30
  MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive not supported
  Full Duplex, 1000Mbps, link type is auto, media type is RJ45
  output flow-control is off, input flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:00, output 00:00:11, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 7000 bits/sec, 4 packets/sec
  5 minute output rate 4000 bits/sec, 4 packets/sec
    22579370 packets input, 8825545968 bytes, 0 no buffer
    Received 67 broadcasts (0 IP multicasts)
    0 runs, 0 giants, 0 throttles
    3612699 input errors, 3612699 CRC, 0 frame, 0 overrun, 0 ignored
    0 watchdog, 10747057 multicast, 0 pause input
    12871167 packets output, 1697953637 bytes, 0 underruns
    0 output errors, 0 collisions, interface resets
    6 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    5 lost carrier, 0 no carrier, 0 pause output
    0 output buffer failures, 0 output buffers swapped out
```

Refer to the exhibit. What is a reason for poor performance on the network interface?

- A. The bandwidth setting of the interface is misconfigured.
- B. The interface is receiving excessive broadcast traffic.
- C. The cable connection between the two devices is faulty.
- D. The interface is operating at a different speed than the connected device.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 113

A Cisco engineer notices that two OSPF neighbors are connected using a crossover Ethernet cable. The neighbors are taking too long to become fully adjacent. Which command must be issued under the interface configuration on each router to reduce the time required for the adjacency to reach the FULL state?

- A. ip ospf dead-interval 40
- B. ip ospf priority 0
- C. ip ospf network broadcast
- D. ip ospf network point-to-point

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 114

Which API is used in controller-based architectures to interact with edge devices?

- A. underlay
- B. southbound
- C. nonhbound
- D. overlay

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 115



```
{
  "interfaces": [ "ethernet0/3", "ethernet0/4", "ethernet0/5" ]
}
```

Refer to the exhibit. Which type of JSON data is shown?

- A. array
- B. Object
- C. Boolean
- D. Key

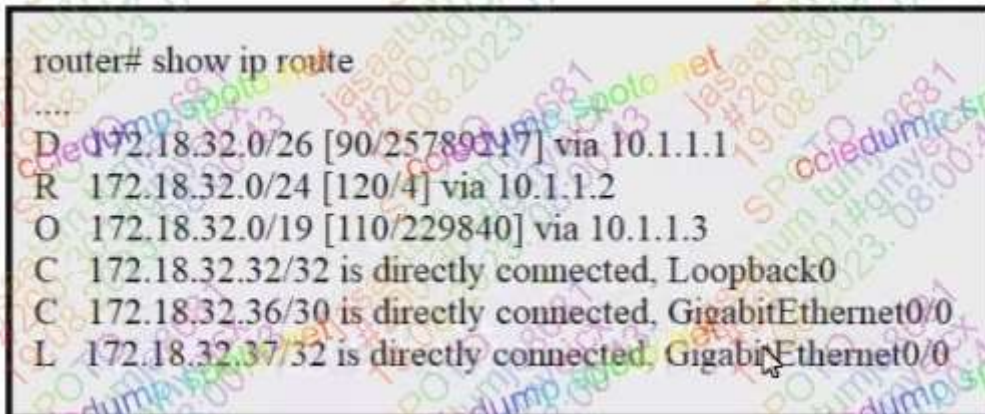
Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 116



```
router# show ip route
....
D 172.18.32.0/26 [90/25789217] via 10.1.1.1
R 172.18.32.0/24 [120/4] via 10.1.1.2
O 172.18.32.0/19 [110/229840] via 10.1.1.3
C 172.18.32.32/32 is directly connected, Loopback0
C 172.18.32.36/30 is directly connected, GigabitEthernet0/0
L 172.18.32.37/32 is directly connected, GigabitEthernet0/0
```

Refer to the exhibit. A packet sourced from 172.18.33.2 is destined for 172.18.32.38. Where does the router forward the packet?

- A. GigabitEthernet0/0
- B. 10.1.1.3
- C. 10.1.1.1
- D. Loopback0

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 117

A DHCP pool has been created with the name CONTROL. The pool uses the next to last usable IP address as the default gateway for the DHCP clients. The server is located at 172.16.32.15. What is the next step in the process for clients on the 192.168.52.0/24 subnet to reach the DHCP server?

- A. ip helper-address 172.16.32.15
- B. ip default-network 192.168.52.253
- C. ip forward-protocol udp 137
- D. ip default-gateway 192.168.52.253

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 118

An engineer must update the configuration on two PCs in two different subnets to communicate locally with each other. One PC is configured with IP address 192.168.25.128/25 and the other with 192.168.25.100/25. Which network mask must the engineer configure on both PCs to enable the communication?

- A. 255.255.255.224
- B. 255.255.255.248
- C. 255.255.255.252
- D. 255.255.255.0

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 119

Which technology allows for multiple operating systems to be run on a single host computer?

- A. virtual device contexts
- B. network port ID virtualization

- C. server virtualization
- D. virtual routing and forwarding

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 120

How do traditional campus device management and Cisco DNA Center device management differ in regards to deployment?

- A. Cisco DNA Center device management can deploy a network more quickly than traditional campus device management.
- B. Cisco DNA Center device management can be implemented at a lower cost than most traditional campus device management options.
- C. Traditional campus device management allows a network to scale more quickly than with Cisco DNA Center device management.
- D. Traditional campus device management schemes can typically deploy patches and updates more quickly than Cisco DNA Center device management.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 121

Which type of address is shared by routers in a HSRP implementation and used by hosts on the subnet as their default gateway address?

- A. multicast address
- B. loopback IP address
- C. virtual IP address
- D. broadcast address

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 122

What event has occurred if a router sends a notice level message to a syslog server?

- A. An interface line has changed status.
- B. An ICMP connection has been built.
- C. A certificate has expired.
- D. A TCP connection has been torn down.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 123

What is an expected outcome when network management automation is deployed?

- A. Custom applications are needed to configure network devices.
- B. Software upgrades are performed from a central controller.
- C. A distributed management plane must be used.
- D. Complexity increases when new device configurations are added.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 124

Which selections must be used on the WLC when implementing a RADIUS server for wireless authentication?

- A. Client Exclusion and SSH
- B. 802.1X and the MAC address of the server
- C. Network Access Control State and SSH
- D. AAA Override and the IP address of the server

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 125

What is one reason to configure a trunk port that connects to a WLC distribution port?

- A. allow multiple VLANs to be used in the data path
- B. permit multiple VLANs to provide out-of-band management
- C. eliminate redundancy with a link failure in the data path
- D. provide redundancy in the event of a link failure for out-of-band management

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 126

Which QoS feature drops traffic that exceeds the committed access rate?

- A. FIFO
- B. Weighted fair queuing
- C. policing
- D. shaping

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 127

What mechanism carries multicast traffic between remote sites and supports encryption?

- A. IPsec over ISATAP
- B. GRE
- C. GRE over IPsec
- D. ISATAP

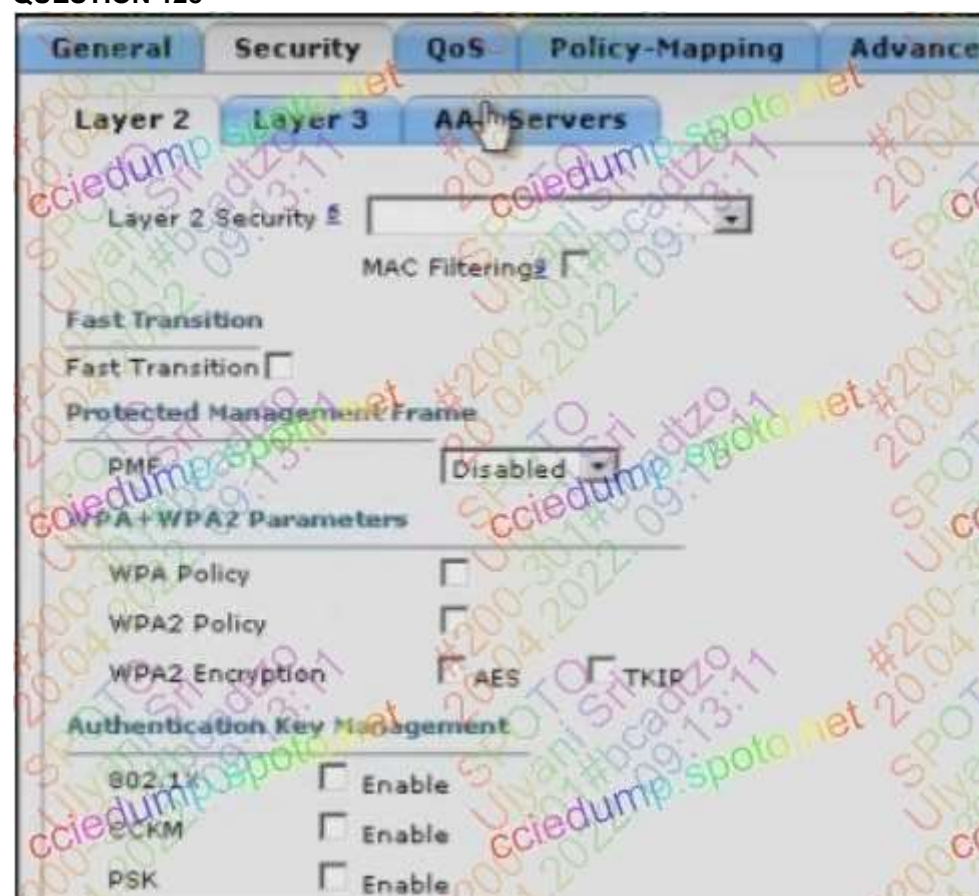
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 128



Refer to the exhibit. What are the two steps an engineer must take to provide the highest encryption and authentication using domain credentials from LDAP? (Choose two.)

- A. Select PSK under Authentication Key Management
- B. Select Static-WEP+802.1X on Layer 2 Security.
- C. Select WPA+WPA2 on Layer 2 Security.
- D. Select 802.1X from under Authentication Key Management.
- E. Select WPA Policy with TKIP Encryption.

Correct Answer: CD

Section: (none)

Explanation

Explanation/Reference:

<https://www.cisco.com/c/en/us/support/docs/wireless-mobility/wireless-lan-wlan/211277-WLC-with-LDAP-Authentication-Configurati.html#anc9>

QUESTION 129

What are two characteristics of a small office /home office connection environment? (Choose two.)

- A. It supports between 50 and 100 users.
- B. It requires 10Gb ports on all uplinks.
- C. It requires a core, distribution, and access layer architecture.
- D. A router port connects to a broadband connection.
- E. It supports between 1 and 50 users.

Correct Answer: DE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 130

What provides connection redundancy, increased bandwidth, and load sharing between a wireless LAN controller and a Layer 2 switch?

- A. VLAN trunking
- B. tunneling
- C. first hop redundancy
- D. link aggregation

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 131

How do AAA operations compare regarding user identification, user services, and access control?

- A. Authorization Identifies users, and authentication provides access control.
- B. Accounting tracks user services, and authentication provides access control.
- C. Authentication identifies users, and accounting tracks user services.
- D. Authorization provides access control, and authentication tracks user services.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 132

Refer to the exhibit. After the configuration is applied, the two routers fail to establish an OSPF neighbor relationship. What is the reason for the problem?

- A. Router2 is using the default hello timer.
- B. The network statement on Router1 is misconfigured.
- C. The OSPF router IDs are mismatched.
- D. The OSPF process IDs are mismatched.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 133

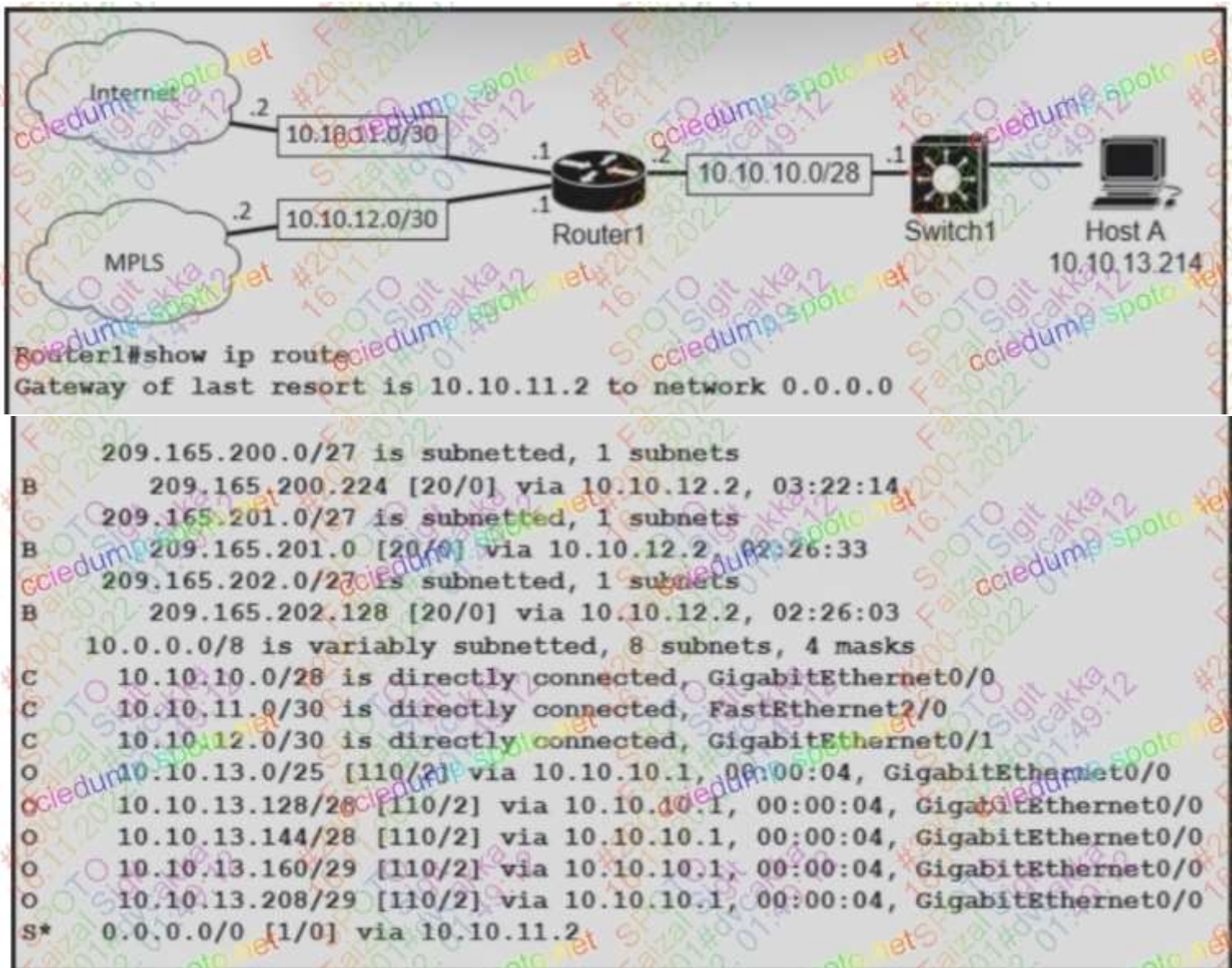
A network engineer must create a diagram of a multivendor network. Which command must be configured on the Cisco devices so that the topology of the network can be mapped?

- A. Device(config)# lldp run
- B. Device(config)# flow-sampler-map topology
- C. Device(config)# cdp run
- D. Device(config-if)# cdp enable

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 134



Refer to the exhibit. What is the prefix length for the route that router1 will use to reach host A?

- A. /25
- B. /29
- C. /28
- D. /27

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 135

After a recent security breach and a RADIUS failure, an engineer must secure the console port of each enterprise router with a local username and password. Which configuration must the engineer apply to accomplish this task?

- A.

```
username localuser secret plaintextpassword
line con 0
no login local
privilege level 15
```
- B.

```
aaa new-model
aaa authorization exec default local
aaa authentication login default radius
username localuser privilege 15 secret plaintextpassword
```
- C.

```
username localuser secret plaintextpassword
line con 0
login authentication default
privilege level 15
```


D.

```
aaa new-model
line con 0
password plaintextpassword
privilege level 15
```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 136

In a cloud-computing environment, what is rapid elasticity?

- A. control and monitoring of resource consumption by the tenant
- B. self-service of computing resources by the tenant
- C. automatic adjustment of capacity based on need
- D. pooling resources in a multitenant model based on need

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 137

When OSPF learns multiple paths to a network, how does it select a route?

- A. It counts the number of hops between the source router and the destination to determine the route with the lowest metric.
- B. It multiplies the active K values by 256 to calculate the route with the lowest metric.
- C. It divides a reference bandwidth of 100 Mbps by the actual bandwidth of the exiting interface to calculate the route with the lowest cost
- D. For each existing interface, it adds the metric from the source router to the destination to calculate the route with the lowest bandwidth

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 138

```
R1# sh ip ospf int gig0/0
Gig0/0 is up, line protocol is up
Internet Address 10.201.24.8/28, Area 1, Attached via Network Statement
Process ID 100, Router ID 100.1.1.1, Network Type BROADCAST, Cost: 1
Topology-MTID      Cost      Disabled      Shutdown      Topology Name
0                  1          no            no            Base
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 192.168.1.1, Interface address 10.201.24.8
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
oob-resp-timeout: 40
R1#
R2# sh ip ospf int gig0/0
gig0/0 is up, line protocol is up
Internet Address 10.201.24.1/28, Area 1
Process ID 100, Router ID 172.16.1.1, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 172.16.1.1, Interface address 10.201.24.1
No backup designated router on this network
Timer intervals configured, Hello 20, Dead 80, Wait 80, Retransmit 5
R2#
```

Refer to the exhibit. What action establishes the OSPF neighbor relationship without forming an adjacency?

- A. modify hello interval
- B. modify priority
- C. modify network type
- D. modify process ID

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 139

```

1 [
2   { "switch": "3750", "port": e2 },
3   { "router": "2951", "port": e20 },
4   { "switch": "3750", "port": e23 }
5 ]

```

Refer to the exhibit. What is represented beginning with line 1 and ending with line 5?

- A. Array
- B. Key
- C. Object
- D. Value

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 140

Which action is taken by a switch port enabled for PoE power classification override?

- A. As power usage on a PoE switch port is checked, data flow to the connected device is temporarily paused
- B. If a switch determines that a device is using less than the minimum configured power, it assumes the device has failed and disconnects it.
- C. Should a monitored port exceed the maximum administrative value for power, the port is shut down and err-disabled.
- D. When a powered device begins drawing power from a PoE switch port, 3 syslog message is generated.

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 141

What are two benefits of FHRPs? (Choose two.)

- A. They enable automatic failover of the default gateway.
- B. They allow encrypted traffic.
- C. They are able to bundle multiple ports to increase bandwidth.
- D. They allow multiple devices to serve as a single virtual gateway for clients in the network.
- E. They prevent loops in the Layer 2 network.

Correct Answer: AD
Section: (none)
Explanation

Explanation/Reference:

QUESTION 142

What is a reason to implement LAG on a Cisco WLC?

- A. Provide link redundancy and load balancing.
- B. Increase security and encrypt management frames.
- C. Allow for stateful and link-state failover.
- D. Enable connected switch ports to fail over and use different VLANs.

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 143



Refer to the exhibit. An engineer must configure a floating static route on an external EIGRP network. The destination subnet is the /29 on the LAN interface of R86. Which command must be executed on R14?

- A. Ip route 10.73.65.66 0.0.0.224 10.80.65.0 255

- B. Ip route 10.80.65.0 255.255.255.240 fa0/1 89
- C. Ip route 10.80.65.0 255.255.255.248 10.73.65.66 171
- D. Ip route 10.80.65.0 255.255.248.0 10.73.65.66 1

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 144

Under which condition is TCP preferred over UDP?

- A. UDP is used when data is highly interactive, and TCP is used when data is time-sensitive.
- B. UDP is used when low latency is optimal, and TCP is used when latency is tolerable.
- C. TCP is used when dropped data is more acceptable, and UDP is used when data is accepted out-of-order.
- D. TCP is used when data reliability is critical, and UDP is used when missing packets are acceptable.

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 145

Which two spanning-tree states are bypassed on an interface running PortFast? (Choose two.)

- A. listening
- B. blocking
- C. forwarding
- D. disabled
- E. learning

Correct Answer: AE
Section: (none)
Explanation

Explanation/Reference:

QUESTION 146

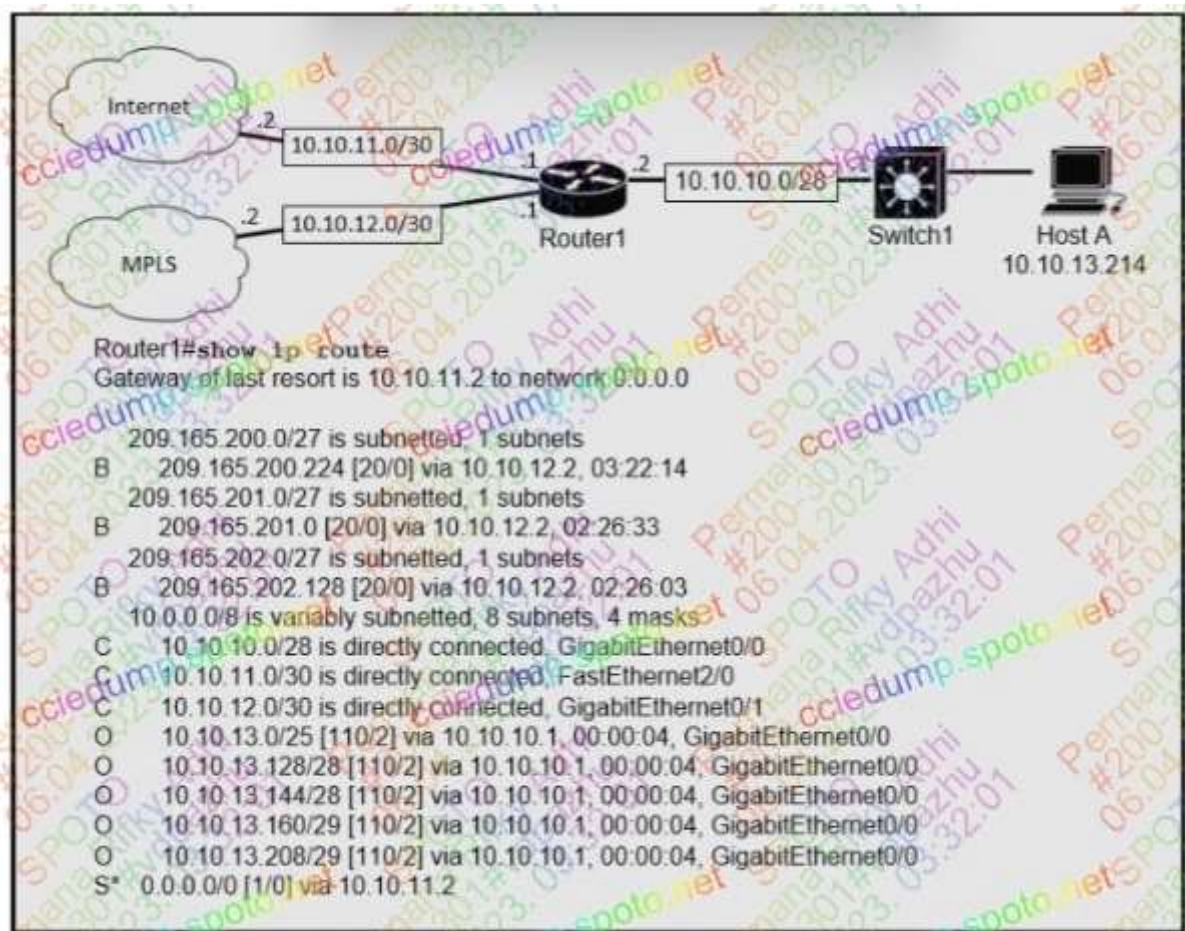
Which level of severity must be set to get informational syslogs?

- A. Alert
- B. Critical
- C. Debug
- D. Notice

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 147



Refer to the exhibit. What is the subnet mask of the route to the 10.10.13.160 prefix?

- A. 255.255.255.128
- B. 255.255.255.248
- C. 255.255.248.0
- D. 255.255.255.240

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 148

What is the primary purpose of private address space?

- A. reduce network complexity
- B. limit the number of nodes reachable via the Internet
- C. simplify the addressing in the network
- D. conserve globally unique address space

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 149

What is a DHCP client?

- A. a server that dynamically assigns IP addresses to hosts.
- B. a host that is configured to request an IP address automatically
- C. a router that statically assigns IP addresses to hosts.
- D. a workstation that requests a domain name associated with its IP address

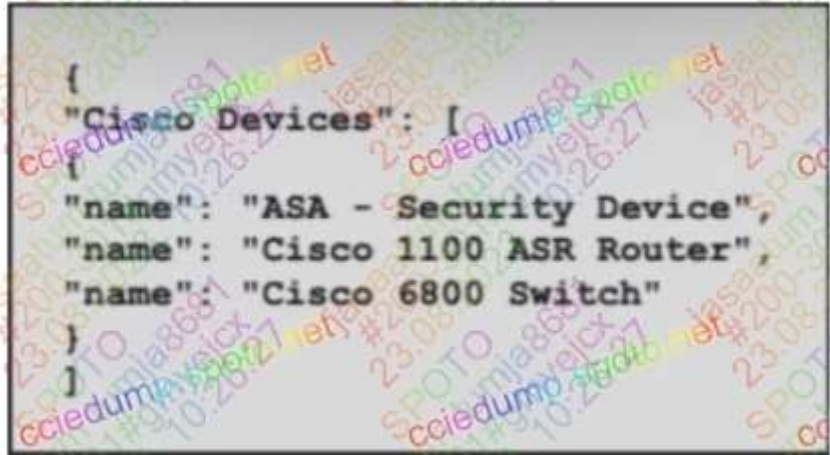
Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 150



Refer to the exhibit. What is missing from this output for it to be executed?

- A. double quotes (" ") around the "Cisco Devices" string
- B. curly brace (}) at the end
- C. square bracket ([) at the beginning
- D. exclamation point (!) at the beginning of each line

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 151

What causes a port to be placed in the err-disabled state?

- A. nothing plugged into the port
- B. link flapping
- C. **shutdown** command issued on the port
- D. latency

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

<https://www.cisco.com/c/en/us/support/docs/lan-switching/spanning-tree-protocol/69980-errdisable-recovery.html#anc8>

QUESTION 152

Which two HTTP methods are suitable for actions performed by REST-based APIs? (Choose two.)

- A. REMOVE
- B. REDIRECT
- C. GET
- D. POP
- E. POST

Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 153

Which mode must be set for APs to communicate to a Wireless LAN Controller using the Control and Provisioning of Wireless Access Points (CAPWAP) protocol?

- A. route
- B. bridge
- C. lightweight
- D. autonomous

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 154

Router R1 must send all traffic without a matching routing-table entry to 192.168.1.1. Which configuration accomplishes this task?

- A.

```
R1#Config t
R1(config)#ip routing
R1(config)#ip route default-route 192.168.1.1
```
- B.

```
R1#Config t
R1(config)#ip routing
R1(config)#ip route 192.168.1.1 0.0.0.0 0.0.0.0
```
- C.

```
R1#Config t
R1(config)#ip routing
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.1
```
- D.

```
R1#Config t
R1(config)#ip routing
R1(config)#ip default-gateway 192.168.1.1
```

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 155

What does the implementation of a first-hop redundancy protocol protect against on a network?

- A. spanning-tree loops
- B. default gateway failure
- C. BGP neighbor flapping
- D. root-bridge loss

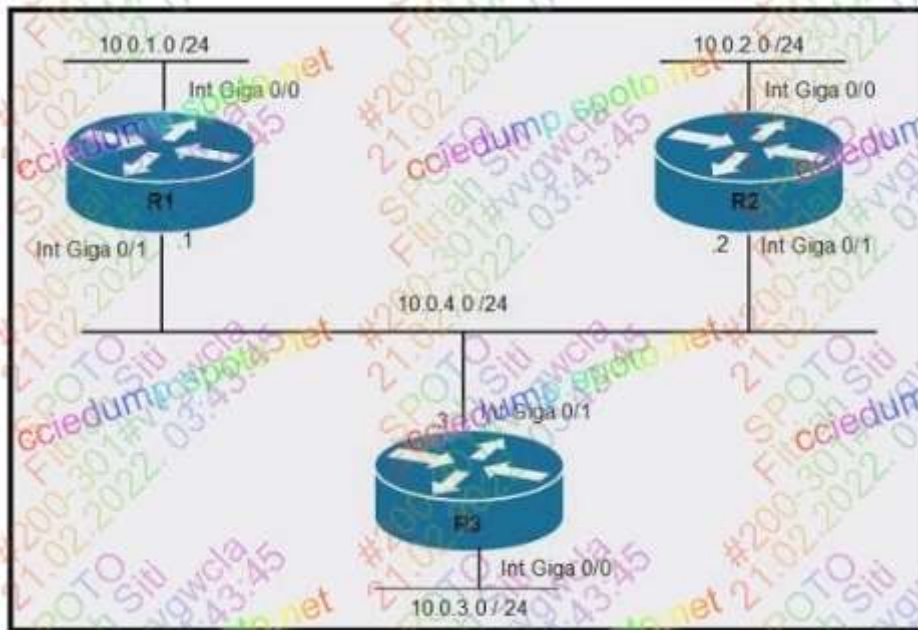
Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 156



Refer to the exhibit. Router R1 must be configured to reach the 10.0.3.0/24 network from the 10.0.1.0/24 segment. Which command must be used to configure the route?

- A. ip route 10.0.3.0 0.255.255.255 10.0.4.2
- B. route add 10.0.3.0 0.255.255.255 10.0.4.2
- C. route add 10.0.3.0 mask 255.255.255.0 10.0.4.3
- D. ip route 10.0.3.0 255.255.255.0 10.0.4.3

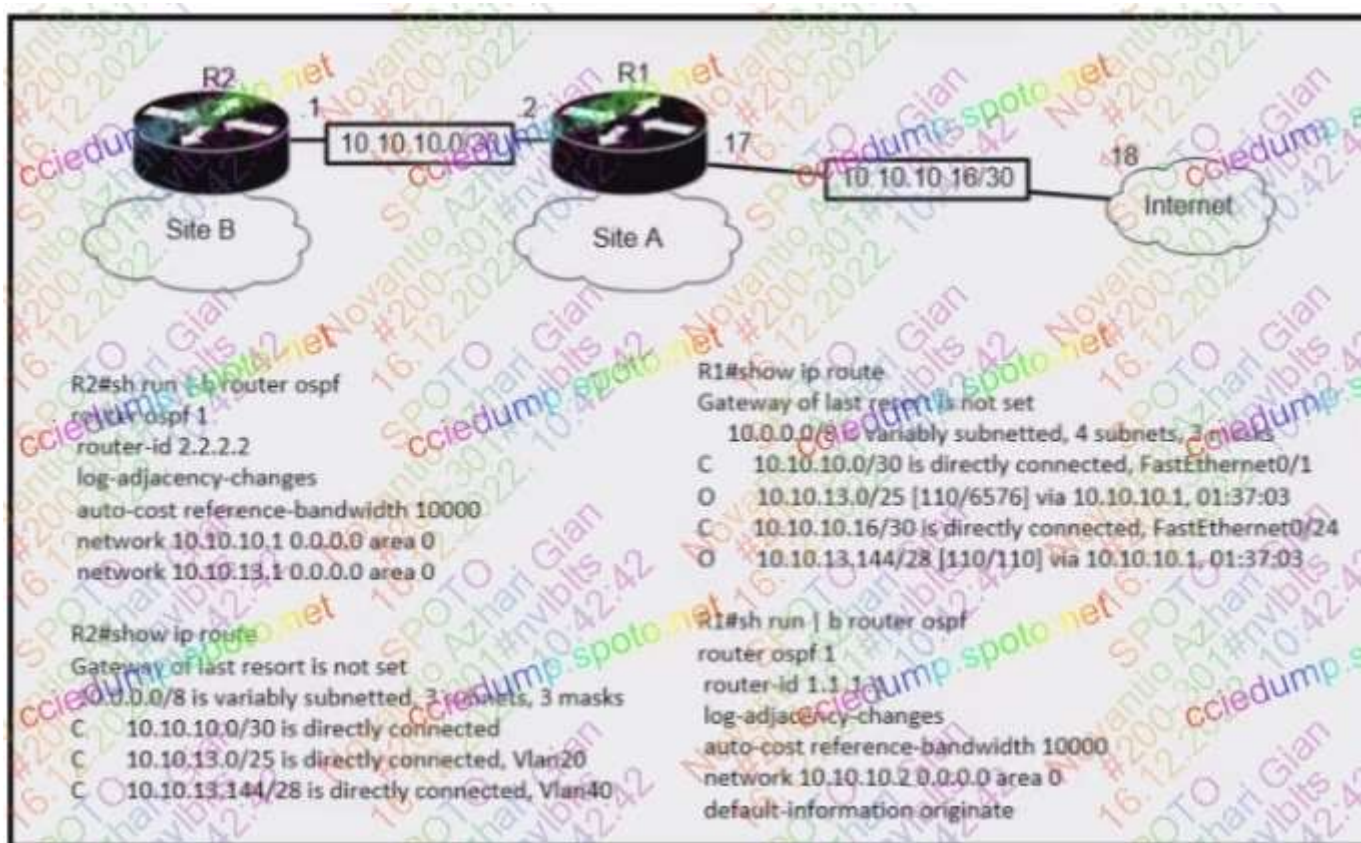
Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 157



Refer to the exhibit. The default-information originate command is configured under the R1 OSPF configuration. After testing, workstations or VLAN 20 at Site B cannot reach a DNS server on the internet. Which action corrects the configuration issue?

- A. Add the default-information originate command on R2
- B. Add the always keyword to the default-information originate command on R1
- C. Configure the ip route 0.0.0.0 0.0.0.0 10.10.10.2 command on R2
- D. Configure the ip route 0.0.0.0 0.0.0.0 10.10.10.18 Command on R1

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 158

What is the function of a controller in a software-defined network?

- A. fragmenting and reassembling packets
- B. forwarding packets
- C. setting packet-handling policies
- D. multicast replication at the hardware level

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 159

What does a switch do when it receives a frame whose destination MAC address is missing from the MAC address table?

- A. It updates the CAM table with the destination MAC address of the frame.
- B. It appends the table with a static entry for the MAC and shuts down the port.
- C. It floods the frame unchanged across all remaining ports in the incoming VLAN.
- D. It changes the checksum of the frame to a value that indicates an invalid frame.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 160



Refer to the exhibit. What commands are needed to add a subinterface to Ethernet0/0 R1 to allow for VLAN 20, with IP address 10.20.20.1/24?

- A. R1(config)# interface ethernet0/0
R1(config)# encapsulation dot1q 20
R1(config)# ip address 10.20.20.1 255.255.255.0
- B. R1(config)# interface ethernet0/0.20
R1(config)# ip address 10.20.20.1 255.255.255.0
- C. R1(config)# interface ethernet0/0.20
R1(config)# encapsulation dot1q 20
R1(config)# ip address 10.20.20.1 255.255.255.0
- D. R1(config)# interface ethernet0/0
R1(config)# ip address 10.20.20.1 255.255.255.0

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 161

Which action must be taken when password protection is implemented?

- A. Include special characters and make passwords as long as allowed.
- B. Use less than eight characters in length when passwords are complex.
- C. Store passwords as contacts on a mobile device with single-factor authentication.
- D. Share passwords with senior IT management to ensure proper oversight

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 162

What is the role of a firewall in an enterprise network?

- A. forwards packets based on stateless packet inspection
- B. processes unauthorized packets and allows passage to less secure segments of the network
- C. explicitly denies all packets from entering an administrative domain
- D. determines which packets are allowed to crass from unsecured to secured networks

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 163

Which two actions are performed by the Weighted Random Early Detection mechanism? (Choose two.)

- A. It drops lower-priority packets before it drops higher-priority packets.
- B. It supports protocol discovery.
- C. It guarantees the delivery of high-priority packets.
- D. It can mitigate congestion by preventing the queue from filling up.
- E. It can identify different flows with a high level of granularity.

Correct Answer: AD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 164

What is the functionality of the Cisco DNA Center?

- A. software-defined controller for automation of devices and services
- B. console server that permits secure access to all network devices
- C. data center network policy controller
- D. IP address pool distribution scheduler

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 165

Refer to the exhibit. Switch AccSw2 has just been added to the network along with PC2. All VLANs have been implemented on AccSw2. How must the ports on AccSw2 be configured to establish Layer 2 connectivity between PC1 and PC2?

- A.

```
interface GigabitEthernet1/2
switchport mode access
switchport access vlan 2
!
interface GigabitEthernet1/24
switchport mode trunk
```
- B.

```
interface GigabitEthernet1/2
switchport mode access
switchport access vlan 12
!
interface GigabitEthernet1/24
switchport mode trunk
switchport trunk allowed vlan 11,12
```
- C.

```
interface GigabitEthernet1/24
switchport mode trunk
switchport trunk allowed vlan 11,12
!
interface GigabitEthernet1/1
switchport access vlan 11
```


D. 

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 166

What is the PUT method within HTTP?

- A. It replaces data at the destination.
- B. It displays a web site
- C. It is a nonidempotent operation
- D. It is a read-only operation

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 167

Which access point mode relies on a centralized controller for management, roaming, and SSID configuration?

- A. Autonomous mode
- B. Repeater mode
- C. Bridge mode
- D. Lightweight mode

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 168

Which action must be taken to assign a global unicast IPv6 address on an interface that is derived from the MAC address of that interface?

- A. explicitly assign a link-local address
- B. enable SLAAC on an interface
- C. disable the EU-64 bit process
- D. configure a stateful DHCPv6 server on the network

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 169

How does Rapid PVST+ create a fast loop-free network topology?

- A. It maps multiple VLANs into the same spanning-tree instance.
- B. It requires multiple links between core switches.
- C. It generates one spanning-tree instance for each VLAN.
- D. It uses multiple active paths between end stations.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 170

Which command must be entered to configure a DHCP relay?

- A. ip dhcp relay
- B. ip helper-address
- C. ip dhcp pool
- D. ip address dhcp

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 171

The clients and DHCP server reside on different subnets. Which command must be used to forward requests and replies between clients on the 10.10.0.1/24 subnet and the DHCP server at 192.168.10.1?

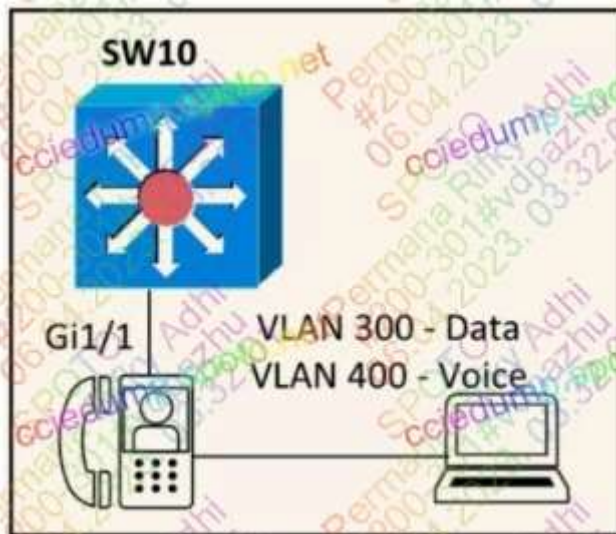
- A. ip default-gateway 192.168.10.1
- B. ip helper-address 192.168.10.1
- C. ip route 192.168.10.1
- D. ip dhcp address 192.168.10.1

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 172

Refer to the exhibit. An engineer must configure GigabitEthernet1/1 to accommodate voice and data traffic. Which configuration accomplishes this

- A.

```
interface gigabitethernet1/1
switchport mode trunk
switchport trunk vlan 300
switchport trunk vlan 400
```
- B.

```
interface gigabitethernet1/1
switchport mode access
switchport access vlan 300
switchport voice vlan 400
```
- C.

```
interface gigabitethernet1/1
switchport mode trunk
switchport trunk vlan 300
switchport voice vlan 400
```
- D.

```
interface gigabitethernet1/1
switchport mode access
switchport voice vlan 300
switchport access vlan 400
```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 173

A router received three destination prefixes: 10.0.0.0/8, 10.0.0.0/16, and 10.0.0.0/24. When the show ip route command is executed, which output does it return?

- A. Gateway of last resort is 172.16.1.1 to network 0.0.0.0
O E2 10.0.0.0/8 [110/5] via 192.168.1.1, 0:01:00, Ethernet0
- B. Gateway of last resort is 172.16.1.1 to network 0.0.0.0
O E2 10.0.0.0/24 [110/5] via 192.168.3.1, 0:01:00, Ethernet2
- C. Gateway of last resort is 172.16.1.1 to network 0.0.0.0
O E2 10.0.0.0/8 [110/5] via 192.168.1.1, 0:01:00, Ethernet0
O E2 10.0.0.0/16 [110/5] via 192.168.2.1, 0:01:00, Ethernet1
O E2 10.0.0.0/24 [110/5] via 192.168.3.1, 0:01:00, Ethernet2
- D. Gateway of last resort is 172.16.1.1 to network 0.0.0.0
O E2 10.0.0.0/16 [110/5] via 192.168.2.1, 0:01:00, Ethernet1
O E2 10.0.0.0/24 [110/5] via 192.168.3.1, 0:01:00, Ethernet2

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 174

Which field within the access-request packet is encrypted by RADIUS?

- A. authorized services
- B. authenticator
- C. username
- D. Password

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 175

Which WLC management connection type is vulnerable to man-in-the-middle attacks?

- A. Telnet
- B. SSH
- C. HTTPS
- D. Console

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 176

What is the definition of backdoor malware?

- A. malicious program that is used to launch other malicious programs
- B. malicious code that infects a user machine and then uses that machine to send spam
- C. malicious code with the main purpose of downloading other malicious code
- D. malicious code that is installed onto a computer to allow access by an unauthorized user

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 177

When a switch receives a frame for an unknown destination MAC address, how is the frame handled?

- A. forwarded to the first available port
- B. broadcast to all ports on the switch
- C. inspected and dropped by the switch
- D. flooded to all ports except the origination port

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 178

Which HTTP status code is returned after a successful REST API request?

- A. 500
- B. 404
- C. 200
- D. 301

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 179

What is the same for both copper and fiber interfaces when using SFP modules?

- A. They accommodate single-mode and multi-mode in a single module
- B. They support an inline optical attenuator to enhance signal strength
- C. They provide minimal interruption to services by being hot-swappable
- D. They offer reliable bandwidth up to 100 Mbps in half duplex mode

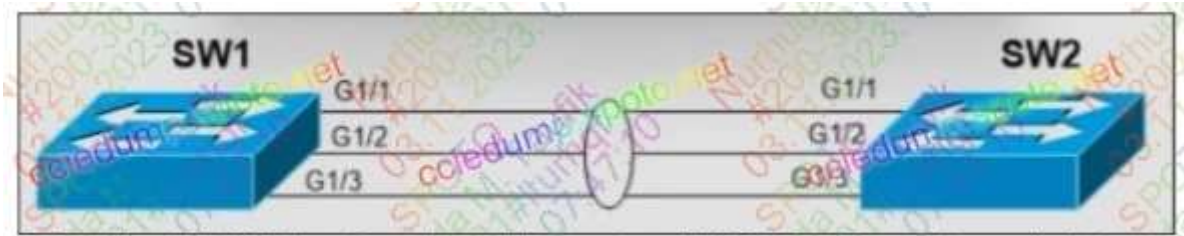
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 180



Refer to the exhibit. Which configuration establishes a Layer 2 LACP EtherChannel when applied to both switches?

- A. interface range G1/1-1/3
switchport mode access
channel-group 1 mode passive
no shutdown
- B. interface range G1/1-1/3
switchport mode access
channel-group 1 mode on
no shutdown
- C. interface range G1/1-1/3
switchport mode trunk
channel-group 1 mode desirable
no shutdown
- D. interface range G1/1-1/3
switchport mode trunk
channel-group 1 mode active
no shutdown

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 181



Refer to the exhibit. An engineer configured the New York router with static routes that point to the Atlanta and Washington sites. Which command must be configured on the Atlanta and Washington routers so that both sites are able to reach the loopback2 interface on the New York router?

- A. ipv6 route ::/0 2000::2
- B. ipv6 route ::/0 Serial 0/0/0
- C. ip route 0.0.0.0.0.0.0.0 Serial 0/0/0
- D. ipv6 route ::/0>Serial 0/0/1
- E. ipv6 route 0/0 Serial 0/0/0

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 182

Which WPA mode uses PSK authentication?

- A. Personal
- B. Local
- C. Client
- D. Enterprise

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 183

What describes a northbound REST API for SDN?

- A. application-facing interface for SNMP GET requests
- B. network-element-facing interface for GET, POST, PUT, and DELETE methods
- C. network-element-facing interface for the control and data planes
- D. application-facing interface for GET, POST, PUT, and DELETE methods

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 184

The address block 192.168.32.0/24 must be subnetted into smaller networks. The engineer must meet these requirements:

1. Create 8 new subnets.
2. Each subnet must accommodate 30 hosts.
3. Interface VLAN 10 must use the last usable IP in the first new subnet.
4. A Layer 3 interface is used.

Which configuration must be applied to the interface?

- A. `no switchport mode access`
`ip address 192.168.32.62 255.255.255.240`
- B. `no switchport`
`ip address 192.168.32.30 255.255.255.224`
- C. `no switchport mode trunk`
`ip address 192.168.32.97 255.255.255.224`
- D. `switchport`
`ip address 192.168.32.65 255.255.255.240`

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 185

Why is UDP more suitable than TCP for applications that require low latency, such as VoIP?

- A. TCP uses congestion control for efficient packet delivery, and UDP uses flow control mechanisms for the delivery of packets.
- B. TCP sends an acknowledgment for every packet that is received, and UDP operates without acknowledgments.
- C. UDP uses sequencing data for packets to arrive in order, and TCP offers the capability to receive packets in random order.
- D. UDP reliably guarantees delivery of all packets, and TCP drops packets under heavy load.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 186

What occurs when overlapping Wi-Fi channels are implemented?

- A. Network communications are open to eavesdropping.
- B. Users experience poor wireless network performance.
- C. Wireless devices are unable to distinguish between different SSIDs.
- D. The wireless network becomes vulnerable to unauthorized access.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 187

What are two benefits for using private IPv4 addressing? (Choose two.)

- A. They offer Internet connectivity to endpoints on private networks
- B. They allow for Internet access from IoT devices.
- C. They supply redundancy in the case of failure.
- D. They provide a layer of security from Internet threats.
- E. They alleviate the shortage of public IPv4 addresses.

Correct Answer: DE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 188

Which group of channels in the 802.11b/g/n/ac 2.4 GHz frequency bands are nonoverlapping channels?

- A. channels 1, 6, and 10
- B. channels 1, 5, and 10

- C. channels 1, 6, and 11
- D. channels 1, 5, and 11

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 189

A manager asks a network engineer to recommend a cloud service model so that employees do not spend time installing, managing, and updating software that is only used occasionally. Which cloud service model does the engineer recommend?

- A. platform-as-a-service
- B. Infrastructure-as-a-service
- C. business process as a service to support different types of service
- D. software-as-a-service

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 190

How is a configuration change made to a wireless AP in lightweight mode?

- A. HTTPS connection directly to the out-of-band address of the AP
- B. CAPWAP/LWAPP connection via the parent WLC
- C. SSH connection to the management IP of the AP
- D. EoIP connection via the parent WLC

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 191

What are two functions of a server on a network? (Choose two)

- A. runs the same operating system in order to communicate with other servers
- B. achieves redundancy by exclusively using virtual server clustering
- C. runs applications that send and retrieve data for workstations that make requests
- D. handles requests from multiple workstations at the same time
- E. housed solely in a data center that is dedicated to a single client

Correct Answer: CD
Section: (none)
Explanation

Explanation/Reference:

QUESTION 192

Which occurs when PortFast is enabled on an interface that is connected to another switch?

- A. Root port choice and spanning tree recalculation are accelerated when a switch link goes down.
- B. Spanning-tree may fail to detect a switching loop in the network that causes broadcast storms.
- C. VTP is allowed to propagate VLAN configuration information from switch to switch automatically.
- D. After spanning tree converges, PortFast shuts down any port that receives BPDUs.

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 193

Which set of actions satisfy the requirement for multifactor authentication?

- A. The user enters a PIN into an RSA token, and then enters the displayed RSA key on a login screen
- B. The user swipes a key fob, then clicks through an email link
- C. The user enters a user name and password, and then re-enters the credentials on a second screen.
- D. The user enters a user name and password, and then clicks a notification an authentication app on a mobile device.

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 194


```

R1# show ip route | begin gateway
Gateway of last resort is not set
  172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
C    172.16.1.0/24 is directly connected, FastEthernet0/0
L    172.16.1.1/32 is directly connected, FastEthernet0/0
EX   172.16.2.0/24 [170/2] via 207.165.200.250, 00:00:25, Serial0/0/0
O    192.168.1.0/24 [110/84437] via 207.165.200.254, 00:00:17, Serial0/0/1
D    192.168.2.0/24 [90/184437] via 207.165.200.254, 00:00:15, Serial0/0/1
E1   192.168.3.0/24 [110/1851437] via 207.165.200.254, 00:00:19, Serial0/0/1
     207.165.200.0/24 is variably subnetted, 4 subnets, 2 masks
C    207.165.200.248/30 is directly connected, Serial0/0/0
L    207.165.200.249/32 is directly connected, Serial0/0/0
C    207.165.200.252/30 is directly connected, Serial0/0/1
L    207.165.200.253/32 is directly connected, Serial0/0/1

```

Refer to the exhibit. Which prefix did router R1 learn from internal EIGRP?

- A. 192.168.3.0/24
- B. 172.16.1.0/24
- C. 192.168.1.0/24
- D. 192.168.2.0/24

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 195

Which QoS traffic handling technique retains excess packets in a queue and reschedules these packets for later transmission when the configured maximum bandwidth has been surpassed?

- A. weighted random early detection
- B. traffic prioritization
- C. traffic policing
- D. traffic shaping

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 196



Refer to the exhibit. An access list is created to deny Telnet access from host PC-1 to RTR-1 and allow access from all other hosts. A Telnet attempt from PC-2 gives this message: "% Connection refused by remote host." Without allowing Telnet access from PC-1, which action must be taken to permit the traffic?

- A. Add the **access-list 10 permit any** command to the configuration.
- B. Remove the **access-class 10 in** command from line vty 0 4.
- C. Remove the **password** command from line vty 0 4.
- D. Add the **ip access-group 10 out** command to interface g0/0

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 197



Refer to the exhibit. Local access for R4 must be established and these requirements must be met:

1. Only Telnet access is allowed.
2. The enable password must be stored securely.
3. The enable password must be applied in plain text.
4. Full access to R4 must be permitted upon successful login

Which configuration script meets the requirements?

- A.
- ```
!
Conf t
!
username test1 password testpass1
enable secret level 15 0 Test123
!
line vty 0 15
login local transport
input telnet
```
- B.
- ```
!
Config t
!
username test1 password testpass1
enable password level 1 7 Test123
!
line vty 0 15
accounting exec default
transport input all
```
- C.
- ```
!
Config t
!
username test1 password testpass1
enable secret level 1 0 Test123
!
line vty 0 15
login authentication
password Test123
transport input telnet
```
- D.
- ```
!
Config t
!
username test1 password testpass1
enable password level 15 0 Test123
!
line vty 0 15
password Test123
transport input all
```

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 198

Which set of methods is supported with the REST API?

- A. GET, PUT, ERASE, CHANGE
- B. GET, POST, MOD, ERASE
- C. GET, PUT, POST, DELETE
- D. GET, POST, ERASE, CHANGE

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 199

Which two characteristics are representative of virtual machines (VMs)? (Choose two.)

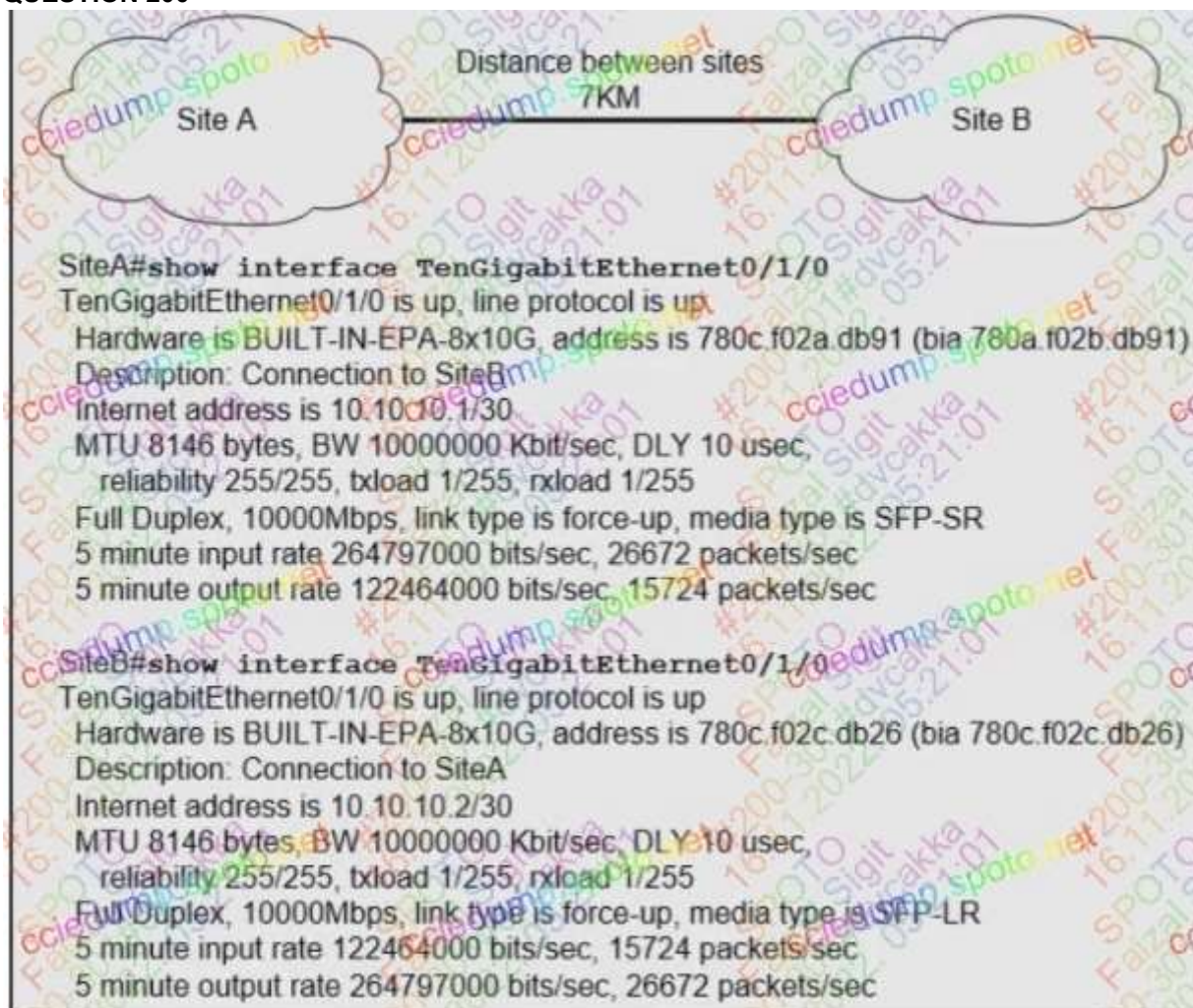
- A. A VM on a hypervisor is automatically interconnected to other VMs.
- B. A VM on an individual hypervisor shares resources equally
- C. Each VM runs independently of any other VM in the same hypervisor.
- D. multiple VMs operate on the same underlying hardware.
- E. Each VMs operating system depends on its hypervisor.

Correct Answer: CD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 200

Refer to the exhibit. Site A was recently connected to site B over a new single-mode fiber path. Users at site A report intermittent connectivity issues with applications hosted at site B. What is the reason for the problem?

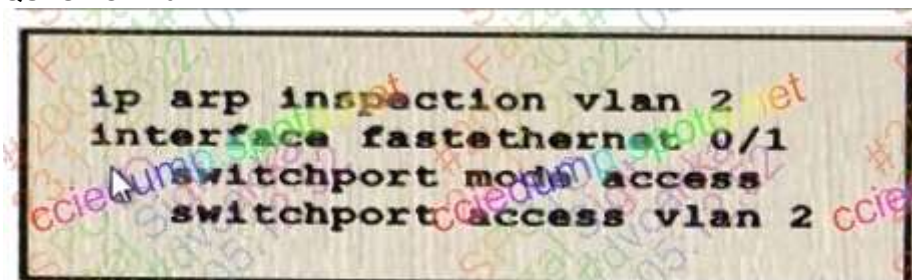
- A. The wrong cable type was used to make the connection.
- B. Heavy usage is causing high latency.
- C. Physical network errors are being transmitted between the two sites.
- D. An incorrect type of transceiver has been inserted into a device on the link.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 201

Refer to the exhibit. What is the effect of this configuration?

- A. The switch port remains administratively down until the interface is connected to another switch
- B. Dynamic ARP Inspection is disabled because the ARP ACL is missing
- C. The switch port remains down until it is configured to trust or untrust incoming packets
- D. The switch port interface trust state becomes untrusted

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 202

A wireless access point is needed and must meet these requirements:

1. "zero-touch" deployed and managed by a WLC
2. process only real-time MAC functionality
3. used in a split-MAC architecture

Which access point type must be used?

- A. autonomous
- B. cloud-based
- C. mesh
- D. lightweight

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 203

Which two transport layer protocols carry syslog messages? (Choose two.)

- A. TCP
- B. RTP
- C. UDP
- D. IP
- E. ARP

Correct Answer: AC

Section: (none)

Explanation

Explanation/Reference:

QUESTION 204

```
Designated Router (ID) 10.11.11.11, Interface address 10.10.10.1
Backup Designated router (ID) 10.3.3.3, Interface address 10.10.10.3
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
oob-resync timeout 40
Hello due in 00:00:08
Supports Link-local Signaling (LLS)
Cisco NSF helper support enabled
IETF NSF helper support enabled
Index 1/1/1, flood queue length 0
Next 0x0(0)/0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 6
Last flood scan time is 0 msec, maximum is 1 msec
Neighbor Count is 3, Adjacent neighbor count is 3
Adjacent with neighbor 10.1.1.4
Adjacent with neighbor 10.2.2.2
Adjacent with neighbor 10.3.3.3 (Backup Designated Router)
Suppress hello for 0 neighbor(s)
```

Refer to the exhibit. The show ip ospf interface command has been executed on R1. How is OSPF configured?

- A. The interface is not participating in OSPF
- B. The default Hello and Dead timers are in use.
- C. There are six OSPF neighbors on this interface
- D. A point-to-point network type is configured

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 205

What facilitates a Telnet connection between devices by entering the device name?

- A. Syslog
- B. SNMP
- C. NTP
- D. DNS lookup

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 206

What are two functions of a Layer 2 switch? (Choose two)

- A. moves packets between different VLANs
- B. acts as a central point for association and authentication servers
- C. selects the best route between networks on a WAN
- D. makes forwarding decisions based on the MAC address of a packet
- E. moves packets within a VLAN

Correct Answer: DE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 207

```
router# show ip route
....
D 172.16.32.0/26 [90/25789317] via 10.0.0.1
R 172.16.32.0/24 [120/4] via 10.0.0.2
O 172.16.32.0/19 [110/229840] via 10.0.0.3
C 172.16.32.32/32 is directly connected, Loopback0
C 172.16.32.4/30 is directly connected, GigabitEthernet0/0
```

Refer to the exhibit. A packet sourced from 172.16.32.254 is destined for 172.16.32.8. What is the subnet mask of the preferred destination route?

- A. 255.255.255.252
- B. 255.255.224.0
- C. 255.255.255.192
- D. 255.255.255.0

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 208

```
R1# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default
       U - per-user static route, o - ODR
Gateway of last resort is not set
C 172.16.0.0/16 is directly connected, Loopback0
  172.16.0/16 is variably subnetted, 4 subnets, 2 masks
O 172.16.1.3/3 [110/100] via 192.168.7.40, 00:39:08, Serial0
  172.16.1.0/24 is directly connected, Serial0
O 172.16.1.184/29 [110/5] via 192.168.7.35, 00:39:08, Serial0
O 172.16.3.0/24 [110/10] via 192.168.7.4, 00:39:08, Gigabit Ethernet 0/0
D 172.16.1.0/28 [90/10] via 192.168.7.7, 00:39:08, Gigabit Ethernet 0/0
```

Refer to the exhibit. Load-balanced traffic is coming in from the WAN destined to a host at 172.16.1.190. Which next-hop is used by the router to forward the request?

- A. 192.168.7.35
- B. 192.168.7.40
- C. 192.168.7.7
- D. 192.168.7.4

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 209

Which plane is centralized in software-defined networking?

- A. services
- B. application
- C. data

D. control

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 210



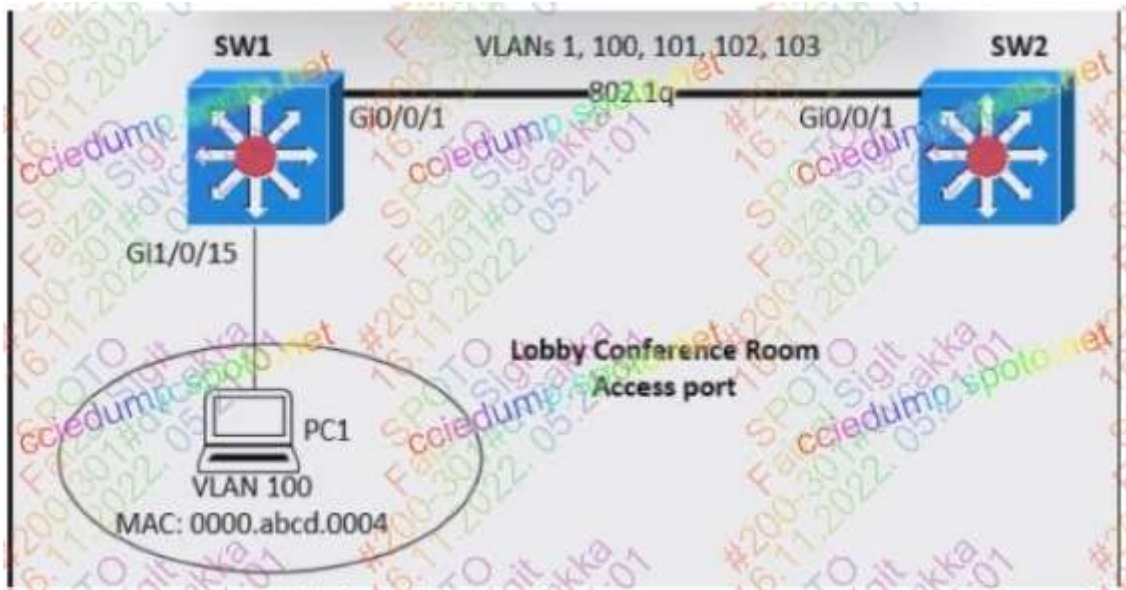
Refer to the exhibit. Which configuration issue is preventing the OSPF neighbor relationship from being established between the two routers?

- A. R1 interface Gi1/0 has a larger MTU size.
- B. R1 has an incorrect network command for interface Gi1/0.
- C. R2 is using the passive-interface default command.
- D. R2 should have its network command in area 1.

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 211



SW1 supports connectivity for a lobby conference room and must be secured. The engineer must limit the connectivity from PC1 to the SW1 and SW2 network. The MAC addresses allowed must be limited to two. Which configuration secures the conference room connectivity?

- A.

```
interface gi1/0/15
switchport port-security mac-address 0000.abcd.0004 vlan 100
```
- B.

```
interface gi1/0/15
switchport port-security mac-address 0000.abcd.0004 vlan 100
interface switchport secure-mac limit 2
```
- C.

```
interface gi1/0/15
switchport port-security
switchport port-security mac-address 0000.abcd.0004 vlan 100
```
- D.

```
interface gi1/0/15
switchport port-security
switchport port-security maximum 2
```

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 212

Why would a network administrator choose to implement automation in a network environment?

- A. to deploy the management plane separately from the rest of the network
- B. to centralize device information storage
- C. to implement centralized user account management
- D. to simplify the process of maintaining a consistent configuration state across all devices

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 213

What is a benefit of VRRP?

- A. It provides traffic load balancing to destinations that are more than two hops from the source.
- B. It allows neighbors to share routing table information between each other.
- C. It prevents loops in a Layer 2 LAN by forwarding all traffic to a root bridge, which then makes the final forwarding decision
- D. It provides the default gateway redundancy on a LAN using two or more routers

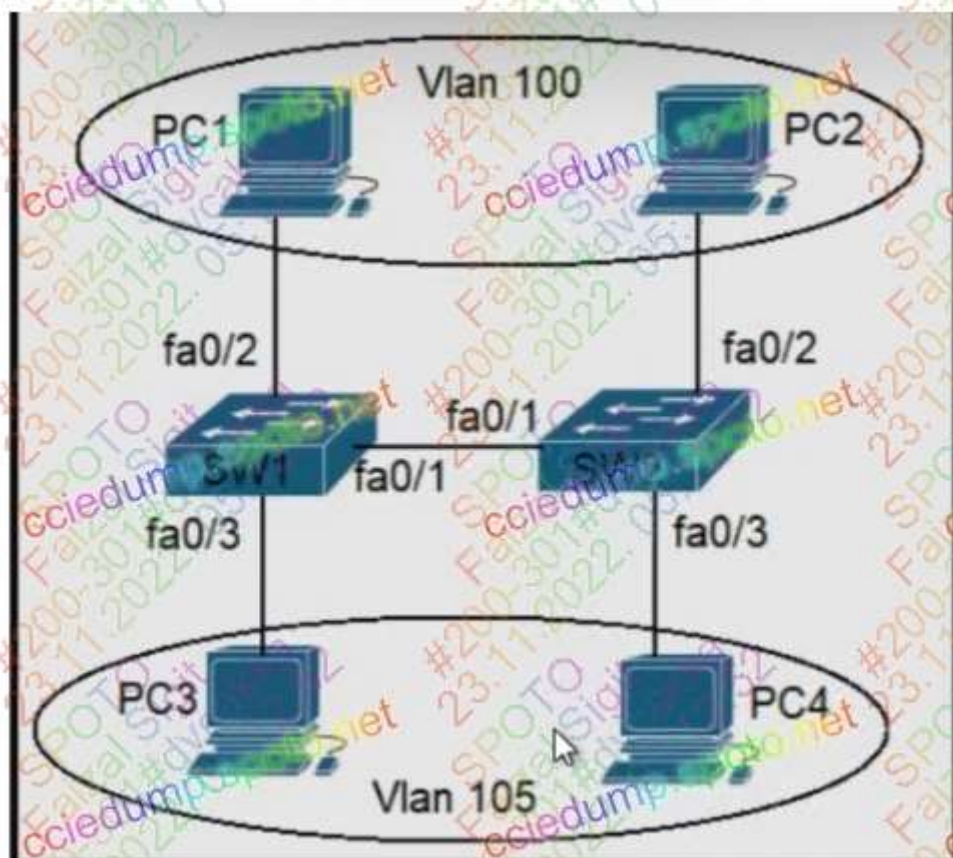
Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 214



Refer to the exhibit. An engineer configures interface fa0/1 on SW1 and SW2 to pass traffic from two different VLANs. For security reasons, company policy requires the native VLAN to be set to a non-default value. Which configuration must be used?

- A. Switch(config-if)# switchport mode access
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# switchport access vlan 100,105
Switch(config-if)# switchport trunk native vlan 3
- B. Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# switchport trunk allowed vlan 100,105
Switch(config-if)# switchport trunk native vlan 3
- C. Switch(config-if)# switchport mode dynamic
Switch(config-if)# switchport access vlan 100,105
Switch(config-if)# switchport trunk native vlan 1
- D. Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk encapsulation isl
Switch(config-if)# switchport trunk allowed vlan 100,105
Switch(config-if)# switchport trunk native vlan 1

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 215

Which type of IPv4 address type helps to conserve the globally unique address classes?

- A. Multicast
- B. Public
- C. private
- D. Loopback

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 216

What is a zero-day exploit?

- A. It is when the network is saturated with malicious traffic that overloads resources and bandwidth.
- B. It is when the perpetrator inserts itself in a conversation between two parties and captures or alters data.
- C. It is when a new network vulnerability is discovered before a fix is available.
- D. It is when an attacker inserts malicious code into a SQL server.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 217



Refer to the exhibit. The DHCP server is configured with a DHCP pool for each of the subnets represented. Which command must be configured on switch SW1 to allow DHCP clients on VLAN 10 to receive dynamic IP addresses from the DHCP server?

- A. SW1(config-if)# ip helper-address 192.168.20.2
- B. SW1(config-if)# ip helper-address 192.168.10.2
- C. SW1(config-if)# ip helper-address 192.168.10.1
- D. SW1(config-if)# ip helper-address 192.168.20.1

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 218



Refer to the exhibit. R1 learns all routes via OSPF. Which command configures a backup static route on R1 to reach the 192.168.20.0/24 network via R3?

- A. R1(config)# ip route 192.168.20.0 255.255.255.0 192.168.30.2
- B. R1(config)# ip route 192.168.20.0 255.255.255.0 192.168.30.2 90
- C. R1(config)# ip route 192.168.20.0 255.255.0.0 192.168.30.2
- D. R1 (config)# ip route 192.168.20.0 255.255.255.0 192.168.30.2 111

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 219

What is the purpose of classifying network traffic in QoS?

- A. configures traffic-matching rules on network devices
- B. services traffic according to its class
- C. writes the class identifier of a packet to a dedicated field in the packet header
- D. identifies the type of traffic that will receive a particular treatment

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 220

What does physical access control regulate?

- A. access to networking equipment and facilities
- B. access to computer networks and file systems
- C. access to servers to prevent malicious activity
- D. access to specific networks based on business function

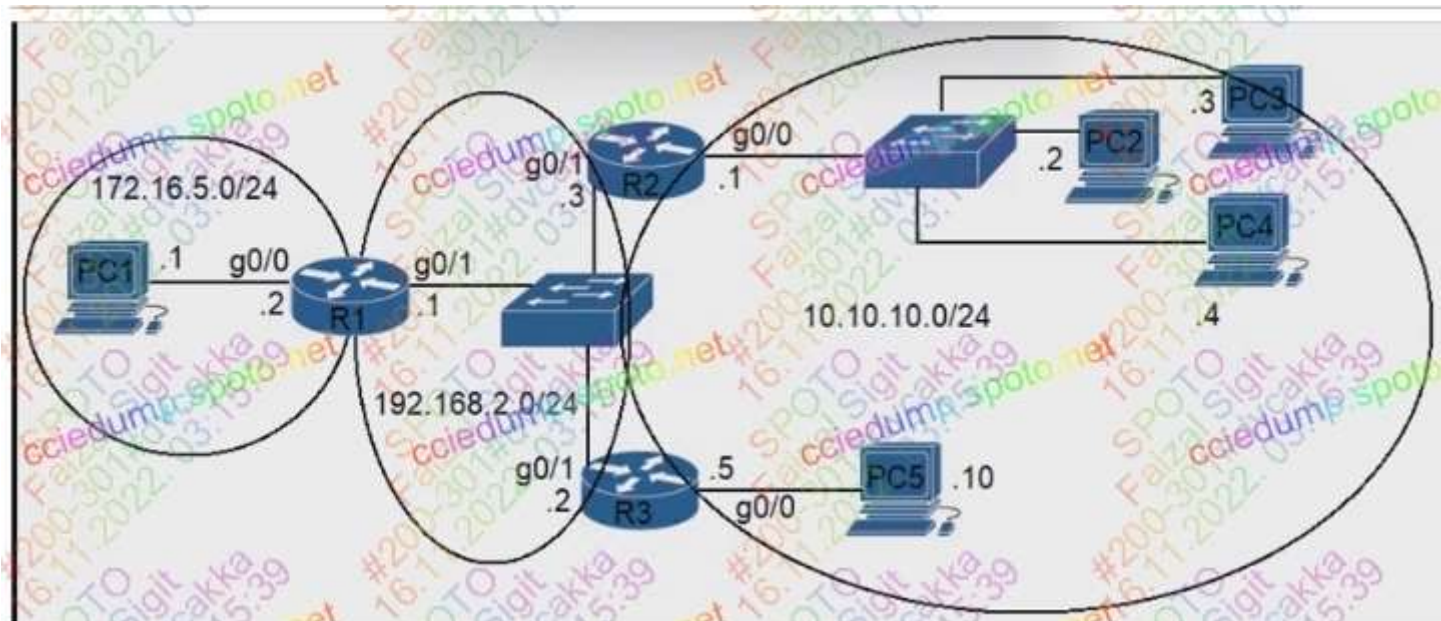
Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 221



Refer to the exhibit The router R1 is in the process of being configured. Routers R2 and R3 are configured correctly for the new environment. Which

two commands must be configured on R1 for PC1 to communicate to all PCs on the 10.10.10.0/24 network?

- A.

```
ip route 10.10.10.0 255.255.255.0 192.168.2.3
```

```
ip route 10.10.10.10 255.255.255.255 192.168.2.2
```
- B.

```
ip route 10.10.10.0 255.255.255.248 192.168.2.2
```

```
ip route 10.10.2.8 255.255.255.252 g0/1
```
- C.

```
ip route 10.10.10.0 255.255.255.0 192.168.2.2
```

```
ip route 10.10.2.2 255.255.255.255 10.10.10.10
```
- D.

```
ip route 10.10.10.0 255.255.255.0 192.168.2.3
```

```
ip route 10.10.10.8 255.255.255.252 g0/0
```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 222

Which WAN access technology is preferred for a small office/home office architecture?

- A. Integrated Services Digital Network switching
- B. broadband cable access
- C. frame-relay packet switching
- D. dedicated point-to-point leased line

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 223

Which protocol does a REST API use to communicate?

- A. HTTP
- B. STP
- C. SSH
- D. SNMP

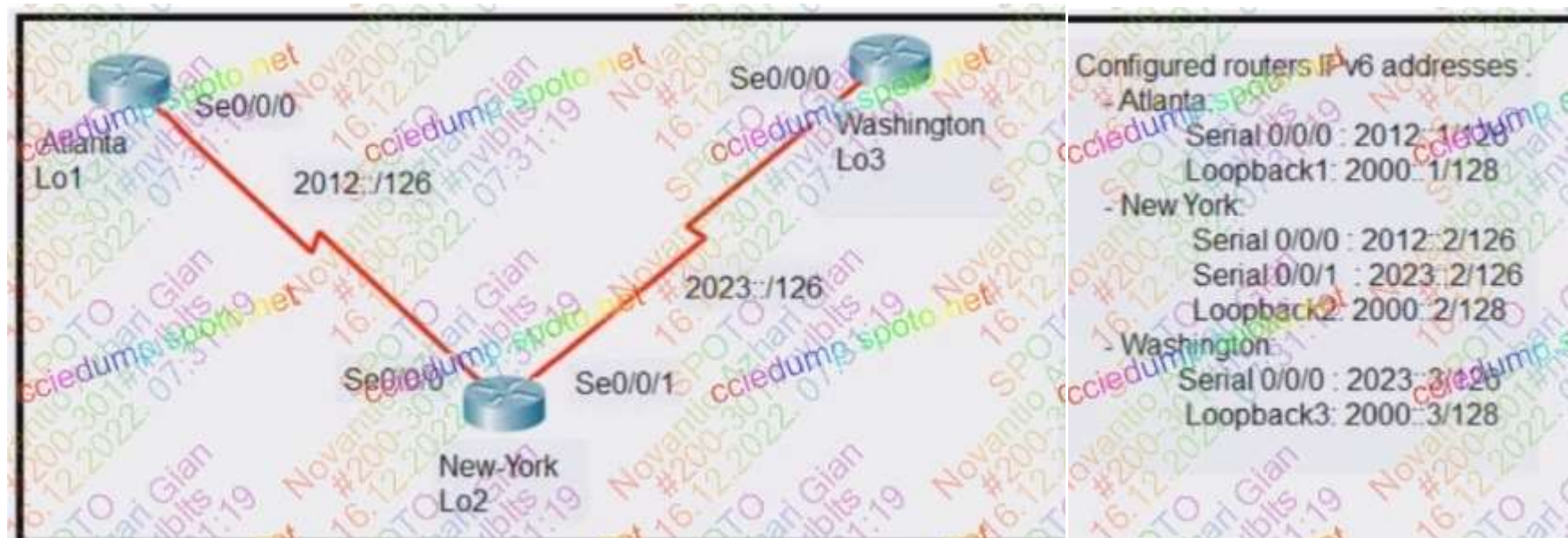
Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 224



Refer to the exhibit. The New York router is configured with static routes pointing to the Atlanta and Washington sites. Which two tasks must be performed so that the Serial0/0/0 interfaces on the Atlanta and Washington routers can reach one another? (Choose two.)

- A. Configure the ipv6 route 2023::/126 2012::1 command on the Atlanta router
- B. Configure the ipv6 route 2023:/126 2012::2 command on the Atlanta router.
- C. Configure the ipv6 route 2012:/126 2023::1 command on the Washington router.
- D. Configure the ipv6 route 2012::/126 s0/0/0 command on the Atlanta router.
- E. Configure the ipv6 route 2012::/126 2023::2 command on the Washington router

Correct Answer: BE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 225

Which feature on the Cisco Wireless LAN Controller when enabled restricts management access from specific networks?

- A. RADIUS
- B. CPU ACL
- C. Flex ACL
- D. TACACS

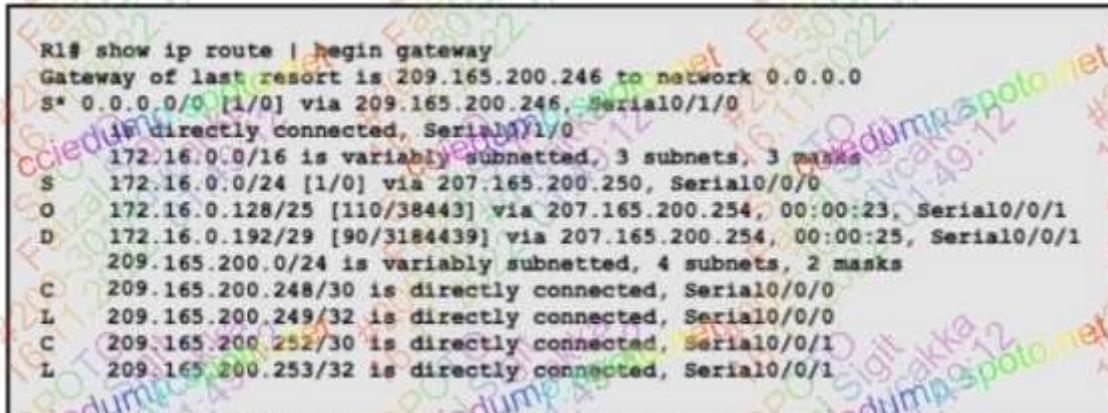
Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 226



```

R1# show ip route | begin gateway
Gateway of last resort is 209.165.200.246 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 209.165.200.246, Serial0/1/0
    is directly connected, Serial0/1/0
S 172.16.0.0/16 is variably subnetted, 3 subnets, 3 masks
S 172.16.0.0/24 [1/0] via 207.165.200.250, Serial0/0/0
O 172.16.0.128/25 [110/38443] via 207.165.200.254, 00:00:23, Serial0/0/1
D 172.16.0.192/29 [90/3184439] via 207.165.200.254, 00:00:25, Serial0/0/1
    209.165.200.0/24 is variably subnetted, 4 subnets, 2 masks
C 209.165.200.248/30 is directly connected, Serial0/0/0
L 209.165.200.249/32 is directly connected, Serial0/0/0
C 209.165.200.252/30 is directly connected, Serial0/0/1
L 209.165.200.253/32 is directly connected, Serial0/0/1
  
```

Refer to the exhibit. With which metric was the route to host 172.16.0.202 learned?

- A. 0
- B. 38443
- C. 3184439
- D. 110

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 227

Which command creates a static NAT binding for a PC address of 10.1.1.1 to the public routable address 209.165.200.225 assigned to the PC?

- A. R1(config)#ip nat outside source static 10.1.1.1 209.165.200.225
- B. R1(config)#ip nat inside source static 209.165.200.225 10.1.1.1
- C. R1(config)#ip nat outside source static 209.165.200.225 10.1.1.1
- D. R1(config)#ip nat inside source static 10.1.1.1 209.165.200.225

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 228

What is a function of Opportunistic Wireless Encryption in an environment?

- A. protect traffic on open networks
- B. provide authentication
- C. offer compression
- D. increase security by using a WEP connection

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 229

What is a capability of FTP in network management operations?

- A. offers proprietary support at the session layer when transferring data
- B. encrypts data before sending between data resources
- C. devices are directly connected and use UDP to pass file information
- D. uses separate control and data connections to move files between server and client

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 230

A network engineer must configure an interface with IP address 10.10.10.145 and a subnet mask equivalent to 11111111.1.11111111.11111111.11111000. Which subnet mask must the engineer use?

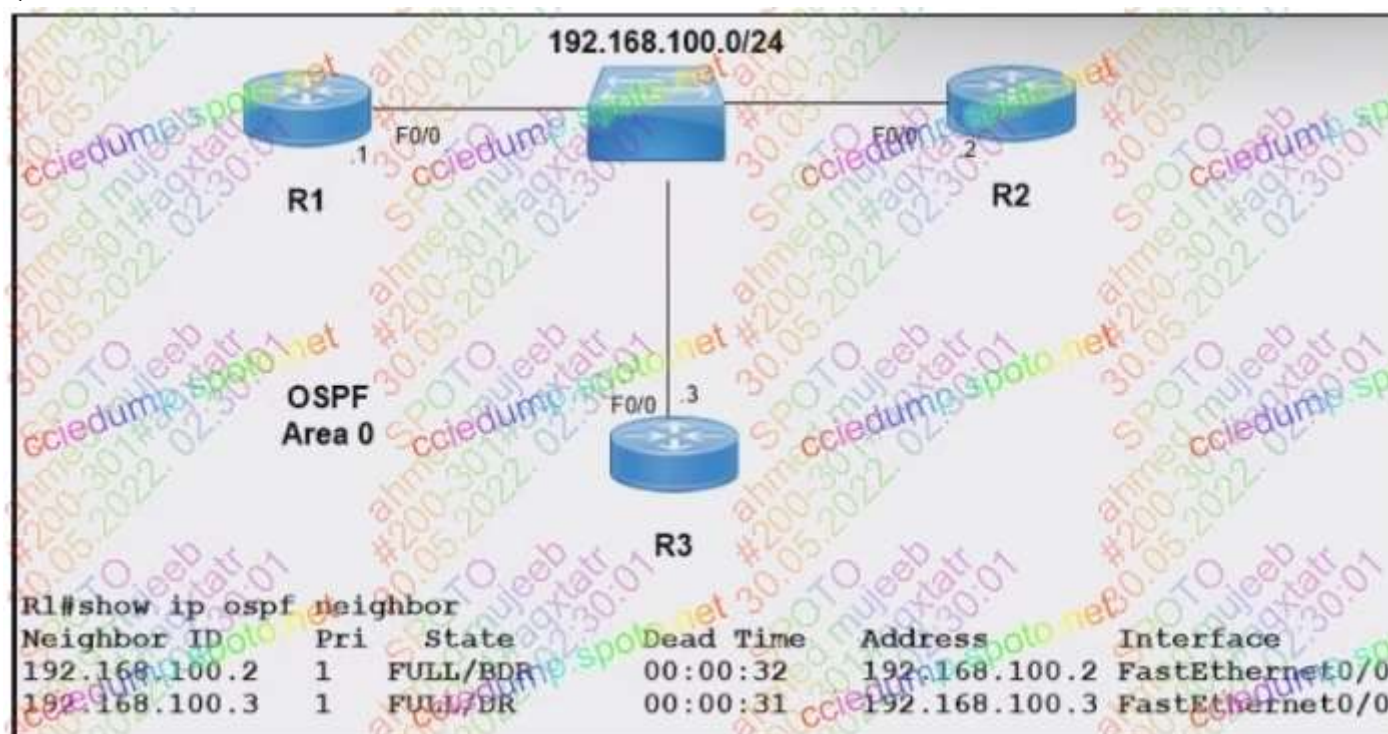
- A. /29
- B. /28
- C. /30
- D. /27

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

11111111.11111111.11111111.11111000 = 255.255.255.248 = /29

QUESTION 231



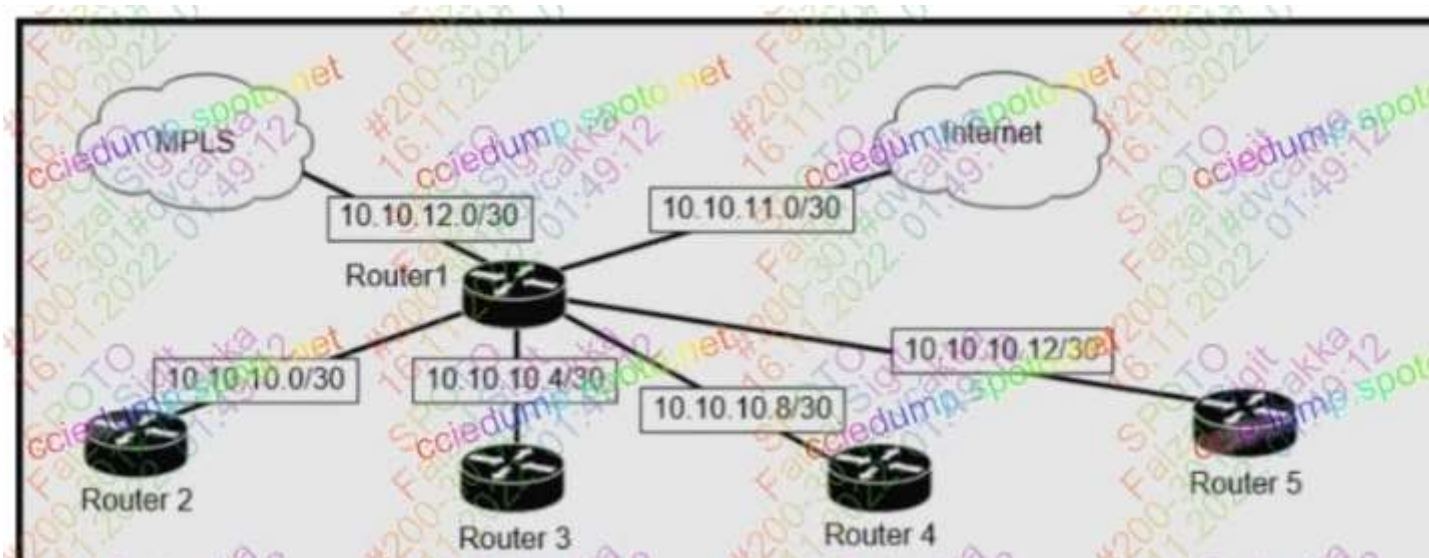
Refer to the exhibit. Which two configurations must the engineer apply on this network so that R1 becomes the DR? (Choose two.)

- A. R1(config)# router ospf 1
R1(config-router)# router-id 192.168.100.1
- B. R1(config)# interface fastethernet 0/0
R1(config-if)# ip ospf priority 0
- C. R1(config)# interface fastethernet 0/0
R1(config-if)# ip ospf priority 200
- D. R3(config)# interface fastethernet 0/0
R3(config-if)# ip ospf priority 200
- E. R3(config)# interface fastethernet 0/0
R3(config-if)# ip ospf priority 0

Correct Answer: CE
Section: (none)
Explanation

Explanation/Reference:

QUESTION 232




```

Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0
209.165.200.0/27 is subnetted, 1 subnets
B    209.165.200.224 [20/0] via 10.10.12.2, 02:26:14
209.165.201.0/27 is subnetted, 1 subnets
B    209.165.201.0 [20/0] via 10.10.12.2, 02:26:03
209.165.202.0/27 is subnetted, 1 subnets
B    209.165.202.128 [20/0] via 10.10.12.2, 02:26:03
10.0.0.0/8 is variably subnetted, 10 subnets, 4 masks
O    10.10.13.0/25 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O    10.10.13.128/28 [110/2] via 10.10.10.5, 00:00:12, GigabitEthernet0/1
O    10.10.13.144/28 [110/2] via 10.10.10.5, 00:01:57, GigabitEthernet0/2
O    10.10.13.160/29 [110/2] via 10.10.10.5, 00:00:12, GigabitEthernet0/1
O    10.10.13.208/29 [110/2] via 10.10.10.13, 00:01:57, GigabitEthernet0/3
S*   0.0.0.0/0 [1/0] via 10.10.11.2

```

Refer to the exhibit. To which device does Router1 send packets that are destined to host 10.10.13.165?

- A. Router5
- B. Router3
- C. Router2
- D. Router4

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 233

An engineer observes high usage on the 2.4GHz channels and lower usage on the 5GHz channels. What must be configured to allow clients to preferentially use 5GHz access points?

- A. Re-Anchor Roamed Clients
- B. 11ac MU-MIMO
- C. EAP Split Tunnel
- D. Client Band Select

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 234

Which component controls and distributes physical resources for each virtual machine?

- A. Hypervisor
- B. CPU
- C. physical enclosure
- D. OS

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 235

```

{
  "Test Questions" : [
    "Automation",
    "Configuration",
  ],
  "Test_Exam_Level" : [
    "CCNA",
    "CCNP",
  ],
  "Test_Response" : [
    "Correct",
    "Incorrect",
  ]
}

```

Refer to the exhibit. How many arrays are present in the JSON data?

- A. nine
- B. One
- C. Three
- D. Six

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 236

Which cable type must be used when connecting a router and switch together using these:

1. Pins 1 and 2 are receivers and pins 3 and 6 are transmitters
2. Auto detection MDI-X is unavailable

- A. rollover
- B. crossover
- C. console
- D. straight-through

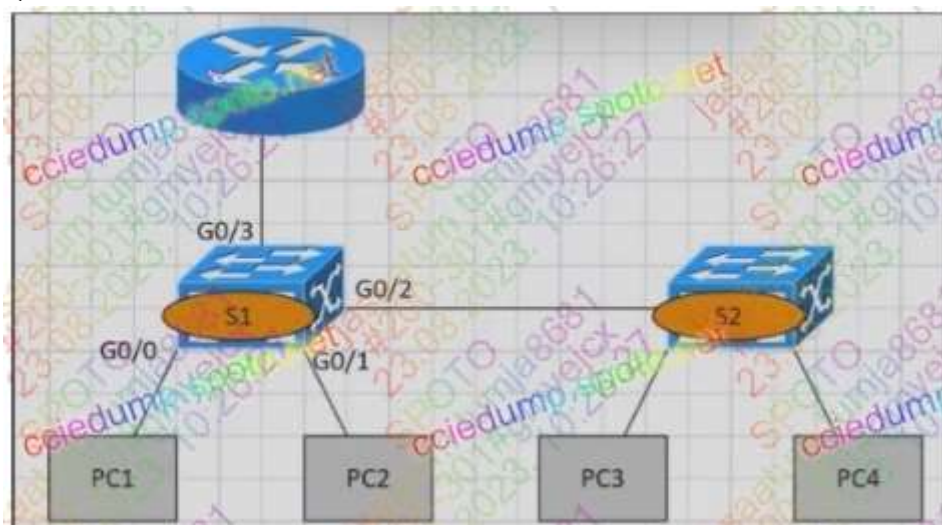
Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 237



Refer to the exhibit. PC1 tries to ping PC3 for the first time and sends out an ARP to S1. Which action is taken by S1?

- A. It forwards it out interface G0/2 only.
- B. It forwards it out G0/3 only.
- C. It drops the frame.
- D. It is flooded out every port except G0/0.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 238

Which two encoding methods are supported by REST APIs? (Choose two.)

- A. EBCDIC
- B. SGML
- C. JSON
- D. XML
- E. YAML

Correct Answer: CD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 239

Which communication interaction takes place when a southbound API is used?

- A. between network applications and switches and routers on the network
- B. between the SDN controller and PCs on the network
- C. between the SDN controller and services and applications on the network
- D. between the SDN controller and switches and routers on the network

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 240


```
Switch(config)#hostname R1
R1(config)#interface FastEthernet0/1
R1(config-if)#no switchport
R1(config-if)#ip address 10.100.20.42 255.255.255.0
R1(config-if)#line vty 0 4
R1(config-line)#login
```

Refer to the exhibit An engineer booted a new switch and applied this configuration via the console port. Which additional configuration must be applied to allow administrators to authenticate directly to enable privilege mode via Telnet using a local username and password?

- A. R1(config)#username admin privilege 15 secret p@ss1234
R1(config-if)#line vty 0 4
R1(config-line)#login local
- B. R1(config)#username admin
R1(config-if)#line vty 0 4
R1(config-line)#password p@ss1234
R1(config-line)#transport input telnet
- C. R1(config)#username admin secret p@ss1234
R1(config-if)#line vty 0 4
R1(config-line)#login local
R1(config)#enable secret p@ss1234
- D. R1(config)#username admin
R1(config-if)#line vty 0 4
R1(config-line)#password p@ss1234

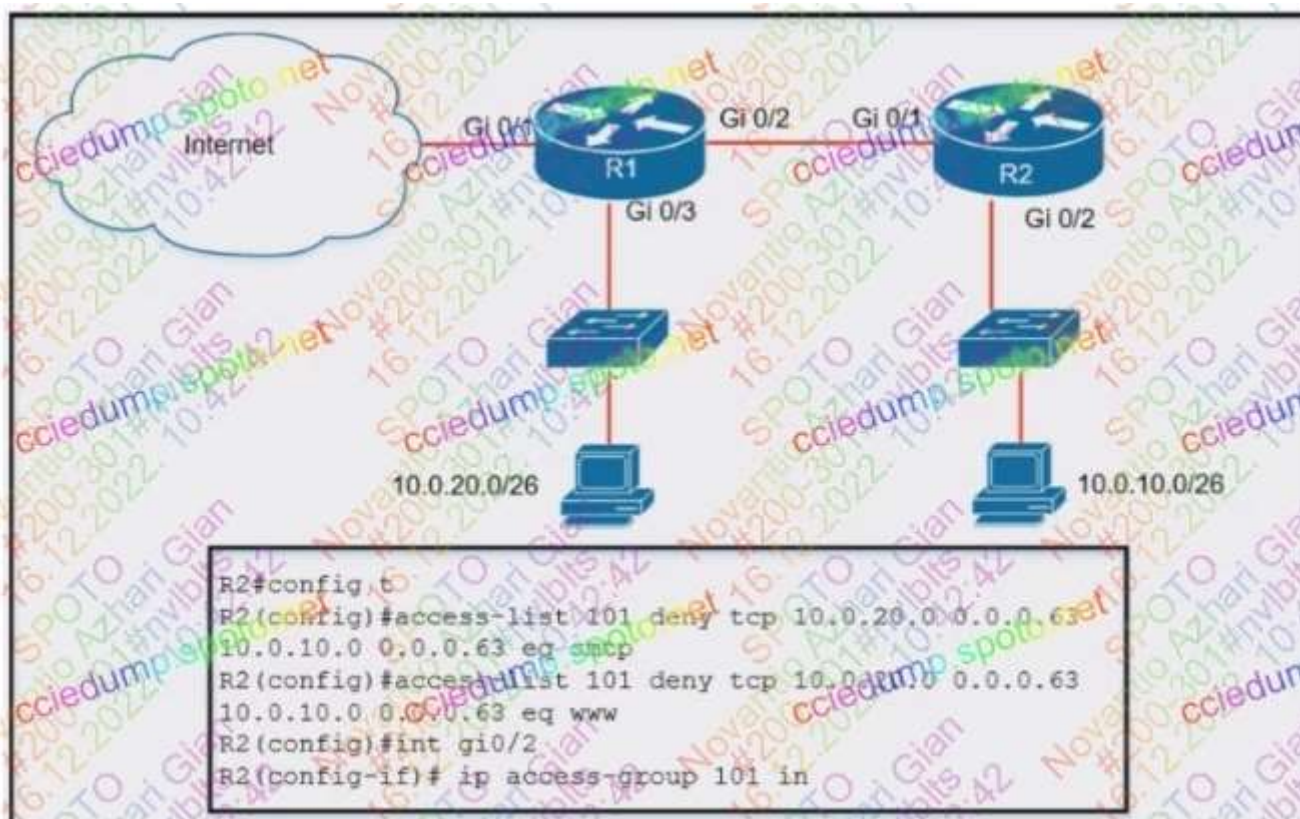
Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 241



Refer to the exhibit. An extended ACL has been configured and applied to router R2. The configuration failed to work as intended. Which two changes stop outbound traffic on TCP ports 25 and 80 to 10.0.20.0/26 from the 10.0.10.0/26 subnet while still allowing all other traffic? (Choose two)

- A. Add a "permit ip any any" statement to the beginning of ACL 101 for allowed traffic.
- B. The ACL must be moved to the Gi0/1 interface outbound on R2.
- C. Add a "permit ip any any" statement at the end of ACL 101 for allowed traffic.
- D. The source and destination IPs must be swapped in ACL 101.
- E. The ACL must be configured the Gi0/2 interface inbound on R1.

Correct Answer: CD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 242

Which Rapid PVST+ feature should be configured on a switch port to immediately send traffic to a connected server as soon as it is active?

- A. portfast
- B. loop guard
- C. uplinkfast
- D. BPDU guard

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 243

Which networking function occurs on the data plane?

- A. forwarding remote client/server traffic
- B. sending and receiving OSPF Hello packets
- C. processing inbound SSH management traffic
- D. facilitates spanning-tree elections

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 244

What is a function of Layer 3 switches?

- A. They transmit broadcast traffic when operating in Layer 3 mode exclusively
- B. They forward Ethernet frames between VLANs using only MAC addresses.
- C. They route traffic between devices in different VLANs.
- D. They move frames between endpoints limited to IP addresses.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 245

Which port type supports the **spanning-tree portfast** command without additional configuration?

- A. Layer 3 subinterfaces
- B. trunk ports
- C. Layer 3 main Interfaces
- D. access ports

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 246

What is a function performed by a web server?

- A. securely store files for FTP access
- B. authenticate and authorize a user's identity
- C. send and retrieve email from client devices
- D. provide an application that is transmitted over HTTP

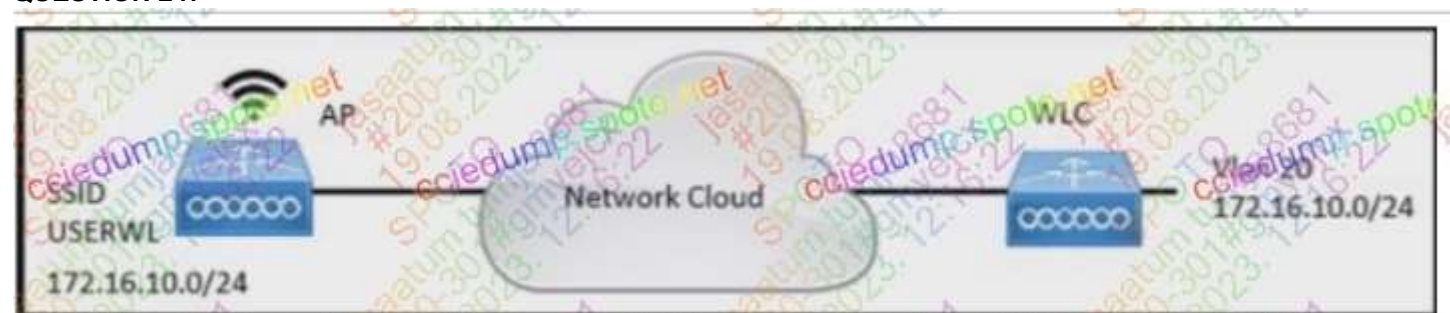
Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 247



Refer to the exhibit. A network engineer is configuring a WLAN to connect with the 172.16.10.0/24 network on VLAN 20. The engineer wants to limit the number of devices that connect to the WLAN on the USERWL SSID to 125. Which configuration must the engineer perform on the WLC?

- A. In the Advanced configuration, set the DTIM value to 125.
- B. In the WLAN configuration, set the Maximum Allowed Clients value to 125.
- C. In the Management Software activation configuration, set the Clients value to 125.
- D. In the Controller IPv6 configuration, set the Throttle value to 125.

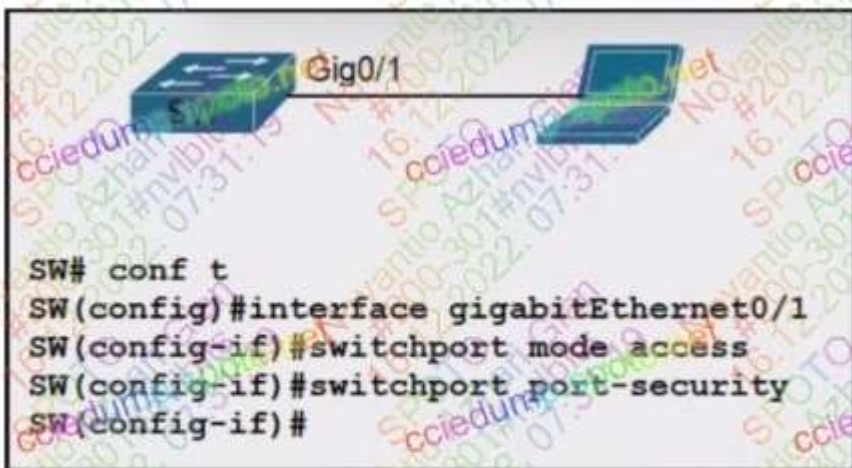
Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 248



Refer to the exhibit. A network engineer started to configure port security on a new switch. These requirements must be met:

1. MAC addresses must be learned dynamically.
2. Log messages must be generated without disabling the interface when unwanted traffic is seen.

Which two commands must be configured to complete this task? (Choose two.)

- A. SW(config-if)#switchport port-security violation shutdown
- B. SW(config-if)#switchport port-security mac-address 0010.XXXX.XXXX
- C. SW(config-if)#switchport port-security maximum 2
- D. SW(config-if)#switchport port-security mac-address sticky
- E. SW(config-if)#switchport port-security violation restrict

Correct Answer: DE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 249

What does an SDN controller use as a communication protocol to relay forwarding changes to a southbound API?

- A. XML
- B. REST
- C. OpenFlow
- D. Java

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 250

Two switches have been implemented and all interfaces are at the default configuration level. A trunk link must be implemented between two switches with these requirements:

1. using an industry-standard trunking protocol
2. permitting VLANs 1-10 and denying other VLANs

How must the interconnecting ports be configured?

- A. switchport mode dynamic
channel-protocol lacp
switchport trunk allowed vlans 1-10
- B. switchport mode trunk
switchport trunk allowed vlans 1-10
switchport trunk native vlan 11
- C. switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlans 1-10
- D. switchport mode dynamic desirable
channel-group 1 mode desirable
switchport trunk encapsulation isl
switchport trunk allowed vlan except 11-4094

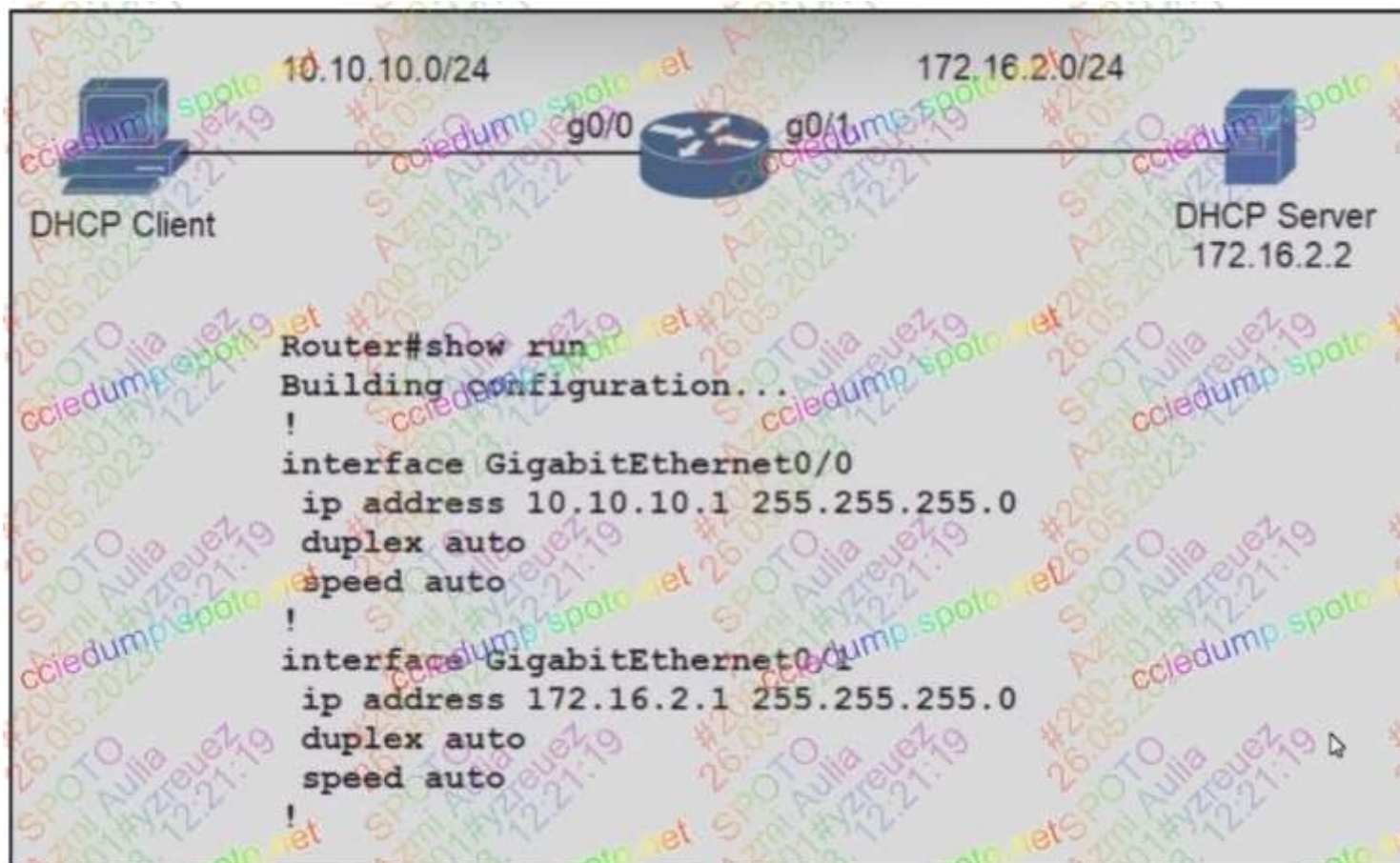
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 251



Refer to the exhibit. An engineer is configuring a new router on the network and applied this configuration. Which additional configuration allows the PC to obtain its IP address from a DHCP server?

- A. Configure the ip helper-address 172.16.2.2 command under interface Gi0/0.
- B. Configure the ip address dhcp command under interface Gi0/0.
- C. Configure the ip dhcp relay information command under interface Gi0/1.
- D. Configure the ip dhcp smart-relay command globally on the router.

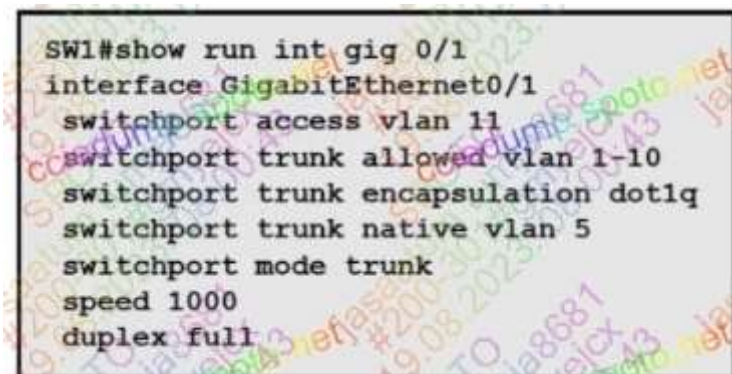
Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 252



Refer to the exhibit. Which action is expected from SW1 when the untagged frame is received on the GigabitEthernet0/1 interface?

- A. The frame is processed in VLAN 1.
- B. The frame is dropped.
- C. The frame is processed in VLAN 11.
- D. The frame is processed in VLAN 5.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 253

What are two facts that differentiate optical-fiber cabling from copper cabling? (Choose two.)

- A. It provides greater throughput options.
- B. It is less expensive when purchasing patch cables.
- C. It has a greater sensitivity to changes in temperature and moisture.
- D. It carries electrical current further distances for PoE devices.
- E. It carries signals for longer distances.

Correct Answer: AE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 254

Which type of security program is violated when a group of employees enters a building using the ID badge of only one person?

- A. physical access control
- B. user awareness
- C. network authorization
- D. intrusion detection

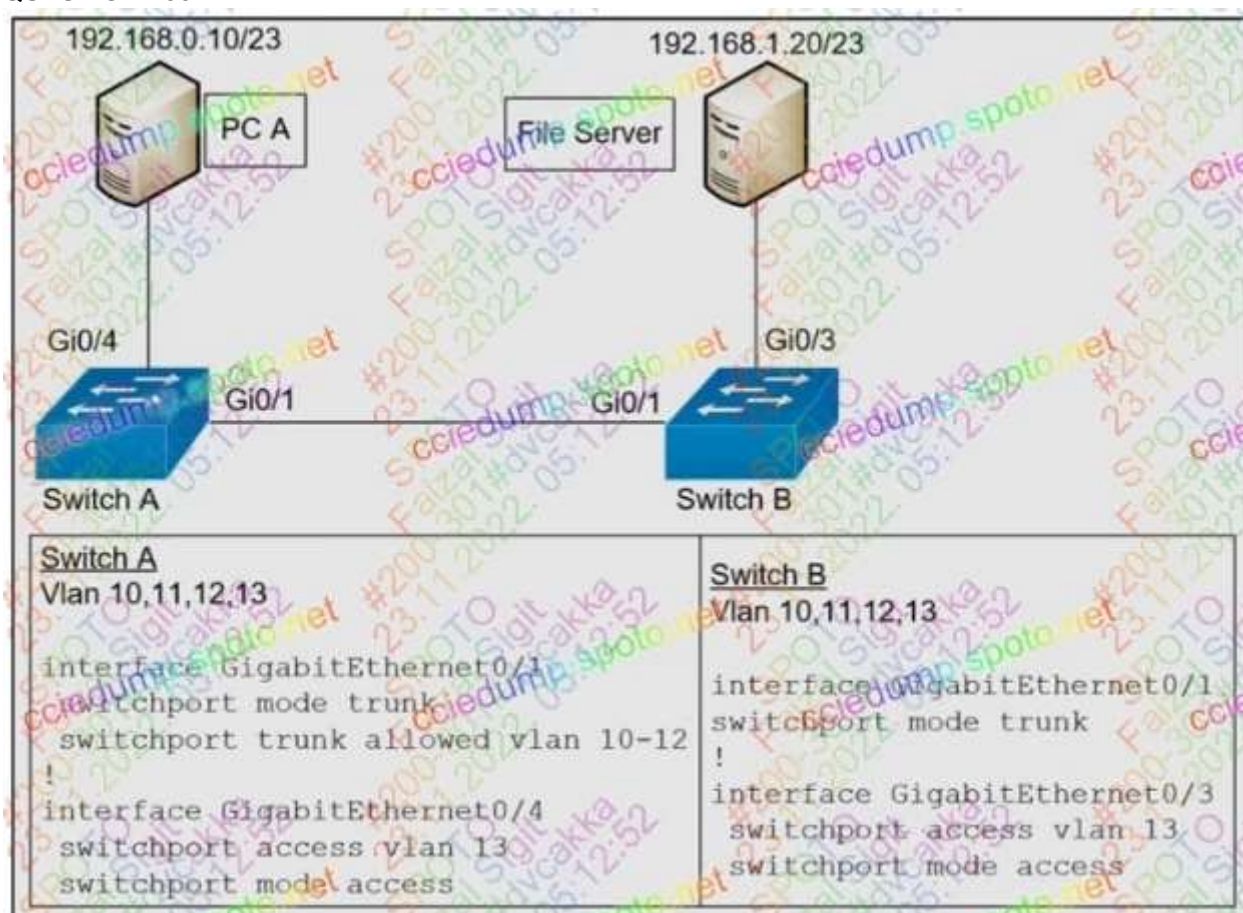
Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 255



Refer to the exhibit. A network engineer must configure communication between PC A and the File Server. To prevent interruption for any other communication which command must be configured?

- A. switchport trunk allowed vlan add 13
- B. switchport trunk allowed vlan 12
- C. switchport trunk allowed vlan none
- D. switchport trunk allowed vlan remove 10-11

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 256

What is the purpose of the **ip address dhcp** command?

- A. to configure an interface as a DHCP relay
- B. to configure an interface as a DHCP client
- C. to configure an interface as a DHCP helper
- D. to configure an interface as a DHCP server

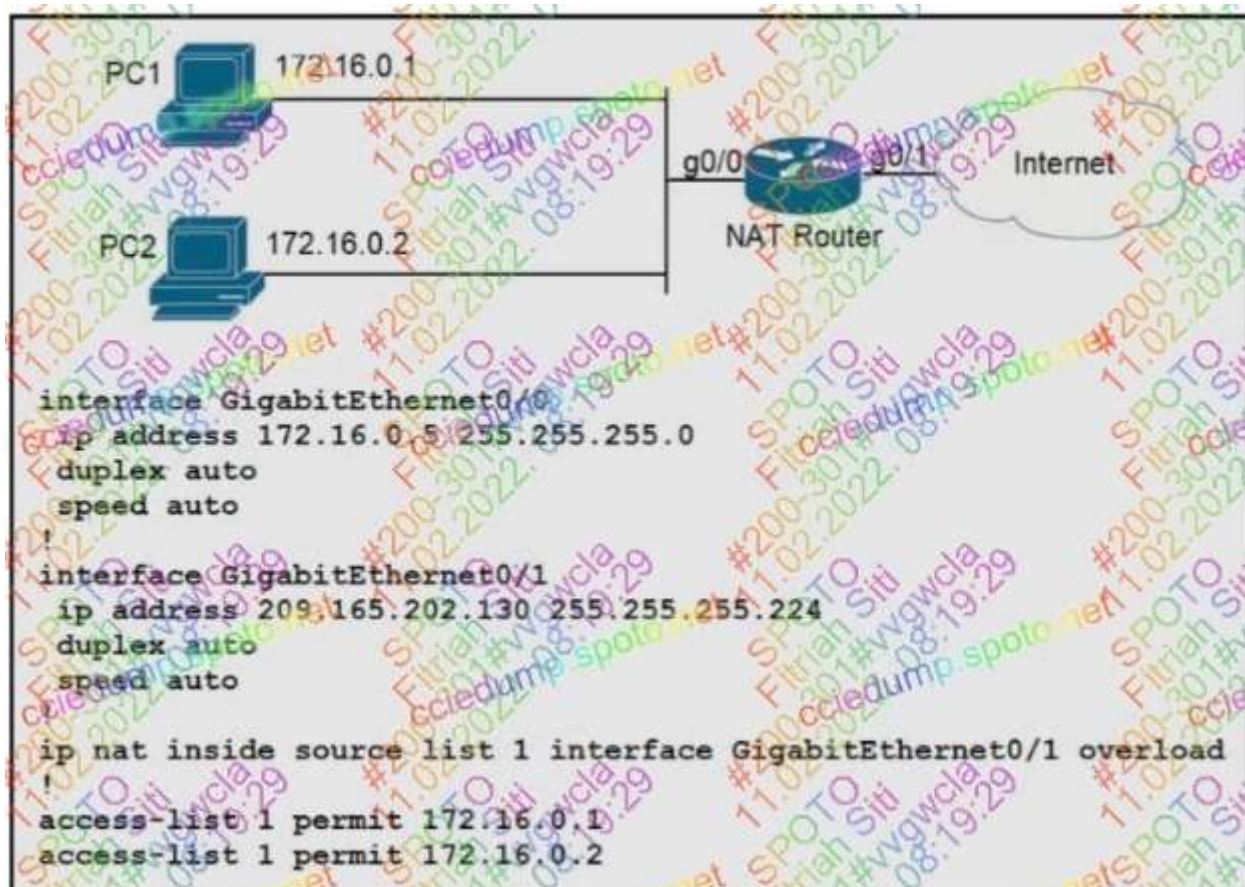
Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 257



Refer to the exhibit. How should the configuration be updated to allow PC1 and PC2 access to the internet?

- A. Add either the **ip nat (inside | outside)** command under both interfaces.
- B. Remove the **overload** keyword from the **ip nat inside source** command.
- C. Modify the configured number of the second access list.
- D. Change the **ip nat inside source** command to use interface GigabitEthernet0/0.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 258

Which technology must be implemented to configure network device monitoring with the highest security?

- A. NetFlow
- B. SNMPv3
- C. Syslog
- D. IP SLA

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 259

Which wireless security protocol relies on Perfect Forward Secrecy?

- A. WPA3
- B. WPA
- C. WPA2
- D. WEP

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 260

Which Cisco proprietary protocol ensures traffic recovers immediately, transparently, and automatically when edge devices or access circuits fail?

- A. SLB
- B. FHRP
- C. VRRP
- D. HSRP

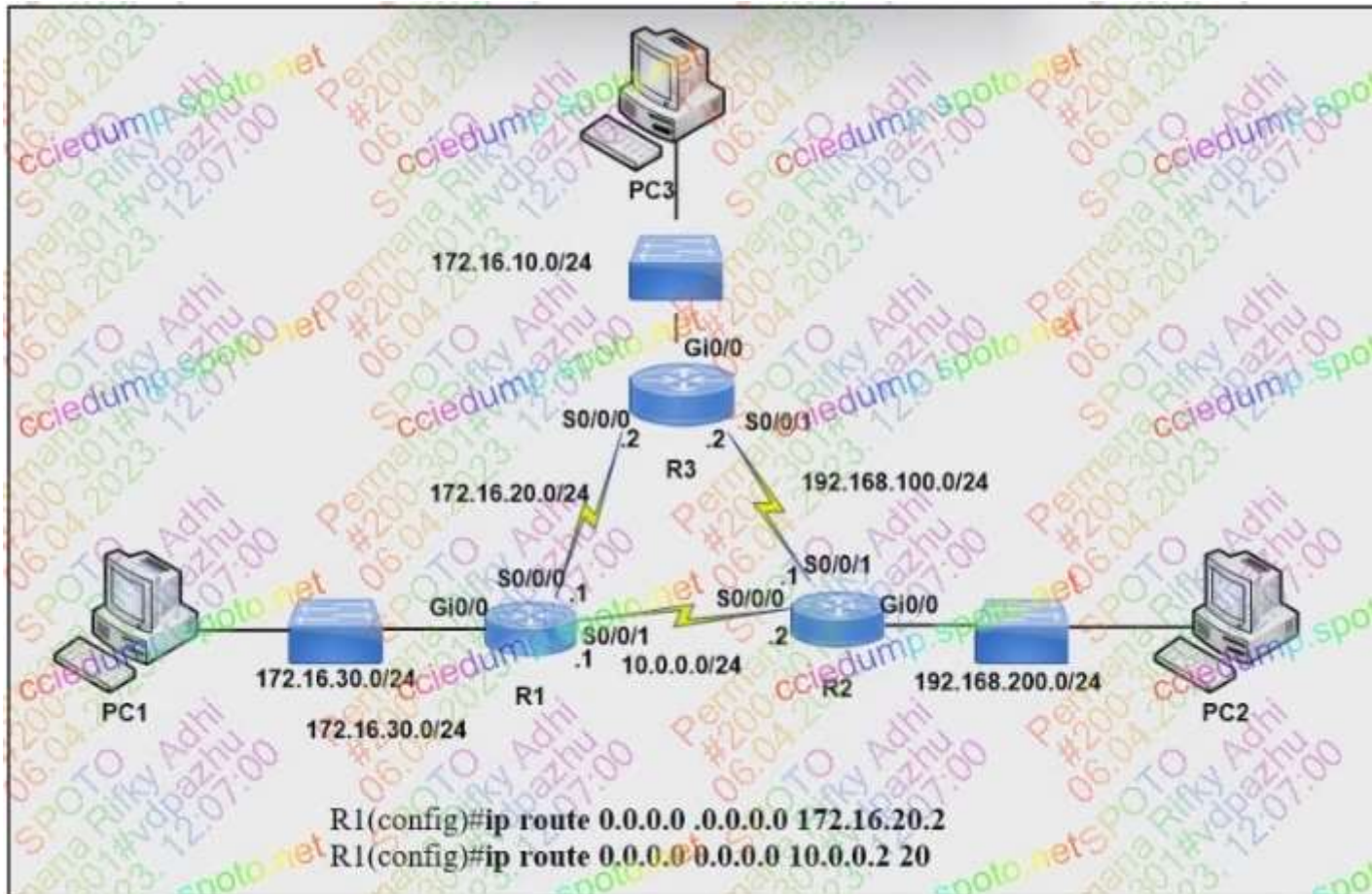
Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 261



Refer to the exhibit. After applying this configuration to router R1, a network engineer is verifying the implementation. If all links are operating normally, and the engineer sends a series of packets from PC1 to PC3, how are the packets routed?

- A. They are distributed sent round robin to interfaces S0/0/0 and S0/0/1.
- B. They are routed to 192.168.100.2.
- C. They are routed to 172.16.20.2.
- D. They are routed to 1.0.0.0.2.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 262

In which circumstance would a network architect decide to implement a global unicast subnet instead of a unique local unicast subnet?

- A. when the subnet must be routable over the internet
- B. when the subnet does not need to be routable
- C. when the subnet must be available only within an organization
- D. when the addresses on the subnet must be equivalent to private IPv4 addresses

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 263

What is the benefit of configuring PortFast on an interface?

- A. Real-time voice and video frames entering the interface are processed faster
- B. After the cable is connected, the interface uses the fastest speed setting available for that cable type.
- C. After the cable is connected, the interface is available faster to send and receive user data.
- D. The frames entering the interface are marked with higher priority and then processed faster by a switch

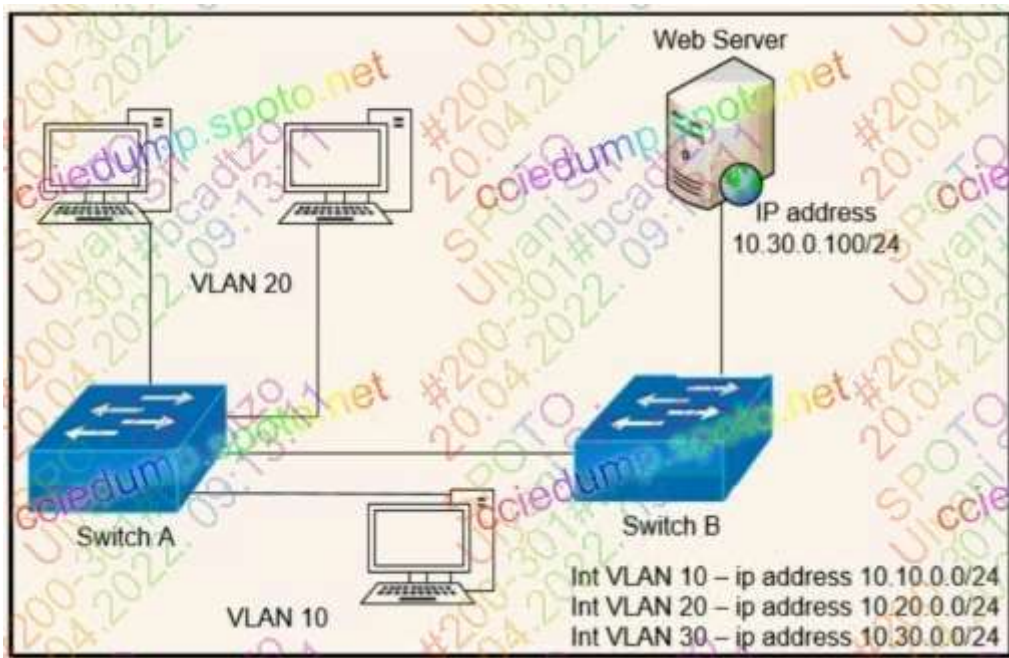
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 264



Refer to the exhibit. A network engineer must block access for all computers on VLAN 20 to the web server via HTTP. All other computers must be able to access the web server. Which configuration when applied to switch A accomplishes this task?

- A.
- ```

config t
ip access-list extended wwwblock
deny tcp any host 10.30.0.100 eq 80
int vlan 10
ip access-group wwwblock in

```
- B.
- ```

config t
ip access-list extended wwwblock
deny tcp any host 10.30.0.100 eq 80
permit ip any any
int vlan 20
ip access-group wwwblock in

```
- C.
- ```

config t
ip access-list extended wwwblock
permit ip any any
deny tcp any host 10.30.0.100 eq 80
int vlan 30
ip access-group wwwblock in

```
- D.
- ```

config t
ip access-list extended wwwblock
permit ip any any
deny tcp any host 10.30.0.100 eq 80
int vlan 20
ip access-group wwwblock in

```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 265

```

access-list 101 permit ospf any any
access-list 101 permit tcp any any eq 179
access-list 101 permit tcp any eq 179 any
access-list 101 permit gre any any
access-list 101 permit esp any any

access-list 101 deny ospf any any
access-list 101 permit tcp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255 eq telnet
access-list 101 permit udp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255 eq 500
access-list 101 permit udp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255 eq 4500
access-list 101 deny ip any any log

interface Ethernet0/0
ip address 10.1.1.25 255.255.255.0
ip access-group 101 in

```

Refer to the exhibit. A network administrator has been tasked with securing VTY access to a router. Which access-list entry accomplishes this task?

- A. access-list 101 permit tcp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255 eq telnet
- B. access-list 101 permit tcp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255 eq scp
- C. access-list 101 permit tcp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255 eq https
- D. access-list 101 permit tcp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255 eq ssh

Correct Answer: D
 Section: (none)
 Explanation

Explanation/Reference:

QUESTION 266



Refer to the exhibit. Which change to the configuration on Switch2 allows the two switches to establish an EtherChannel?

- A. Change the protocol to PAgP and use auto mode.
- B. Change the LACP mode to desirable.
- C. Change the LACP mode to active.
- D. Change the protocol to EtherChannel mode on.

Correct Answer: C
 Section: (none)
 Explanation

Explanation/Reference:

QUESTION 267

What is the collapsed layer in collapsed core architectures?

- A. distribution and access
- B. core and distribution
- C. core and WAN
- D. access and WAN

Correct Answer: B
 Section: (none)
 Explanation

Explanation/Reference:

QUESTION 268

Which network plane is centralized and manages routing decisions?

- A. policy plane
- B. data plane
- C. management plane
- D. control plane

Correct Answer: D
 Section: (none)
 Explanation

Explanation/Reference:

QUESTION 269

Which two protocols must be disabled to increase security for management connections to a Wireless LAN Controller? (Choose two)

- A. SSH
- B. TFTP
- C. HTTPS
- D. Telnet
- E. HTTP

Correct Answer: DE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 270

What uses HTTP messages to transfer data to applications residing on different hosts?

- A. OpenFlow
- B. REST
- C. OpenStack
- D. OpFlex

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 271

Which purpose does a northbound API serve in a controller-based networking architecture?

- A. reports device errors to a controller
- B. communicates between the controller and the physical network hardware
- C. generates statistics for network hardware and traffic
- D. facilitates communication between the controller and the application

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 272

A network administrator enabled port security on a switch interface connected to a printer. What is the next configuration action in order to allow the port to learn the MAC address of the printer and insert it into the table automatically?

- A. implement auto MAC address learning
- B. implement static MAC addressing
- C. enable dynamic MAC address learning
- D. enable sticky MAC addressing

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 273

What are two benefits of network automation? (Choose two.)

- A. reduced operational costs
- B. faster changes with more reliable results
- C. increased network security
- D. fewer network failures
- E. reduced hardware footprint

Correct Answer: AB

Section: (none)

Explanation

Explanation/Reference:

QUESTION 274

A corporate office uses four floors in a building.

- 1. Floor 1 has 24 users.
- 2. Floor 2 has 29 users.
- 3. Floor 3 has 28 users.
- 4. Floor 4 has 22 users.

Which subnet summarizes and gives the most efficient distribution of IP addresses for the router configuration?

- A. 192.168.0.0/23 as summary and 192.168.0.0/25 for each floor
- B. 192.168.0.0/24 as summary and 192.168.0.0/28 for each floor
- C. 192.168.0.0/26 as summary and 192.168.0.0/29 for each floor

D. 192.168.0.0/25 as summary and 192.168.0.0/27 for each floor

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 275

A network administrator needs to aggregate 4 ports into a single logical link which must negotiate layer 2 connectivity to ports on another switch. What must be configured when using active mode on both sides of the connection?

- A. LACP
- B. 802.1q trunks
- C. LLDP
- D. Cisco vPC

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 276

What is a function of Cisco Advanced Malware Protection for a Next-Generation IPS?

- A. authenticating end users
- B. inspecting specific files and file types for malware
- C. URL filtering
- D. authorizing potentially compromised wireless traffic

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 277



Refer to the exhibit. The router has been configured with a supernet to accommodate the requirement for 380 users on a subnet. The requirement already considers 30% future growth. Which configuration verifies the IP subnet on router R4?

- A. Subnet: 10.7.54.0
Subnet mask: 255.255.255.0
Broadcast address: 10.7.54.255
Usable IP address range: 10.7.54.1 - 10.7.55.254
- B. Subnet: 10.7.54.0
Subnet mas: 255.255.128.0
Broadcast address: 10.7.55.255
Usable IP address range: 10.7.54.1 - 10.7.55.254
- C. Subnet: 10.7.54.0
Subnet mas: 255.255.254.0
Broadcast address: 10.7.55.255
Usable IP address range: 10.7.54.1 - 10.7.55.254
- D. Subnet: 10.7.54.0
Subnet mas: 255.255.254.0
Broadcast address: 10.7.54.255
Usable IP address range: 10.7.54.1 - 10.7.55.254

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 278

What is a characteristic of a collapsed-core network topology?

- A. It allows the core and distribution layers to run as a single combined layer.
- B. It enables the core and access layers to connect to one logical distribution device over an EtherChannel.
- C. It allows wireless devices to connect directly to the core layer, which enables faster data transmission.
- D. It enables all workstations in a SOHO environment to connect on a single switch with internet access.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 279

```
C:\Users\ciscoadmin>ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-480J88T
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : arcep.se


Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . . . . :
Description . . . . . : Realtek PCIe GBE Family
Controller
Physical Address. . . . . : 3C-52-82-33-F3-8F
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes


Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . . . . . : arcep.se
Description . . . . . : Intel(R) Dual Band Wireless-AC 7265
Physical Address. . . . . : C8-21-58-B4-F3-8F
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::45a1:b3fa:2f37:bf37%2 (Preferred)
IPv4 Address. . . . . : 192.168.1.226 (Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained . . . . . : October 3, 2019 12:28:08 PM
Lease Expires . . . . . : October 3, 2019 7:18:37 PM
Default Gateway . . . . . : 192.168.1.100
DHCP Server . . . . . : 192.168.1.254
DHCPv6 IAID . . . . . : 46670168
DHCPv6 Client DUID . . . . . : 00-01-00-01-20-FF-05-55-3C-52-82-33-D3-84
DNS Servers . . . . . : 192.168.1.253
NetBIOS over Tcpip. . . . . : Enabled
Connection-specific DNS Suffix Search List :
arcep.se
```

Refer to the exhibit. The given Windows PC is requesting the IP address of the host at www.cisco.com. To which IP address is the request sent?

- A. 192.168.1.226
- B. 192.168.1.100
- C. 192.168.1.254
- D. 192.168.1.253

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 280

Why is a first-hop redundancy protocol implemented?

- A. to enable multiple switches to operate as a single unit
- B. to prevent loops in a network
- C. to protect against default gateway failures
- D. to provide load-sharing for a multilink segment

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 281

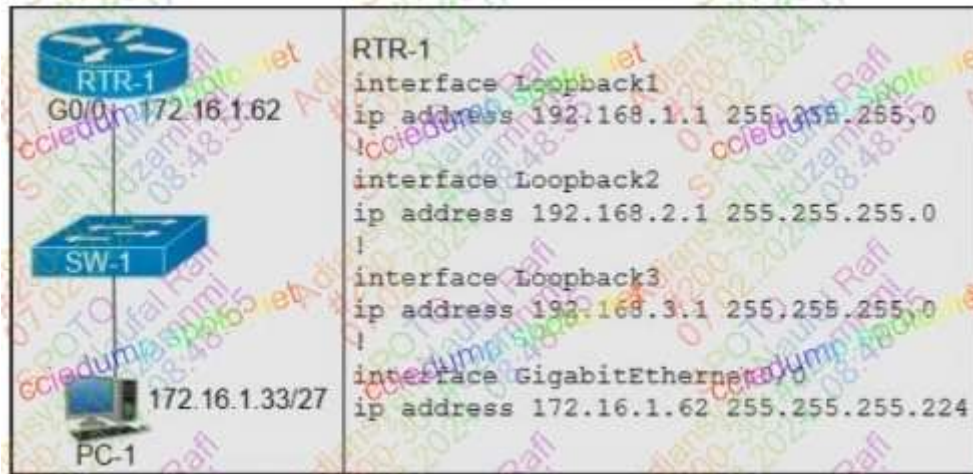
Which Layer 2 switch function encapsulates packets for different VLANs so that the packets traverse the same port and maintain traffic separation between the VLANs?

- A. VLAN numbering
- B. VLAN DSCP
- C. VLAN marking
- D. VLAN tagging

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 282



Refer to the exhibit. Which configuration on RTR-1 denies SSH access from PC-1 to any RTR-1 interface and allows all other traffic?

- A.

```
access-list 100 deny tcp host 172.16.1.33 any eq 23
access-list 100 permit ip any any

line vty 0 15
access-class 100 in
```
- B.

```
access-list 100 deny tcp host 172.16.1.33 any eq 22
access-list 100 permit ip any any

line vty 0 15
access-class 100 in
```
- C.

```
access-list 100 deny tcp host 172.16.1.33 any eq 22
access-list 100 permit ip any any

interface GigabitEthernet0/0
ip access-group 100 in
```
- D.

```
access-list 100 deny tcp host 172.16.1.33 any eq 23
access-list 100 permit ip any any

interface GigabitEthernet0/0
ip access-group 100 in
```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 283

What are two purposes of HSRP? (Choose two.)

- A. It provides a mechanism for diskless clients to autoconfigure their IP parameters during boot
- B. It helps hosts entire network to reach remote subnets without a default gateway.
- C. It improves network availability by providing redundant gateways.
- D. It passes configuration information to hosts in a TCP/IP network
- E. It groups two or more routers to operate as one virtual router.

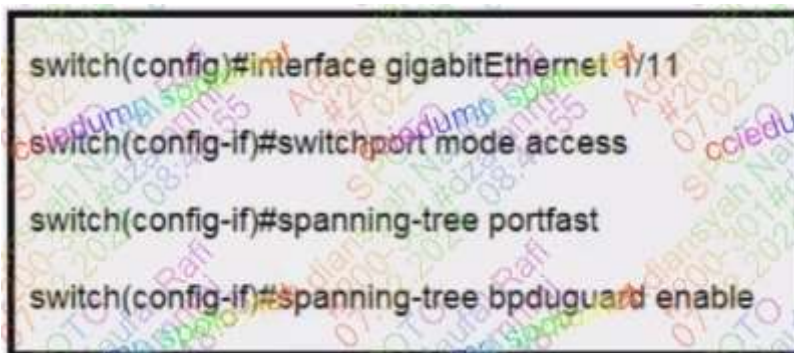
Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 284



Refer the exhibit. What is the result if Gig1/11 receives an STP BPDU?

- A. The port goes into error-disable state.
- B. The port transitions to the root port.
- C. The port transitions to STP blocking.
- D. The port immediately transitions to STP forwarding.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 285

What describes the functionality of southbound APIs?

- A. They convey information from the controller to the SDN applications.
- B. They communicate with the management plane.
- C. They enable communication between the controller and the network device.
- D. They use HTTP messages to communicate.

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 286



Refer to the exhibit. An engineer must configure the interface that connects to PC1 and secure it in a way that only PC1 is allowed to use the port. No VLAN tagging can be used except for a voice VLAN. Which command sequence must be entered to configure the switch?

- A. SW1(config-if)# switchport mode access
SW1(config-if)# switchport port-security
SW1(config-if)# switchport port-security mac-address 0050.7966.6800
- B. SW1(config-if)# switchport mode nonegotiate
SW1(config-if)# switchport port-security
SW1(config-if)# switchport port-security maximum 1
- C. SW1(config-if)# switchport mode dynamic auto
SW1(config-if)# switchport port-security
SW1(config-if)# switchport port-security violation restrict
- D. SW1(config-if)# switchport mode dynamic desirable
SW1(config-if)# switchport port-security mac-address 0050.7966.6800
SW1(config-if)# switchport port-security mac-address sticky

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 287

Which command enables HTTP access to the Cisco WLC?

- A. config network telnet enable
- B. config certificate generate webadmin
- C. config network webmode enable
- D. config network secureweb enable

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 288

A Cisco engineer is configuring a factory-default router with these three passwords:

1. The user EXEC password for console access is p4ssw0rd1.
2. The user EXEC password for Telnet access is s3cr3t2.
3. The password for privileged EXEC mode is priv4t3p4ss.

Which command sequence must the engineer configure?

- A. enable secret priv4t3p4ss
!
line con 0
password login p4ssw0rd1
!
line vty 0 15
password login s3cr3t2
login
- B. enable secret priv4t3p4ss
!
line con0
password p4ssword1
!
line vty0 15
password s3cr3t2

- C. enable secret privilege 15 priv4t3p4ss
!
line con0
password p4ssword1
login
!
line vty 0 15
password s3cr3t2
login
- D. enable secret priv4t3p4ss
!
line con0
password p4ssword1
login
!
line vty 0 15
password s3cr3t2
login

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 289

```
hostname CPE
service password-encryption
ip domain name ccna.cisco.com
ip name-server 198.51.100.210

crypto key generate rsa modulus 1024

username admin privilege 15 secret $0mls3cr3t

line vty 0 4
transport input ssh
login local
```

Refer to the exhibit. An engineer executed the script and added commands that were not necessary for SSH and now must remove the commands. Which two commands must be executed to correct the configuration? (Choose two.)

- A. no login local
- B. no ip domain name ccna.cisco.com
- C. no ip name-server 198.51.100.210
- D. no service password-encryption
- E. no hostname CPE

Correct Answer: CD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 290

Which syslog severity level is considered the most severe and results in the system being considered unusable?

- A. Emergency
- B. Alert
- C. Error
- D. Critical

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 291



Refer to the exhibit. The network engineer is configuring a new WLAN and is told to use a setup password for authentication instead of the RADIUS servers. Which additional set of tasks must the engineer perform to complete the configuration?

- A. Disable PMF
Enable PSK
Enable 802.1x
- B. Select WPA2 Policy
Disable PMF
Enable PSK
- C. Select WPA
Policy Select WPA
Policy Enable FT PSK
- D. Disable PMF
Enable PSK
Enable 802.1x

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 292

Why choose Cisco DNA Center for automated lifecycle management?

- A. to allow SSH access to all nodes in the network
- B. to perform upgrades without service interruption
- C. to provide software redundancy in the network
- D. to provide fast and accurate deployment of patches and updates

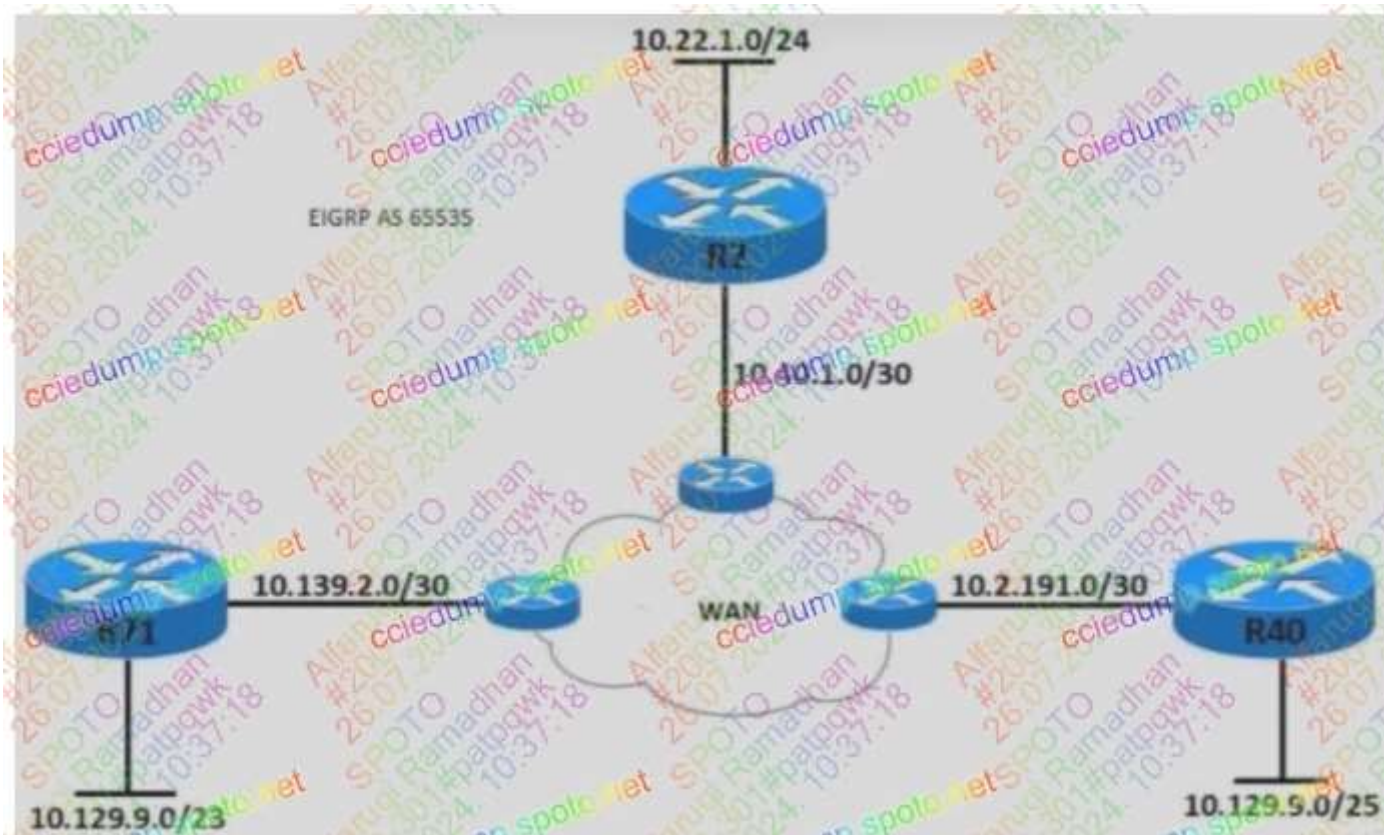
Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 293



Refer to the exhibit. All routers in the network are configured correctly, and the expected routes are being exchanged among the routers. Which set of routes are learned from neighbors and installed on router 2?

- A. 10.40.1.0/30
10.139.2.0/30
10.2.191.0/30
10.129.9.0/25
- B. 10.129.9.0/23
10.139.2.0/30
10.2.191.0/30
10.129.9.0/25
- C. 10.129.9.0/23
10.40.1.0/30
10.2.191.0/30
10.129.9.0/25
- D. 10.129.9.0/23
10.139.2.0/30
10.129.9.0/25
10.22.1.0/24

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 294



Refer to the exhibit. The switches are connected via a Cat5 Ethernet cable that was successfully tested. The interfaces are configured as access ports and are both in a "down" status. What is the cause of this issue?

- A. The distance between the two switches is not supported by Cat5.
- B. The switches are configured with incompatible duplex settings.
- C. The portfast command is missing from the configuration.
- D. The speed settings on the switches are mismatched.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 295

What is the difference between controller-based networks and traditional networks as they relate to control-plane and/or data-plane functions?

- A. Traditional networks centralize all important data-plane functions, and controller-based networks distribute data-plane functions.
- B. Controller-based networks centralize all important control-plane functions, and traditional networks distribute control-plane functions.
- C. Controller-based networks centralize all important data-plane functions, and traditional networks distribute data-plane functions.
- D. Traditional networks centralize all important control-plane functions, and controller-based networks distribute control-plane functions.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 296

```
Router-WAN1#show interface g0/0
GigabitEthernet0/0 is up, line protocol is up
Hardware is C3K NIC, address is 5000.0001.0000 (bia 5000.0001.0000)
Interface address is 192.168.0.8/31
MTU 1500 bytes, BW 1000000 Kbit/sec, DLR 10 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full Duplex, 1000Mbps, link type is auto, media type is RTC
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00
Last input never, output 00:00:03, output hang never
Last clearing of "show interface" counters never
Input queue: 0/25/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate: 1000 bps/sec, 0 packets/sec
5 minute output rate: 2000 bits/sec, 1 packets/sec
0 packets input, 0 bytes, 0 no buffer
Received 110 broadcasts (0 IP multicasts)
0 runs, 0 giants, 0 throttles
100 input errors, 100 CRC, 100 frame, 0 overrun, 0 ignored
0 watchdog, 0 multicast, 0 pause input
260 packets output, 83070 bytes, 0 underruns
Output 0 broadcasts (0 IP multicasts)
0 output errors, 100 collisions, interface resets
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
1 lost carrier, 0 no carrier, 0 pause output
```

Refer to the exhibit. Router-WAN1 has a new connection via Gi0/0 to the ISP. Users running the web applications indicate that connectivity is unstable to the internet What is causing the interface issue?

- A. Frames are discarded due to a half-duplex negotiation.
- B. The receive buffer is full due to a broadcast storm.
- C. Small frames less than 64 bytes are rejected due to size.
- D. Broadcast packets are rejected because ARP timeout is enabled.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 297

A switch is forwarding a frame out of all interfaces except the interface that received the frame. What is the technical term for this process?

- A. Multicast
- B. GDP
- C. Flooding
- D. ARP

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 298

What must a network administrator consider when deciding whether to configure a new wireless network with APs in autonomous mode or APs running in cloud-based mode?

- A. Autonomous mode APs are less dependent on an underlay but more complex to maintain than APs in cloud-based mode.
- B. Cloud-based mode APs are easy to deploy but harder to automate than APs in autonomous mode.
- C. Cloud-based mode APs rely on underlays and are more complex to maintain than APs in autonomous mode.
- D. Autonomous mode APs are easy to deploy and automate than APs in cloud-based mode.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 299

A network engineer is configuring a new router at a branch office. The router is connected to an upstream WAN network that allows the branch to communicate with the head office. The central time server with IP address 172.24.54.8 is located behind a firewall at the head office. Which command must the engineer configure so that the software clock of the new router synchronizes with the timeserver?

- A. ntp master 172.24.54.8
- B. ntp peer 172.24.54.8
- C. ntp client 172.24.54.8
- D. ntp server 172.24.54.8

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 300

What is the RFC 4627 default encoding for JSON text?

- A. GB18030
- B. Hex
- C. UTF-8
- D. UCS-2

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 301

Which solution is appropriate when mitigating password attacks where the attacker was able to sniff the clear-text password of the system administrator?

- A. next-generation firewall to keep stateful packet inspection
- B. ACL to restrict incoming Telnet sessions "admin" accounts
- C. IPS with a block list of known attack vectors
- D. multifactor authentication using two separate authentication sources

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 302

Which mechanism allows WPA3 to provide a higher degree of security than its predecessors?

- A. special-character support in preshared keys
- B. certificate-based authentication
- C. automatic device pairing
- D. SAE password-based key exchange

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 303

Which protocol does Ansible use to push modules to nodes in a network?

- A. SNMP
- B. Telnet
- C. Kerberos
- D. SSH

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 304



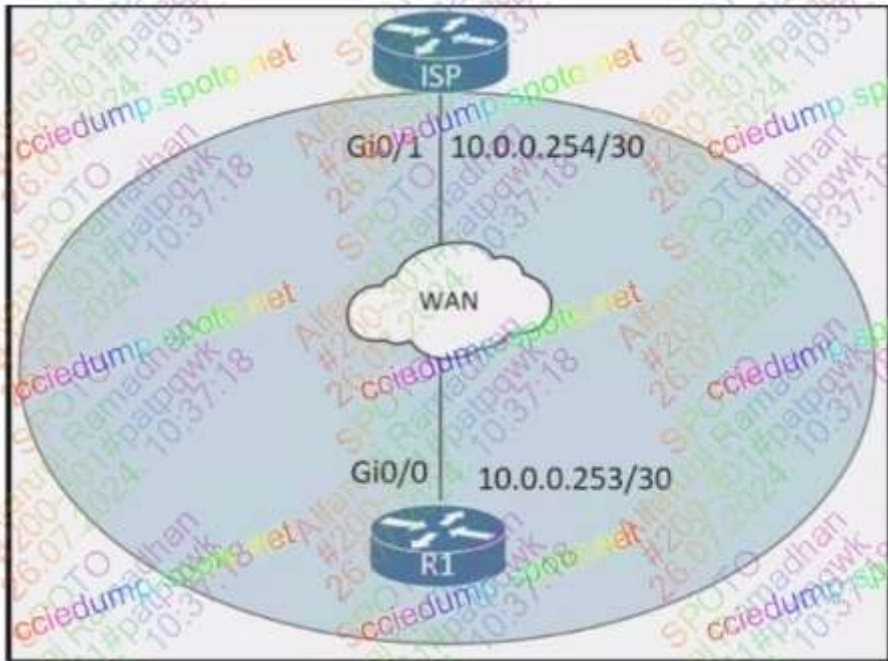
Refer to the exhibit. If the network environment is operating normally, which type of device must be connected to interface fastethernet 0/1?

- A. DHCP client
- B. PC
- C. router
- D. access point

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 305



Refer to the exhibit. An administrator must turn off the Cisco Discovery Protocol on the port configured with address last usable address in the 10.0.0.0/30 subnet. Which command set meets the requirement?

- A. `interface gi0/0`
`no cdp run`
- B. `interface gi0/1`
`no cdp enable`
- C. `interface gi0/1`
`clear cdp table`
- D. `interface gi0/0`
`no cdp advertise-v2`

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 306



Refer to the exhibit. Users need to connect to the wireless network with IEEE 802.11r-compatible devices. The connection must be maintained as users travel between floors or to other areas in the building. What must be the configuration of the connection?

- A. Disable AES encryption.
- B. Enable Fast Transition and select the FT PSK option.
- C. Enable Fast Transition and select the FT 802.1X option.
- D. Select the WPA Policy option with the CCKM option.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 307



Refer to the exhibit. For security reasons, automatic neighbor discovery must be disabled on the R5 Gi0/1 interface. These tasks must be completed:

- Disable all neighbor discovery methods on R5 interface Gi0/1.
- Permit neighbor discovery on R5 interface Gi0/2.
- Verify there are no dynamically learned neighbors on R5 interface Gi0/1.
- Display the IP address of R6's interface Gi0/2.

Which configuration must be used?

- A.

```
R5(config)# int Gi0/1
R5(config-if)# no cdp run
R5(config-if)# exit
R5(config)# lldp run
R5(config)# cdp enable
R5# sh cdp neighbor
R5# sh lldp neighbor
```
- B.

```
R5(config)# int Gi0/1
R5(config-if)# no cdp enable
R5(config-if)# exit
R5(config)# no lldp run
R5(config)# cdp run
R5# sh cdp neighbor detail
R5# sh lldp neighbor
```
- C.

```
R5(config)# int Gi0/1
R5(config-if)# no cdp enable
R5(config-if)# exit
R5(config)# lldp run
R5(config)# no cdp run
R5# sh cdp neighbor detail
R5# sh lldp neighbor
```
- D.

```
R5(config)# int Gi0/1
R5(config-if)# no cdp enable
R5(config-if)# exit
R5(config)# no lldp run
R5(config)# cdp run
R5# sh cdp neighbor
R5# sh lldp neighbor
```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 308



Refer to the exhibit. An engineer is asked to configure router R1 so that it forms an OSPF single-area neighbor relationship with R2. Which command sequence must be implemented to configure the router?

- A.

```
router ospf 10
network 10.0.0.0 0.0.0.3 area 0
network 10.0.2.0 0.0.0.255 area 0
```
- B.

```
router ospf 10
network 10.0.0.0 0.0.0.3 area 0
network 10.0.1.0 0.0.0.255 area 0
```
- C.

```
router ospf 100
network 10.0.0.0 0.0.0.3 area 0
network 10.0.2.0 255.255.255.0 area 0
```

D. `router ospf 100`
`network 10.0.0.0 0.0.0.252 area 0`
`network 10.0.1.0 0.0.0.255 area 0`

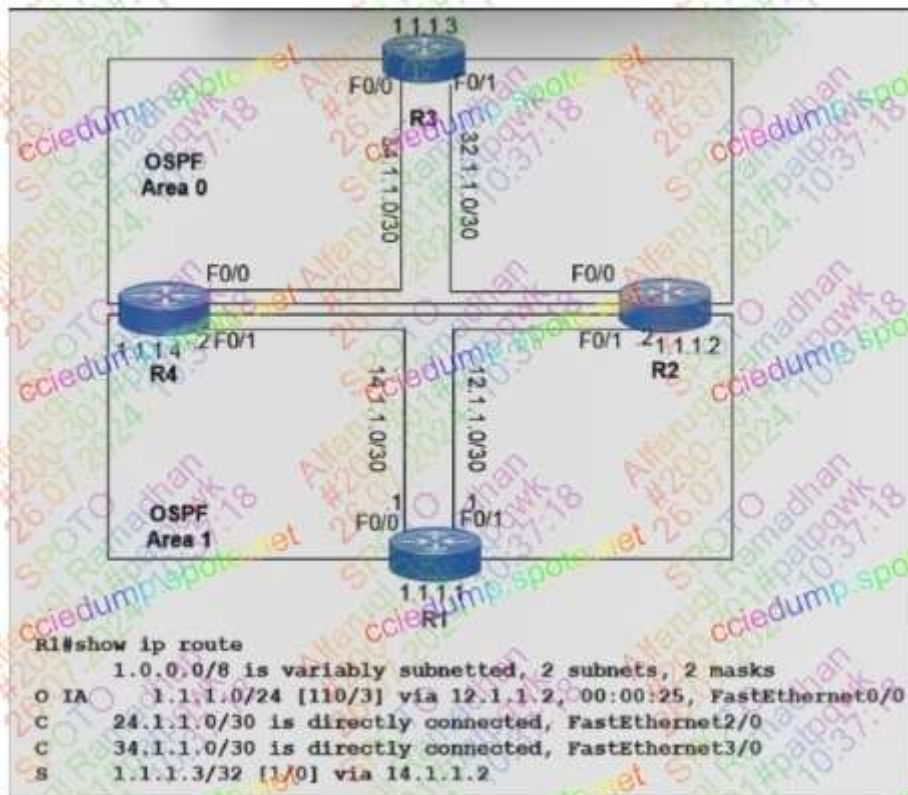
Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 309



Refer to the exhibit. Which two values does router R1 use to identify valid routes for the R3 loopback address 1.1.1.3/32? (Choose two.)

- A. lowest cost to reach the next hop
- B. lowest administrative distance
- C. lowest metric
- D. highest metric
- E. highest administrative distance

Correct Answer: BC

Section: (none)

Explanation

Explanation/Reference:

QUESTION 310

An engineer must configure R1 for a new user account. The account must meet these requirements:

1. It must be configured in the local database.
 2. The username is engineer2.
 3. It must use the strongest password configurable.
- Which command must the engineer configure on the router?

- A. `R1(config)# username engineer2 algorithm-type scrypt secret test2021`
- B. `R1(config)# username engineer2 privilege 1 password 7 test2021`
- C. `R1(config)# username engineer2 secret 5 password XXXXXXXXXXXX`
- D. `R1(config)# username engineer2 secret 4 XXXXXXXXXXXX`

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 311

What is an Ansible inventory?

- A. collection of actions to perform on target devices, expressed in YAML format
- B. file that defines the target devices upon which commands and tasks are executed
- C. device with Ansible installed that manages target devices
- D. unit of Python code to be executed within Ansible

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 312

Which protocol is implemented when an organization must verify network performance, troubleshoot issues, and use an agent to communicate between monitoring tools and devices?

- A. SNMP
B. NFS
C. FTP
D. NTP

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 313



Refer to the exhibit. An engineer is creating a secure preshared key based SSID using WPA2 for a wireless network running on 2.4 GHz and 5 GHz. Which two tasks must the engineer perform to complete the process? (Choose two.)

- Select the AES option for Auth Key Management.
- Select the PSK option for Auth Key Management.
- Select the 802.1X option for Auth Key Management.
- Select the AES (CCMP128) option for WPA2/WPA3 Encryption.
- Select the WPA Policy option.

Correct Answer: BD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 314



Refer to the exhibit. An administrator received a call from a branch office regarding poor application performance hosted at the headquarters. Ethernet 1 is connected between Router1 and the LAN switch. What identifies the issue?

- A. The QoS policy is dropping traffic.
- B. There is a duplex mismatch.
- C. The MTU is not set to the default value.
- D. The link is over utilized.

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

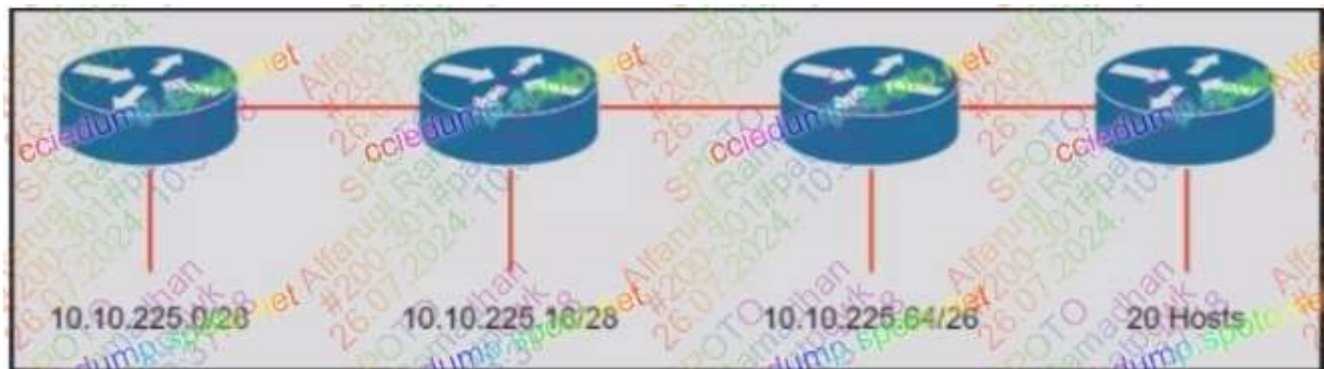
QUESTION 315
 What device segregates a network into separate zones that have their own security policies?

- A. firewall
- B. access poin
- C. IPS
- D. Switch

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 316



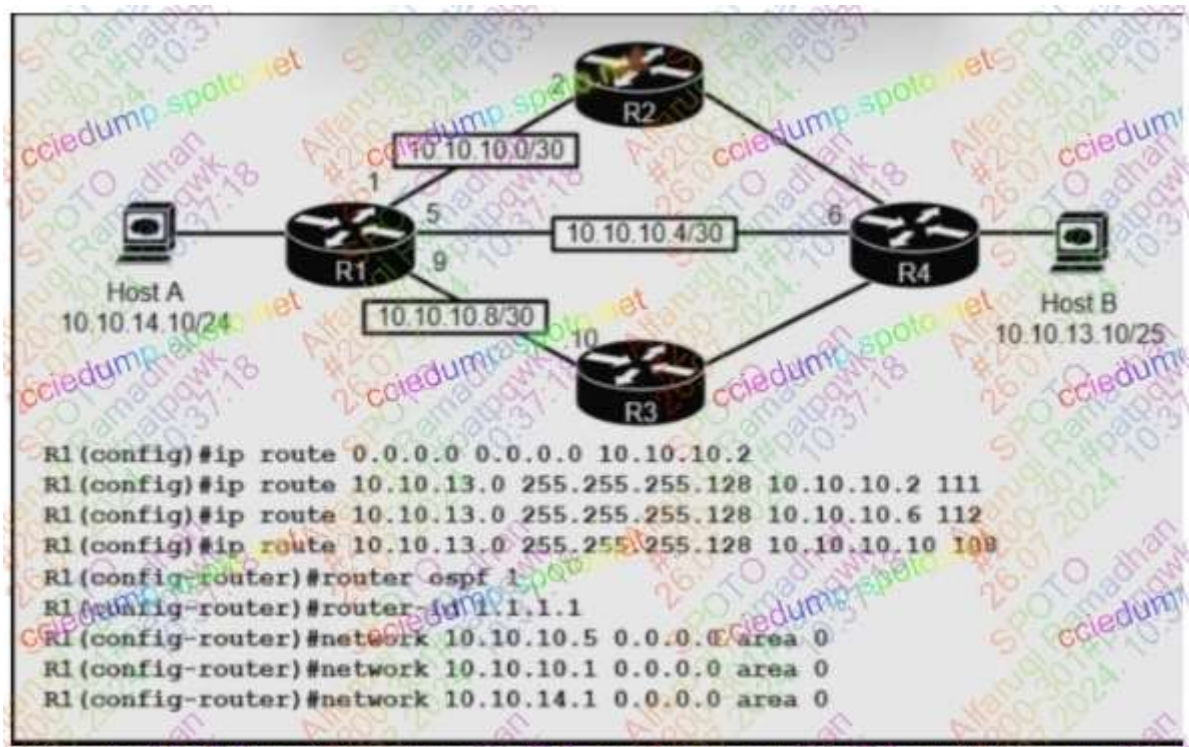
Refer to the exhibit. An engineer must add a subnet for a new office that will add 20 users to the network. Which IPv4 network and subnet mask combination does the engineer assign to minimize wasting addresses?

- A. 10.10.225.32 255.255.255.240
- B. 10.10.225.48 255.255.255.240
- C. 10.10.225.48 255.255.255.224
- D. 10.10.225.32 255.255.255.224

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 317



Refer to the exhibit. R1 has just received a packet from host A that is destined to host B. Which route in the routing table is used by R1 to reach host B?

- A. 10.10.13.0/25 [110/2] via 10.10.10.2
- B. 10.10.13.0/25 [110/2] via 10.10.10.6
- C. 10.10.13.0/25 [108/0] via 10.10.10.10

D. 10.10.13.0/25 [1/0] via 10.10.10.2

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 318

What is the advantage of separating the control plane from the data plane within an SDN network?

- A. reduces cost
- B. limits data queries to the control plane
- C. decreases overall network complexity
- D. offloads the creation of virtual machines to the data plane

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 319

A router has two static routes to the same destination network under the same OSPF process. How does the router forward packets to the destination if the next-hop devices are different?

- A. The router load-balances traffic over all routes to the destination.
- B. The router chooses the next hop with the lowest IP address.
- C. The router chooses the next hop with the lowest MAC address.
- D. The router chooses the route with the oldest age.

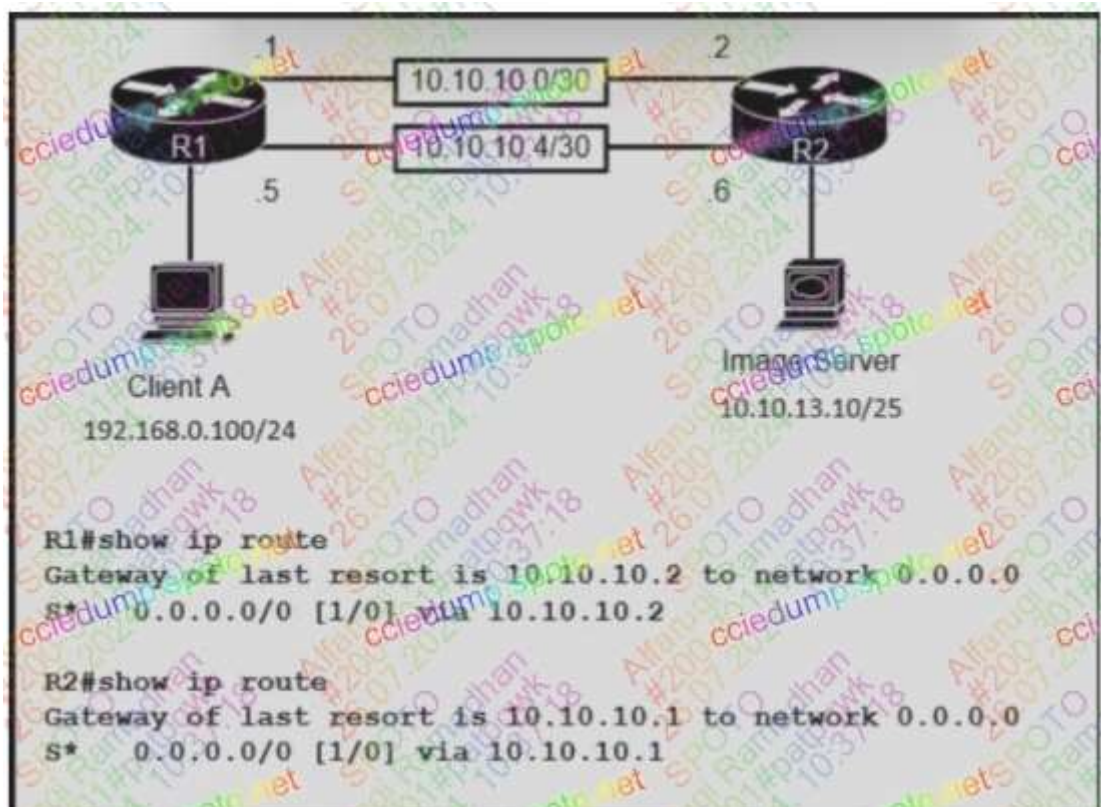
Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 320



Refer to the exhibit. The image server and client A are running an application that transfers an extremely high volume of data between the two. An engineer is configuring a dedicated circuit between routers R1 and R2. Which set of commands must the engineer apply to the two routers so that only traffic between the image server and client A is forced to use the new circuit?

- A. R1(config)#ip route 10.10.13.10 255.255.255.252 10.10.10.6
R2(config)#ip route 192.168.0.100 255.255.255.252 10.10.10.5
- B. R1(config)#ip route 10.10.13.10 255.255.255.255 10.10.10.2
R2(config)#ip route 192.168.0.100 255.255.255.255 10.10.10.1
- C. R1(config)#ip route 10.10.13.10 255.255.255.255 10.10.10.6
R2(config)#ip route 192.168.0.100 255.255.255.255 10.10.10.5
- D. R1(config)#ip route 10.10.13.10 255.255.255.128 10.10.10.6
R2(config)#ip route 192.168.0.100 255.255.255.0 10.10.10.5

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 321



Refer to the exhibit. The My_WLAN wireless LAN was configured with WPA2 Layer 2 PSK security .Which additional configuration must the administrator perform to allow users to connect to this WLAN on a different subnet called Data?

- A. Enable Status and enable Broadcast SSID.
- B. Enable Status and set the NAS-ID to data.
- C. Enable Broadcast SSID and select data from the Interface Interface Group drop-down list.
- D. Enable Status and select data from the Interface/Interface Group drop-down list.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 322

Which IPsec mode encapsulates the entire IP packet?

- A. SSL VPN
- B. transport
- C. Q-in-Q
- D. tunnel

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 323



Refer to the exhibit. A network engineer started to change default settings on SW1 to allow remote access and has entered the following in the configuration mode.

```
SW1#(config)#line vty 015
SW1#(config-line)#password Labtest32!
```

Which set of commands are needed to allow only SSH access and hide passwords in the running configuration?

- A. SW1#(config-line)#transport input ssh
SW1#(config-line)#exit
SW1(config)#service password-encryption
- B. SW1#(config-line)#exit
SW1(config)#aaa new-model
- C. SW1#(config-line)#login local
SW1#(config-line)#exit
SW1(config)#crypto key generate rsa


```
D. SW1#(config-line)#login local
SW1#(config-line)#exit
SW1(config)#enable secret test12E
```

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 324



Refer to the exhibit. Clients on the WLAN are required to use 802.11r. What action must be taken to meet the requirement?

- A. Set the Fast Transition option and the WPA gtk-randomize State to disable.
- B. Under Protected Management Frames, set the PMF option to Required.
- C. Set the Fast Transition option to Enable and enable FT 802.1X under Authentication Key Management.
- D. Enable CCKM under Authentication Key Management.

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 325

What is the benefit of using FHRP?

- A. higher degree of availability
- B. reduced management overhead on network routers
- C. balancing traffic across multiple gateways in proportion to their loads
- D. reduced ARP traffic on the network

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 326



Which configuration parameter is preventing host C from reaching the internet?

- A. automatic DNS
- B. default gateway
- C. IP network mask
- D. IP address assignment

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 327

```
cat9k-acc-1# show interfaces gigabitEthernet 1/0/1
GigabitEthernet1/0/1 is up, line protocol is up
Hardware is GigabitEthernet, address is aa00.0400.0134 (via 0000.0c00.4369)
MTU 1500 bytes, BW 1000 Kbit, DLY 1000 usec, rely 255/255, load 1/255
Encapsulation ARPA, loopback not set, keepalive s/w (10 sec)
ARP type: ARPA, PROBE, ARP Timeout 4:00:00
Last input 0:00:00, output 0:00:00, output hang never
Output queue 1/1, 1 drops; input queue 0/0, 0 drops
Five minute input rate 61000 bits/sec, 200 packets/sec
Five minute output rate 1000 bits/sec, 200 packets/sec
2295197 packets input, 305539992 bytes, 0 no buffer
Received 1935500 broadcasts, 0 runts, 0 giants
0 input errors, 1790 CRC, 1790 frame, 0 overrun, 0 ignored, 0 abort
0 input packets with dribble condition detected
3594664 packets output, 436549843 bytes, 1 underruns
0 output errors, 1 collisions, 1 interface resets, 0 restarts
```

Refer to the exhibit. The switch cat9k-acc-1 connects users to the campus LAN. Printing services are inaccessible through the network. Which interface issue is causing the connectivity problems?

- A. The interface output queue cannot process the Ethernet frames.
- B. Excessive collisions are causing dropped frames.
- C. A large number of broadcast packets are resulting in a port reset.
- D. A bad checksum is causing Ethernet frames to drop.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 328

```
100.0.0.0/8 is variably subnetted, 4 subnets, 4 masks
R    100.0.0.0/8 [120/2] via 192.168.3.1, 00:00:13, Ethernet0/3
S    100.100.0.0/16 [1/0] via 192.168.4.1
D    100.100.100.0/24 [90/435200] via 192.168.2.1, 00:00:13, Ethernet0/2
O    100.100.100.100/32 [110/21] via 192.168.1.1, 00:05:57, Ethernet0/1
```

Refer to the exhibit. How will the device handle a packet destined to IP address 100.100.100.100?

- A. It will choose the route with the lowest metric.
R 100.0.0.0/8 [120/2] via 192.168.3.1, 00:00:13, Ethernet0/3.
- B. It will choose the route with the highest metric.
D 100.100.100.0/24 [90/435200] via 192.168.2.1, 00:00:13, Ethernet0/2.
- C. It will always prefer the static route over dynamic routes and choose the route
S 100.100.0.0/16 [1/0] via 192.168.4.1.

- D. It will choose the route with the longest match.
O 100.100.100.100/32 [110/21] via 192.168.1.1,00:05:57, Ethern0/1.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 329

`["red", "one"]`

Refer to the exhibit. Which type of JSON data is represented?

- A. string
- B. number
- C. object
- D. array

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 330



Refer to the exhibit. A network engineer is configuring a WLAN to use a WPA2 PSK and allow only specific clients to join. Which two actions must be taken to complete the process?
(Choose two.)

- A. Enable the CCKM option for Authentication Key Management.
- B. Enable the WPA2 Policy option.
- C. Enable the MAC Filtering option.
- D. Enable the OSEN Policy option.
- E. Enable the 802.1X option for Authentication Key Management.

Correct Answer: BC

Section: (none)

Explanation

Explanation/Reference:

QUESTION 331

Which protocol should be used to transfer large files on a company intranet that allows TCP 20 and 21 through the firewall?

- A. SMTP
- B. FTP
- C. TFTP
- D. REST API

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 332

What is the function of a northbound API?

- A. It provides a path between an SDN controller and network applications.
- B. It upgrades software and restores files.

- C. It supports distributed processing for configuration.
- D. It relies on global provisioning and configuration.

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 333

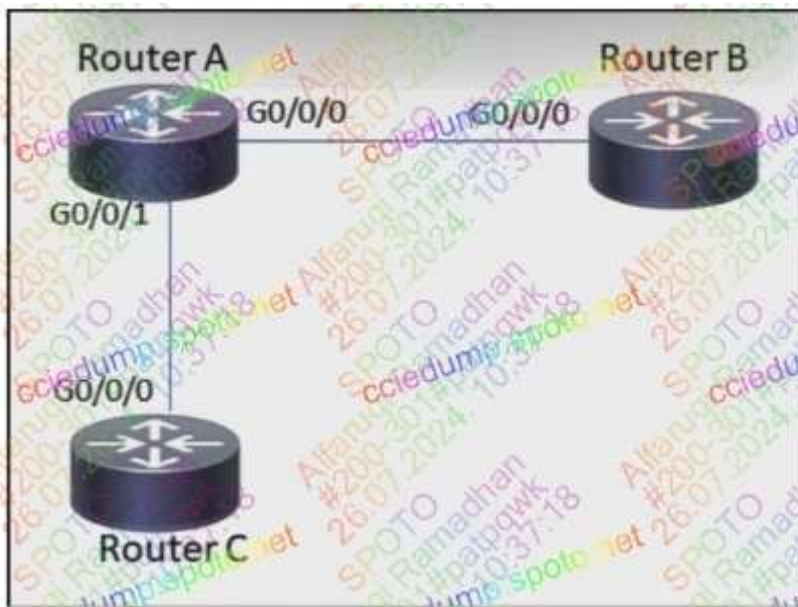
How does CAPWAP communicate between an access point in local mode and a WLC?

- A. The access point must directly connect to the WLC using a copper cable.
- B. The access point has the ability to link to any switch in the network, assuming connectivity to the WLC.
- C. The access point must be connected to the same switch as the WLC.
- D. The access point must not be connected to the wired network, as it would create a loop.

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 334



Refer to the exhibit. How must router A be configured so that it only sends Cisco Discovery Protocol information to router C?

- A.

```
#config t
Router A(config)# cdp run
Router A(config) # interface gi0/0/0
Router A(config-if)# no cdp enable
```
- B.

```
#config t
Router A(config)# cdp run
Router A(config)# interface gi0/0/0
Router A(config-if)# cdp enable
```
- C.

```
#config t
Router A(config)# no cdp run
Router A(config)# interface gi0/0/1
Router A(config-if)# cdp enable
```
- D.

```
#config t
Router A(config)# cdp run
Router A(config)# interface gi0/0/1
Router A(config-if)# cdp enable
```

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 335

Which two protocols are supported on service-port interfaces? (Choose two.)

- A. Telnet
- B. TACACS+
- C. SSH
- D. RADIUS
- E. SCP

Correct Answer: AC
Section: (none)
Explanation

Explanation/Reference:

QUESTION 336



Refer to the exhibit. Routers R1, R2, and R3 use a protocol to identify their neighbors' IP addresses, hardware platforms, and software versions. A network engineer must configure R2 to avoid sharing any neighbor information with R3, and maintain its relationship with R1. What action meets this requirement?

- A. Configure the `no lldp receive` command on g0/1.
- B. Configure the `no cdp run` command globally.
- C. Configure the `no lldp run` command globally.
- D. Configure the `no cdp enable` command on g0/2.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 337

Which is a fact related to FTP?

- A. It always operates without user authentication.
- B. It uses block numbers to identify and mitigate data-transfer errors.
- C. It uses two separate connections for control and data traffic.
- D. It relies on the well-known UDP port 69.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 338



Refer to the exhibit. A network engineer configures the CCNA WLAN so that clients must reauthenticate hourly and to limit the number of simultaneous connections to the WLAN to

10. Which two actions complete this configuration? (Choose two.)

- A. Set the Maximum Allowed Clients Per AP Radio value to 10.
- B. Enable the Enable Session Timeout option and set the value to 3600.
- C. Enable the Wi-Fi Direct Clients Policy option.
- D. Enable the Client Exclusion option and set the value to 3600.
- E. Set the Maximum Allowed Clients value to 10.

Correct Answer: BE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 339

What are two differences between optical-fiber cabling and copper cabling? (Choose two.)

- A. The data can pass through the cladding.
- B. Light is transmitted through the core of the fiber.
- C. A BNC connector is used for fiber connections.
- D. Fiber connects to physical interfaces using RJ-45 connections.
- E. The glass core component is encased in a cladding.

Correct Answer: BE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 340

```
{"Employee's name": "Arthur"}
```

Refer to the exhibit. Which type of JSON data is shown?

- A. array
- B. Object
- C. Boolean
- D. Key

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 341

```
Cat9300# show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled
```

Refer to the exhibit. Which action must be taken so that neighboring devices rapidly discover switch Cat9300?

- A. Configure the **cdp timer 10** command on the neighbors of switch Cat9300.
- B. Enable **portfast** on the ports that connect to neighboring devices.
- C. Configure the **cdp timer 10** command on switch Cat9300.
- D. Configure the **cdp holdtime 10** command on switch Cat9300.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 342

What is a characteristic of a SOHO network?

- A. connects each switch to every other switch in the network
- B. provides high throughput access for 1000 or more users
- C. enables multiple users to share a single broadband connection
- D. includes at least three tiers of devices to provide load balancing and redundancy

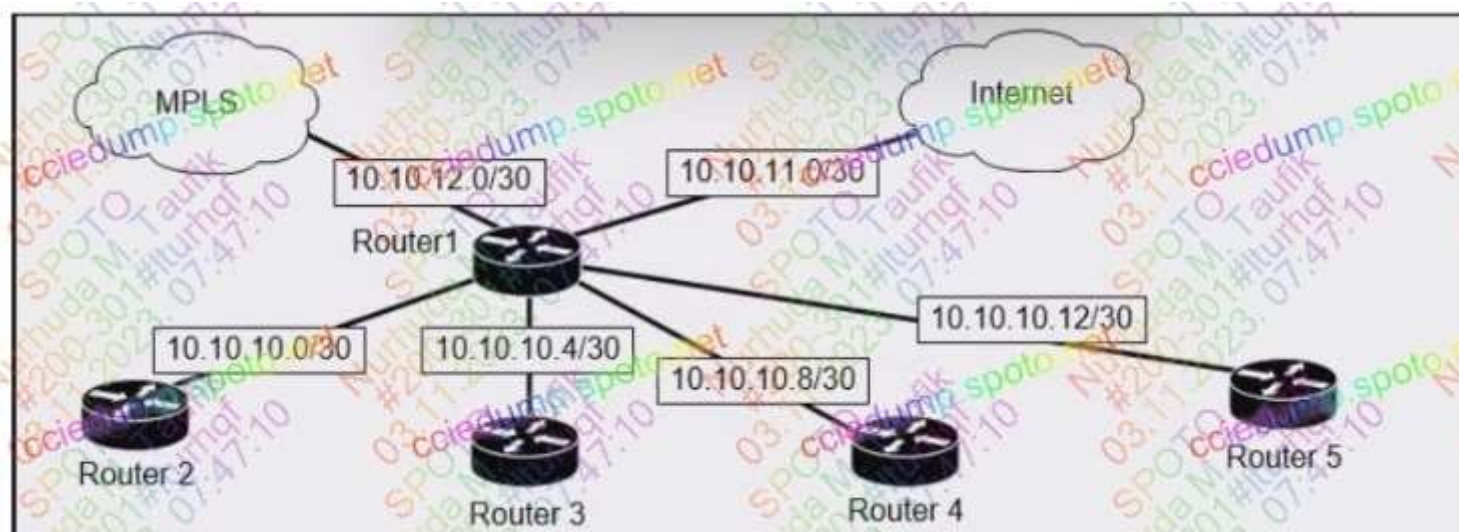
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 343




```

Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0
 209.165.200.0/27 is subnetted, 1 subnets
B    209.165.200.224 [20/0] via 10.10.12.2, 03:22:14
 209.165.201.0/27 is subnetted, 1 subnets
B    209.165.201.0 [20/0] via 10.10.12.2, 02:26:33
 209.165.202.0/27 is subnetted, 1 subnets
B    209.165.202.128 [20/0] via 10.10.12.2, 02:26:03
10.0.0.0/8 is variably subnetted, 10 subnets, 4 masks
O    10.10.13.0/25 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O    10.10.13.128/28 [110/2] via 10.10.10.5, 00:00:12, GigabitEthernet0/1
O    10.10.13.144/28 [110/2] via 10.10.10.9, 00:01:57, GigabitEthernet0/2
O    10.10.13.160/29 [110/2] via 10.10.10.5, 00:00:12, GigabitEthernet0/1
O    10.10.13.208/29 [110/2] via 10.10.10.13, 00:01:57, GigabitEthernet0/3
S*   0.0.0.0/0 [1/0] via 10.10.11.2

```

Refer to the exhibit. Which next-hop IP address does Router1 use for packets destined to host 10.10.13.158?

- A. 10.10.10.9
- B. 10.10.10.5
- C. 10.10.12.2
- D. 10.10.11.2

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 344

Which SDN plane forwards user-generated traffic?

- A. control plane
- B. management plane
- C. data plane
- D. policy plane

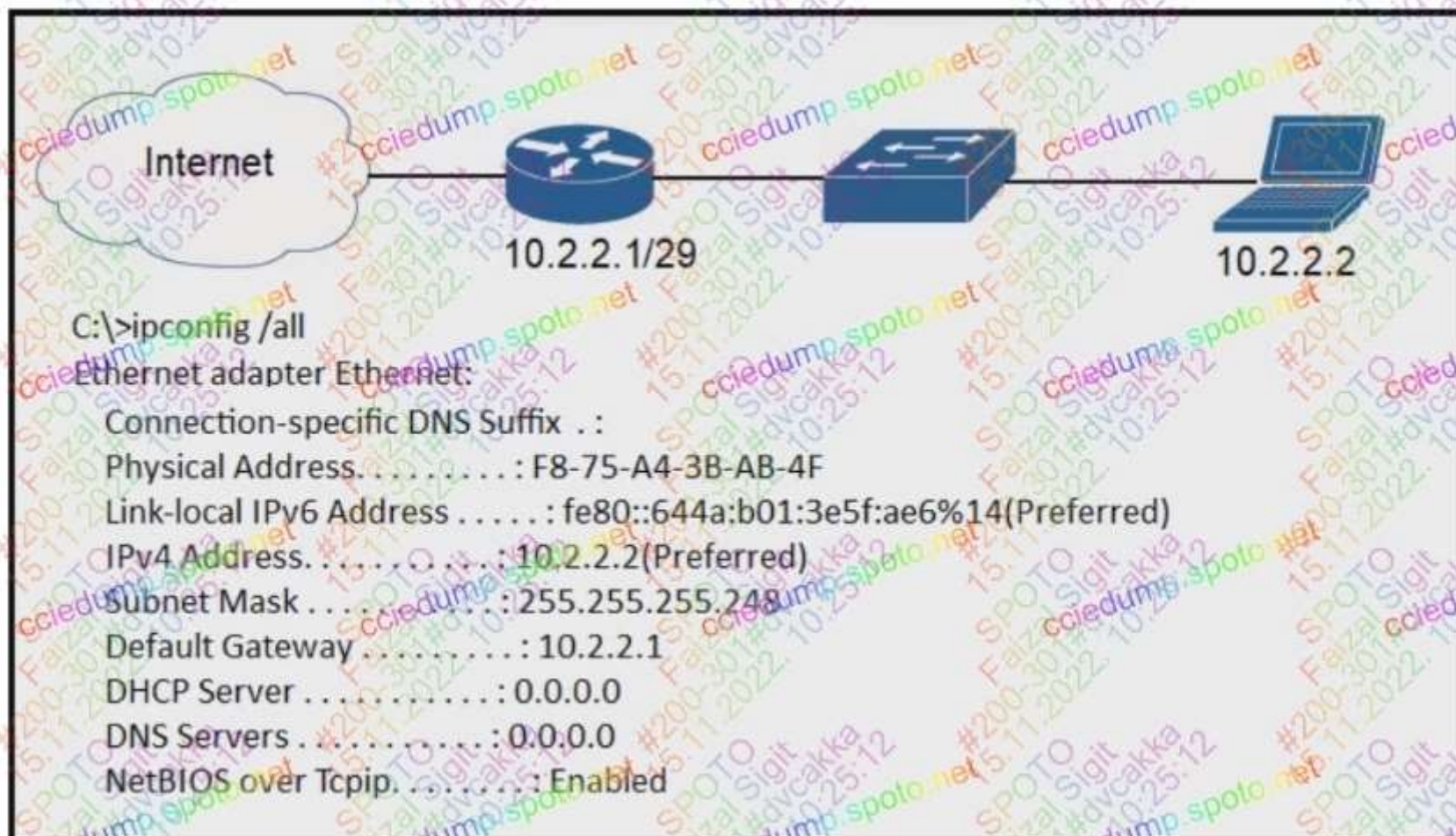
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 345



Refer to the exhibit. A newly configured PC fails to connect to the Internet by using TCP port 80 to www.cisco.com. Which setting must be modified for the connection to work?

- A. Default Gateway
- B. DHCP Servers
- C. Subnet Mask
- D. DNS Servers

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 346

Which WLC interface provides out-of-band management in the Cisco Unified Wireless Network Architecture?

- A. AP-Manager
- B. virtual
- C. service port
- D. Dynamic

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 347

A network engineer must configure an access list on a new Cisco IOS router. The access list must deny HTTP traffic to network 10.125.128.32/27 from the 192.168.240.0/20 network, but it must allow the 192.168.240.0/20 network to reach the rest of the 10.0.0.0/8 network Which configuration must the engineer apply?

- A.

```
ip access-list extended deny_outbound
10 deny tcp 192.168.240.0 0.0.15.255 any eq 80
20 deny tcp 192.168.240.0 0.0.15.255 10.125.128.32 0.0.0.31 eq 80
30 permit ip 192.168.240.0 0.0.15.255 10.0.0.0 0.255.255.255
```
- B.

```
ip access-list extended deny_outbound
10 deny tcp 192.168.240.0 0.0.15.255 10.125.128.32 0.0.0.31 eq 80
20 permit ip 192.168.240.0 0.0.15.255 10.0.0.0 0.255.255.255
30 deny ip any any log
```
- C.

```
ip access-list extended deny_outbound
10 deny tcp 10.125.128.32 255.255.255.224 192.168.240.0 255.255.240.0 eq 443
20 deny tcp 192.168.240.0 255.255.240.0 10.125.128.32 255.255.255.224 eq 443
30 permit ip 192.168.240.0 255.255.240.0 10.0.0.0 255.0.0.0
```
- D.

```
ip access-list extended deny_outbound
10 permit ip 192.168.240.0 255.255.240.0 10.0.0.0 255.0.0.0
20 deny tcp 192.168.240.0 255.255.240.0 10.125.128.32 255.255.255.224 eq 443
30 permit ip any any
```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 348

Which functionality is provided by the console connection on a Cisco WLC?

- A. unencrypted in-band connectivity for file transfers
- B. out-of-band management
- C. secure in-band connectivity for device administration
- D. HTTP-based GUI connectivity

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 349



Refer to the exhibit. A network engineer updates the existing configuration on interface FastEthernet1/1 switch SW1. It must establish an EtherChannel by using the same group designation with another vendor switch. Which configuration must be performed to complete the process?

- A.

```
interface port-channel 2
channel-group 2 mode desirable
```
- B.

```
interface fastethernet 1/1
channel-group 2 mode on
```
- C.

```
interface port-channel 2
channel-group 2 mode auto
```
- D.

```
interface fastethernet 1/1
channel-group 2 mode active
```


Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 350

Which enhancements were implemented as part of WPA3?

- A. forward secrecy and SAE in personal mode for secure initial key exchange
- B. AES-64 in personal mode and AES-128 in enterprise mode
- C. TKIP encryption improving WEP and per-packet keying
- D. 802.1x authentication and AES-128 encryption

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 351

Which enhancements were implemented as part of WPA3?

- A. forward secrecy and SAE in personal mode for secure initial key exchange
- B. AES-64 in personal mode and AES-128 in enterprise mode
- C. TKIP encryption improving WEP and per-packet keying
- D. 802.1x authentication and AES-128 encryption

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 352

Which enhancement is implemented in WPA3?

- A. protects against brute force attacks
- B. applies 802.1x authentication
- C. employs PKI to identify access points
- D. uses TKIP

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 353

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#ip domain-name CC-Net.com
R1(config)#enable secret Passfornewuser
R1(config)#line vty 0 15
R1(config-line)#transport input ssh
R1(config-line)#login local
```

Refer to the exhibit. A network administrator is configuring a router for user access via SSH. The service-password encryption command has been issued. The configuration must meet these requirements:

1. Create the username as CCUser.
2. Create the password as NA!2\$cc.
3. Encrypt the user password.

What must be configured to meet the requirements?

- A. username CCUser password NA!2\$cc
enable password level 5 NA!2\$cc
- B. username CCUser privilege 15 password NA!2\$cc
enable secret 0 NA!2\$cc
- C. username CCUser privilege 10 password NA!2\$cc
- D. username CCUser secret NA!2\$cc

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 354



Refer to the exhibit. An engineer is configuring an EtherChannel using LACP between Switches 1 and 2. Which configuration must be applied so that only Switch 1 sends LACP initiation packets?

- A. Switch1(config-if)#channel-group 1 mode passive
Switch2(config-if)#channel-group 1 mode active
- B. Switch1(config-if)#channel-group 1 mode on
Switch2(config-if)#channel-group 1 mode active
- C. Switch1(config-if)#channel-group 1 mode on
Switch2(config-if)#channel-group 1 mode passive
- D. Switch1(config-if)#channel-group 1 mode active
Switch2(config-if)#channel-group 1 mode passive

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 355

What is an appropriate use for private IPv4 addressing?

- A. on hosts that communicates only with other internal hosts
- B. on the public-facing interface of a firewall
- C. on internal hosts that stream data solely to external resources
- D. to allow hosts inside to communicate in both directions with hosts outside the organization

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 356

Which authentication method requires the user to provide a physical attribute to authenticate successfully?

- A. multifactor
- B. biometric
- C. password
- D. certificate

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 357

What is a reason to implement IPv4 private addressing?

- A. Reduce the size of the forwarding table on network routers.
- B. Reduce the risk of a network security breach.
- C. Comply with PCI regulations.
- D. Comply with local law.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 358


```

R1#show ip ospf interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Internet address is 192.168.1.2/24, Area 0
Process ID 1, Router ID 192.168.1.2, Network Type POINT-TO-POINT, Cost: 1
Transmit Delay is 1 sec, State POINT-TO-POINT,
Timer intervals configured, Hello 15, Dead 20, Wait 20, Retransmit 5
Hello due in 00:00:08
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Suppress hello for 0 neighbor(s)

R2#show ip ospf interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Internet address is 192.168.1.1/24, Area 0
Process ID 1, Router ID 10.1.1.1, Network Type POINT-TO-POINT, Cost: 1
Transmit Delay is 1 sec, State POINT-TO-POINT,
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:11
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Suppress hello for 0 neighbor(s)

```

Refer to the exhibit. The network engineer is configuring router R2 as a replacement router on the network. After the initial configuration is applied, it is determined that R2 failed to show R1 as a neighbor. Which configuration must be applied to R2 to complete the OSPF configuration and enable it to establish the neighbor relationship with R1?

- A. R2(config)# router ospf 1
R2(config-router)# network 192.168.1.0 255.255.255.0 area 2
R2(config-router)# network 10.1.1.0 255.255.255.255 area 2
- B. R2(config)# router ospf 1
R2(config-router)# router-id 192.168.1.2
- C. R2(config)# interface g0/0/0
R2(config-if)# ip ospf hello-interval 15
R2(config-if)# ip ospf dead-interval 20
- D. R2(config)# interface g0/0/0
R2(config-if)# ip ospf dead-interval 20

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 359

An engineer must configure a core router with a floating static default route to the backup router at 10.200.0.2. Which command meets the requirements?

- A. ip route 0.0.0.0 0.0.0.0 10.200.0.2
- B. ip route 0.0.0.0 0.0.0.0 10.200.0.2
- C. ip route 0.0.0.0 0.0.0.0 10.200.0.2 floating
- D. ip route 0.0.0.0 0.0.0.0 10.200.0.2 10

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 360

Which function generally performed by a traditional network device is replaced by a software-defined controller?

- A. changing the source or destination address during NAT operations
- B. building route tables and updating the forwarding table
- C. encapsulation and decapsulation of packets in a data-link frame
- D. encryption and decryption for VPN link processing

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 361

An office has 8 floors with approximately 30-40 users per floor. One subnet must be used. Command must be configured on the router Switched Virtual interface to use address efficiently?

- A. ip address 192.168.0.0 255.255.254.0
- B. ip address 192.168.0.0 255.255.255.224
- C. ip address 192.168.0.0 255.255.255.128
- D. ip address 192.168.0.0 255.255.0.0

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 362

An engineer is configuring NAT to translate the source subnet of 10.10.0.0/24 to any one of three addresses:192.168.3.1, 192.168.3.2, or 192.168.3.3. Which configuration should be used?

- A.


```
enable
configure terminal
ip nat pool mypool 192.168.3.1 192.168.3.3 prefix-length 30
access-list 1 permit 10.10.0.0 0.0.0.255
ip nat outside destination list 1 pool mypool
interface g1/1
ip nat inside
interface g1/2
ip nat outside
```
- B.


```
enable
configure terminal
ip nat pool mypool 192.168.3.1 192.168.3.3 prefix-length 30
access-list 1 permit 10.10.0.0 0.0.0.254
ip nat inside source list 1 pool mypool
interface g1/1
ip nat inside
interface g1/2
ip nat outside
```
- C.


```
enable
configure terminal
ip nat pool mypool 192.168.3.1 192.168.3.3 prefix-length 30
access-list 1 permit 10.10.0.0 0.0.0.255
ip nat inside source list 1 pool mypool
interface g1/1
ip nat inside
interface g1/2
ip nat outside
```
- D.


```
enable
configure terminal
ip nat pool mypool 192.168.3.1 192.168.3.3 prefix-length 30
route-map permit 10.10.0.0 255.255.255.0
ip nat outside destination list 1 pool mypool
interface g1/1
ip nat inside
interface g1/2
ip nat outside
```

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 363

While examining excessive traffic on the network, it is noted that all incoming packets on an interface appear to be allowed even though an IPv4 ACL is applied to the interface. Which two misconfigurations cause this behavior? (Choose two.)

- A. A matching permit statement is too broadly defined.
- B. A matching permit statement is too high in the access list.
- C. The packets fail to match any permit statement.
- D. The ACL is empty.
- E. A matching deny statement is too high in the access list.

Correct Answer: AB

Section: (none)

Explanation

Explanation/Reference:

QUESTION 364

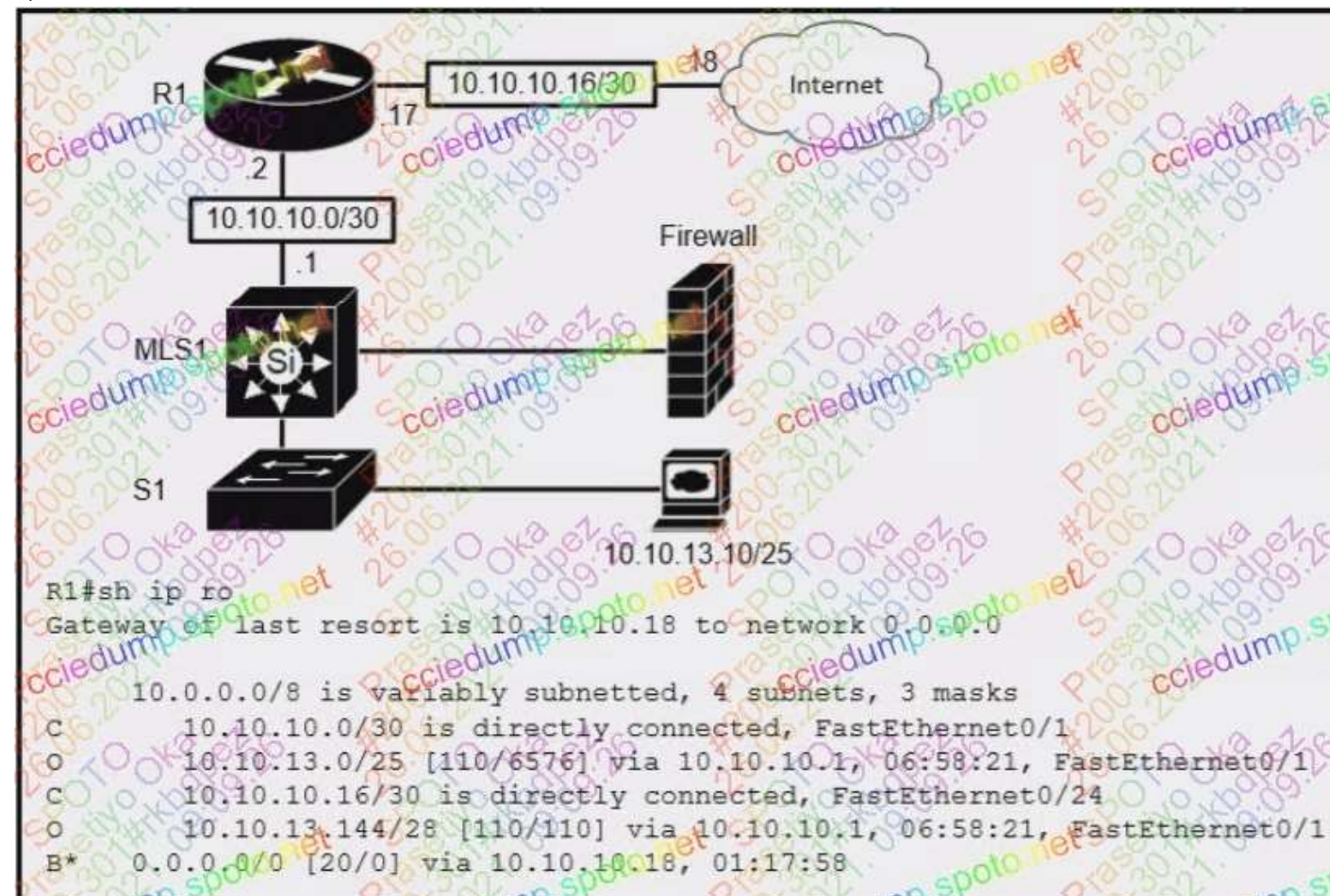
What is a purpose of traffic shaping?

- A. It provides best-effort service.
- B. It enables dynamic flow identification.
- C. It limits bandwidth usage.
- D. It enables policy-based routing.

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 365



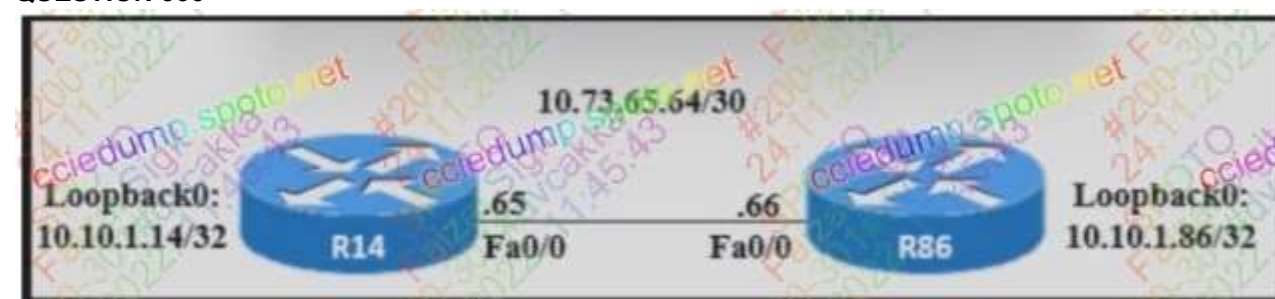
Refer to the exhibit. Which route type is configured to reach the internet?

- A. network route
- B. floating static route
- C. default route
- D. host route

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 366



Refer to the exhibit. A static route must be configured on R14 to forward traffic for the 172.21.34.0/25 network that resides on R86. Which command must be used to fulfill the request? .

- A. ip route 172.21.34.0 255.55.255.128 10.73.65.66
- B. ip route 172.21.34.0 255.255.128.0 10.73.65.64
- C. ip route 172.21.34.0 255.255.255.0 10.73.65.65
- D. ip route 172.21.34.0 255.255.255.192 10.73.65.65

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 367



Refer to the exhibit. All routers in the network are configured. R2 must be the DR. After the engineer connected the devices, R1 was elected as the DR. Which command sequence must be configured on R2 to be elected as the DR in the network?

- A. R2(config)# router ospf 1
R2(config-router)# router-id 10.100.100.100
- B. R2(config)# interface gi0/0
R2(config-if)# ip ospf priority 1
- C. R2(config)# router ospf 1
R2(config-router)# router-id 192.168.2.7
- D. R2(config)# interface gi0/0
R2(config-if)# ip ospf priority 100

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 368

Which interface is used for out-of-band management on a WLC?

- A. dynamic
- B. service port
- C. Virtual
- D. Management

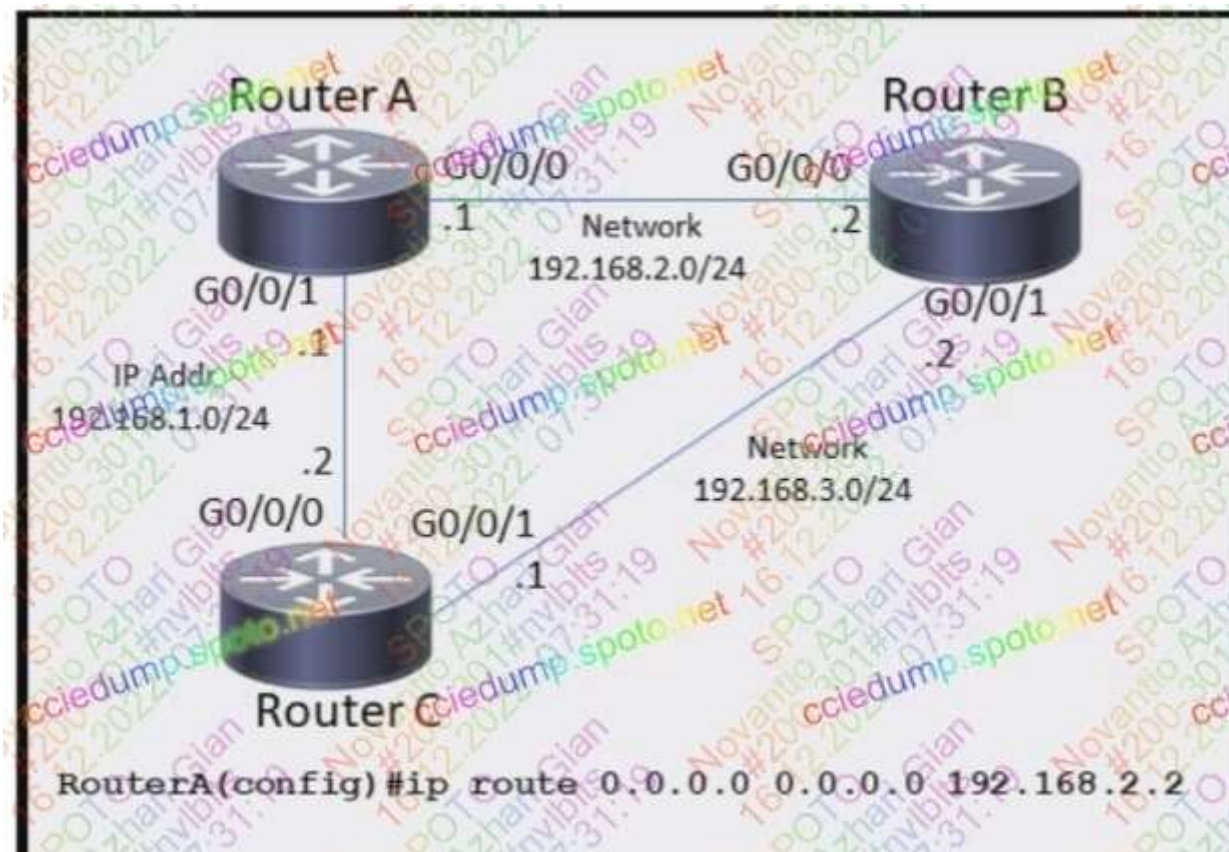
Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 369



Refer to the exhibit. Which command must be issued to enable a floating static default route on router A?

- A. ip route 0.0.0.0 0.0.0.0 192.168.1.2
- B. ip default-gateway 192.168.2.1
- C. ip route 0.0.0.0 0.0.0.0 192.168.1.2 10
- D. ip route 0.0.0.0 0.0.0.0 192.168.2.1 10

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

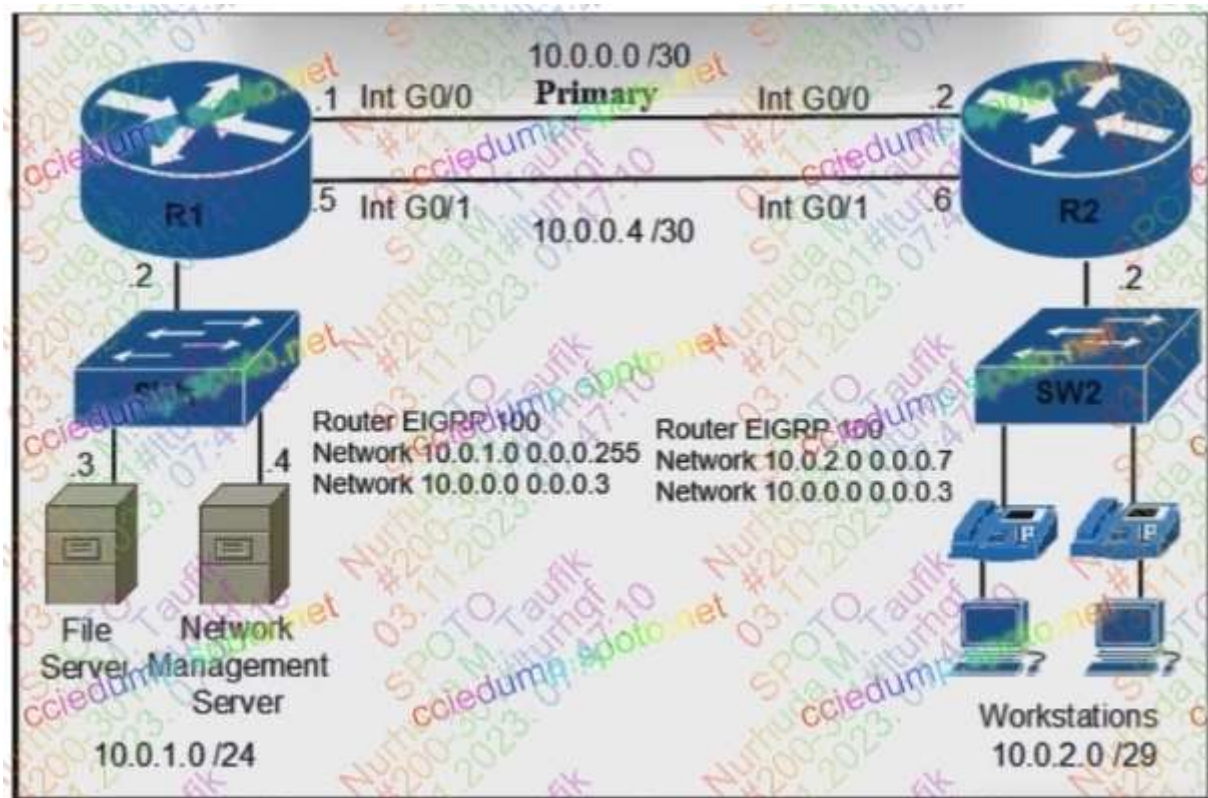
QUESTION 370
 Which cipher is supported for wireless encryption only with the WPA2 standard?

- A. AES256
- B. RC4
- C. SHA
- D. AES

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 371



Refer to the exhibit. A secondary route is required on router R1 to pass traffic to the LAN network on R2 if the primary link fails. Which command must be entered to configure the router?

- A. ip route 10.0.2.0 255.255.255.240 10.0.0.7 92
- B. ip route 10.0.2.0 255.255.255.248 10.0.0.6 91
- C. ip route 10.0.2.0 255.255.255.248 null0 93
- D. ip route 10.0.2.0 255.255.255.240 10.0.0.6 91

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 372



Refer to the exhibit. This ACL is configured to allow client access only to HTTP, HTTPS, and DNS services via UDP. The new administrator wants to add TCP access to the DNS service. Which continuation updates the ACL efficiently?

- A. ip access-list extended Services
35 permit tcp 10.0.0.0 0.255.255.255 host 198.51.100.11 eq domain
- B. ip access-list extended Services
permit tcp 10.0.0.0 0.255.255.255 host 198.51.100.11 eq domain

- C. no ip access-list extended Services
ip access-list extended Services
permit udp 10.0.0.0 0.255.255.255 any eq 53
permit tcp 10.0.0.0 0.255.255.255 host 198.51.100.11 eq domain
deny ip any any log
- D. no ip access-list extended Services
ip access-list extended Services
30 permit tcp 10.0.0.0 0.255.255.255 host 198.51.100.11 eq domain

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 373

```
SW1#show spanning-tree vlan 30
VLAN0030
Spanning tree enabled protocol rstp
Root ID    Priority
Address    0025.63e9.c800
Cost       19
Port       1 (FastEthernet 2/1)
Hello Time  2 sec
Max Age    30 sec
Forward Delay 20 sec
[Output suppressed]
```

Refer to the exhibit. What two conclusions should be made about this configuration? (Choose two.)

- A. The spanning-tree mode is PVST+
- B. The root port is FastEthernet 2/1.
- C. The spanning-tree mode is Rapid PVST+.
- D. This is a root bridge.
- E. The designated port is FastEthernet 2/1.

Correct Answer: BC

Section: (none)

Explanation

Explanation/Reference:

QUESTION 374

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, ! - next hop override, p - overrides from PFR
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
S* 0.0.0.0/0 is directly connected, Null0
C 10.0.0.0/8 is variably subnetted, 6 subnets, 1 masks
C 10.0.12.0/24 is directly connected, GigabitEthernet0/1
L 10.0.12.1/32 is directly connected, GigabitEthernet0/1
C 10.0.13.0/24 is directly connected, GigabitEthernet0/2
C 10.0.13.1/32 is directly connected, GigabitEthernet0/2
C 10.0.14.0/24 is directly connected, GigabitEthernet0/3
L 10.0.14.1/32 is directly connected, GigabitEthernet0/3
D 192.168.0.0/16 [90/130816] via 10.0.13.3, 00:10:09, GigabitEthernet0/2
O 192.168.0.0/23 [110/2] via 10.0.14.4, 00:00:46, GigabitEthernet0/3
S 192.168.0.0/24 [100/0] via 10.0.12.2
```

Refer to the exhibit. Which interface is chosen to forward traffic to the host at 192.168.0.55?

- A. GigabitEthernet0/2
- B. Null0
- C. GigabitEthernet0/3
- D. GigabitEthernet0/1

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 375

Which 802.11 frame type is indicated by a probe response after a client sends a probe request?

- A. Action
- B. Management
- C. Control

D. Data

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Table 4-9. Class 1 frames		
Control	Management	Data
Request to Send (RTS)	Probe Request	Any frame with ToDS and FromDS false (0)
Clear to Send (CTS)	Probe Response	
Acknowledgment (ACK)	Beacon	
CF-End	Authentication	
CF-End+CF-Ack	Deauthentication	
	Announcement Traffic Indication Message (ATIM)	

QUESTION 376

```
Router1#show ip route
Gateway of last resort is not set
209.165.200.0/27 is subnetted, 1 subnets
B 209.165.200.224 [20/0] via 10.10.10.1, 00:09:57
10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C 10.10.10.0/28 is directly connected, GigabitEthernet0/0
C 10.10.11.0/30 is directly connected, FastEthernet2/0
O 10.10.13.0/24 [110/2] via 10.10.10.1, 00:08:34, GigabitEthernet0/0
C 10.10.12.0/30 is directly connected, GigabitEthernet0/1
```

Refer to the exhibit. Which action is taken by the router when a packet is sourced from 10.10.10.2 and destined for 10.10.10.16?

- A. It queues the packets waiting for the route to be learned.
- B. It discards the packets.
- C. It uses a route that is similar to the destination address.
- D. It floods packets to all learned next hops.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 377

What is the function of a controller in controller-based networking?

- A. It is a pair of core routers that maintain all routing decisions for a campus.
- B. It centralizes the data plane for the network.
- C. It serves as the centralized management point of an SDN architecture.
- D. It is the card on a core router that maintains all routing decisions for a campus.

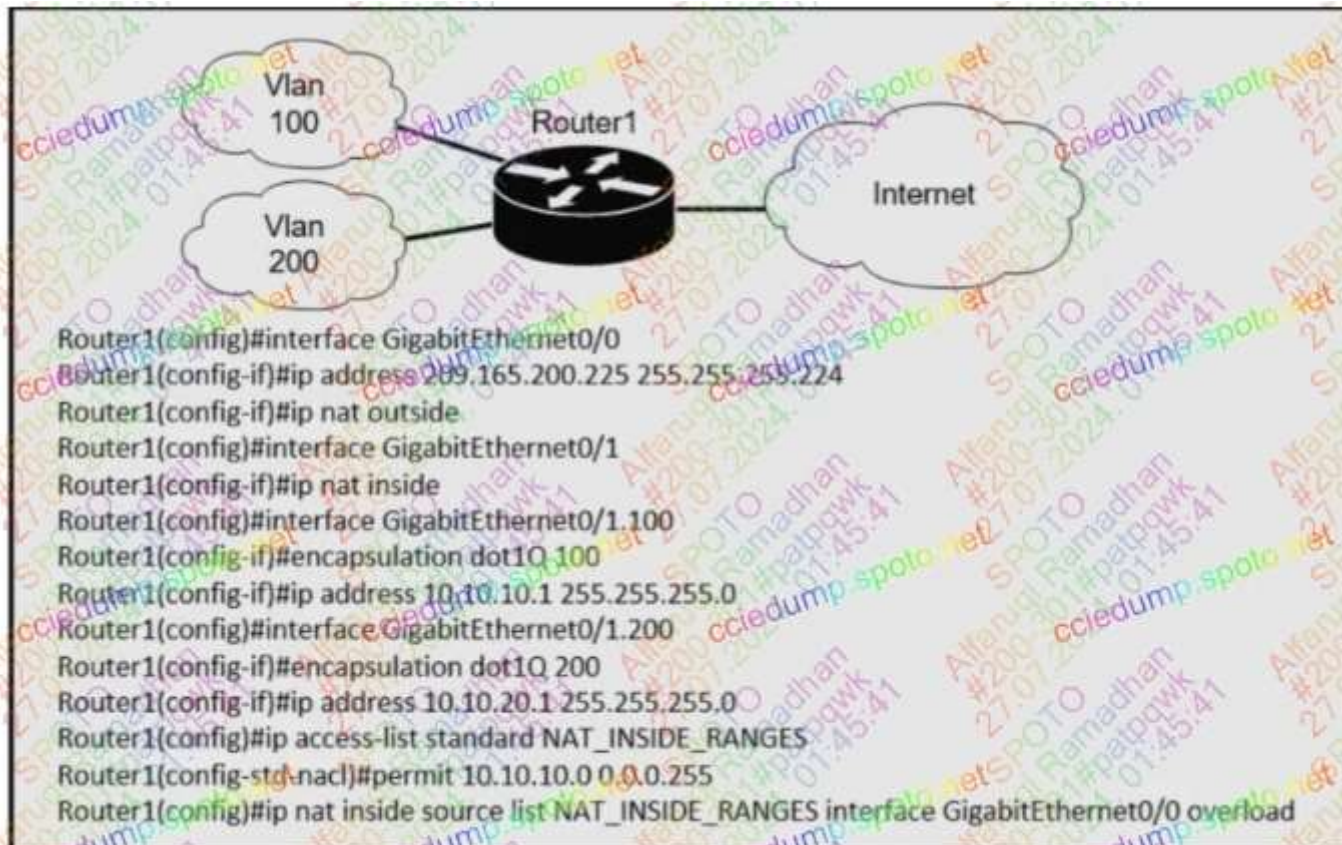
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 378



Refer to the exhibit. Users on existing VLAN 100 can reach sites on the Internet. Which action must the administrator take to establish connectivity to the Internet for users in VLAN 200?

- A. Configure static NAT translations for VLAN 200.
- B. Define a NAT pool on the router.
- C. Update the NAT_INSIDE_RANGES ACL.
- D. Configure the **ip not outside** command on another interface for VLAN 200.

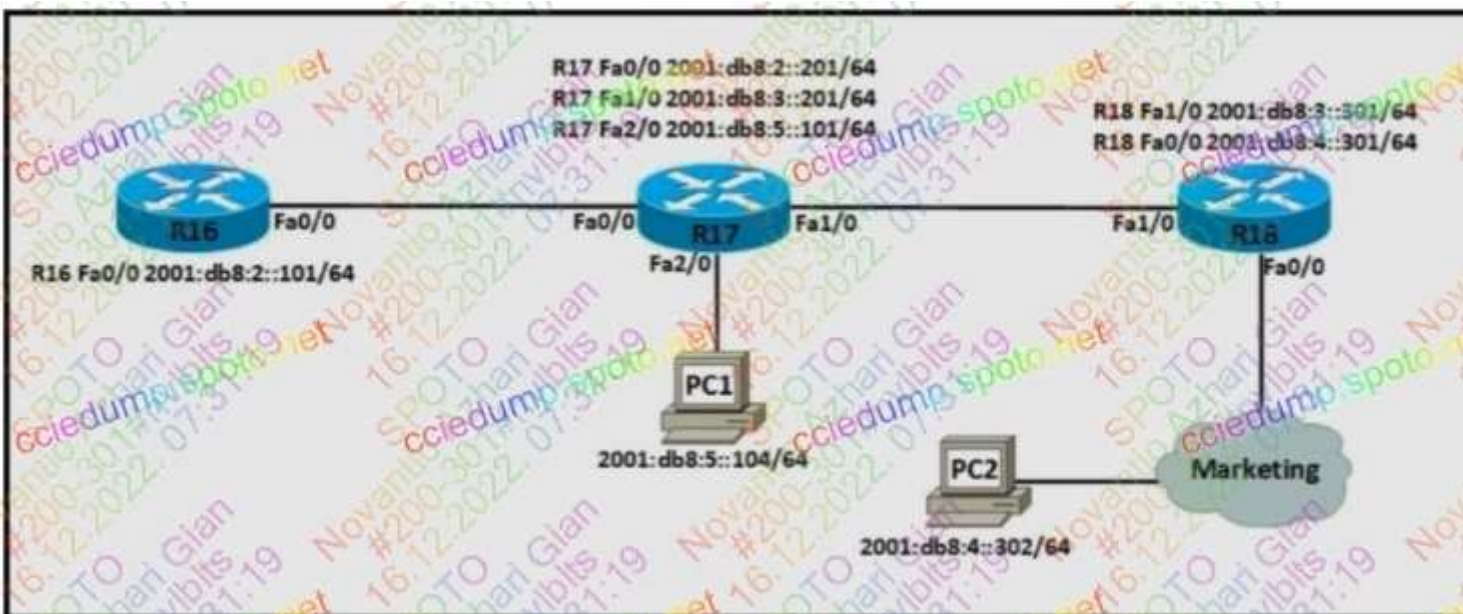
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 379



Refer to the exhibit Which IPv6 configuration is required for R17 to successfully ping the WAN interface on R18?

- A.


```

R17#
!
no ip domain lookup
ip cef
ipv6 unicast-routing
!
interface FastEthernet0/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:2::201/64
!
interface FastEthernet1/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:3::201/64
!
no cdp log mismatch duplex
ipv6 route 2001:DB8:4::/64 2001:DB8:3::301
      
```


- B. R17#
!
no ip domain lookup
ip cef
ipv6 cef
!
interface FastEthernet0/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:2::201/64
!
interface FastEthernet1/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:3::201/64
!
no cdp log mismatch duplex
ipv6 route 2001:DB8:4::/64 2001:DB8:4::302
- C. R17#
!
no ip domain lookup
ip cef
ipv6 unicast-routing
!
interface FastEthernet0/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:2::201/64
!
interface FastEthernet1/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:3::201/64
!
no cdp log mismatch duplex
ipv6 route 2001:DB8:4::/64 2001:DB8:2::201
- D. R17#
!
no ip domain lookup
ip cef
!
interface FastEthernet0/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:3::201/64
!
interface FastEthernet1/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:2::201/64
!
no cdp log mismatch duplex
ipv6 route 2001:DB8:4::/64 2001:DB8:5::101

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 380

A client experiences slow throughput from a server that is directly connected to the core switch in a data center. A network engineer finds minimal latency on connections to the server, but data transfers are unreliable, and the output of the show interfaces counters errors command shows a high FCS-Err count on the interface that is connected to the server.

What is the cause of the throughput issue?

- A. high bandwidth usage
- B. a physical cable fault
- C. a cable that is too long
- D. a speed mismatch

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 381

Which Rapid PVST+ port state does a port operate in without receiving BPDUs from neighbors or updating the address database?

- A. blocking
- B. disabled
- C. listening
- D. forwarding

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 382

What is a characteristic of RSA?

- A. It requires both sides to have identical keys.
- B. It is a public-key cryptosystem.
- C. It is a private-key encryption algorithm.
- D. It uses preshared keys for encryption.

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 383



```
TenGigabitEthernet0/0/0 is up, line protocol is up
Hardware is BUILT-IN-2T+6X1GE, address is 74a0.2f7a.0123 (bia 74a0.2f7a.0123)
Description: Uplink
Internet address is 10.1.1.1/24
MTU 1500 bytes, BW 10000000 Kbit/sec, DLY 10 usec,
  reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive not supported
Full Duplex, 10000Mbps, link type is force-up, media type is unknown media type
output flow-control is on, input flow-control is on
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:00, output 00:05:40, output hang never
Last clearing of "show interface" counters never
Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
  5 minute input rate 6160000 bits/sec, 1111 packets/sec
  5 minute output rate 11213000 bits/sec, 1553 packets/sec
  12662416065 packets input, 12607032232894 bytes, 0 no buffer
  Received 14117163 broadcasts (0 IP multicasts)
  0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
  0 watchdog, 26271385 multicast, 0 pause input
  7907779058 packets output, 5073750426832 bytes, 0 underruns
  0 output errors, 8662416065 collisions, 1 interface resets
  0 unknown protocol drops
  0 babbles, 0 late collision, 0 deferred
  0 lost carrier, 0 no carrier, 0 pause output
  0 output buffer failures, 0 output buffers swapped out
  1 carrier transitions
```

Refer to the exhibit. Traffic that is flowing over interface TenGigabitEthernet0/0 experiences slow transfer speeds. What is the reason for the issue?

- A. a duplex incompatibility
- B. queuing drops
- C. a speed conflict
- D. heavy traffic congestion

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 384

What is a function of a southbound API?

- A. Automate configuration changes between a server and a switching fabric.
- B. Facilitate the information exchange between an SDN controller and application.
- C. Manage flow control between an SDN controller and a switching fabric.
- D. Use orchestration to provision a virtual server configuration from a web server.

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

QUESTION 385

What is the purpose of the URI string in a REST request?

- A. to transport data or payload to a remote resource
- B. to specify the way in which a remote resource is modified
- C. to respond with the data content encoding for a request
- D. to identify a resource on a target server

Correct Answer: A
Section: (none)

Explanation

Explanation/Reference:

QUESTION 386

Which IPv6 address block forwards packets to a multicast address rather than a unicast address?

- A. FF00::/12
- B. 2000::/3
- C. FC00::/7
- D. FE80::/10

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 387

A network administrator wants the syslog server to filter incoming messages into different files based on their importance. Which filtering criteria must be used?

- A. message body
- B. facility
- C. level
- D. process ID

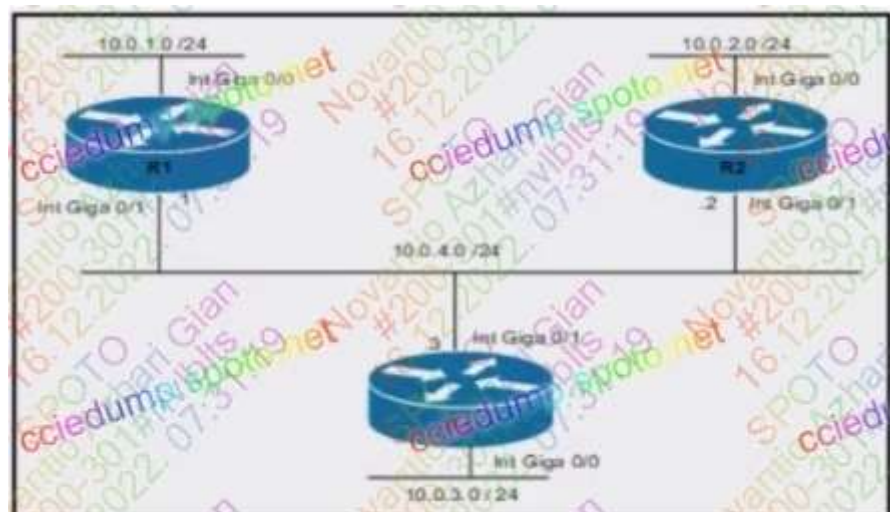
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 388



Refer to the exhibit. Routers R1 and R3 have the default configuration. The router R2 priority is set to 99. Which commands on R3 configure it as the DR in the 10.0.4.0/24 network?

- A. R3(config)# interface Gig0/1
R3(config-if)# ip ospf priority 0
- B. R3(config)# interface Gig0/0
R3(config-if)# ip ospf priority 1
- C. R3(config)# interface Gig0/0
R3(config-if)# ip ospf priority 100
- D. R3(config)# interface Gig0/1
R3(config-if)# ip ospf priority 100

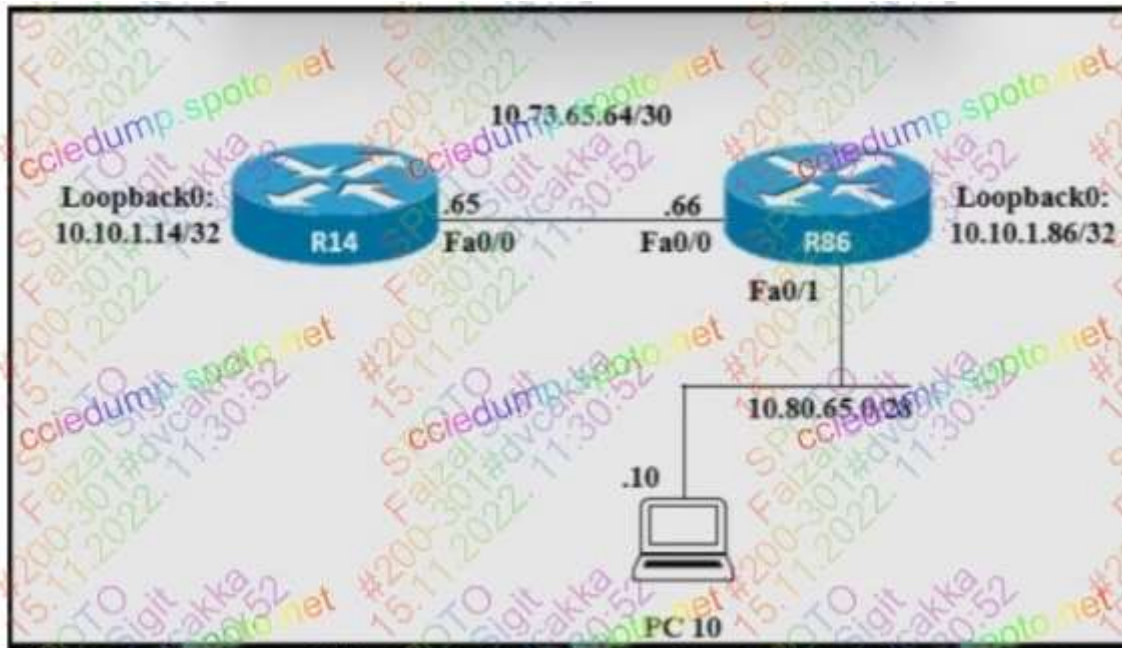
Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 389



Refer to the exhibit. Router R14 is in the process of being configured. Which configuration must be used to establish a host route to PC 10?

- A. ip route 10.73.65.65 255.0.0.0 10.80.65.10
- B. ip route 10.73.65.66 0.0.0.255 10.80.65.10
- C. ip route 10.80.65.10 255.255.255.255 10.73.65.66
- D. ip route 10.80.65.10 255.255.255.254 10.80.65.1

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 390

Which type of traffic is sent with pure IPsec?

- A. broadcast packets from a switch that is attempting to locate a MAC address at one of several remote sites
- B. unicast messages from a host at a remote site to a server at headquarters.
- C. multicast traffic from a server at one site to hosts at another location
- D. spanning-tree updates between switches that are at two different sites

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 391

Aside from discarding, which two states does the switch port transition through while using RSTP (802.1w)? (Choose two.)

- A. speaking
- B. blocking
- C. forwarding
- D. listening
- E. learning

Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 392

Which key function is provided by the data plane?

- A. exchanging routing table data
- B. originating packets
- C. forwarding traffic to the next hop
- D. making routing decisions

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 393



Refer to the exhibit. The primary route across Gi0/0 is configured on both routers. A secondary route must be configured to establish connectivity between the workstation networks. Which command set must be configured to complete this task?

- A. R1
ip route 172.16.2.0 255.255.255.224 172.16.0.6 111
R2
ip route 172.16.1.0 255.255.255.0 172.16.0.5 112
- B. R1
ip route 172.16.2.0 255.255.255.248 172.16.0.5 110
R2
ip route 172.16.1.0 255.255.255.0 172.16.0.6 110
- C. R1
ip route 172.16.2.0 255.255.255.240 172.16.0.2 113
R2
ip route 172.16.1.0 255.255.255.0 172.16.0.1 114
- D. R1
ip route 172.16.2.0 255.255.255.240 172.16.0.5 89
R2
ip route 172.16.1.0 255.255.255.0 172.16.0.6 89

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 394

An administrator must secure the WLC from receiving spoofed association requests. Which steps must be taken to configure the WLC to restrict the requests and force the user to wait 10 ms to retry an association request?

- A. Enable MAC filtering and set the SA Query timeout to 10.
- B. Enable the Protected Management Frame service and set the Comeback timer to 10.
- C. Enable 802.1X Layer 2 security and set the Comeback timer to 10.
- D. Enable Security Association Teardown Protection and set the SA Query timeout to 10.

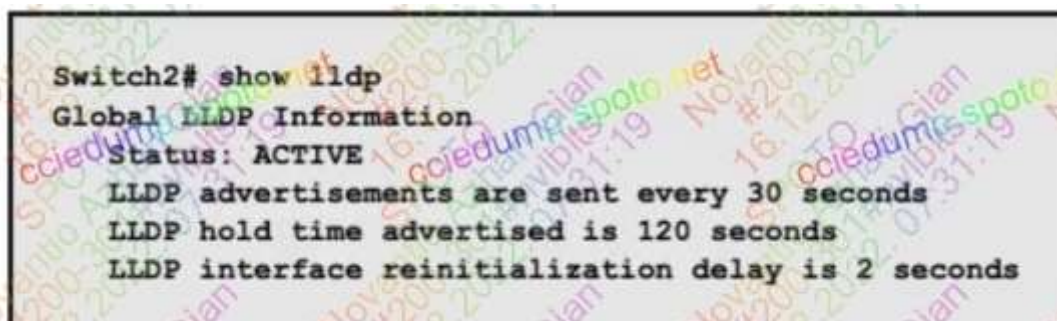
Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 395



Refer to the exhibit. A network engineer must update the configuration on Switch2 so that it sends LLDP packets every minute and the information sent via LLDP is refreshed every 3 minutes. Which configuration must the engineer apply?

- A. Switch2(config)# lldp timer 1
Switch2(config)# lldp tlv-select 3
- B. Switch2(config)# lldp timer 60
Switch2(config)# lldp holdtime 180
- C. Switch2(config)# lldp timer 1

```
Switch2(config)# lldp holdtime 3
D. Switch2(config)# lldp timer 60
Switch2(config)# lldp tlv-select 180
```

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

<https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst4500/12-2/46sg/configuration/guide/Wrapper-46SG/swlldp.html#wp1084239>

QUESTION 396



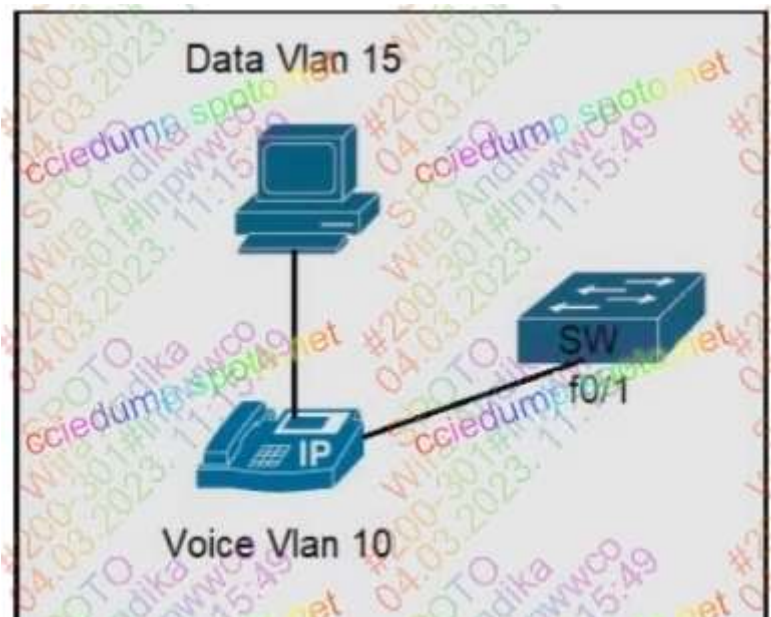
Refer to the exhibit. Switch A is newly configured. All VLANs are present in the VLAN database. The IP phone and PC A on Gi0/1 must be configured for the appropriate VLANs to establish connectivity between the PCs Which command set fulfills the requirement? .

- A. SwitchA(config-if)# switchport mode trunk
SwitchA(config-if)# switchport trunk allowed vlan add 50,51
SwitchA(config-if)# switchport voice vlan dot1p
- B. SwitchA(config-if)# switchport mode access
SwitchA(config-if)# switchport access vlan 50
SwitchA(config-if)# switchport voice vlan untagged
- C. SwitchA(config-if)# switchport mode trunk
SwitchA(config-if)# switchport trunk allowed vlan 50,51
SwitchA(config-if)# mls qos trust cos
- D. SwitchA(config-if)# switchport mode access
SwitchA(config-if)# switchport access vlan 50
SwitchA(config-if)# switchport voice vlan 51

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 397




```
SW#show run
Building configuration...
!
interface FastEthernet0/1
  switchport access vlan 15
!
end
```

Refer to the exhibit. All VLANs are present in the VLAN database. Which command sequence must be applied to complete the configuration?

- A. Interface FastEthernet0/1
switchport trunk allowed vlan add 10
vlan 10
private-vlan isolated
- B. interface FastEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,15
- C. interface FastEthernet0/1
switchport trunk native vlan 10
switchport trunk allowed vlan 10,15
- D. interface FastEthernet0/1
switchport mode access
switchport voice vlan 10

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 398

A Cisco engineer at a new branch office is configuring a wireless network with access points that connect to a controller that is based at corporate headquarters. Wireless client traffic must terminate at the branch office and access-point survivability is required in the event of a WAN outage. Which access point mode must be selected?

- A. FlexConhect with local switching enabled
- B. Local with AP fallback enabled
- C. Lightweight with local switching disabled
- D. OfficeExtend with high availability disabled

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 399



Refer to the exhibit. An engineer is using the Cisco WLC GUI to configure a WLAN for WPA2 encryption with AES and preshared key Cisc0123456. After the engineer selects the WPA WPA2 option from the Layer 2 Security drop-down list, which two tasks must they perform to complete the process? (Choose two.)

- A. Select the WPA2 Policy and AES check boxes.

- B. Select CCKM from the Auth Key Mgmt drop-down list, set the PSK Format to Hex, and enter the key.
- C. Select ASCII from the PSK Format drop-down list, enter the key, and leave the Auth Key Mgmt setting blank.
- D. Select the WPA2 Policy, AES, and TKIP check boxes.
- E. Select PSK from the Auth Key Mgmt drop-down list, set the PSK Format to ASCII, and enter the key.

Correct Answer: AE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 400

Which two statements distinguish authentication from accounting? (Choose two.)

- A. Only authentication provides supporting information for billing users.
- B. Only authentication records the duration of a user's connection.
- C. Only authentication validates "who you are."
- D. Only authentication challenges users for their credentials and returns a response.
- E. Only authentication supports user-activity audits.

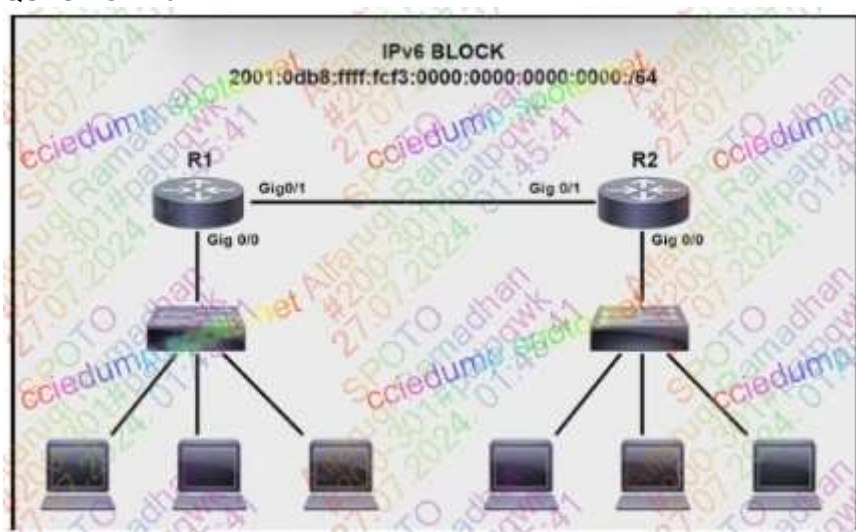
Correct Answer: CD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 401



Refer to the exhibit. IPv6 is being implemented within the enterprise. The command `ipv6 unicast-routing` is configured. Interface Gig0/0 on R1 must be configured to provide a dynamic assignment using the assigned IPv6 block. Which command accomplishes this task?

- A. `ipv6 address autoconfig 2001:DB8:FFFF:FCF2::/64`
- B. `ipv6 address 2001:DB8:FFFF:FCF3::/64 eui-64`
- C. `ipv6 address 2001:DB8:FFFF:FCF3::/64 link-local`
- D. `ipv6 address 2001:DB8:FFFF:FCF3::1/64`

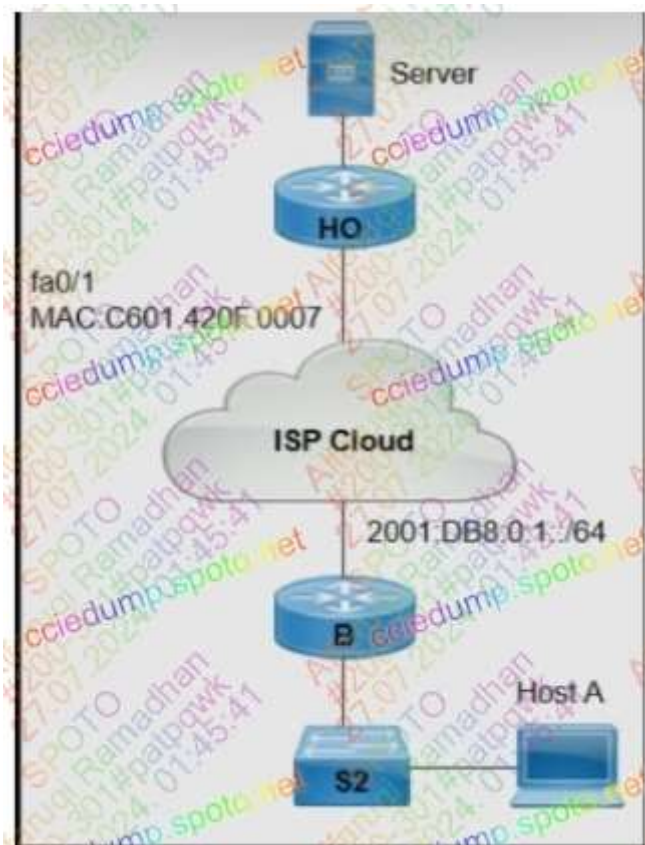
Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 402



Refer to the exhibit. An engineer is configuring the HO router. Which IPv6 address configuration must be applied to the router fa0/1 interface for the router to assign a unique 64-bit IPv6 address to itself?

- A. ipv6 address 2001:DB8:0:1::C601:42FE:800F::7/64
- B. ipv6 address 2001:DB8:0:1::FFFF::C601:420F::7/64
- C. ipv6 address 2001:DB8:0:1::C601:42FF:FE0F::7/64
- D. ipv6 address 2001:DB8:0:1::FE80::C601:420F::7/64

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 403

A network administrator must enable DHCP services between two sites. What must be configured for the router to pass DHCPDISCOVER messages on to the server?

- A. DHCP Snooping
- B. a DHCP Pool
- C. DHCP Binding
- D. a DHCP Relay Agent

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 404

Which condition must be met before an NMS handles an SNMP trap from an agent?

- A. The NMS software must be loaded with the MIB associated with the trap.
- B. The NMS must receive the same trap from two different SNMP agents to verify that it is reliable.
- C. The NMS must receive a trap and an inform message from the SNMP agent within a configured interval.
- D. The NMS must be configured on the same router as the SNMP agent.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 405

What is a benefit of using private IPv4 addressing?

- A. Communication to the internet is reachable without the use of NAT.
- B. Direct connectivity is provided to internal hosts from outside an enterprise network.
- C. All external hosts are provided with secure communication to the internet.
- D. Multiple companies can use the same addresses without conflicts.

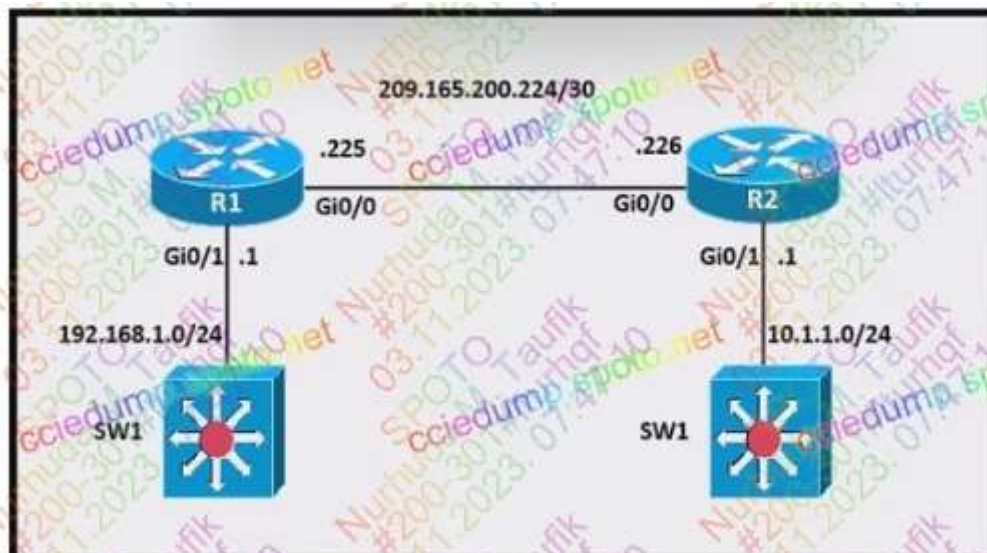
Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 406



Refer to the exhibit. A network engineer is in the process of establishing IP connectivity between two sites. Routers R1 and R2 are partially configured with IP addressing. Both routers have the ability to access devices on their respective LANs. Which command set configures the IP connectivity between devices located on both LANs in each site?

- A. R1
ip route 192.168.1.0 255.255.255.0 GigabitEthernet0/0
R2
ip route 10.1.1.1 255.255.255.0 GigabitEthernet0/0
- B. R1
ip route 192.168.1.1 255.255.255.0 GigabitEthernet0/1
R2
ip route 10.1.1.1 255.255.255.0 GigabitEthernet0/1
- C. R1
ip route 0.0.0.0 0.0.0.0 209.165.200.225
R2
ip route 0.0.0.0 0.0.0.0 209.165.200.226
- D. R1
ip route 0.0.0.0 0.0.0.0 209.165.200.226
R2
ip route 0.0.0.0 0.0.0.0 209.165.200.225

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 407

Which alternative to password authentication is implemented to allow enterprise devices to log in to the corporate network?

- A. digital certificates
- B. magic links
- C. 90-day renewal policie
- D. sone-time passwords

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 408

```
R19#sh int fa0/0
FastEthernet0/0 is up, line protocol is up
Hardware is DEC21140, address is ca02.7788.0000 (bia ca02.7788.0000)
Description: SALES_SUBNET
Internet address is 10.32.102.2/30
MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (60 sec)
Full-duplex, 100Mb/s, 100BaseTX/FR
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:01, output 00:00:00, output hang never
Last clearing of "show interface" counters never
Input queue: 0/300/0/0 (size/max/drops/flushes); Total output drops:
135298429
```



```

Queueing strategy: fifo
Output queue: 0/300 (size/max)
30 second input rate 0 bits/sec, 0 packets/sec
30 second output rate 0 bits/sec, 0 packets/sec
73310 packets input, 7101162 bytes
Received 73115 broadcasts (0 IP multicasts)
0 runs, 0 giants, 0 throttles
0 input errors, 4 CRC, 0 frame, 0 overrun, 0 ignored
0 watchdog
0 input packets with dribble condition detected
3927513096455 packets output, 14404034810952 bytes, 0 underruns
0 output errors, 11 collisions, 0 interface resets
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out

```

Refer to the exhibit. What is the cause of poor performance on router R19?

- A. excessive collisions
- B. port oversubscription
- C. speed and duplex mismatch
- D. excessive CRC errors

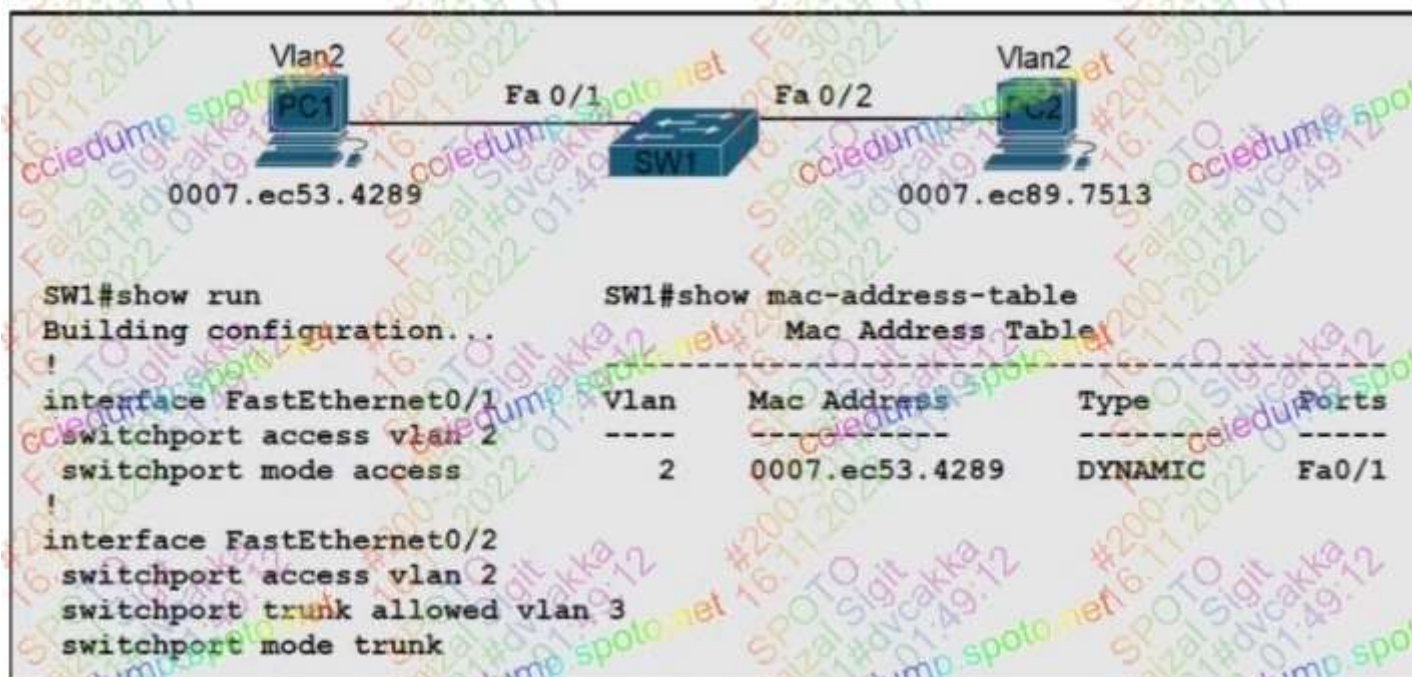
Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 409



Refer to the exhibit. An engineer has started to configure replacement switch SW1. To verify part of the configuration, the engineer issued the commands as shown and noticed that the entry for PC2 is missing. Which change must be applied to SW1 so that PC1 and PC2 communicate normally? .

- A. SW1(config)#interface fa0/1
SW1(config-if)#no switchport access vlan 2
SW1(config-if)#switchport trunk native vlan 2
SW1(config-if)#switchport trunk allowed vlan 3
- B. SW1(config)#interface fa0/1
SW1(config-if)#no switchport access vlan 2
SW1(config-if)#switchport access vlan 3
SW1(config-if)#switchport trunk allowed vlan 2
- C. SW1(config)#interface fa0/2
SW1(config-if)#no switchport mode trunk
SW1(config-if)#no switchport trunk allowed vlan 3
SW1(config-if)#switchport mode access
- D. SW1(config)#interface fa0/2
SW1(config-if)#no switchport access vlan 2
SW1(config-if)#no switchport trunk allowed vlan 3
SW1(config-if)#switchport trunk allowed vlan 2

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 410

How do UTP and STP cables compare?

- A. UTP cables provide faster and more reliable data transfer rates and STP cables are slower and less reliable.
- B. STP cables are shielded and protect against electromagnetic interference and UTP lacks the same protection against electromagnetic interference.
- C. STP cables are cheaper to procure and easier to install and UTP cables are more expensive and harder to install.
- D. UTP cables are less prone to crosstalk and interference and STP cables are more prone to crosstalk and interference.

Correct Answer: B
 Section: (none)
 Explanation

Explanation/Reference:

QUESTION 411

```
SW1#show run
Building configuration...
!
hostname SW1
!
ip domain-name CCNA-test
!
username CCNA privilege 1 password 0 cisco123
!
interface FastEthernet0/1
 switchport access vlan 10
!
interface Vlan10
 ip address 192.168.1.2 255.255.255.0
!
line vty 0 4
 login local
 transport input telnet
line vty 5 15
 login local
 transport input telnet

SW1#show crypto key mypubkey rsa
* Key pair was generated at: 0:1:23 UTC Mar 1 2020
Key name: SW1.CCNA-test
```

Refer to the exhibit. An engineer is updating the management access configuration of switch SW1 to allow secured, encrypted remote configuration. Which two commands or command sequences must the engineer apply to the switch? (Choose two.)

- A. SW1(config)# crypto key generate rsa
- B. SW1(config)# interface f0/1
SW1(config-if)# switchport mode trunk
- C. SW1(config)# enable secret ccna Test123
- D. SW1(config)# line vty 0 15
SW1(config-line)# transport input ssh
- E. SW1(config)# username NEW secret R3mote123

Correct Answer: CD
 Section: (none)
 Explanation

Explanation/Reference:

QUESTION 412



Refer to the exhibit. Which switch in this configuration will be elected as the root bridge?

```
SW1: 0C:0E:15:XX:XX:XX
SW2: 0C:C4:XX:XX:XX:XX
SW3: 0C:0E:18:XX:XX:XX
SW4: 0C:0C:XX:XX:XX:XX
```

- A. SW1
- B. SW2
- C. SW4
- D. SW3

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 413

What is an advantage of Cisco DNA Center versus traditional campus device management?

- A. It is designed primarily to provide network assurance.
- B. It enables easy autodiscovery of network elements in a brownfield deployment.
- C. It supports numerous extensibility options, including cross-domain adapters and third-party SDKs.
- D. It supports high availability for management functions when operating in cluster mode.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 414

What is an enhancement implemented in WPA3?

- A. applies 802.1X authentication and AES-128 encryption
- B. uses TKIP and per-packet keying
- C. defends against deauthentication and disassociation attacks
- D. employs PKI and RADIUS to identify access points

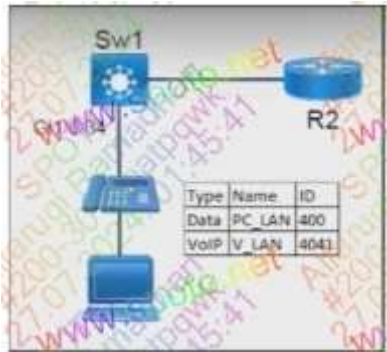
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 415



Refer to the exhibit. Network services must be enabled on interface Gi1/0/34. Which configuration meets the needs for this implementation?

- A.

```
interface Gi1/0/34
switchport mode access
switchport access vlan 4041
switchport voice vlan 400
```
- B.

```
interface Gi1/0/34
switchport mode trunk
switchport trunk allowed vlan 400, 4041
switchport voice vlan 4041
```
- C.

```
interface Gi1/0/34
switchport mode trunk
switchport trunk allowed native vlan 400
switchport voice vlan 4041
```
- D.

```
interface Gi1/0/34
switchport mode access
switchport access vlan 400
switchport voice vlan 4041
```

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 416

Which cable type must be used when connecting two like devices together using these criteria?

1. Pins 1 to 3 and 2 to 6 are required
2. Auto detection MDI-X is unavailable

- A. crossover
- B. straight-through
- C. rollover
- D. console

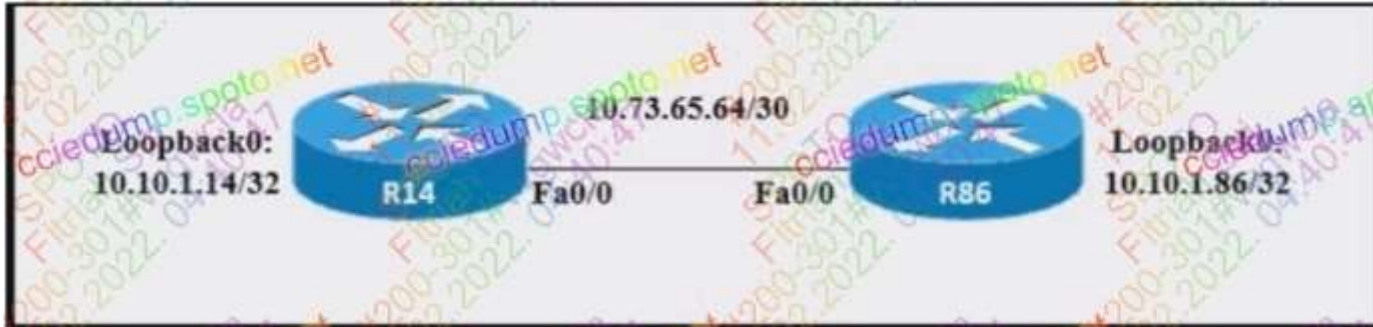
Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 417



Refer to the exhibit. All interfaces are configured with duplex auto and ip ospf network broadcast. Which configuration allows routers R14 and R86 to form an OSPFv2 adjacency while acting as a central point for exchanging OSPF information between routers?

- A.
- ```

R14#
interface FastEthernet0/0
ip address 10.73.65.65 255.255.255.252
ip ospf network broadcast
ip ospf priority0
ip mtu 1400

router ospf 10
router-id 10.10.1.14
network 10.10.1.14 0.0.0.0 area 0
network 10.73.65.64 0.0.0.3 area 0

R86#
interface Loopback0
ip address 10.10.1.86 255.255.255.255

interface FastEthernet0/0
ip address 10.73.65.66 255.255.255.252
ip ospf network broadcast
ip mtu 1500

router ospf 10
router-id 10.10.1.86
network 10.10.1.86 0.0.0.0 area 0
network 10.73.65.64 0.0.0.3 area 0

```



B. R14#  
interface FastEthernet0/0  
ip address 10.73.65.65 255.255.255.252  
ip ospf network broadcast  
ip ospf priority 255  
ip mtu 1500  
  
router ospf 10  
router-id 10.10.1.14  
network 10.10.1.14 0.0.0.0 area 0  
network 10.73.65.64 0.0.0.3 area 0

R86#  
interface FastEthernet0/0  
ip address 10.73.65.66 255.255.255.252  
ip ospf network broadcast  
ip mtu 1500  
  
router ospf 10  
router-id 10.10.1.86  
network 10.10.1.86 0.0.0.0 area 0  
network 10.73.65.64 0.0.0.3 area 0

C. R14#  
interface Loopback0  
ip ospf 10 area 0  
  
interface FastEthernet0/0  
ip address 10.73.65.65 255.255.255.252  
ip ospf network broadcast  
ip ospf 10 area 0  
ip mtu 1500  
  
router ospf 10  
ip ospf priority 255  
router-id 10.10.1.14  
  
R86#  
interface Loopback0  
ip ospf 10 area 0  
  
interface FastEthernet0/0  
ip address 10.73.65.66 255.255.255.252  
ip ospf network broadcast  
ip ospf 10 area 0  
ip mtu 1500  
  
router ospf 10  
router-id 10.10.1.86

D. R14#  
interface FastEthernet0/0  
ip address 10.73.65.65 255.255.255.252  
ip ospf network broadcast  
ip ospf priority 255  
ip mtu 1500  
  
router ospf 10  
router-id 10.10.1.14  
network 10.10.1.14 0.0.0.0 area 0  
network 10.73.65.64 0.0.0.3 area 0  
  
R86#  
interface FastEthernet0/0  
ip address 10.73.65.66 255.255.255.252  
ip ospf network broadcast  
ip mtu 1400  
  
router ospf 10  
router-id 10.10.1.86  
network 10.10.1.86 0.0.0.0 area 0  
network 10.73.65.64 0.0.0.3 area 0

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 418

Which action is taken by the data plane within a network device?

- A. constructs a routing table based on a routing protocol
- B. provides CLI access to the network device
- C. looks up an egress interface in the forwarding information base
- D. forwards traffic to the next hop

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**



Exam B

QUESTION 1

Drag and drop the statements about AAA services from the left to the corresponding AAA services on the right. Not all options are used.

Select and Place:

|                                                               | Authentication |
|---------------------------------------------------------------|----------------|
| It grants access to network assets, such as FTP servers.      |                |
| It restricts the CLI commands that a user is able to perform. |                |
| It performs user validation Via TACACS+.                      |                |
| It records the duration of each connection.                   |                |
| It supports User Access Reporting.                            |                |
| It verifies "who you are".                                    |                |

Correct Answer:

|                                             | Authentication                           |
|---------------------------------------------|------------------------------------------|
|                                             | It performs user validation Via TACACS+. |
|                                             | It verifies "who you are".               |
|                                             |                                          |
|                                             |                                          |
| It records the duration of each connection. |                                          |
| It supports User Access Reporting.          |                                          |
|                                             |                                          |

|  | Authorization                                                 |
|--|---------------------------------------------------------------|
|  | It grants access to network assets, such as FTP servers.      |
|  | It restricts the CLI commands that a user is able to perform. |

Section: (none)

Explanation

Explanation/Reference:

QUESTION 2

Drag and drop the Cisco Wireless LAN Controller security settings from the left onto the correct security mechanism categories on the right.

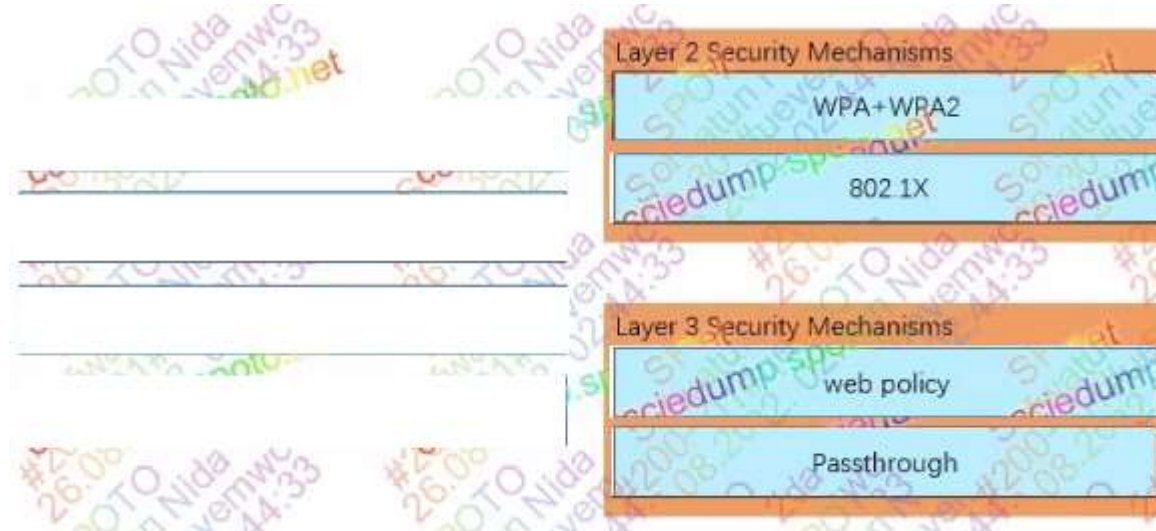
Select and Place:

|             | Layer 2 Security Mechanisms |
|-------------|-----------------------------|
| web policy  |                             |
| Passthrough |                             |
| WPA+WPA2    |                             |
| 802.1X      |                             |

|  | Layer 3 Security Mechanisms |
|--|-----------------------------|
|  |                             |
|  |                             |

Correct Answer:





Section: (none)  
Explanation

Explanation/Reference:

### QUESTION 3

```
R1# show ip route | begin gateway
Gateway of last resort is not set
172.16.0.0/16 is variably subnetted, 5 subnets, 5 masks
O 172.16.2.128/25 [110/3184437] via 207.165.200.250, 00:00:25, Serial0/0/0
O 172.16.3.64/27 [110/3184437] via 207.165.200.250, 00:00:25, Serial0/0/0
O 172.16.3.128/28 [110/3184437] via 207.165.200.250, 00:00:25, Serial0/0/0
O 172.16.3.192/29 [110/3184437] via 207.165.200.250, 00:00:25, Serial0/0/0
O 172.16.4.0/23 [110/3184437] via 207.165.200.250, 00:00:25, Serial0/0/0
207.165.200.0/24 is variably subnetted, 4 subnets, 2 masks
C 207.165.200.248/30 is directly connected, Serial0/0/0
L 207.165.200.249/32 is directly connected, Serial0/0/0
C 207.165.200.252/30 is directly connected, Serial0/0/1
L 207.165.200.253/32 is directly connected, Serial0/0/1
```

Refer to the exhibit. Drag and drop the learned prefixes from the left onto the subnet masks on the right.

Select and Place:

|              |                 |
|--------------|-----------------|
| 172.16.3.128 | 255.255.254.0   |
| 172.16.3.64  | 255.255.255.128 |
| 172.16.2.128 | 255.255.255.224 |
| 172.16.3.192 | 255.255.255.240 |
| 172.16.4.0   | 255.255.255.248 |

Correct Answer:

|  |              |
|--|--------------|
|  | 172.16.4.0   |
|  | 172.16.2.128 |
|  | 172.16.3.64  |
|  | 172.16.3.128 |
|  | 172.16.3.192 |

Section: (none)  
Explanation



Explanation/Reference:

QUESTION 4

```
Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0
S* 209.165.200.0/27 is subnetted, 1 subnets
S 209.165.200.224 [20/0] via 10.10.12.2, 06:09:59
B 209.165.201.0/27 is subnetted, 1 subnets
B 209.165.201.0 [20/0] via 10.10.12.2, 05:13:18
B 209.165.202.0/27 is subnetted, 1 subnets
B 209.165.202.128 [20/0] via 10.10.12.2, 05:12:48
C 10.0.0.0/8 is variably subnetted, 9 subnets, 4 masks
C 10.10.10.0/28 is directly connected, GigabitEthernet0/0
C 10.10.11.0/30 is directly connected, FastEthernet2/0
C 10.10.12.0/30 is directly connected, GigabitEthernet0/1
O 10.10.13.0/25 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O 10.10.13.128/28 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O 10.10.13.144/28 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O 10.10.13.160/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O 10.10.13.208/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O 10.10.13.252/30 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
S* 0.0.0.0/0 [1/0] via 10.10.11.2
```

Refer to the exhibit. Drag and drop the subnet masks from the left onto the corresponding subnets on the right. Not all subnet masks are used.

Select and Place:

|                 |              |
|-----------------|--------------|
| 255.255.248.0   | 10.10.13.0   |
| 255.255.255.128 | 10.10.13.128 |
| 255.255.255.224 | 10.10.13.160 |
| 255.255.255.240 | 10.10.13.252 |
| 255.255.255.248 |              |
| 255.255.255.252 |              |

Correct Answer:

|                 |                 |
|-----------------|-----------------|
| 255.255.248.0   | 255.255.255.128 |
|                 | 255.255.255.240 |
| 255.255.255.224 | 255.255.255.248 |
|                 | 255.255.255.252 |
|                 |                 |
|                 |                 |

Section: (none)  
Explanation

Explanation/Reference:

QUESTION 5

Drag and drop the statements about networking from the left onto the corresponding networking types on the right.

Select and Place:



|                                                                                         |
|-----------------------------------------------------------------------------------------|
| This type allows better control over how networks work and how networks are configured. |
| This type enables networks to integrate with applications through APIs.                 |
| New devices are configured using the physical infrastructure.                           |
| This type provisions resources from a centralized location.                             |
| This type requires a distributed control plane.                                         |

| Traditional Networking      |
|-----------------------------|
|                             |
|                             |
| Controller-Based Networking |
|                             |
|                             |
|                             |

Correct Answer:

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |

| Traditional Networking                                                                  |
|-----------------------------------------------------------------------------------------|
| New devices are configured using the physical infrastructure.                           |
| This type requires a distributed control plane.                                         |
| Controller-Based Networking                                                             |
| This type allows better control over how networks work and how networks are configured. |
| This type enables networks to integrate with applications through APIs.                 |
| This type provisions resources from a centralized location.                             |

Section: (none)  
Explanation

Explanation/Reference:

QUESTION 6



```

C:\>ipconfig/all

Windows IP Configuration

Host Name : Inspiron15
Primary Dns Suffix :
Node Type : Mixed
IP Routing Enabled. : No
WINS Proxy Enabled. : No

Wireless LAN adapter Local Area Connection* 12:

Media State : Media disconnected
Connection-specific DNS Suffix :
Description : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. : 1A-76-3F-7C-57-DF
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix :
Description : Dell Wireless 1703 802.11b/g/n (2.4GHz)
Physical Address. : B8-76-3F-7C-57-DF
DHCP Enabled. : No
Autoconfiguration Enabled : Yes
Link-local IPv6 Address : fe80::e09f:9839:6e86:f755%12(Preferred)
IPv4 Address. : 192.168.1.20(Preferred)
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1
DHCPv6 IAID : 263742135
DHCPv6 Client DUID. : 00-01-00-01-18-E6-32-43-B8-76-3F-7C-57-DF

NetBIOS over Tcpip. : Enabled

```

Refer to the exhibit. An engineer is tasked with verifying network configuration parameters on a client workstation to report back to the team lead. Drag and drop the node identifiers from the left onto the network parameters on the right. Not all values are used.

Select and Place:

|                   |                                          |
|-------------------|------------------------------------------|
| 192.168.1.1       | Broadcast address                        |
| 192.168.1.20      | Default gateway                          |
| 192.168.1.254     | Host IP address                          |
| 192.168.1.255     | Fast assignable IP address in the subnet |
| B8-76-3F-7C-57-DF | MAC address                              |

Correct Answer:

|  |                   |
|--|-------------------|
|  | 192.168.1.255     |
|  | 192.168.1.1       |
|  | 192.168.1.20      |
|  | 192.168.1.254     |
|  | B8-76-3F-7C-57-DF |

Section: (none)  
Explanation

Explanation/Reference:

QUESTION 7

Drag and drop the management connection types from the left onto the definitions on the right

Select and Place:



|         |                                                                         |
|---------|-------------------------------------------------------------------------|
| console | supports clear-text connections to the controller CLI                   |
| HTTPS   | supports encrypted access to CLI and a secure channel for data transfer |
| SSH     | supports physical connections over a serial cable                       |
| Telnet  | supports secure web access for management of the device                 |

Correct Answer:

|  |         |
|--|---------|
|  | Telnet  |
|  | SSH     |
|  | console |
|  | HTTPS   |

Section: (none)  
Explanation

Explanation/Reference:

#### QUESTION 8

Drag and drop the statements about networking from the left onto the corresponding networking types on the right.

Select and Place:

|                                                                  |                             |
|------------------------------------------------------------------|-----------------------------|
| This type implements changes individually at each device.        | Traditional Networking      |
| This type leverages controllers to handle network management.    |                             |
| Maintenance costs are higher than with other networking options. | Controller-Based Networking |
| This type provides a centralized view of the network.            |                             |

Correct Answer:

|  |                                                                  |
|--|------------------------------------------------------------------|
|  | Traditional Networking                                           |
|  | This type implements changes individually at each device.        |
|  | Maintenance costs are higher than with other networking options. |
|  | Controller-Based Networking                                      |
|  | This type leverages controllers to handle network management.    |
|  | This type provides a centralized view of the network.            |

Section: (none)  
Explanation



Explanation/Reference:

QUESTION 9

Drag and drop the AAA terms from the left onto the descriptions on the right.

Select and Place:

|                |                            |
|----------------|----------------------------|
| accounting     | tracks activity            |
| authentication | updates session attributes |
| authorization  | verifies access rights     |
| CoA            | verifies identity          |

Correct Answer:

|  |                |
|--|----------------|
|  | accounting     |
|  | CoA            |
|  | authorization  |
|  | authentication |

Section: (none)  
Explanation

Explanation/Reference:

QUESTION 10

Drag and drop the TCP or UDP details from the left onto their corresponding protocols on the right.

Select and Place:

|                                                           |     |
|-----------------------------------------------------------|-----|
| used to reliably share files between devices              | TCP |
| appropriate for streaming operations with minimal latency |     |
| provides best-effort service                              |     |
| supports reliable data transmission                       | UDP |
|                                                           |     |

Correct Answer:



|  |                                                           |
|--|-----------------------------------------------------------|
|  | TCP                                                       |
|  | used to reliably share files between devices              |
|  | supports reliable data transmission                       |
|  | UDP                                                       |
|  | appropriate for streaming operations with minimal latency |
|  | provides best-effort service                              |

Section: (none)

Explanation

Explanation/Reference:

#### QUESTION 11

Drag and drop the descriptions of AAA services from the left onto the corresponding services on the right

Select and Place:

|                                          |                |
|------------------------------------------|----------------|
| allows the user to change to enable mode | Accounting     |
| limits the user's access permissions     |                |
| logs session statistics                  |                |
| records user commands                    | Authentication |
| secures access to routers                |                |
| validates user credentials               | Authorization  |
|                                          |                |

Correct Answer:



|  |                                          |
|--|------------------------------------------|
|  | Accounting                               |
|  | logs session statistics                  |
|  | records user commands                    |
|  | Authentication                           |
|  | secures access to routers                |
|  | validates user credentials               |
|  | Authorization                            |
|  | allows the user to change to enable mode |
|  | limits the user's access permissions     |

Section: (none)

Explanation

Explanation/Reference:

#### QUESTION 12

Drag and drop the characteristics of network architectures from the left onto the type of architecture on the right.

Select and Place:

|                                                             |                |
|-------------------------------------------------------------|----------------|
| single device handles the core and the distribution layer   | Collapsed Core |
| enhances network availability                               |                |
| more cost-effective than other options                      |                |
| most appropriate for small network designs                  | Three-Tier     |
| separate devices handle the core and the distribution layer |                |
|                                                             |                |

Correct Answer:



|  |                                                             |
|--|-------------------------------------------------------------|
|  | Collapsed Core                                              |
|  | single device handles the core and the distribution layer   |
|  | more cost-effective than other options                      |
|  | most appropriate for small network designs                  |
|  | Three-Tire                                                  |
|  | enhances network availability                               |
|  | separate devices handle the core and the distribution layer |

Section: (none)  
Explanation

Explanation/Reference:

### QUESTION 13

Drag and drop the DHCP snooping terms from the left onto the descriptions on the right

Select and Place:

|                           |                                                                            |
|---------------------------|----------------------------------------------------------------------------|
| DHCP server               | list of hosts on the network that are unknown to the administrative domain |
| snooping binding database | network component that propagates IP addresses to hosts on the network     |
| spurious DHCP server      | internal device under the control of the network administrator             |
| trusted                   | unknown DHCP server within an administrative domain                        |
| untrusted                 | default state of all interfaces                                            |

Correct Answer:

|  |                           |
|--|---------------------------|
|  | snooping binding database |
|  | DHCP server               |
|  | trusted                   |
|  | spurious DHCP server      |
|  | untrusted                 |

Section: (none)  
Explanation

Explanation/Reference:

### QUESTION 14

Drag and drop the characteristics of northbound APIs from the left onto any position on the right. Not all characteristics are used.

Select and Place:



|                                                                   |   |
|-------------------------------------------------------------------|---|
| supports automation                                               | 1 |
| communicates between the SDN controller and the application plane | 2 |
| communicates between the SDN controller and the data plane        | 3 |
| supports data sharing between systems                             | 4 |
| supports network virtualization protocols                         |   |
| supports REST-based requirements                                  |   |
| uses OpenFlow to interface between the data and control planes    |   |

Correct Answer:

|                                                                |                                                                   |
|----------------------------------------------------------------|-------------------------------------------------------------------|
|                                                                | supports automation                                               |
|                                                                | communicates between the SDN controller and the application plane |
| communicates between the SDN controller and the data plane     | supports data sharing between systems                             |
|                                                                | supports REST-based requirements                                  |
| supports network virtualization protocols                      |                                                                   |
|                                                                |                                                                   |
| uses OpenFlow to interface between the data and control planes |                                                                   |

Section: (none)  
Explanation

Explanation/Reference:

#### QUESTION 15

Drag and drop the device behaviors from the left onto the matching HSRP state on the right.

Select and Place:

|                                                                                      |         |
|--------------------------------------------------------------------------------------|---------|
| has heard from the neighbor device and is receiving hello packets                    | Active  |
| is forwarding packets                                                                | Learn   |
| is ready to forward packets if the device that is currently forwarding packets fails | Listen  |
| is transmitting and receiving hello packets                                          | Speak   |
| is waiting to hear from the neighbor device                                          | Standby |

Correct Answer:



|  |                                                                                      |
|--|--------------------------------------------------------------------------------------|
|  | is forwarding packets                                                                |
|  | has heard from the neighbor device and is receiving hello packets                    |
|  | is waiting to hear from the neighbor device                                          |
|  | is transmitting and receiving hello packets                                          |
|  | is ready to forward packets if the device that is currently forwarding packets fails |

Section: (none)  
Explanation

Explanation/Reference:

#### QUESTION 16

Drag the descriptions of device management from the left onto the types of device management on the right.

Select and Place:

|                                                                                                                       |                                    |
|-----------------------------------------------------------------------------------------------------------------------|------------------------------------|
| implements changes via an SSH terminal                                                                                | Cisco DNA Center Device Management |
| manages device configurations on a per-device basis                                                                   |                                    |
| monitors the cloud for software updates                                                                               |                                    |
| security is managed near the perimeter of the network with firewalls, VPNs, and IPS                                   | Traditional Device Management      |
| uses CLI templates to apply a consistent configuration to multiple devices at an individual location                  |                                    |
| uses NetFlow to analyze potential security threats throughout the network and take appropriate action on that traffic |                                    |

Correct Answer:

|  |                                                                                                                       |
|--|-----------------------------------------------------------------------------------------------------------------------|
|  | Cisco DNA Center Device Management                                                                                    |
|  | monitors the cloud for software updates                                                                               |
|  | uses CLI templates to apply a consistent configuration to multiple devices at an individual location                  |
|  | uses NetFlow to analyze potential security threats throughout the network and take appropriate action on that traffic |
|  | Traditional Device Management                                                                                         |
|  | implements changes via an SSH terminal                                                                                |
|  | manages device configurations on a per-device basis                                                                   |
|  | security is managed near the perimeter of the network with firewalls, VPNs, and IPS                                   |

Section: (none)  
Explanation

Explanation/Reference:

#### QUESTION 17

Drag and drop the IPv6 address type characteristics from the left to the right.



Select and Place:

|                                                                  |                      |
|------------------------------------------------------------------|----------------------|
| addresses with prefix FC00::7                                    | Link-Local Address   |
| configured only once per interface                               |                      |
| attached to a single subnet                                      |                      |
| addressing for exclusive use internally without Internet routing | Unique Local Address |
|                                                                  |                      |
|                                                                  |                      |

Correct Answer:

|  |                                                                  |
|--|------------------------------------------------------------------|
|  | Link-Local Address                                               |
|  | configured only once per interface                               |
|  | attached to a single subnet                                      |
|  | Unique Local Address                                             |
|  | addresses with prefix FC00::7                                    |
|  | addressing for exclusive use internally without Internet routing |

Section: (none)  
Explanation

Explanation/Reference:

#### QUESTION 18

An engineer is configuring an encrypted password for the **enable** command on a router where the local user database has already been configured. Drag and drop the configuration commands from the left into the correct sequence on the right. Not all commands are used.

Select and Place:

|                             |        |
|-----------------------------|--------|
| configure terminal          | first  |
| enable                      | second |
| enable secret sh1l@4fs      | third  |
| exit                        | fourth |
| line vty 0 4                |        |
| service password-encryption |        |

Correct Answer:



enable

configure terminal

enable secret shfl@4fs

exit

line vty 0 4

service password-encryption

Section: (none)

Explanation

Explanation/Reference:

#### QUESTION 19

```

R1# show ip route | begin gateway
Gateway of last resort is not set
172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
192.168.1.0/24 is directly connected, FastEthernet0/0
172.16.1.1/32 is directly connected, FastEthernet0/0
172.16.2.0/24 [120/2] via 207.165.200.250, 00:00:25, Serial0/0/0
192.168.1.0/24 [110/84437] via 207.165.200.254, 00:00:17, Serial0/0/1
192.168.2.0/24 [90/3184437] via 207.165.200.254, 00:00:15, Serial0/0/1
207.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
207.165.200.244/30 [1/1] via 207.165.200.254, Serial0/0/1
207.165.200.248/30 is directly connected, Serial0/0/0
207.165.200.249/32 is directly connected, Serial0/0/0
207.165.200.252/30 is directly connected, Serial0/0/1
207.165.200.253/32 is directly connected, Serial0/0/1

```

Refer to the exhibit. Drag and drop the learned prefixes from the left onto the preferred route methods from which they were learned on the right.

Select and Place:

172.16.2.0/24

192.168.1.0/24

192.168.2.0/24

207.165.200.244/30

207.165.200.248/30

static

EIGRP

OSPF

RIP

Correct Answer:

207.165.200.244/30

192.168.2.0/24

192.168.1.0/24

172.16.2.0/24

207.165.200.248/30

Section: (none)

Explanation

Explanation/Reference:

#### QUESTION 20



Drag and drop the statements about networking from the left onto the corresponding networking types on the right. Not all statements are used.

Select and Place:

This type deploys a consistent configuration across multiple devices.

A distributed control plane is needed.

This type requires a distributed management plane.

Southbound APIs are used to apply configurations.

Northbound APIs interact with end devices.

Controller-Based Networking

Traditional Networking

Correct Answer:

Northbound APIs interact with end devices.

Controller-Based Networking

This type deploys a consistent configuration across multiple devices.

Southbound APIs are used to apply configurations.

Traditional Networking

This type requires a distributed management plane.

A distributed control plane is needed.

Section: (none)  
Explanation

Explanation/Reference:

QUESTION 21  
Drag and drop the Ansible features from the left to the right. Not all features are used.

Select and Place:

executes modules via SSH by default

uses the YAML language

uses agents to manage hosts

pushes configurations to the client

requires clients to pull configurations from the server

operates without agents

feature

feature

feature

feature

Correct Answer:



|                                                         |                                     |
|---------------------------------------------------------|-------------------------------------|
|                                                         | executes modules via SSH by default |
|                                                         | uses the YAML language              |
| uses agents to manage hosts                             | pushes configurations to the client |
|                                                         | operates without agents             |
| requires clients to pull configurations from the server |                                     |
|                                                         |                                     |

Section: (none)

Explanation

Explanation/Reference:

#### QUESTION 22

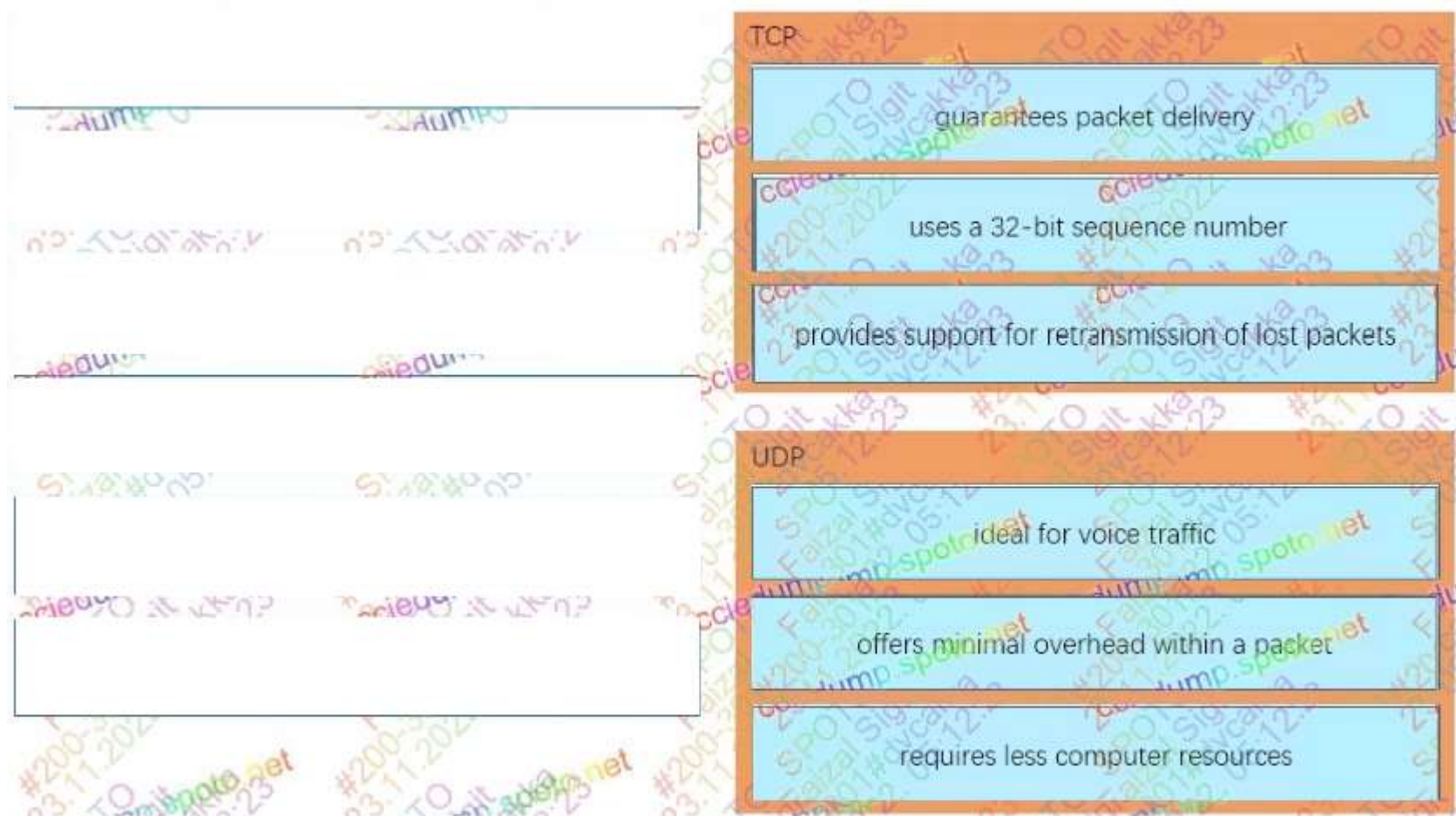
Drag and drop the characteristics of transport layer protocols from the left onto the corresponding protocols on the right.

Select and Place:

|                                                     |     |
|-----------------------------------------------------|-----|
| guarantees packet delivery                          | TCP |
| uses a 32-bit sequence number                       |     |
| ideal for voice traffic                             |     |
| provides support for retransmission of lost packets |     |
| offers minimal overhead within a packet             | UDP |
| requires less computer resources                    |     |

Correct Answer:





Section: (none)  
Explanation

Explanation/Reference:

#### QUESTION 23

Drag and drop the IPv6 addresses from the left onto the corresponding address types on the right

Select and Place:

|                                      |                    |
|--------------------------------------|--------------------|
| 2001:db8:600d:cafe::123              | Global Unicast     |
| fcba:926a:e8e:7a25:b1:c6d2:1a76:8fdc | Link-Local Unicast |
| fe80:a00:27ff:feeb:89aa              | Multicast          |
| ff05::1:3                            | Unique Local       |

Correct Answer:



|  |                                      |
|--|--------------------------------------|
|  | Global Unicast                       |
|  | 2001:db8:600d:cafe::123              |
|  | Link-Local Unicast                   |
|  | fe80:a00:27ff:feeb:89aa              |
|  | Multicast                            |
|  | ff05::1:3                            |
|  | Unique Local                         |
|  | fcba:926a:e8e:7a25:b1:c6d2:1a76:8fdc |

Section: (none)

Explanation

Explanation/Reference:

#### QUESTION 24

Drag and drop the attack-mitigation techniques from the left onto the types of attack that they mitigate on the right.

Select and Place:

|                                                     |                                           |
|-----------------------------------------------------|-------------------------------------------|
| configure 802.1x authentication                     | 802.1q double-tagging VLAN-hopping attack |
| configure DHCP snooping                             | MAC flooding attack                       |
| configure the native VLAN with a nondefault VLAN ID | man-in-the-middle spoofing attack         |
| disable DTP                                         | switch-spoofing VLAN-hopping attack       |

Correct Answer:

|  |                                                     |
|--|-----------------------------------------------------|
|  | configure the native VLAN with a nondefault VLAN ID |
|  | configure 802.1x authentication                     |
|  | configure DHCP snooping                             |
|  | disable DTP                                         |

Section: (none)

Explanation

Explanation/Reference:



## Exam C

### QUESTION 1

#### Guidelines

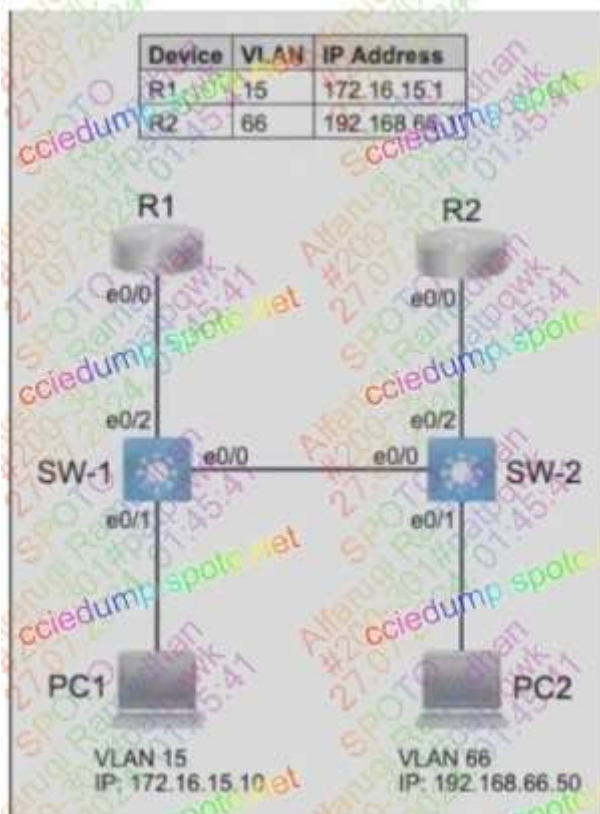
This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened.

#### Tasks

R1 and R2 are pre-configured with all the necessary commands. All physical cabling is in place and verified. Connectivity for PC1 and PC2 must be established to the switches; each port must only allow one VLAN and be operational.

1. Configure SW-1 with VLAN 15 and label it exactly as OPS
2. Configure SW-2 with VLAN 66 and label it exactly as ENGINEERING
3. Configure the switch port connecting to PC1
4. Configure the switch port connecting to PC2
5. Configure the E0/2 connections on SW-1 and SW-2 for neighbor discovery using the vendor-neutral standard protocol and ensure that e0/0 on both switches uses the Cisco proprietary protocol



A. SW-1:

```
SW-1>en
SW-1#conf t
SW-1(config)#lldp run
SW-1(config)#vlan 15
SW-1(config-vlan)#name OPS
SW-1(config-vlan)#int e0/1
SW-1(config-if)#switchport mode access
SW-1(config-if)#switchport access vlan 15
SW-1(config)#int e0/2
SW-1(config-if)#no cdp enable
SW-1(config-if)#lldp transmit
SW-1(config-if)#lldp receive
SW-1(config-if)#int e0/0
SW-1(config-if)#switchport trunk encapsulation isl
SW-1(config-if)#switchport mode trunk
SW-1(config-if)#end
SW-1#wr
```

```
SW-2:
SW-2(config)#lldp run
SW-2(config)#vlan 66
SW-2(config-vlan)#name ENGINEERING
SW-2(config-vlan)#int e0/1
SW-2(config-if)#switchport mode access
SW-2(config-if)#switchport access vlan 66
SW-2(config)#int e0/2
SW-2(config-if)#no cdp enable
SW-2(config-if)#lldp transmit
SW-2(config-if)#lldp receive
SW-2(config-if)#int e0/0
SW-2(config-if)#switchport trunk encapsulation isl
SW-2(config-if)#switchport mode trunk
SW-2(config-if)#end
SW-2#wr
```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:



## QUESTION 2

### Guidelines

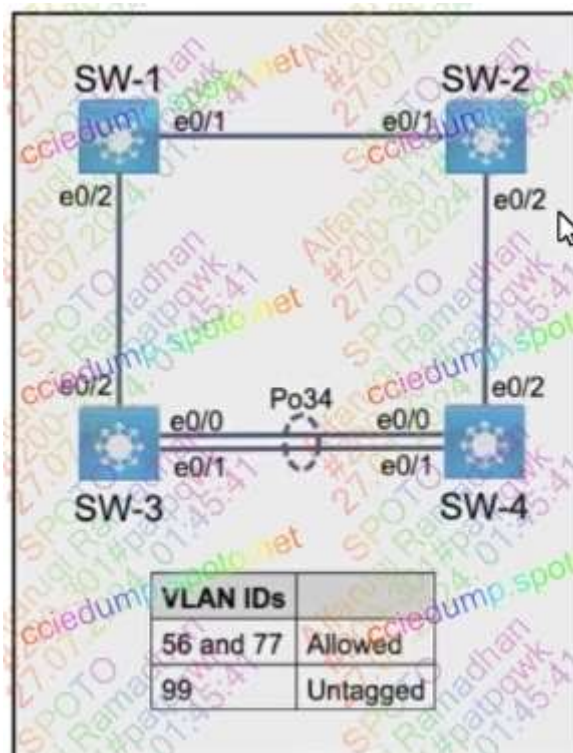
This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened.

### Tasks

All physical cabling is in place and verified. Connectivity between all four switches must be established and operational. All ports are pre-configured as 802.1q trunks.

1. Configure both SW-1 and SW-2 ports e0/1 and e0/2 to permit only the allowed VLANs.
2. Configure both SW-3 and SW-4 ports e0/2 to permit only the allowed VLANs.
3. Configure both SW-1 and SW-2 e0/1 ports to send and receive untagged traffic over VLAN 99.
4. Configure both SW-3 and SW-4 ports e0/0 and e0/1 for link aggregation using the industry standard protocol. All ports must immediately negotiate the link aggregation.
5. Permit only the allowed VLANs on the new link.



A.

SW-1:

```
SW-1>en
SW-1#config t
SW-1(config)#int range e0/1,e0/2
SW-1(config-if)#switchport trunk allowed vlan 56,77,99
SW-1(config-if)#exit
SW-1(config)#int e0/1
SW-1(config-if)#switchport trunk native vlan 99
SW-1(config-if)#end
SW-1#wr
```

SW-2:

```
SW-2>en
SW-2#config t
SW-2(config)#int range e0/1,e0/2
SW-2(config-if)#switchport trunk allowed vlan 56,77,99
SW-2(config-if)#exit
SW-2(config)#int e0/1
SW-2(config-if)#switchport trunk native vlan 99
SW-2(config-if)#end
SW-2#wr
```

SW-3:

```
SW-3>en
SW-3#config t
SW-3(config)#int range e0/0,e0/1
SW-3(config-if-range)#channel-group 34 mode active
SW-3(config-if-range)#exit
SW-3(config)#int range e0/2,p034
SW-3(config-if)#switchport trunk allowed vlan 56,77,99
SW-3(config-if)#end
SW-3#wr
```

SW-4:

```
SW-4>en
SW-4#config t
SW-4(config)#int range e0/0,e0/1
SW-4(config-if-range)#channel-group 34 mode active
SW-4(config-if-range)#exit
SW-4(config)#int range e0/2,p034
SW-4(config-if)#switchport trunk allowed vlan 56,77,99
SW-4(config-if)#end
SW-4#wr
```



Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

### QUESTION 3

#### Guidelines

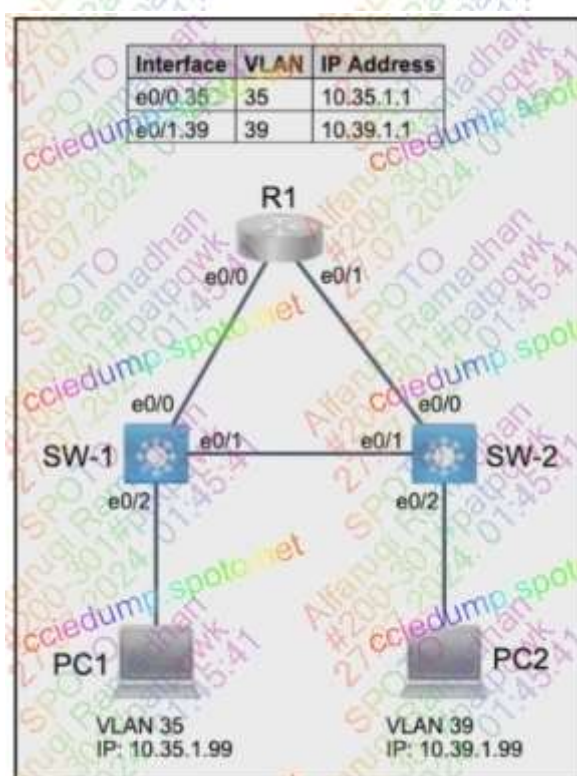
This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened

#### Tasks

R1 has been pre-configured with all the necessary commands. All physical cabling is in place and verified. Connectivity for PC1 and PC2 must be established to the switches, and each port must only allow one VLAN.

1. Configure SW-1 with VLAN 35 and label it exactly as SALES
2. Configure SW-2 with VLAN 39 and label it exactly as MARKETING
3. Configure the switch port connecting to PC1
4. Configure the switch port connecting to PC2
5. Configure SW-1 and SW-2 for universal neighbor discovery using the industry standard protocol and disable it on the interface connecting to PC1



A. SW-1:

```
SW-1>en
SW-1#config t
SW-1(config)#no cdp run
SW-1(config)#lldp run
SW-1(config)#vlan 35
SW-1(config-vlan)#name SALES
SW-1(config-vlan)#int e0/2
SW-1(config-if)#switchport mode access
SW-1(config-if)#switchport access vlan 35
SW-1(config-if)#no lldp transmit
SW-1(config-if)#no lldp receive
SW-1(config-if)#end
SW-1#wr

SW-2:
SW-2(config)#no cdp run
SW-2(config)#lldp run
SW-2(config)#vlan 39
SW-2(config-vlan)#name MARKETING
SW-2(config-vlan)#int e0/2
SW-2(config-if)#switchport mode access
SW-2(config-if)#switchport access vlan 39
SW-2(config-if)#end
SW-2#wr
```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

### QUESTION 4



### Guidelines

This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened

### Tasks

Refer to the topology. All physical cabling is in place. Configure local user account, configure a Named ACL (NACL), and Dynamic Arp Inspection.

1. Configure a local account on Sw3 with telnet access only on virtual ports 0-4. Use the following information:

Username: tech12

Password: load1key

Algorithm type: md5

Privilege level: Exec mode

2. Configure and apply a NACL on R1 to control network traffic towards ISP:

Name: ISP\_ACL

Restrict RFC 1918 class A and B addresses

Allow all other addresses

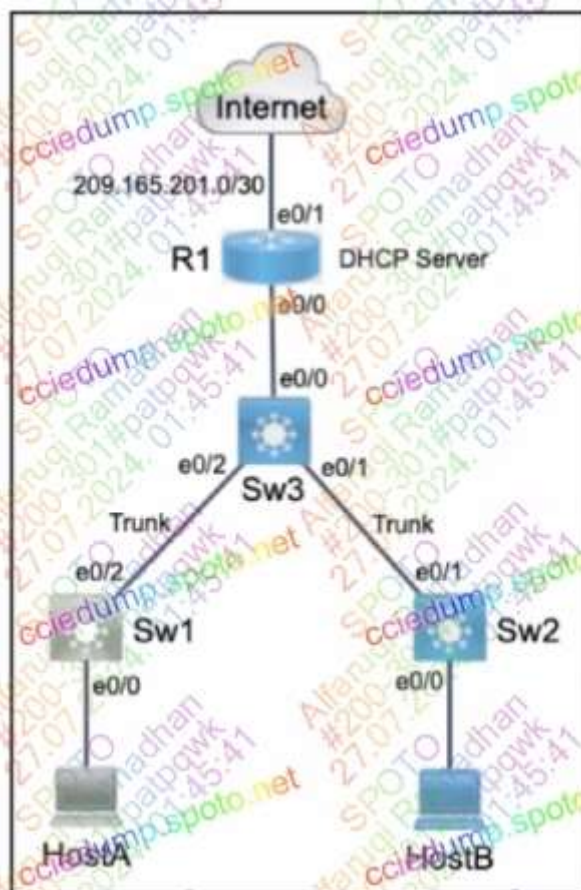
3. A DHCP IP Pool is preconfigured on R1 for VLAN 5, and DHCP Snooping is configured on Sw2. Configure on Sw2:

Dynamic Arp Inspection for VLAN 5

Enable validation of the ARP packet destination MAC address

Enable validation of the ARP packet source MAC address

Enable validation of the ARP Packet IP address



A. R1:

```
R1>en
R1#config t
R1(config)#ip access-list extended ISP_ACL
R1(config-ext-nacl)#deny ip 10.0.0.0 0.255.255.255 any
R1(config-ext-nacl)#deny ip 172.16.0.0 0.15.255.255 any
R1(config-ext-nacl)#permit ip any any
R1(config-ext-nacl)#exit
R1(config)#int e0/1
R1(config-if)#ip access-group ISP_ACL out
R1(config-if)#end
R1#wr
```

Sw2:

```
Sw2>en
Sw2#config t
Sw2(config)#ip arp inspection vlan 5
Sw2(config)#ip arp inspection validate ip src-mac dst-mac
Sw2(config)#end
Sw2#wr
```

Sw3:

```
Sw3>en
Sw3#config t
Sw3(config)#username tech12 privilege 15 algorithm-type md5 secret load1key
Sw3(config)#line vty 0 4
Sw3(config-line)#login local
Sw3(config-line)#transport input telnet
Sw3(config-if-range)#end
Sw3#wr
```



Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

#### QUESTION 5

##### Guidelines

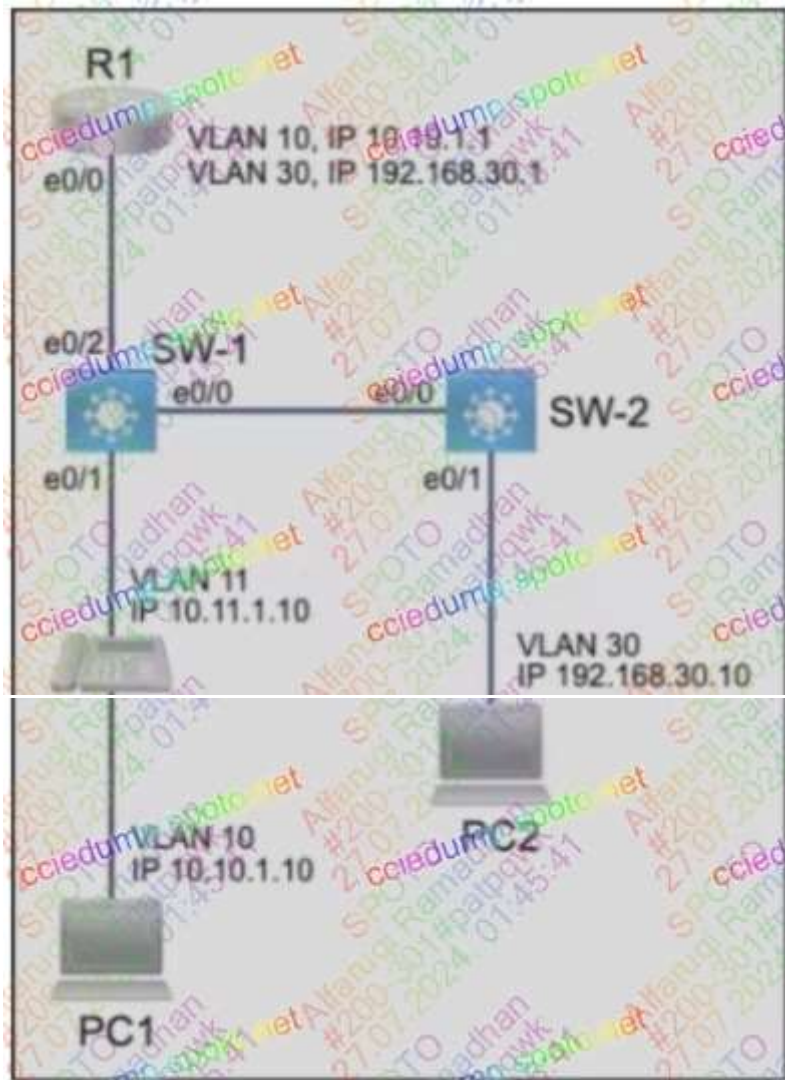
This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened.

##### Tasks

R1 has been pre-configured with all the necessary commands. All physical cabling is in place and verified. Connectivity to the end devices must be configured.

1. Configure SW-1 switch port E0/1 to carry traffic for the Cisco IP phone and PC2.
2. Configure SW-2 E0/1 to carry traffic for PC2.
3. Configure VLAN 10 with the name "Engineering" on SW-1.
4. Configure the link between SW-1 and SW-2 to use the vendor neutral neighbor discovery protocol
5. Configure the link on SW-1 to R1 so that it does not allow the Cisco neighbor discovery protocol to pass



A. SW-1:

```
SW-1>en
SW-1#config t
SW-1(config)#vlan 10
SW-1(config-vlan)#name Engineering
SW-1(config-vlan)#int e0/0
SW-1(config-if)#lldp transmit
SW-1(config-if)#lldp receive
SW-1(config-if)#int e0/1
SW-1(config-if)#switchport mode access
SW-1(config-if)#switchport access vlan 10
SW-1(config-if)#switchport voice vlan 11
SW-1(config-if)#int e0/2
SW-1(config-if)#no cdp enable
SW-1(config-if)#end
SW-1#wr
```

SW-2:

```
SW-2>en
SW-2#config t
SW-2(config)#int e0/0
SW-2(config-if)#lldp transmit
SW-2(config-if)#lldp receive
SW-2(config-if)#int e0/1
SW-2(config-if)#switchport mode access
SW-2(config-if)#switchport access vlan 30
SW-2(config-if)#end
SW-2#wr
```



Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

#### QUESTION 6

##### Guidelines

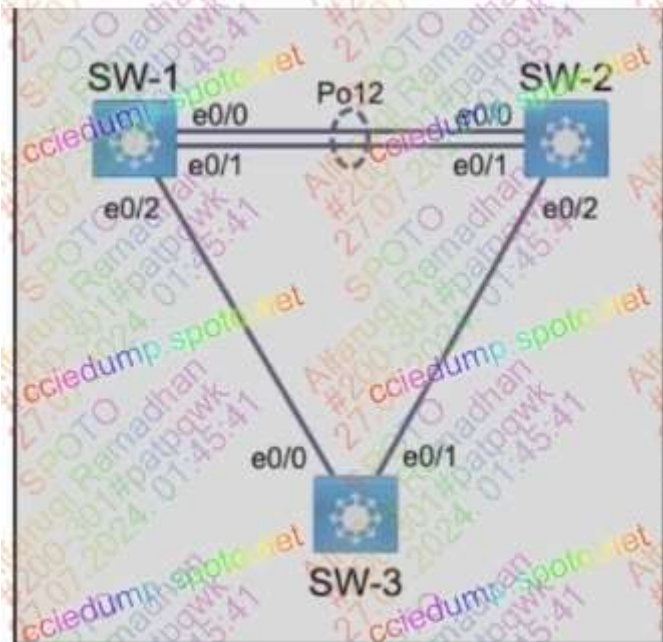
This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened

##### Tasks

VLANs 35 and 45 have been configured in all three switches. All physical connectivity has been installed and verified. All inter-switch links must be operational.

1. Configure SW-1 and SW-2 switch ports e0/0 and e0/1 for 802.1q trunking allowing all VLANs.
2. Configure the inter-switch links on SW-1 e0/2, SW-2 e0/2, and SW-3 e0/0 and e0/1 to use native VLAN 35.
3. Configure SW-1 and SW-2 switch ports e0/0 and e0/1 for link aggregation. SW-1 should immediately negotiate LACP and SW-2 must only respond to LACP requests.



A. SW1:

```
SW1>en
SW1#config t
SW1(config)#int range e0/0-2
SW1(config-if-range)#switchport trunk encapsulation dot1q
SW1(config-if-range)#switchport mode trunk
SW1(config-if-range)#int range e0/0-1
SW1(config-if-range)#channel-group 12 mode active
SW1(config-if-range)#int e0/2
SW1(config-if)#switchport trunk native vlan 35
SW1(config-if)#end
SW1wr
```

SW2:

```
SW2>en
SW2#config t
SW2(config)#int range e0/0-2
SW2(config-if-range)#switchport trunk encapsulation dot1q
SW2(config-if-range)#switchport mode trunk
SW2(config-if-range)#int range e0/0-1
SW2(config-if-range)#channel-group 12 mode passive
SW2(config-if-range)#int e0/2
SW2(config-if)#switchport trunk native vlan 35
SW2(config-if)#end
SW2wr
```

SW3:

```
SW3>en
SW3#config t
SW3(config)#int range e0/0-1
SW3(config-if-range)#switchport trunk encapsulation dot1q
SW3(config-if-range)#switchport mode trunk
SW3(config-if)#switchport trunk native vlan 35
SW3(config-if-range)#end
SW3#wr
```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

#### QUESTION 7



### Guidelines

This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened

### Tasks

All physical cabling is in place and verified. Router R1 is configured and passing traffic for VLANs 5 and 6. All relevant ports are pre-configured as 802.1q trunks.

1. Configure SW-1 port E0/0 to permit only VLANs 5 and 6
2. Configure both SW-1 and SW-2's E0/1 ports to send and receive untagged traffic over VLAN 77
3. Configure SW-2 E0/2 port to permit only VLAN 6
4. Configure both SW-3 and SW-4 ports e0/0 and e0/1 for link aggregation using the industry standard protocol with the following requirements:
  - SW-3 ports must immediately negotiate the aggregation protocol
  - SW-4 ports must not initiate the negotiation for the aggregation protocol
  - Use the designated number assignment



A. SW-1:

```
SW-1>en
SW-1#config t
SW-1(config)#int e0/0
SW-1(config-if)#switchport trunk allowed vlan 5,6
SW-1(config-if)#int e0/1
SW-1(config-if)#switchport trunk native vlan 77
SW-1(config-if)#end
SW-1#wr
```

SW-2:

```
SW-2>en
SW-2#config t
SW-2(config)#int e0/1
SW-2(config-if)#switchport trunk native vlan 77
SW-2(config-if)#int e0/2
SW-2(config-if)#switchport trunk allowed vlan 6
SW-2(config-if)#end
SW-2#wr
```

SW-3:

```
SW-3>en
SW-3#config t
SW-3(config)#int range e0/0-1
SW-3(config-if-range)#channel-group 34 mode active
SW-3(config-if-range)#end
SW-3#wr
```

SW-4:

```
SW-4>en
SW-4#config t
SW-4(config)#int range e0/0-1
SW-4(config-if-range)#channel-group 34 mode passive
SW-4(config-if-range)#end
SW-4#wr
```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 8



### Guidelines

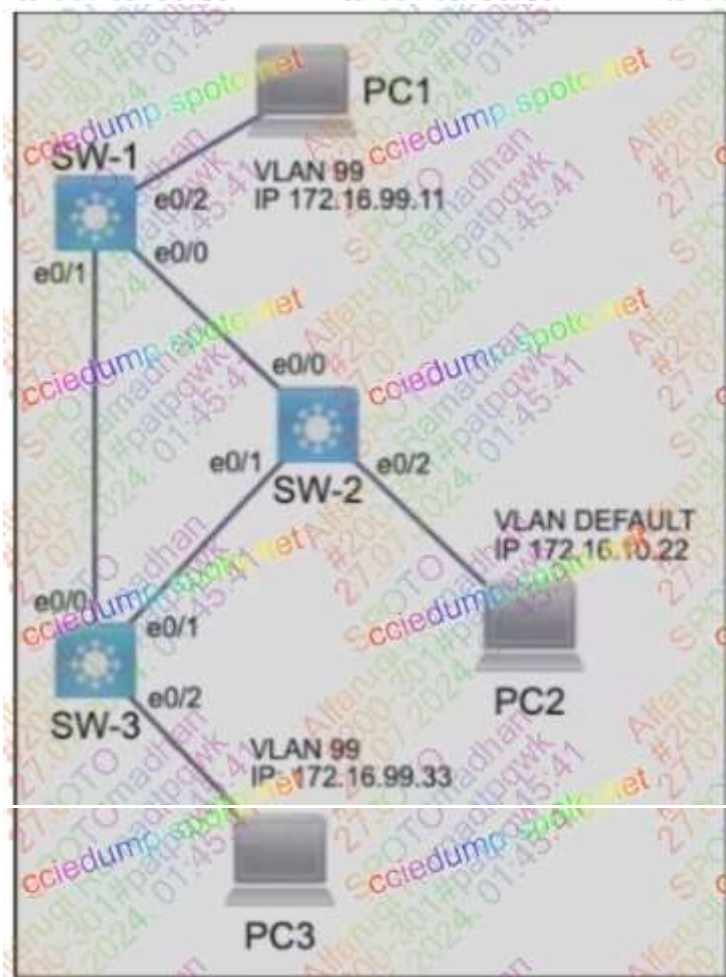
This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened.

### Tasks

All physical cabling is in place and verified. Connectivity for PC1, PC2 and PC3 must be established to the switches. Each port connecting to the PCs must be configured as an end-user port and only allow the designated VLAN.

1. Configure VLAN 99 on all three switches and label it exactly as FINANCIAL
2. Configure the switch ports connecting to PC1, PC2 and PC3
3. Cisco's neighbor discovery protocol has been disabled on SW-1 and must be re-enabled
4. PC1 must not be able to discover SW-1



A. SW-1:

```
SW-1>en
SW-1#config t
SW-1(config)#cdp run
SW-1(config)#vlan 99
SW-1(config-vlan)#name FINANCIAL
SW-1(config-vlan)#exit
SW-1(config-if)#int range e0/0-1
SW-1(config-if-range)#switchport trunk encapsulation dot1q
SW-1(config-if-range)#switchport mode trunk
SW-1(config-if-range)#no shutdown
SW-1(config-if-range)#int e0/2
SW-1(config-if)#switchport mode access
SW-1(config-if)#switchport access vlan 99
SW-1(config-if)#no cdp enable
SW-1(config-if)#no shutdown
SW-1(config-if)#end
SW-1#wr
```

SW-2:

```
SW-2>en
SW-2#config t
SW-2(config)#vlan 99
SW-2(config-vlan)#name FINANCIAL
SW-2(config-vlan)#exit
SW-2(config-if)#int range e0/0-1
SW-2(config-if-range)#switchport trunk encapsulation dot1q
SW-2(config-if-range)#switchport mode trunk
SW-2(config-if-range)#no shutdown
SW-2(config-if-range)#int e0/2
SW-2(config-if)#switchport mode access
SW-2(config-if)#no shutdown
SW-2(config-if)#end
SW-2#wr
```



```

SW-3
SW-3>en
SW-3#config t
SW-3(config)#vlan 99
SW-3(config-vlan)#name FINANCIAL
SW-3(config-vlan)#exit
SW-3(config-if)#int range e0/0-1
SW-3(config-if-range)#switchport trunk encapsulation dot1q
SW-3(config-if-range)#switchport mode trunk
SW-3(config-if-range)#no shutdown
SW-3(config-if-range)#int e0/2
SW-3(config-if)#switchport mode access
SW-3(config-if)#switchport access vlan 99
SW-3(config-if)#no shutdown
SW-3(config-if)#end
SW-3#wr

```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

#### QUESTION 9

##### Guidelines

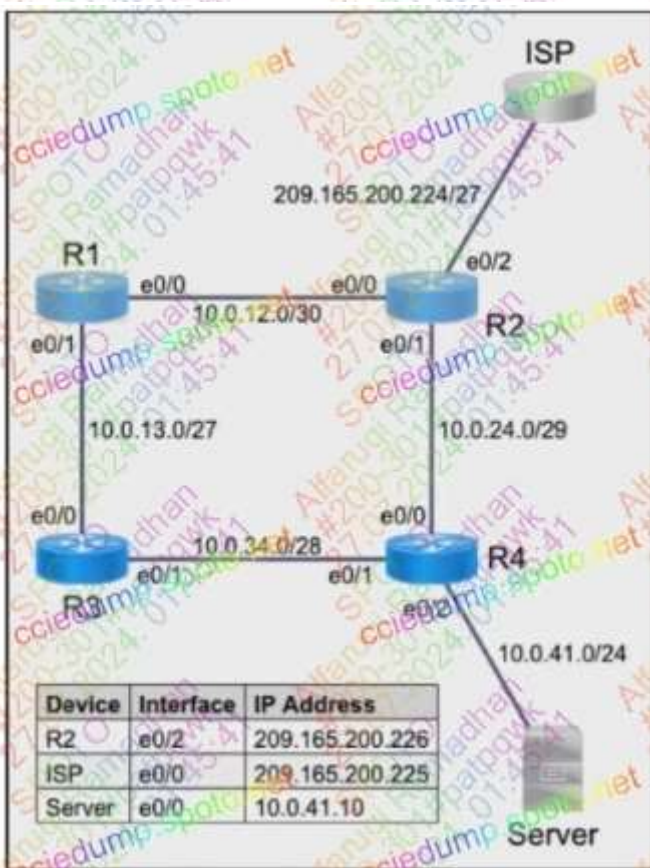
This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened.

##### Tasks

Refer to the topology. All physical cabling is in place. Routers R3 and R4 are fully configured and inaccessible. Configure static routes for various connectivity to the ISP and the LAN, which resides on R4.

1. Configure a default route on R2 to the ISP
2. Configure a default route on R1 to the ISP
3. Configure R2 with a route to the Server at 10.0.41.10
4. Configure R1 with a route to the LAN that prefers R3 as the primary path to the LAN



A.

```

R1:
R1>en
R1#config t
R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.200.225
R1(config)#ip route 10.0.41.0 255.255.255.0 e0/1
R1(config)#end
R1#wr

R2:
R2>en
R2#config t
R2(config)#ip route 0.0.0.0 0.0.0.0 209.165.200.225
R2(config)#ip route 10.0.41.0 255.255.255.0 e0/1
R2(config)#end
R2#wr

```

Correct Answer: A



Section: (none)

Explanation

Explanation/Reference:

#### QUESTION 10

##### Guidelines

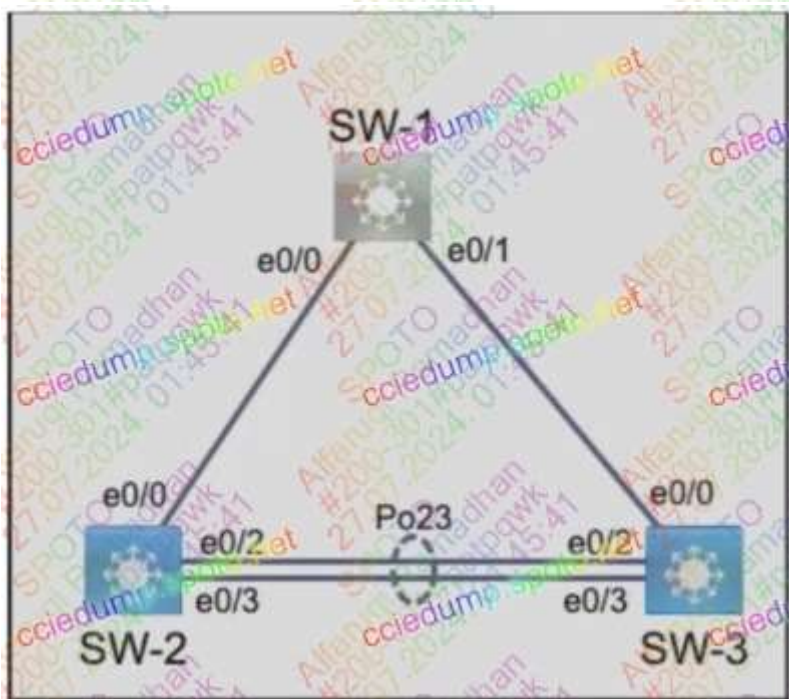
This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened

##### Tasks

All physical cabling is in place and verified. Switch SW-1 is pre-configured and inaccessible. SW-2 and SW-3 ports must be configured and operational to complete the configuration.

1. Configure SW-2 and SW-3 ports E0/0 to use the industry standard encapsulation method for trunking and only tag VLAN 10
2. Configure SW-2 and SW-3 ports E0/0 to send and receive untagged traffic over VLAN 11
3. Configure SW-2 and SW-3 ports E0/2 and E0/3 to use the industry standard encapsulation method for trunking and tag all VLANS
4. Configure SW-2 and SW-3 ports E0/2 and E0/3 for link aggregation using the industry standard protocol with the following requirements:
  - SW-2 ports must not initiate the negotiation for the aggregation protocol
  - SW-3 ports must immediately negotiate the aggregation protocol
  - Use the designated number assignment



A. SW-2:

```
SW-2>en
SW-2#conf t
SW-2(config)#int range e0/0,e0/2-3
SW-2(config-if-range)#switchport trunk encapsulation dot1q
SW-2(config-if-range)#switchport mode trunk
SW-2(config-if-range)#int e0/0
SW-2(config-if)#switchport trunk allowed vlan 10
SW-2(config-if)#switchport trunk native vlan 11
SW-2(config-if)#int range e0/2-3
SW-2(config-if-range)#channel-group 23 mode passive
SW-2(config-if-range)#end
SW-2#wr
```

SW-3:

```
SW-3>en
SW-3#conf t
SW-3(config)#int range e0/0,e0/2-3
SW-3(config-if-range)#switchport trunk encapsulation dot1q
SW-3(config-if-range)#switchport mode trunk
SW-3(config-if-range)#int e0/0
SW-3(config-if)#switchport trunk allowed vlan 10
SW-3(config-if)#switchport trunk native vlan 11
SW-3(config-if)#int range e0/2-3
SW-3(config-if-range)#channel-group 23 mode active
SW-3(config-if-range)#end
SW-3#wr
```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

#### QUESTION 11



### Guidelines

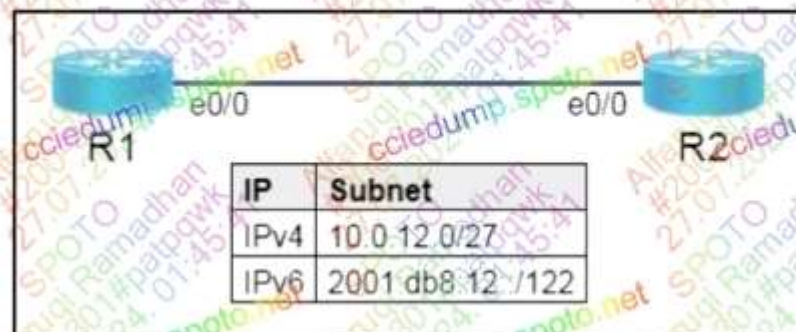
This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened

### Tasks

Physical connectivity between R1 and R2 has been established. IP connectivity must be established between both routers. Ping must be successful between both routers on IPv4 and IPv6.

1. Configure IPv4 Addressing between R1 and R2 using the first usable host IP on R1 and last usable host IP on R2.
2. Configure IPv6 addressing between R1 and R2 using the subnet provided and the X host addressing derived from within IPv4.



A. R1:

```
R1>en
R1#config t
R1(config)#int e0/0
R1(config-if)#ip address 10.0.12.1 255.255.255.224
R1(config-if)#ipv6 enable
R1(config-if)#ipv6 address 2001:db8:12::1/122
R1(config-if)#end
R1#wr
```

R2:

```
R2>en
R2#config t
R2(config)#int e0/0
R2(config-if)#ip address 10.0.12.31 255.255.255.224
R2(config-if)#ipv6 enable
R2(config-if)#ipv6 address 2001:db8:12::31/122
R2(config-if)#end
R2#wr
```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

### QUESTION 12

#### Guidelines

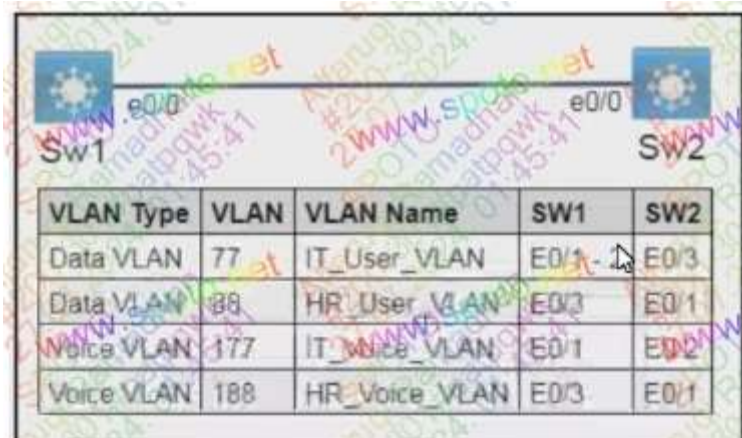
This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened

#### Tasks

All physical cabling is in place and verified. Connectivity for the Switches on ports E0/1, E0/2, and E0/3 must be configured and available for voice and data capabilities.

1. Configure Sw1 and Sw2 with the VLAN naming as indicated.
2. Assign the VLANs to the appropriate interfaces and set a non-trunking, non-tagged, single-VLAN for each interface according to the topology.
3. Configure both switches to use the L2 vendor-neutral discovery protocol to broadcast device information, including the native VLAN across the e0/0 interfaces.





A. Sw1:

```

Sw1>en
Sw1#config t
Sw1(config)#vlan 77
Sw1(config-vlan)#name IT_User_VLAN
Sw1(config-vlan)#vlan 88
Sw1(config-vlan)#name HR_User_VLAN
Sw1(config-vlan)#vlan 177
Sw1(config-vlan)#name IT_Voice_VLAN
Sw1(config-vlan)#vlan 188
Sw1(config-vlan)#name HR_Voice_VLAN
Sw1(config-vlan)#exit
Sw1(config)#int range e0/1-2
Sw1(config-if-range)#switchport mode access
Sw1(config-if-range)#switchport access vlan 77
Sw1(config-if-range)#int e0/1
Sw1(config-if)#switchport voice vlan 177
Sw1(config-if)#int e0/3
Sw1(config-if)#switchport mode access
Sw1(config-if)#switchport access vlan 88
Sw1(config-if)#switchport voice vlan 188
Sw1(config-if)#int e0/0
Sw1(config-if)#lldp receive
Sw1(config-if)#lldp transmit
Sw1(config-if)#end
Sw1#wr

```

Sw2:

```

Sw2>en
Sw2#config t
Sw2(config)#vlan 77
Sw2(config-vlan)#name IT_User_VLAN
Sw2(config-vlan)#vlan 88
Sw2(config-vlan)#name HR_User_VLAN
Sw2(config-vlan)#vlan 177
Sw2(config-vlan)#name IT_Voice_VLAN
Sw2(config-vlan)#vlan 188
Sw2(config-vlan)#name HR_Voice_VLAN
Sw2(config-vlan)#exit
Sw2(config)#int e0/1
Sw2(config-if)#switchport mode access
Sw2(config-if)#switchport access vlan 88
Sw2(config-if)#switchport voice vlan 188
Sw2(config-if)#int e0/2
Sw2(config-if)#switchport mode access
Sw2(config-if)#switchport voice vlan 177
Sw2(config-if)#int e0/3
Sw2(config-if)#switchport mode access
Sw2(config-if)#switchport access vlan 77
Sw2(config-if)#int e0/0
Sw2(config-if)#lldp receive
Sw2(config-if)#lldp transmit
Sw2(config-if)#end
Sw2#wr

```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

### QUESTION 13

#### Guidelines

This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened.

#### Tasks

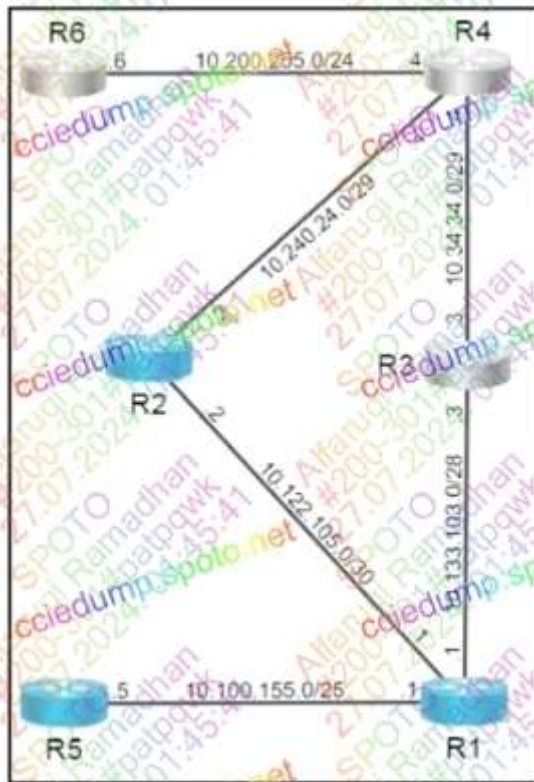
##### Task 1

1. Configure a host route on R5 for the destination of the WAN IP on R6.
2. Configure a static default route on R1 preferring the path through R3 towards R6.
3. From R5, use traceroute and ping to verify the path towards and reachability of R6.

##### Task 2

1. Configure a floating static default route on R1, preferring the path through R2 towards R6 if the link to R3 should fail.
2. Configure the administrative distance for 200.
3. Configure a static route on R2 to forward the return traffic towards 10.100.155.0/25.
4. After shutting interface e0/1 on R1, use traceroute and ping from R5 to verify path towards and reachability of R6.





A.

```

R5>en
R5#config t
R5(config)#ip route 10.200.255.6 255.255.255.255 10.100.155.1
R5(config)#end
R5#wr

R1:

R1>en
R1#config t
R1(config)#ip route 10.200.255.6 255.255.255.255 10.133.103.3
R1(config)#ip route 10.200.255.6 255.255.255.255 10.122.105.2 200
R1(config)#end
R1#wr

R2:

R2>en
R2#config t
R2(config)#ip route 10.100.155.0 255.255.255.128 10.122.105.1
R2(config)#end
R2#wr

```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

#### QUESTION 14

##### Guidelines

This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened

##### Tasks

Refer to the topology. All physical cabling is in place. Routers 2 and 3 are inaccessible. Configure OSPF routing for the network and ensure R1 has joined Area 0 without using network statements.

##### Task 1

1. Configure OSPF on R1 with a process ID and router-ID only as follows:
2. use process ID 33
3. use E0/1 IP as the router ID

##### Task 2

1. Configure R1 to establish neighbor adjacencies with R2 and R3. The network statement under the OSPF process must not be used.
2. Configure R1 to always become the DR for Area 0





A. R1  
 R1>en  
 R1#config t  
 R1(config)#router ospf 33  
 R1(config-router)#router-id 10.0.33.1  
 R1(config-router)#exit  
 R1(config)#int range e0/0-1  
 R1(config-if-range)#ip ospf priority 255  
 R1(config-if-range)#ip ospf 1 area 0  
 R1(config-if-range)#end  
 R1#wr

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

## QUESTION 15

### Guidelines

This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened.

### Tasks

#### Task 1

Configure the trunks between Sw1 and Sw2 on ports E0/0 and E0/1 using the IEEE standard frame tagging method.

1. Only the VLANs for the PCs should be permitted across the trunks.
2. Routers are simulated as PCs and are preconfigured with IP Addresses.
3. PC configurations must remain unchanged.

#### Task 2

On Sw1 and Sw2, use IEEE 802.3ad link aggregation.

1. Assign number 10 to the link.
2. Combine E0/0 and E0/1 into a single logical link.
3. Both links must negotiate aggregation.





A. Sw1/Sw2:

```
Switch>en
Switch#config t
Switch(config)#int range e0/0-1
Switch(config-if-range)#switchport trunk encapsulation dot1q
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#switchport trunk allowed vlan 10,30
Switch(config-if-range)#channel-group 10 mode active
Switch(config-if-range)#end
Switch#wr
```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

## QUESTION 16

### Guidelines

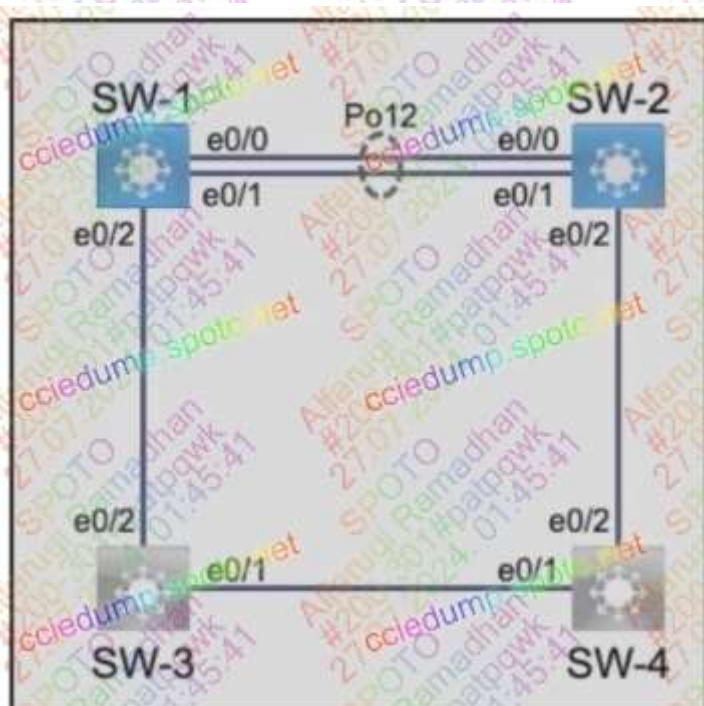
This is a lab item in which tasks will be performed on virtual devices.

1. Refer to the Tasks tab to view the tasks for this lab item.
2. Refer to the Topology tab to access the device console(s) and perform the tasks.
3. Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
4. All necessary preconfigurations have been applied.
5. Do not change the enable password or hostname for any device.
6. **Save your configurations** to NVRAM before moving to the next item.
7. Click **Next** at the bottom of the screen to submit this lab and move to the next question.
8. When **Next** is clicked, the lab closes and cannot be reopened

### Tasks

SW-3 and SW-4 are preconfigured with all necessary commands. All physical cabling is in place and verified. All connectivity must be operational.

1. Configure both SW-1 and SW-2 switch ports e0/0 and e0/1 for 802.1q trunking with only VLANS 1, 12, and 22 permitted
2. Configure SW-1 port e0/2 for 802.1q trunking and include only VLANS 12 and 22
3. Configure both SW-1 and SW-2 switch ports e0/0 and e0/1 for link aggregation using the industry standard protocol. All ports must be configured so that they immediately negotiate the link



A. SW-1:

```
SW-1>en
SW-1#config t
SW-1(config)#int range e0/0-1
SW-1(config-if-range)#switchport trunk encapsulation dot1q
SW-1(config-if-range)#switchport mode trunk
SW-1(config-if-range)#switchport trunk allowed vlan 1,12,22
SW-1(config-if-range)#channel-group 12 mode active
SW-1(config-if-range)#int e0/2
SW-1(config-if)#switchport trunk allowed vlan 12,22
SW-1(config-if)#end
SW-1#wr
```

SW-2:

```
SW-2>en
SW-2#config t
SW-2(config)#int range e0/0-1
SW-2(config-if-range)#switchport trunk encapsulation dot1q
SW-2(config-if-range)#switchport mode trunk
SW-2(config-if-range)#switchport trunk allowed vlan 1,12,22
SW-2(config-if-range)#channel-group 12 mode active
SW-2(config-if-range)#end
SW-2#wr
```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference: